DEPARTMENT OF SCIENCE AND TECHNOLOGY
FOOD AND NUTRITION RESEARCH INSTITUTE

DOST-FNRI

NATIONAL NUTRITION SUMMIT



Halfway Point to 2030: Key Findings of the 2023 National Nutrition Survey

Part 2

JUNE 05, 2025
DUSIT THANI MANILA (MAYUREE 1)

2023 NATIONAL NUTRITION SURVEY

OVERVIEW and METHODOLOGY

LEGAL BASIS

Executive Order 128 Section 22 s. 1987

The DOST-Food and Nutrition Research Institute is **mandated to undertake research** that defines the citizenry's nutritional status, with reference particularly to the malnutrition problem, its causes and effects, and identify alternative solutions to them.

Executive Order 352 s. 1996

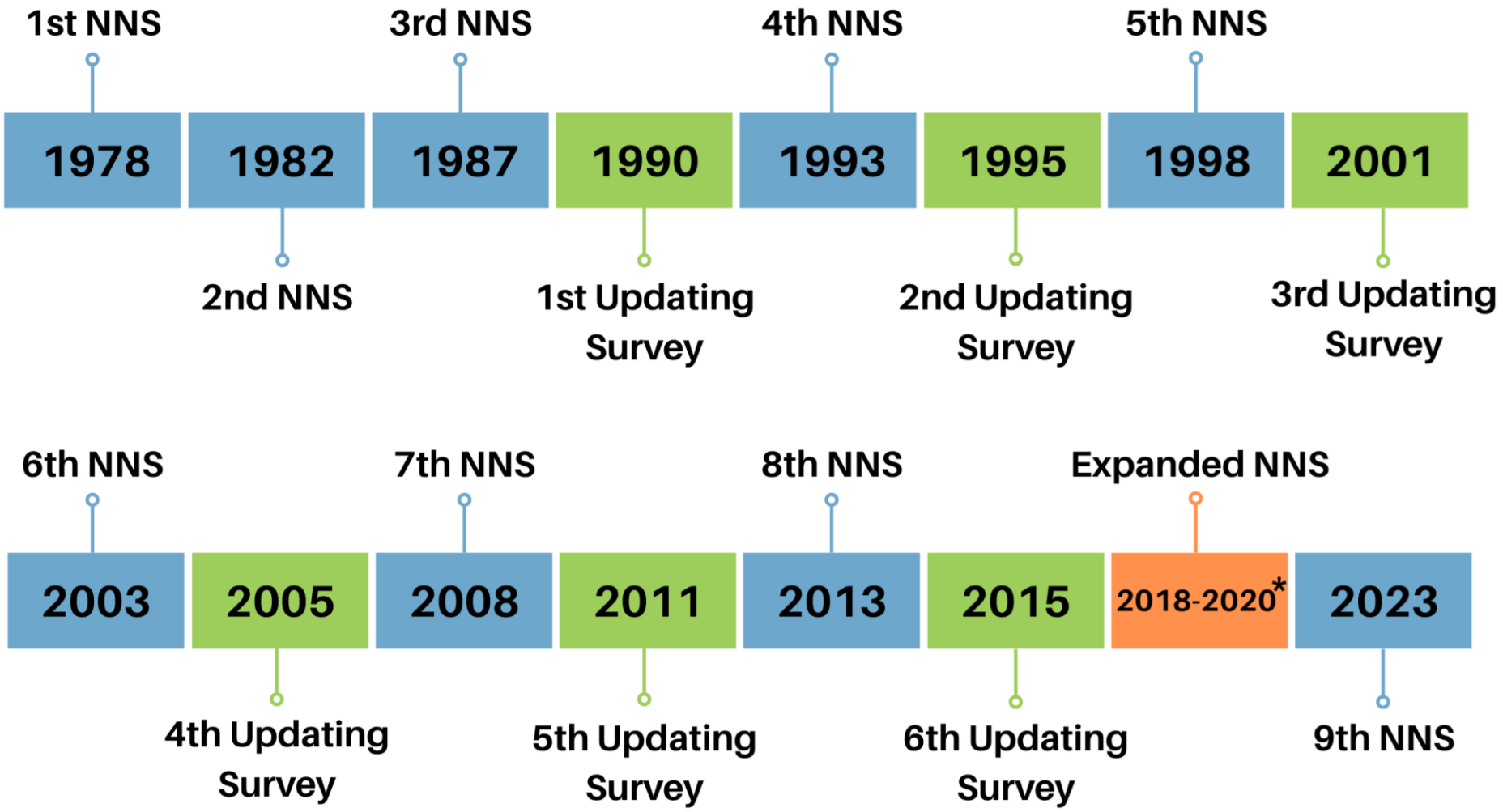
The national nutrition survey is a **designated statistical activity** that will generate critical data for the decision-making of the government and the private sector.

PHILIPPINE STATISTICAL SYSTEM DESIGNATED STATISTICAL ACTIVITIES (Executive Order 352 s. 1996)

- The **National Nutrition Survey (NNS)** is a comprehensive survey conducted every five years

- In-between the NNS, the **Updating of the Nutritional Status of Filipino Children and Other Population Groups (Updating Survey)** is conducted to rapidly assess the nutrition situation of Filipinos.

HISTORICAL BACKGROUND OF THE SURVEYS



*discontinued in 2020 due to the COVID-19 pandemic, continued in 2021

2013 Master Sample (Sampling Design)

Sampling domains:

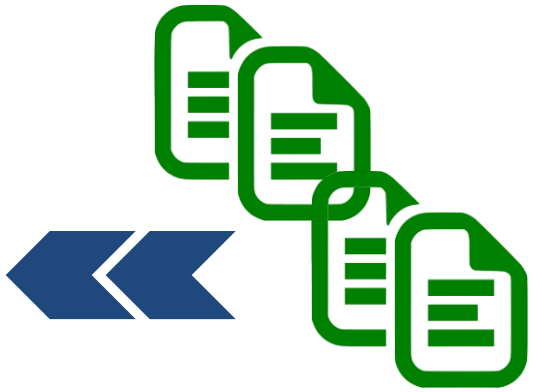
- 81 provinces
- 33 HUCs
- 3 other urban areas
  - Pateros
  - Isabela City
  - Cotabato City

- 100% of HH for Anthropometry, Socio-economic, Food Security, Government Program Participation, IYCF, Maternal, questionnaires of Clinical on NCDs, and Biochemical
- 50% of HH for Dietary



Source: Wikimedia Commons

2-Stage Sampling Design



Four (4) sample replicates are drawn from each domain

1st Stage - PSUs

PSU size ranges from 100 to 400 households

2nd Stage

12 - 16 Households or 384 per Province/HUCs from 4 replicates



DEFINITION:

Primary Sampling Units (PSUs)

- Exhaustive and non-overlapping area segments with about 100 to 400 households
- It can be a barangay/Enumeration Area (EA) or a portion of a large barangay or two or more adjacent small barangays/EAs.

GENERAL OBJECTIVE OF THE 2023 NNS



To provide empirical data on the nutritional and health status of Filipinos for planning nutrition and development programs for timely policy decisions at the **national** and **sub-national/regional** levels.

2023 NNS COMPONENTS



ANTHROPOMETRY



BIOCHEMICAL



DIETARY



CLINICAL AND HEALTH



SOCIO-ECONOMIC



GOVERNMENT PROGRAM PARTICIPATION

3 Components in 1978 NNS

9 Components in 2013 NNS

11 Components in 2023 NNS

FOOD SECURITY



MATERNAL HEALTH AND NUTRITION



INFANT AND YOUNG CHILD FEEDING (IYCF)



FOOD ENVIRONMENT



MENTAL HEALTH AND NUTRITION

2023 NNS COMPONENTS



ANTHROPOMETRY



BIOCHEMICAL



CLINICAL AND HEALTH

FOOD SECURITY



SOCIO-ECONOMIC



GOVERNMENT PROGRAM PARTICIPATION

PART 1 Dissemination

(Held last December 10, 2024)



Home | Facts & Figures | Presentation | Monograph | Public Use File | Data Access Policy | Privacy Policy | About NNS | Contact Us | Infographics

Presentations

Nutrition Surveys

- 2023 National Nutrition Survey
- 2022 Expanded National Nutrition Survey
- 2021 Expanded National Nutrition Survey
- 2020 Rapid Nutrition Assessment Survey
- 2019 Expanded National Nutrition Survey
- 2018 Expanded National Nutrition Survey
- 2015 Updating Survey
- 2013 National Nutrition Survey
- 2008 National Nutrition

https://enutrition.fnri.dost.gov.ph/home.php

2023 NNS - REGIONAL ESTIMATES of SELECTED INDICATORS

2023 NNS - TRENDS AND TRACKING OF NUTRITION TARGETS

2023 NNS - NUTRITIONAL STATUS OF OLDER PERSONS (60 YEARS OLD AND ABOVE)

2023 NNS - NUTRITIONAL STATUS OF ADULTS (20 TO 59 YEARS OLD)

Part 1 Dissemination Results
https://enutrition.fnri.dost.gov.ph/

DOST-FNRI

2023 NNS COMPONENTS



FOOD ENVIRONMENT



DIETARY



MATERNAL HEALTH AND NUTRITION



INFANT AND YOUNG CHILD FEEDING (IYCF)



BIOCHEMICAL



MENTAL HEALTH AND NUTRITION



CLINICAL AND HEALTH

PART 2 Dissemination

FOOD ENVIRONMENT

Assess the household's perception on the dimensions of food environment



Face-to-face interview using e-DCS

DIETARY

Determine the quality and quantity of food and nutrient intake



One-Day Food Weighing (Household Level)



24-hour Food Recall (Individual Level)

MATERNAL HEALTH AND NUTRITION

Determine the health and nutrition of currently pregnant women and their health-seeking behaviors



Face-to-face interview using e-DCS

INFANT AND YOUNG CHILD FEEDING PRACTICES

Update information on the feeding practices of Filipino Children, 0-23 months old, and other related factors affecting feeding practices



Face-to-face interview using e-DCS

BIOCHEMICAL

Assess micronutrient deficiencies such as iodine deficiency, prevalence of vitamin A deficiency, and anemia





Urine Collection

- **Iodine**

Salt Collection

- **Iodine in Salt**

Blood Collection

- **Vitamin A**

CLINICAL AND HEALTH

Assess the prevalence of elevated blood pressure, high fasting blood glucose, and dyslipidemia, as well as behavioral risk factors



Blood Collection

- **Fasting Blood Glucose**
- **Lipids**

MENTAL HEALTH AND NUTRITION

Assess mental well-being and resilience of adolescents (13-17.9 years old) and adults (18 years old and above)

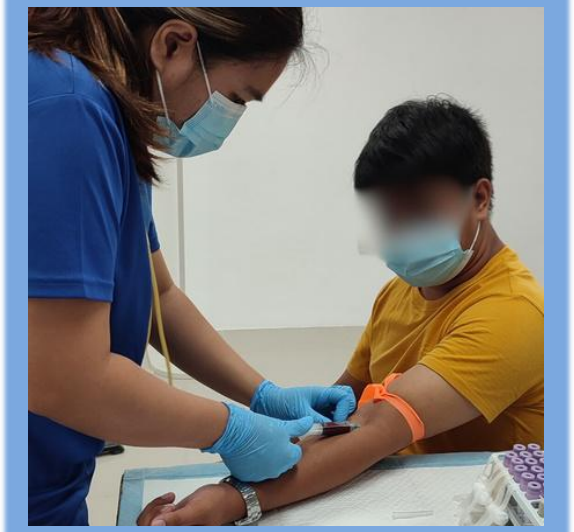


Self-administered paper-based Questionnaire

METHODS OF DATA COLLECTION



Face-to-face interview using e-DCS and self-administered questionnaires



Blood and Urine Collection



One-Day Food Weighing using e-DCS and Salt Collection



24-hour Food Recall

2023 NNS COMPONENTS



FOOD ENVIRONMENT



DIETARY



MATERNAL HEALTH AND NUTRITION



INFANT AND YOUNG CHILD FEEDING (IYCF)



BIOCHEMICAL



MENTAL HEALTH AND NUTRITION



CLINICAL AND HEALTH

PART 2 Dissemination

RESPONSE RATES OF THE 2023 NNS





	Eligible	Response	Response Rate
Households	42,729	36,703	**85.9%**
Individuals	148,907	115,651	**77.7%**

USES OF SURVEY RESULTS

- Craft meaningful policies
- Basis for nutrition interventions
- Serve as a guide to ongoing and future programs
- Monitoring and Evaluation



2023 NATIONAL NUTRITION SURVEY

HOUSEHOLD FOOD ENVIRONMENT



HOUSEHOLD FOOD ENVIRONMENT

Refers to the context in which consumers interact with their food system to acquire, prepare, and consume food.
(Downs & Demmler, 2020)



It encompasses:

- physical space where food is obtained
- the resources
- infrastructure facilitating access

2023 NNS FOOD ENVIRONMENT COMPONENT

- Institutionally developed and validated questionnaire for profiling the household food environment in the Philippines.
- Aimed to assess the **typology**, **availability**, **food acquisition frequency** and **expenditure**, **accessibility**, **usual foods available**, **food retail,** and **food choice considerations.**
- Household **meal planners** are the main respondents.
- Profile of respondents:

National



n=35,399

Rural



n=19,830

Urban



n=15,569

Typology of Food Environments

NATURAL FOOD ENVIRONMENT

Wild (forest, sea, open areas)






Cultivated (farm and garden)






TRADITIONAL FOOD STORES

Talipapa/Wet Market



Mobile Vendor





Carinderia/Turo-turo



Main Public Market



Street Food Vendor





Sari-sari store & other small retailers



MODERN FOOD STORES

Supermarket / Hypermarket



Convenience and Grocery Stores





Fast-food restaurants



All other restaurants



Single Food Item/Specialty Store





Online Vendors



OTHER SOURCES

Institutions and Public Procurement





Food transfer (food aid, food gifts)





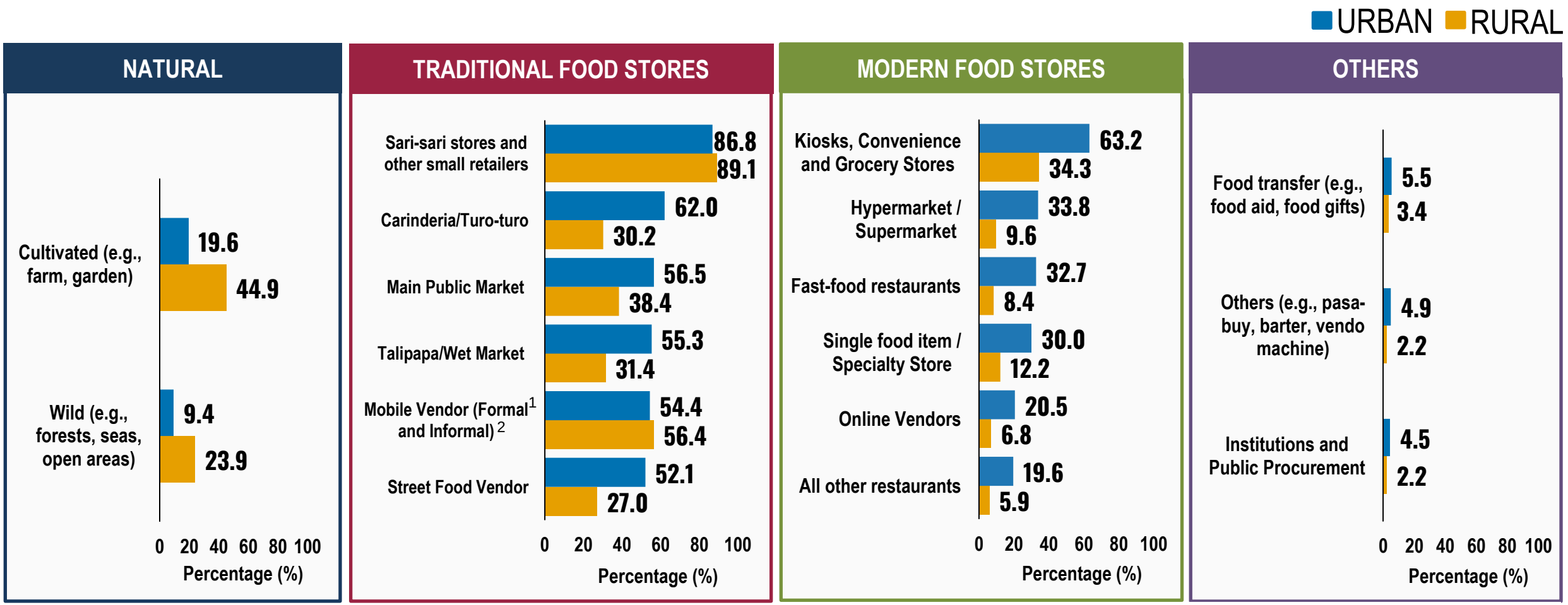
Others (pasa-buy, barter, vendo machine





Availability of Types of Food Environment among Households

Nearest* food environment types among households by place of residence: Philippines, 2023 (n=35,399)



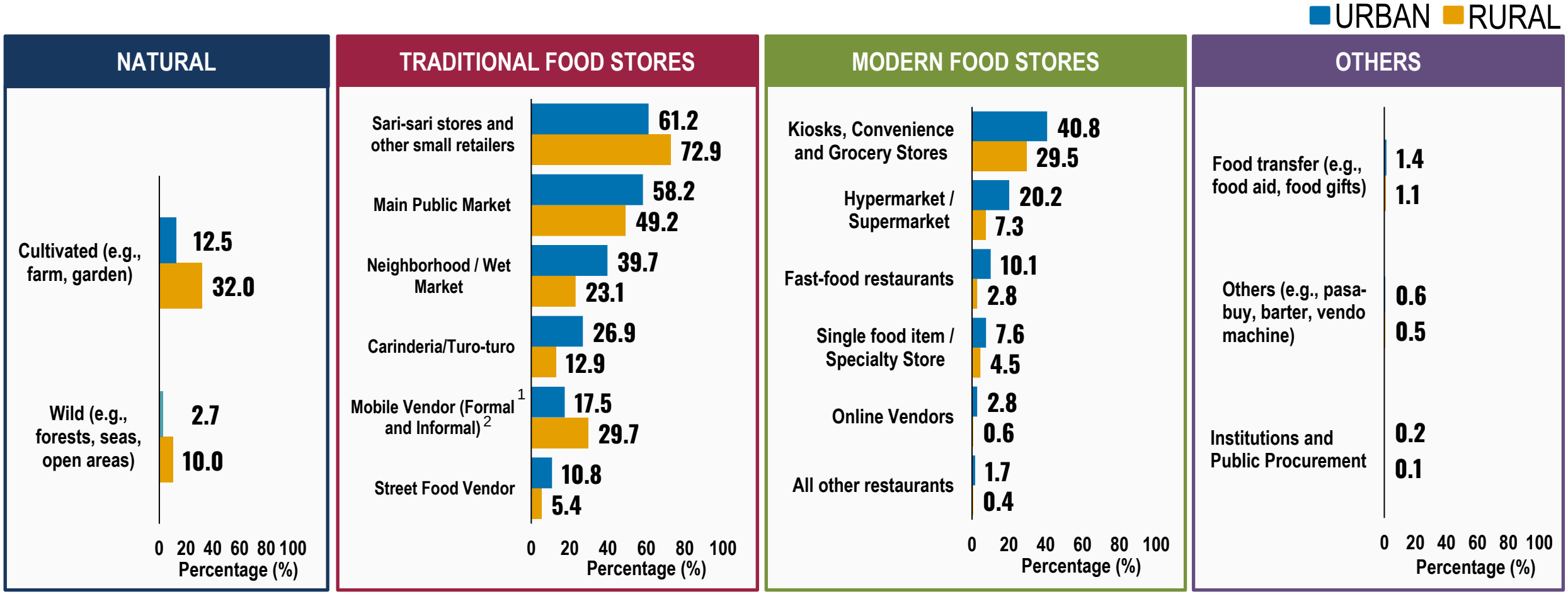
NOTE: *This is a multiple-response question.*

*Within an approximate 20-minute walk from households

[1]*Formal Mobile Vendor: food trucks with license to operate.*

[2]*Informal Mobile Vendor: vendors that do not have permanent infrastructure at the location and travels by vehicle or by walking.*

Types of food environments where households typically buy or gather foods and meals by place of residence: Philippines, 2023 (n=35,399)



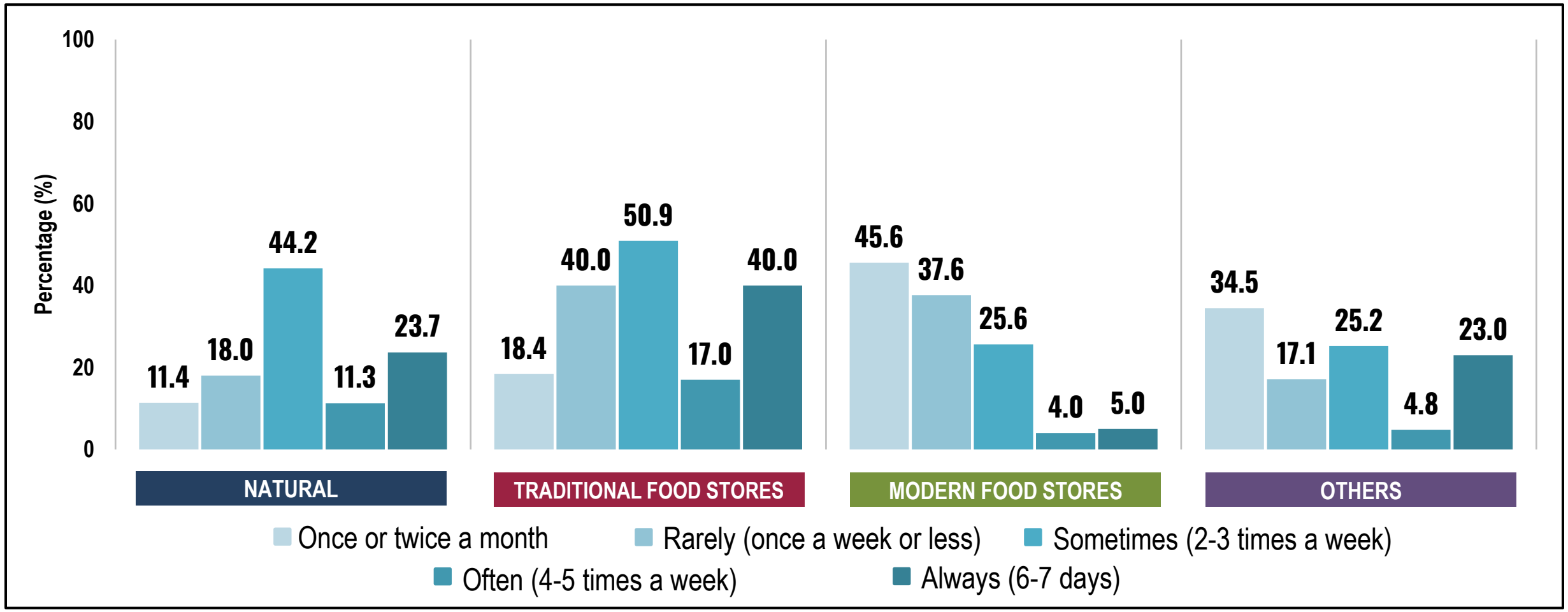
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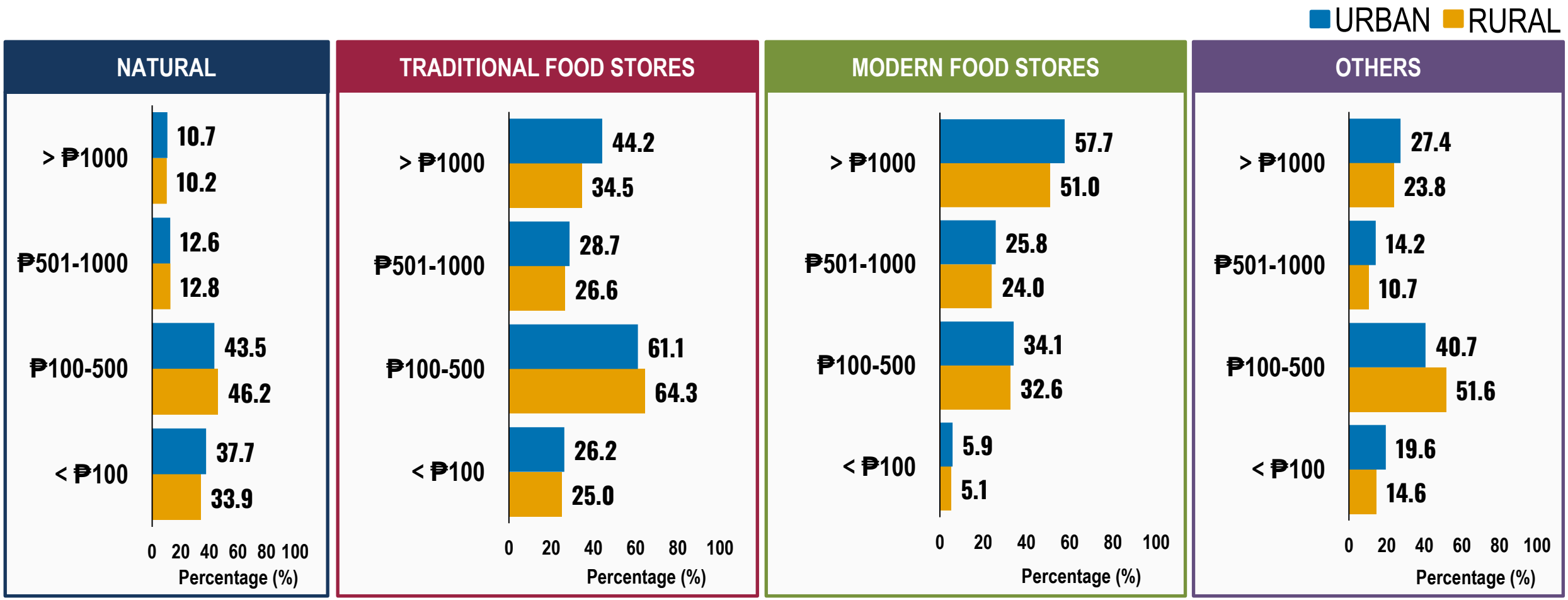
Food Acquisition Frequency and Expenditure of Filipino Households Among Different Food Environments

Frequency of food gathering and purchasing based on food environment: Philippines, 2023



NOTE: *This is a multiple-response question.*
Total sample ≈ 35,399 households; item-level sample size (n) varies

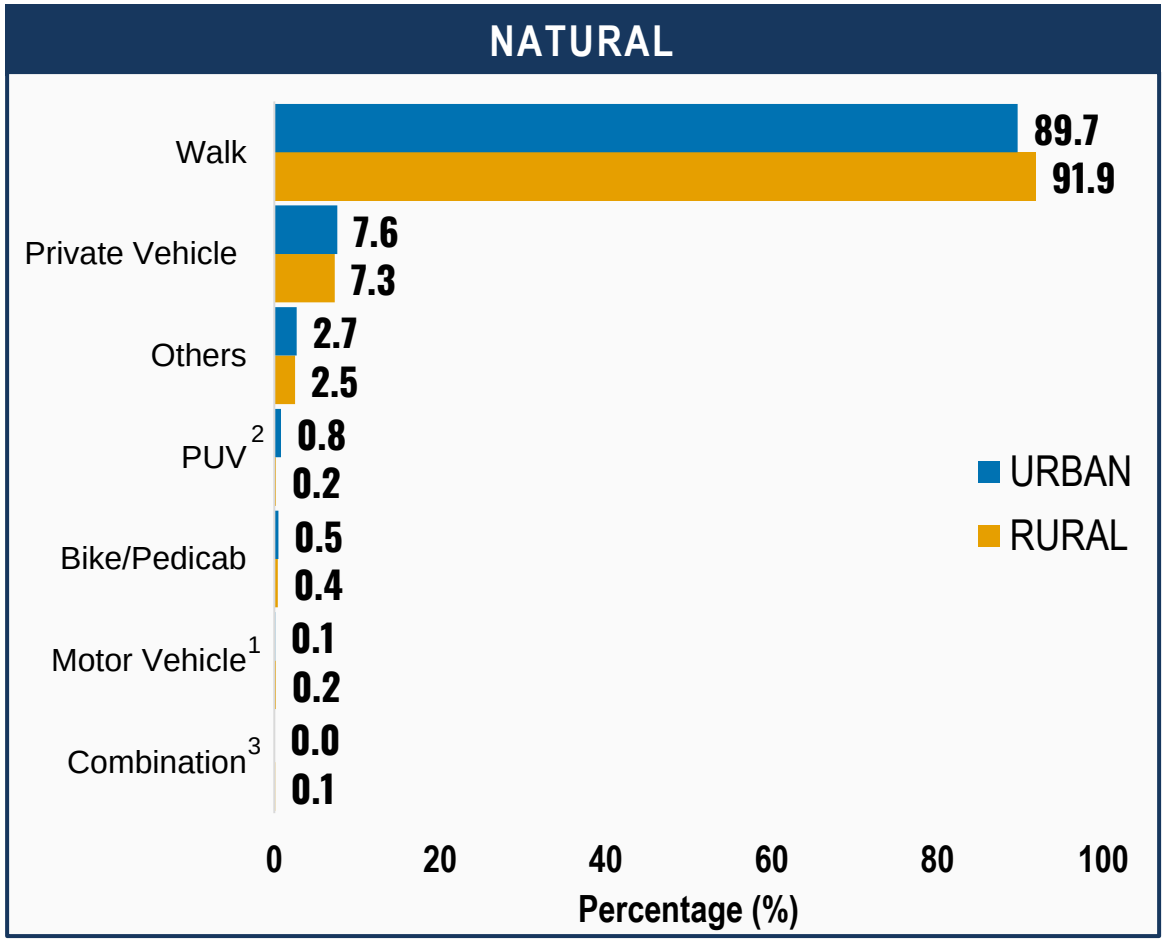
Average food expenditure (PHP) of households by type of food environment by place of residence: Philippines, 2023

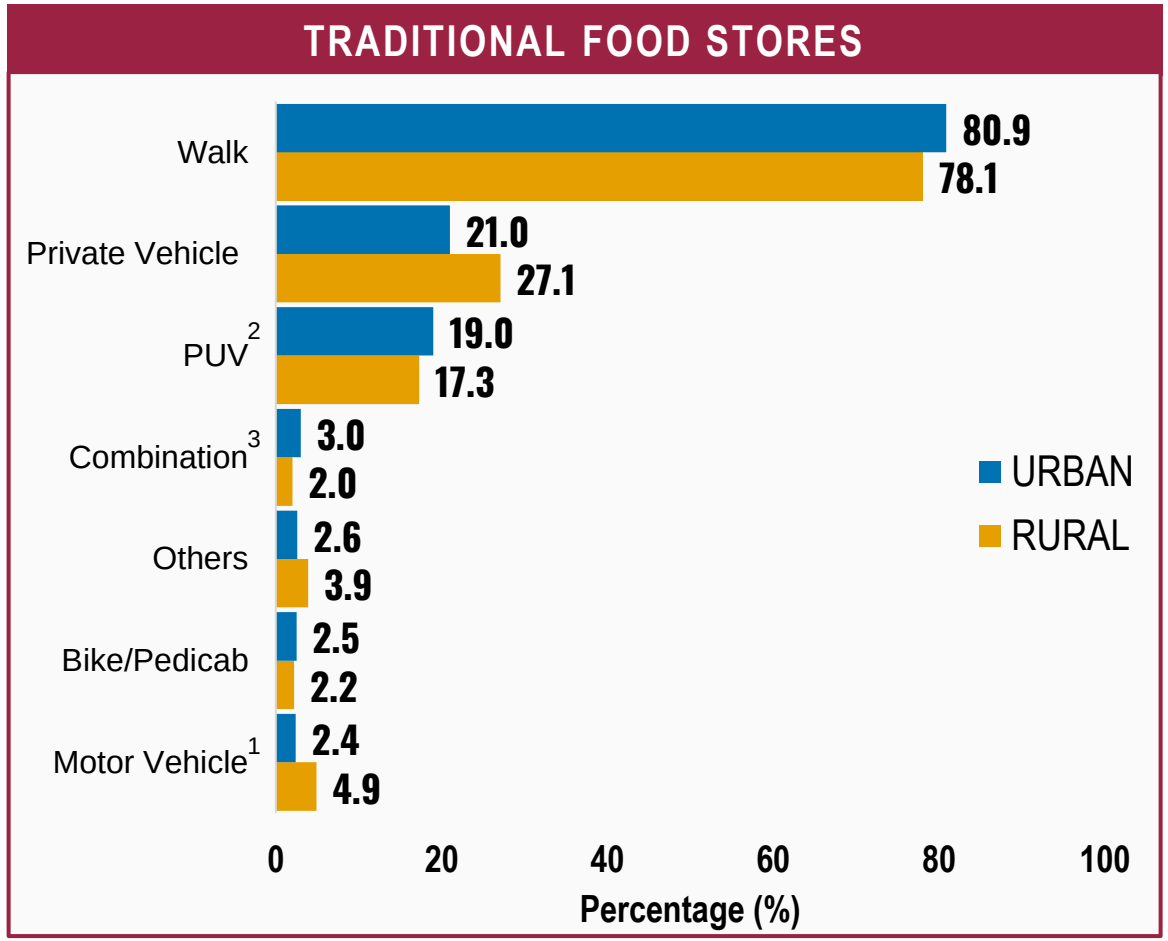


NOTE: *This is a multiple-response question.*
Total sample ≈ 35,399 households; item-level sample size (n) varies

Accessibility and Mobility Factor of Filipino Households among Food Environment

Modes of access to various food environments by place of residence: Philippines, 2023 (n=35,399)

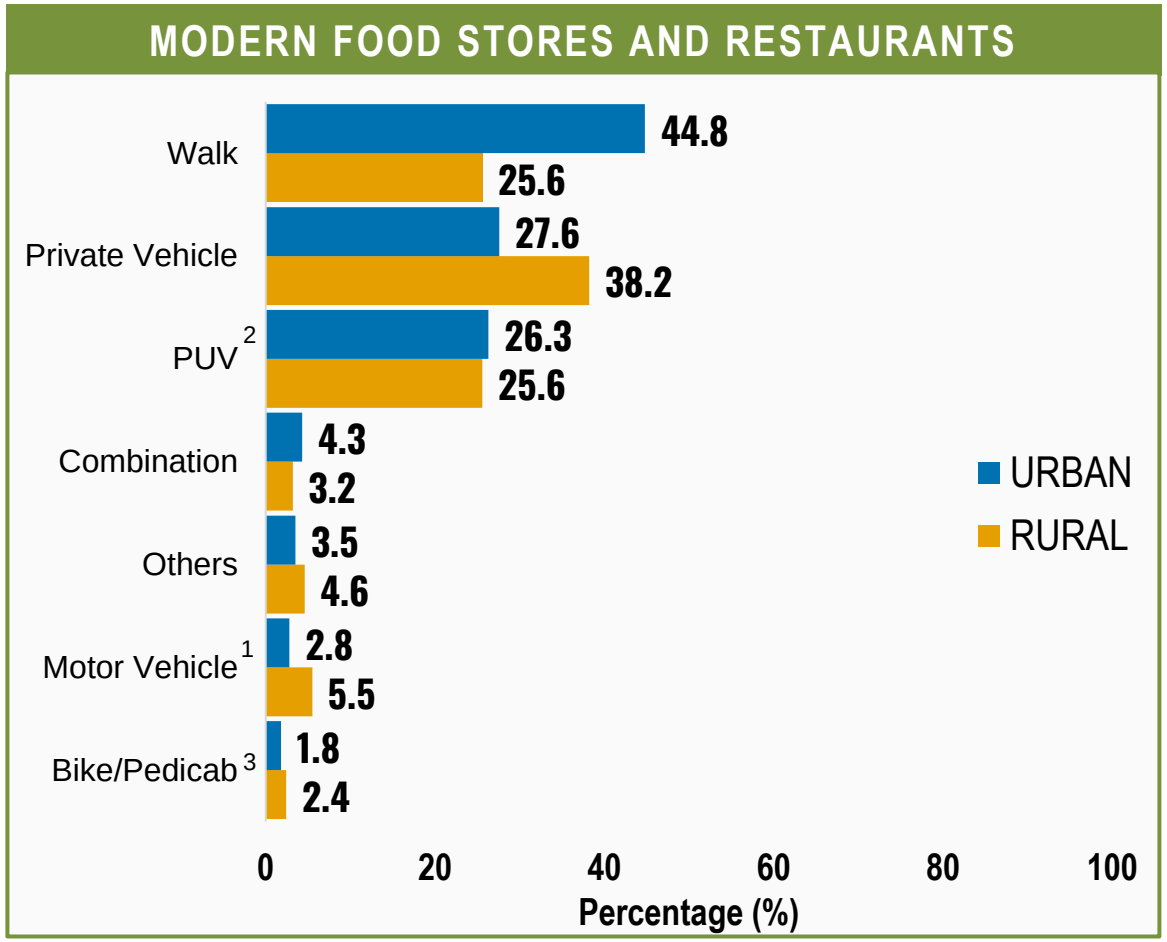


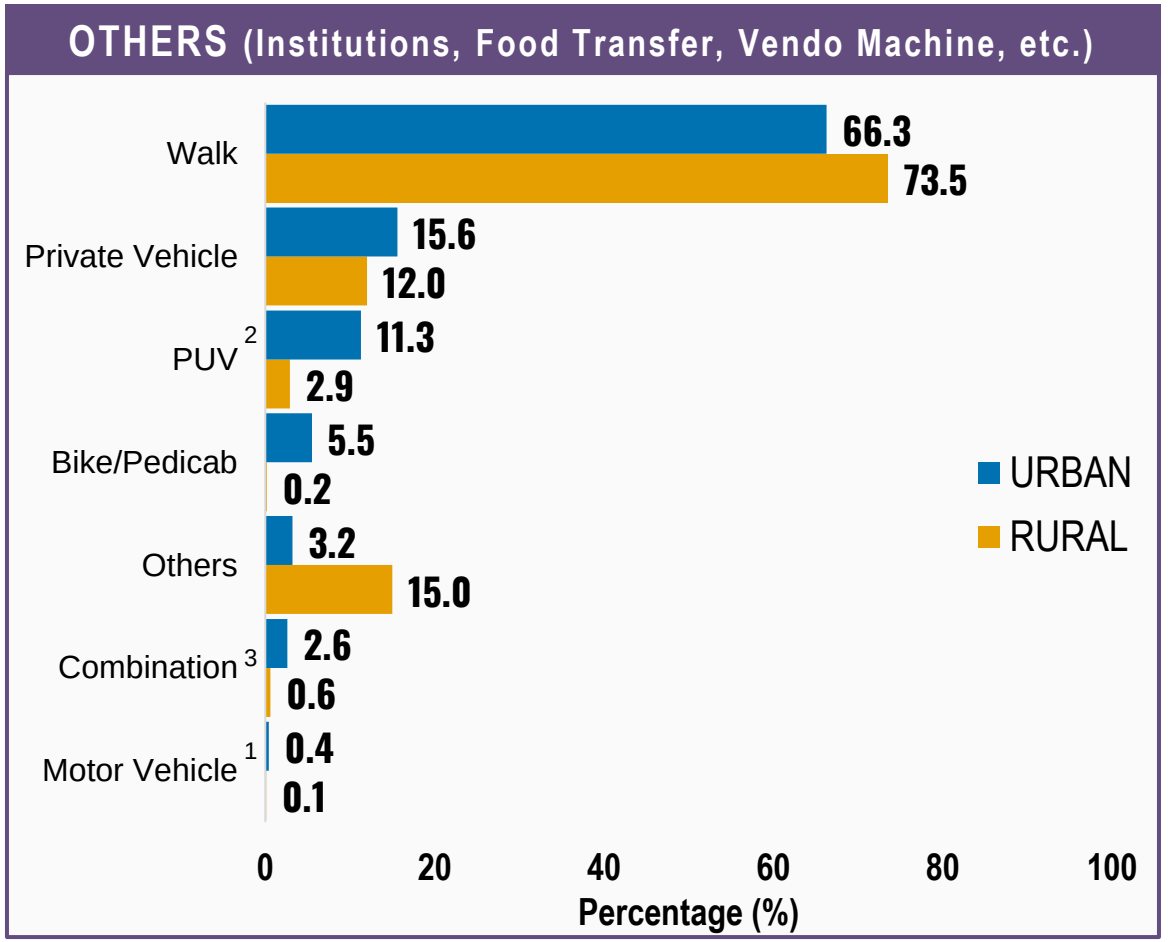


NOTE: *This is a multiple-response question.*

[1]*Angkas, Joyride, Habal, etc.*
[2]*Public Utility Vehicle (e.g. tricycle, jeep, bus, pump boats)*
[3]*Combination of walking and transportation fare*

Modes of access to various food environments by place of residence: Philippines, 2023 (n=35,399)

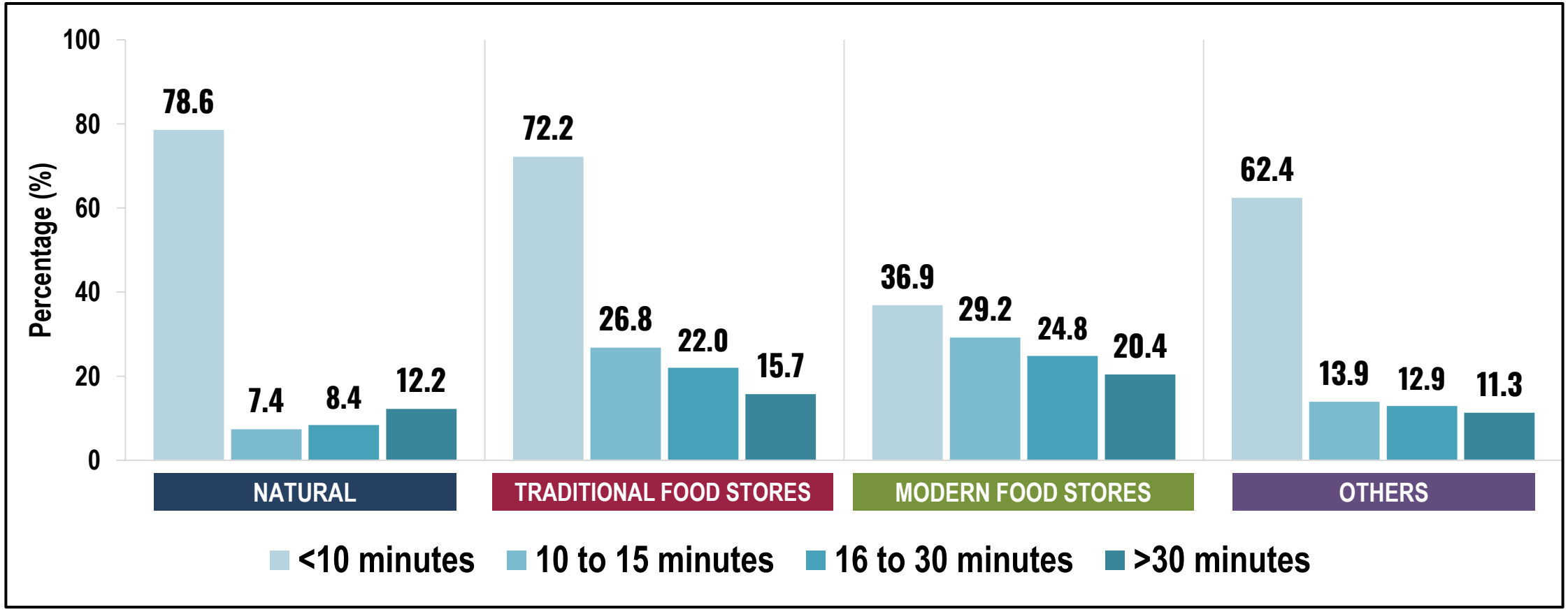




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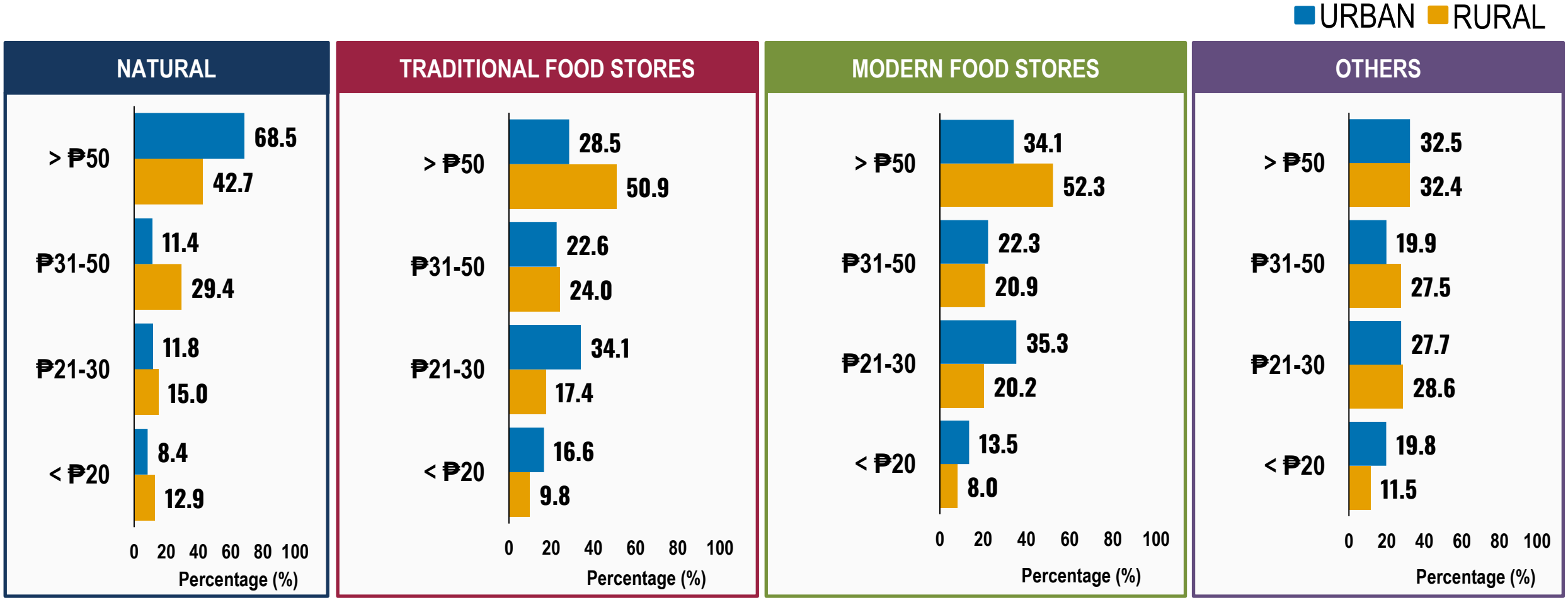
[1]*Angkas, Joyride, Habal, etc.*
[2]*Public Utility Vehicle (e.g. tricycle, jeep, bus, pump boats)*
[3]*Combination of walking and transportation fare*

Travel time of households to and from available food environments: Philippines, 2023 (n=35,399)



NOTE: *This is a multiple-response question.*

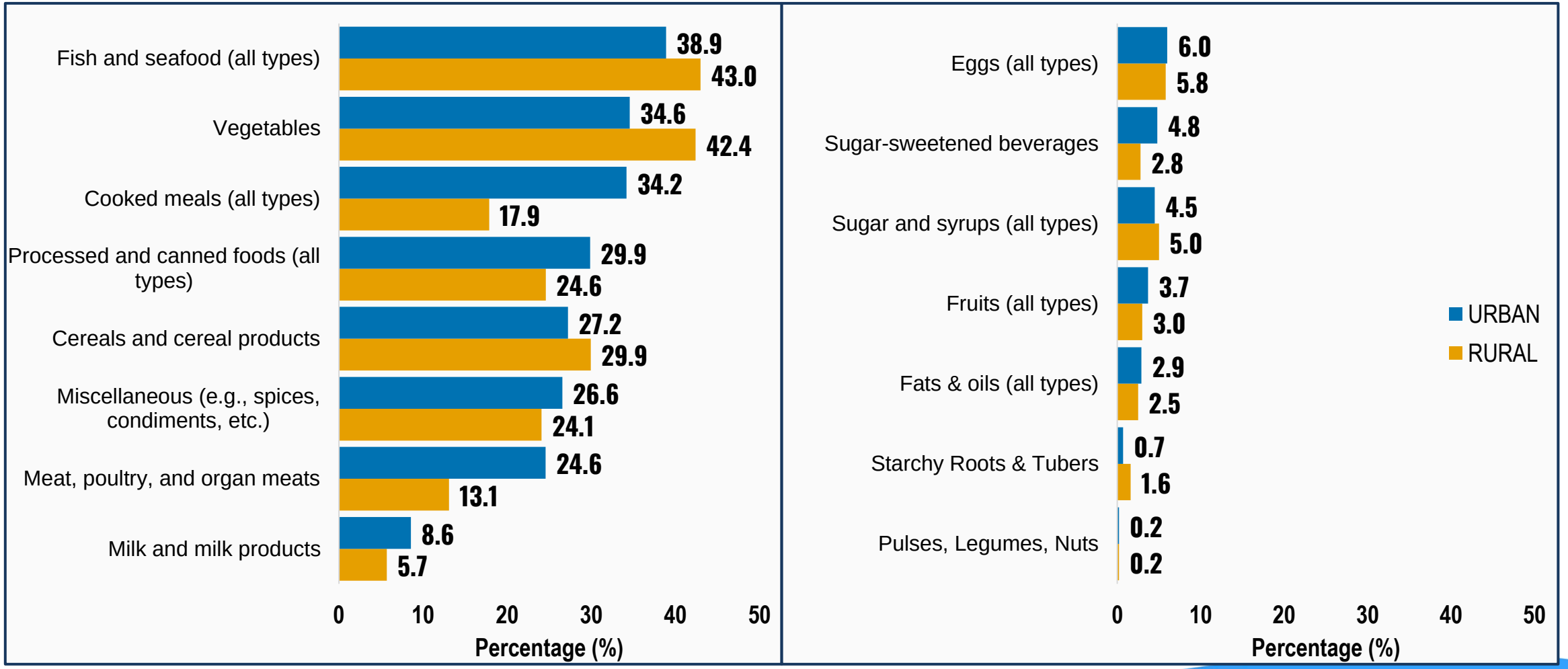
Transportation costs for round-trip travel to and from available food environments by place of residence: Philippines, 2023 (n=35,399)



NOTE: *This is a multiple-response question.*
Responses are based if they spend on transportation

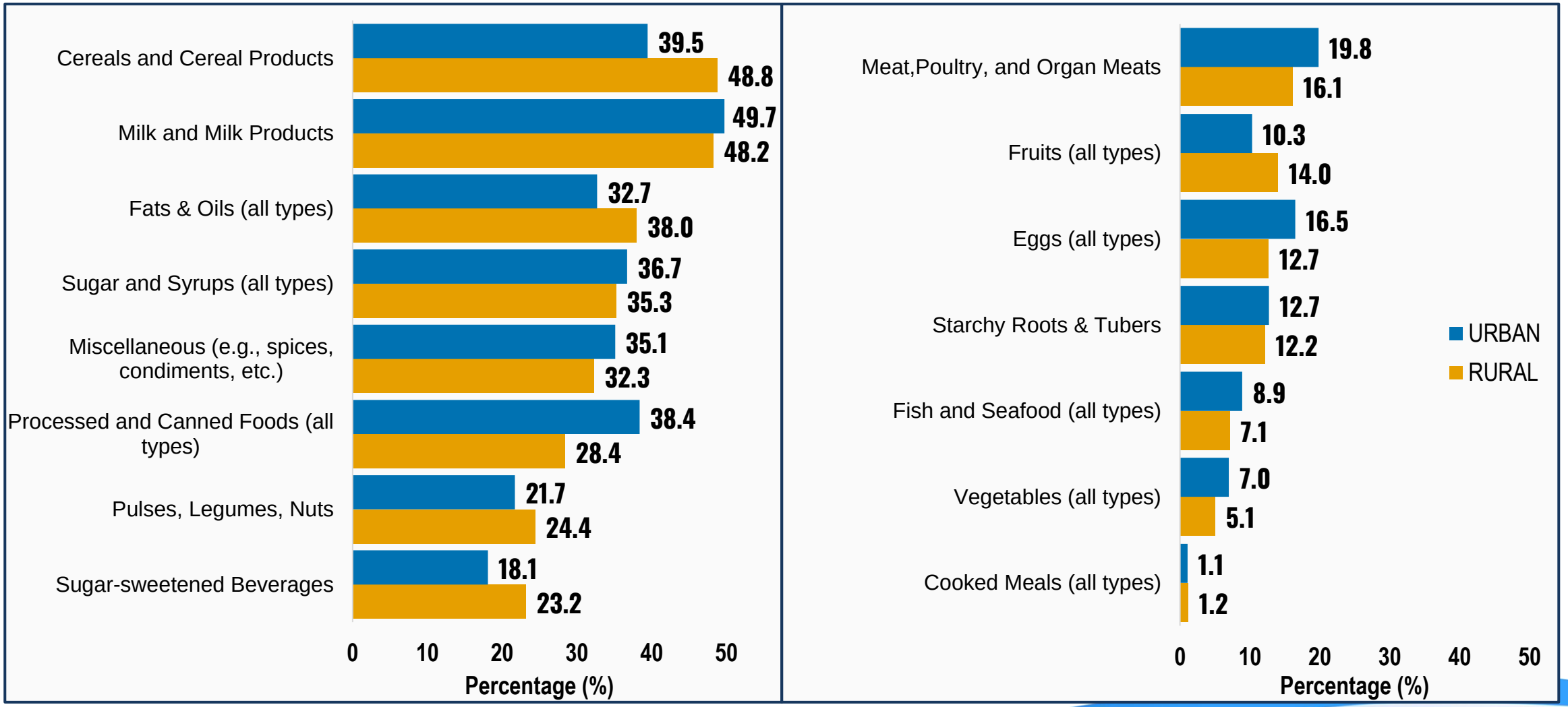
Foods Commonly Available in Filipino Households

Commonly purchased or gathered foods/meals from food environments by place of residence: Philippines, 2023 (n=35,399)



NOTE: *This is a multiple-response question.*

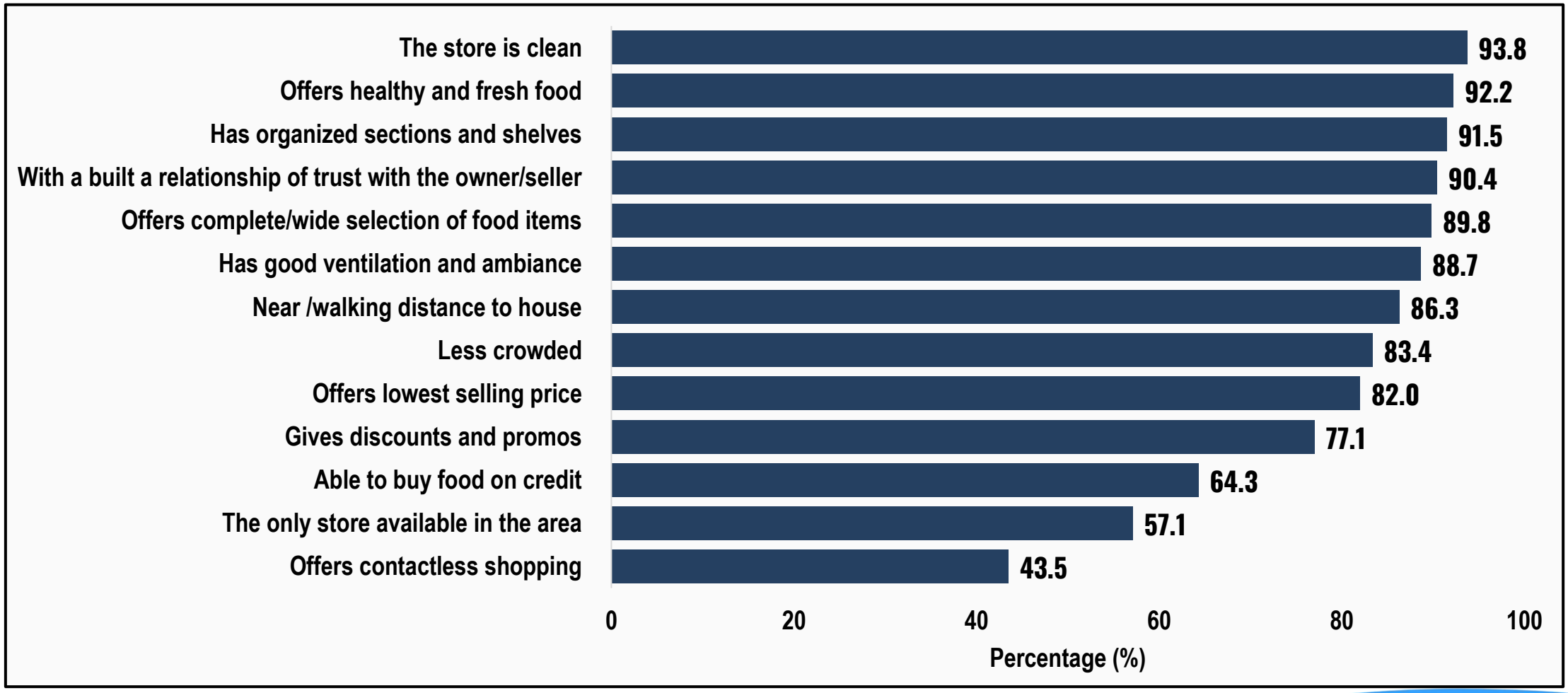
Most stocked food groups among households for 1 week or more across food environments by place of residence : Philippines, 2023



NOTE: Total sample ≈ 35,399 households; item-level sample size (n) varies

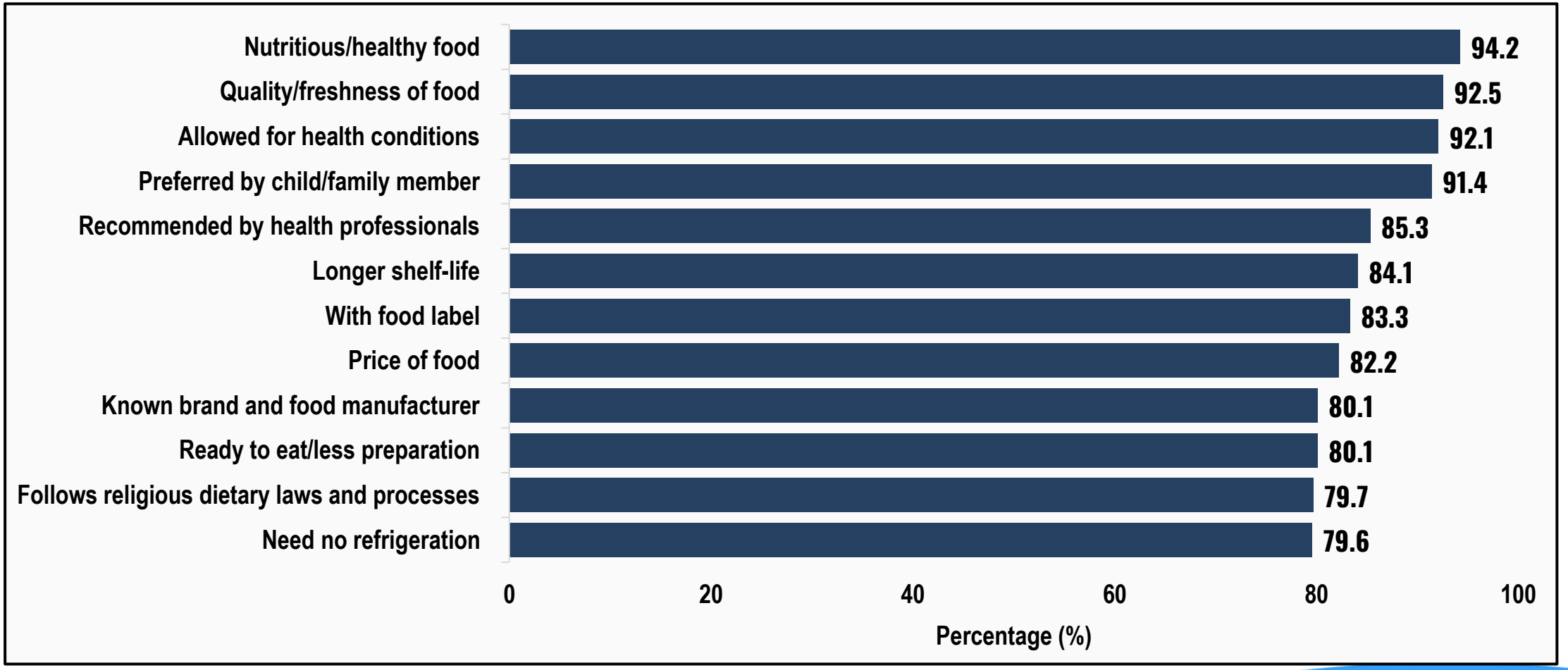
Food Retail and Food Choice Considerations of Filipino Households

Characteristics of food retail outlets considered by households when purchasing food: Philippines, 2023



NOTE: *This is a multiple-response question.*
Total sample ≈ 35,399 households; item-level sample size (n) varies

Food characteristics considered by households when purchasing food: Philippines, 2023



NOTE: *This is a multiple-response question.*
Total sample ≈ 35,399 households; item-level sample size (n) varies

HIGHLIGHTS

AVAILABILITY OF TYPES OF FOOD ENVIRONMENTS TO HOUSEHOLDS

An approximate of 20-minute walk from households

- **Cultivated farms and gardens** are shown as most available natural food environments in rural areas.
- **Sari-sari stores and other small retailers** are widely available and accessible traditional food stores for both urban and rural households.
- **Kiosks, convenience and grocery stores** are the most prevalent modern food outlets, especially in urban areas.
- **Food transfers** (e.g., food aid and gifts) are the most reported source among other sources, but with very low percentages.



HIGHLIGHTS

AFFORDABILITY OF FOOD ENVIRONMENTS TO FILIPINO HOUSEHOLDS

Food Acquisition Frequency and Expenditure

- Households most commonly **spend ₱100-500** in both natural food environment and traditional food stores for **2-3 times a week**.
- Households gather and purchase foods from modern food stores and other sources for **once or twice a month**, **spending more than ₱1000** in modern food stores and **₱100-500** in other sources.



HIGHLIGHTS

ACCESSSIBILITY OF FOOD ENVIRONMENTS TO FILIPINO HOUSEHOLDS

Accessibility and Mobility Factor



Walking is the most common mode of transportation to reach food environments.



The average travel time for households to reach food environments is **less than 10 minutes.**



The average transportation cost for households is **more than ₱50.**

HIGHLIGHTS

FOOD CHOICE AND SHOPPING BEHAVIOR

Food Retail and Food Choice Characteristics for Consideration



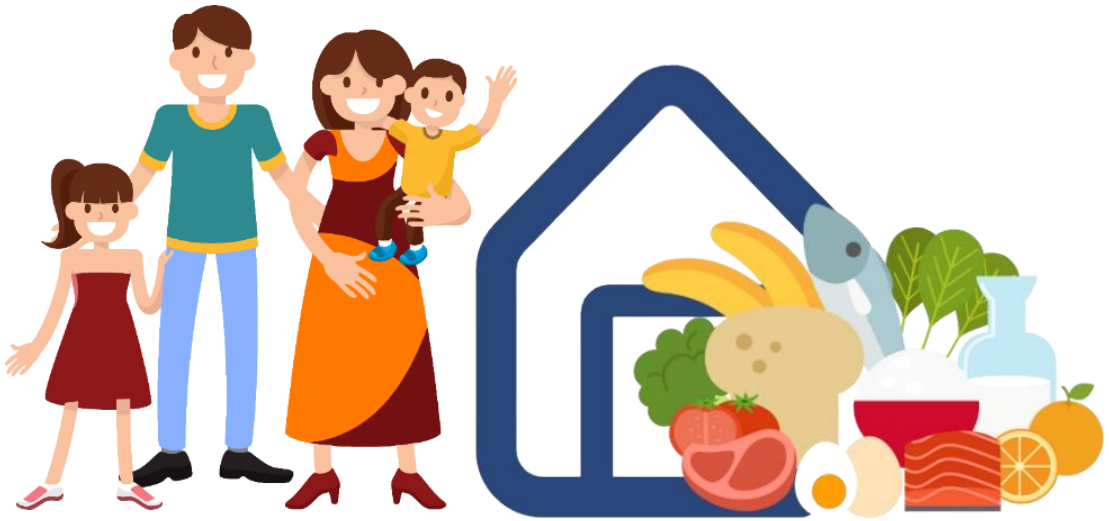
Households primarily consider the **cleanliness** of food retail outlets when buying food.



Households primarily prioritize purchasing **nutritious and healthy foods** among other characteristics.

National Nutrition Survey 2023

FOOD CONSUMPTION SURVEY:



Household Level

FCS: HOUSEHOLD LEVEL

OBJECTIVES



Evaluate the current food intake pattern of households



Reflect the foods available in the households



Examine the trends and changes in the per capita consumption of food and nutrients over time



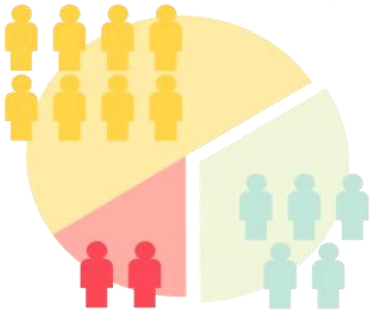
Monitor the trends in the proportion of households meeting energy & nutrient adequacy relative to SDG#2 Zero Hunger

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FCS: HOUSEHOLD LEVEL

SIGNIFICANCE

The **official statistical source of nationally-representative data** on food consumption and nutrient intakes in the Philippines.



Provides insights on the changing dietary pattern of the population.





Aids in the policy and program development for poverty alleviation, food security, and health and nutrition strategies of the government and private organizations.

FCS: HOUSEHOLD LEVEL

HOUSEHOLD DIETARY DATA COLLECTION



Food Inventory





One-day Food Weighing

FCS: HOUSEHOLD LEVEL

HOUSEHOLD DIETARY DATA COLLECTION



Given-out foods



Leftover foods



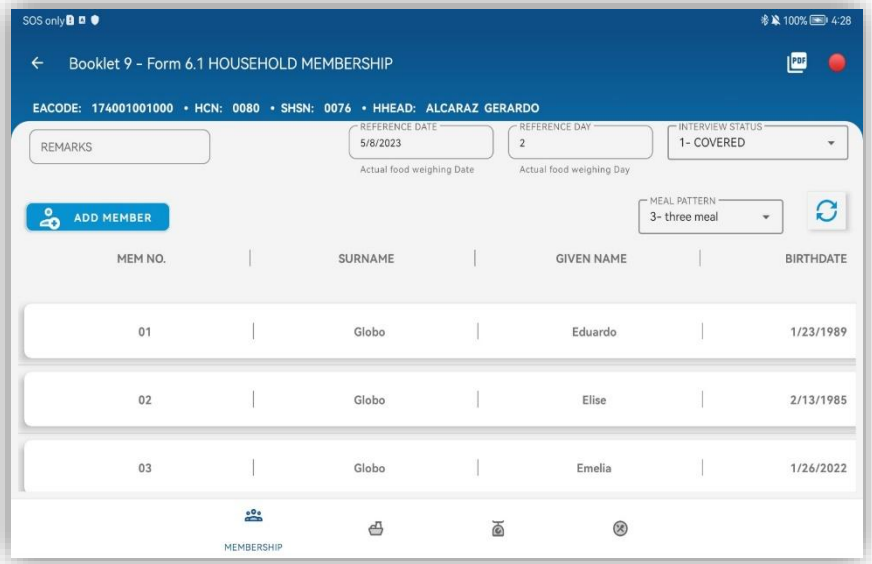
Plate waste

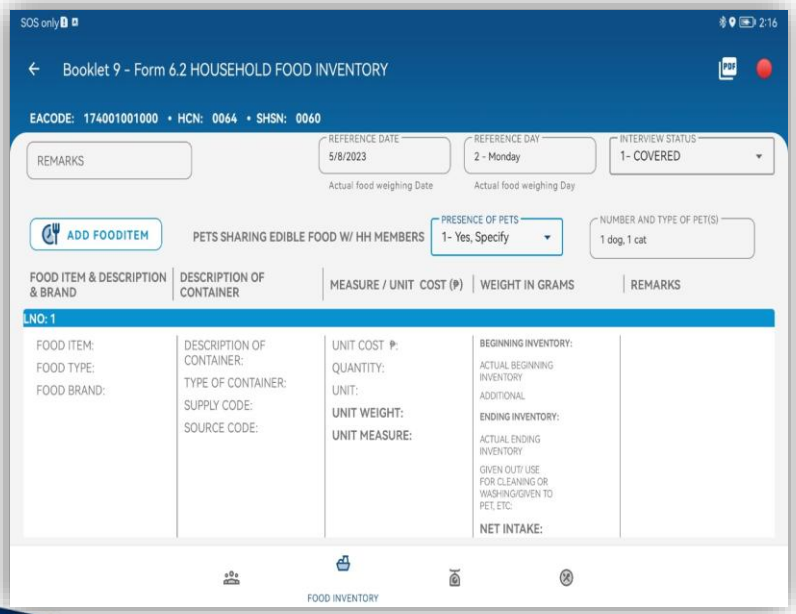
DDCS

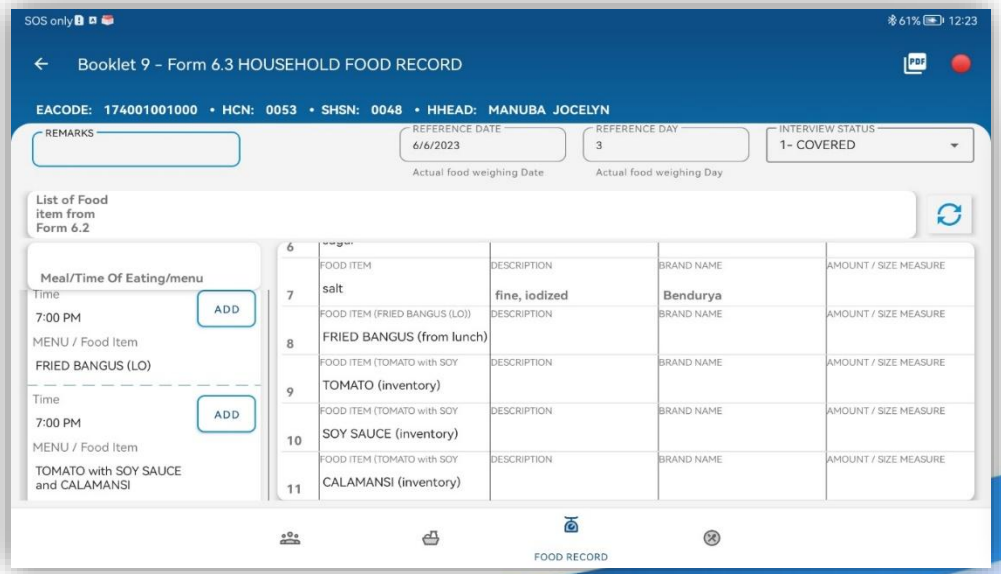
Dietary
Data Collection System



Food Weighing







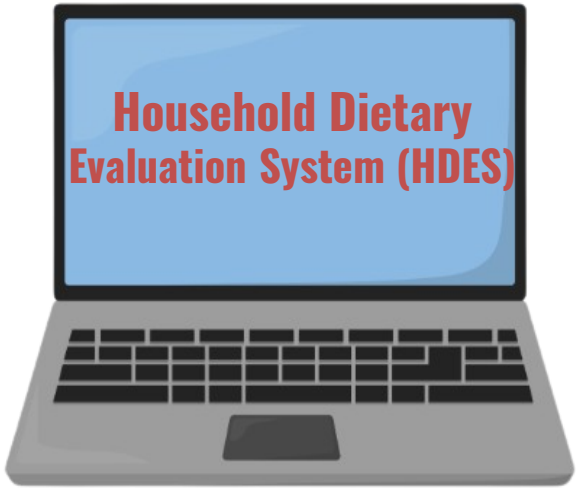


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FOOD AND NUTRITION RESEARCH INSTITUTE
DOST-FNRI

FCS: HOUSEHOLD LEVEL

HOUSEHOLD DIETARY DATA ANALYSIS







1 **Editing and validation:**

- food item code & description
- amount of food in raw as purchased form
- amount of cooked food
- amount of leftovers, given-out, plate waste

2 **Calculating amount of food intake**

- converting cooked food items into raw as purchased form
- converting raw edible food items into raw as purchased form

3 **Calculating energy and nutrient for:**

- each food item and ingredient consumed
- sum of energy and nutrient intake of all foods consumed by each household

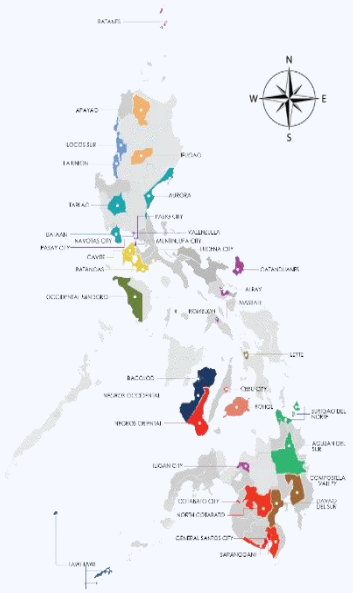
4 **Evaluating energy and nutrient intake adequacy**

FCS: HOUSEHOLD LEVEL

	Eligible	Covered	Response (%)
Households	21,433	15,891	74.1

Sampling

50% of sample households or an average of 192 per provinces / HUCs



Respondents

Household member responsible for the **food preparation**



National Nutrition Survey 2023

FOOD CONSUMPTION SURVEY



Results
Household Level

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Mean one-day household food intake by food group and percent contribution to total intake : Philippines, 2023



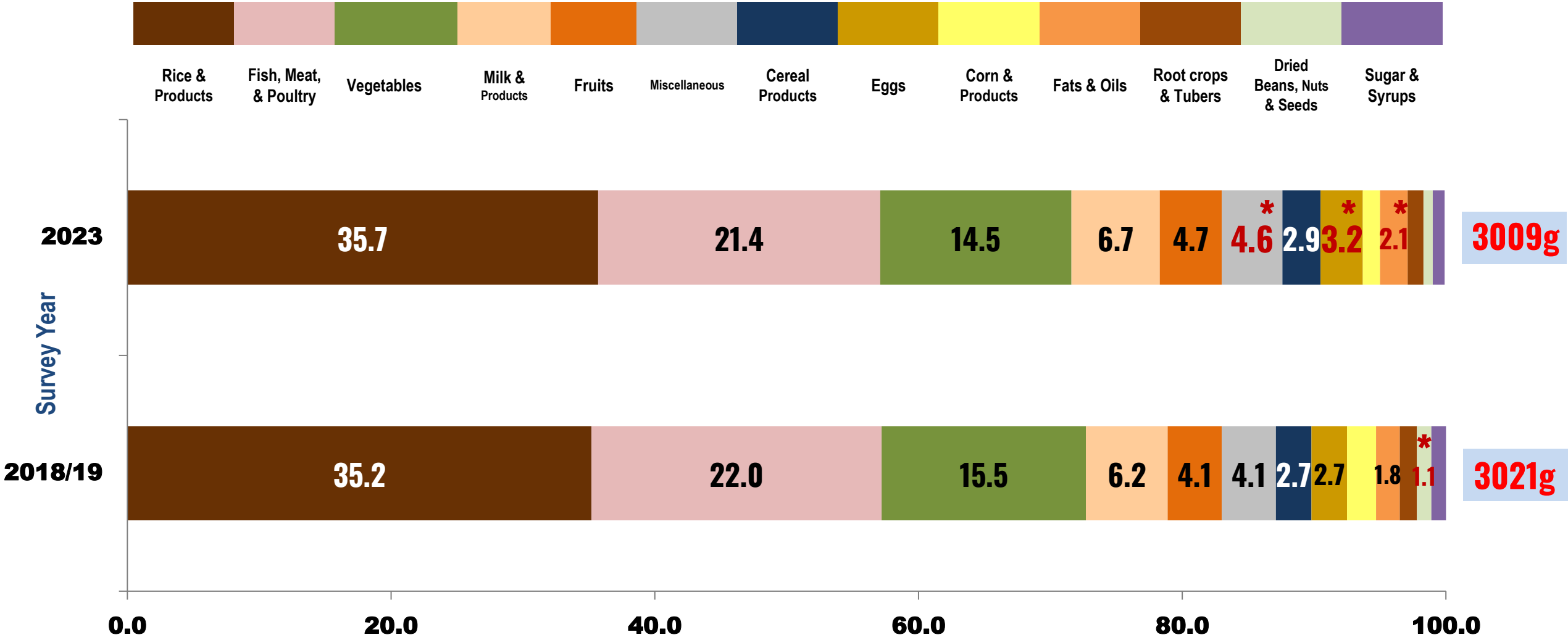
Fats and Oils, 62g, 2.1%
Sugar and Syrups, 28g, 0.9%
Eggs, 97g, 3.2%
Dried Beans, Nuts and Seeds, 22.4g, 0.7%
Poultry, 123g, 4.1%
Miscellaneous, 139g, 4.6%
Fruits, 140g, 4.7%
Rice and Products, 1075g, 35.7%
Meat and Products, 195g, 6.5%
Milk and Products, 201g, 6.7%
Corn and Products, 40g, 1.3%
Cereal Products, 87g, 2.9%
Root crops and Tubers, 37g, 1.2%
Fish and Products, 326g, 10.8%
Vegetables, 438g, 14.5%

Mean one-day per capita intake of food groups: Philippines, 2023



Fats and Oils, 17g, 2.1%
Sugar and Syrups, 8g, 0.9%
Eggs, 27g, 3.3%
Dried Beans, Nuts and Seeds, 7g, 0.8%
Poultry, 34g, 4.1%
Fruits, 41g, 4.9%
Rice and Products, 287g, 34.6%
Miscellaneous, 41g, 4.9%
Milk and Products, 54g, 6.5%
Meat and Products, 54g, 6.5%
Corn and Products, 11g, 1.3%
Cereal Products, 25g, 3.0%
Root crops and Tubers, 10g, 1.3%
Fish and Products, 92g, 11.1%
Vegetables, 123g, 14.8%

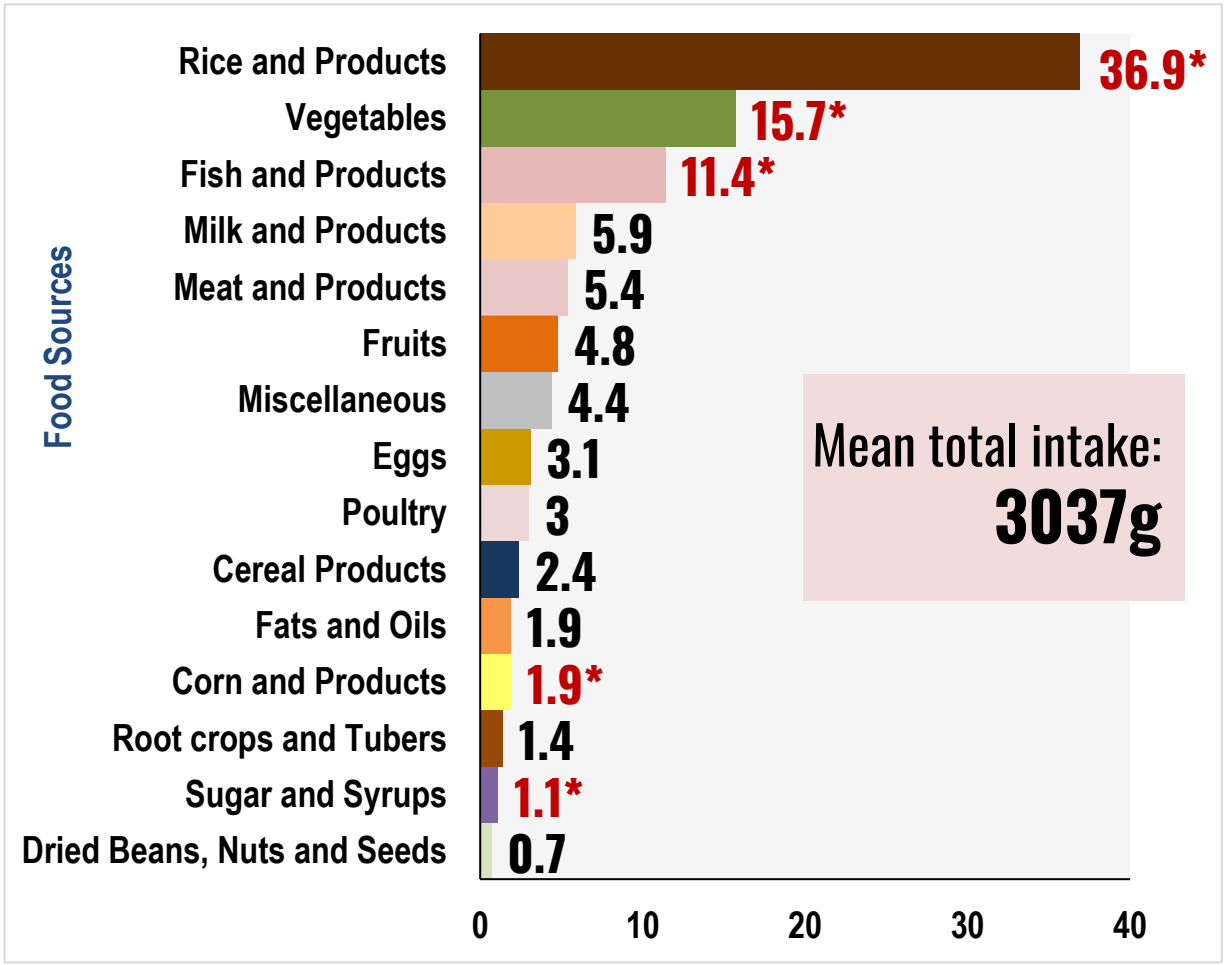
Percent contribution of food groups to total household food intake: Philippines, 2018-2019 and 2023

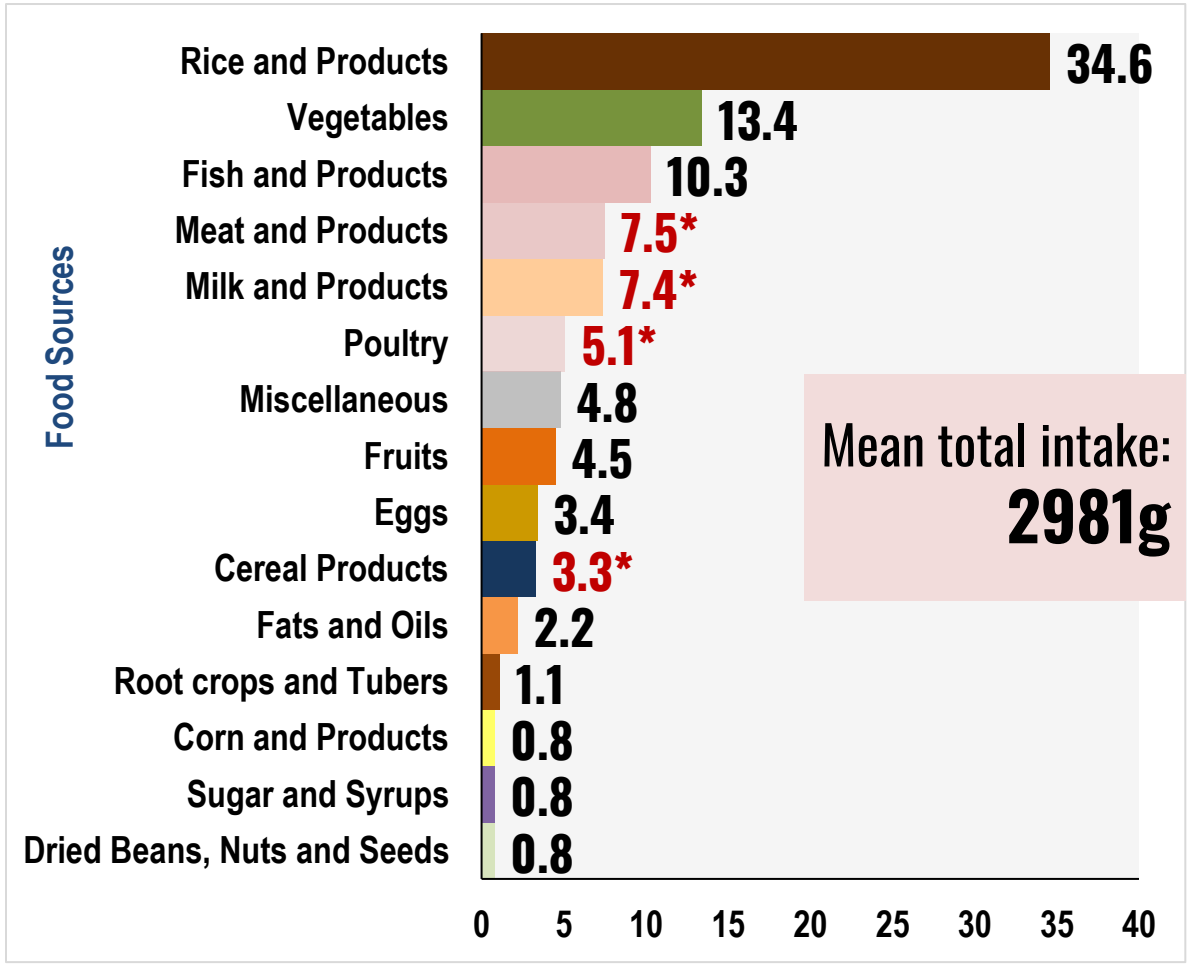


* Significantly different at p < 0.05 level of significance

Note: All other Food Groups without values are <2%.

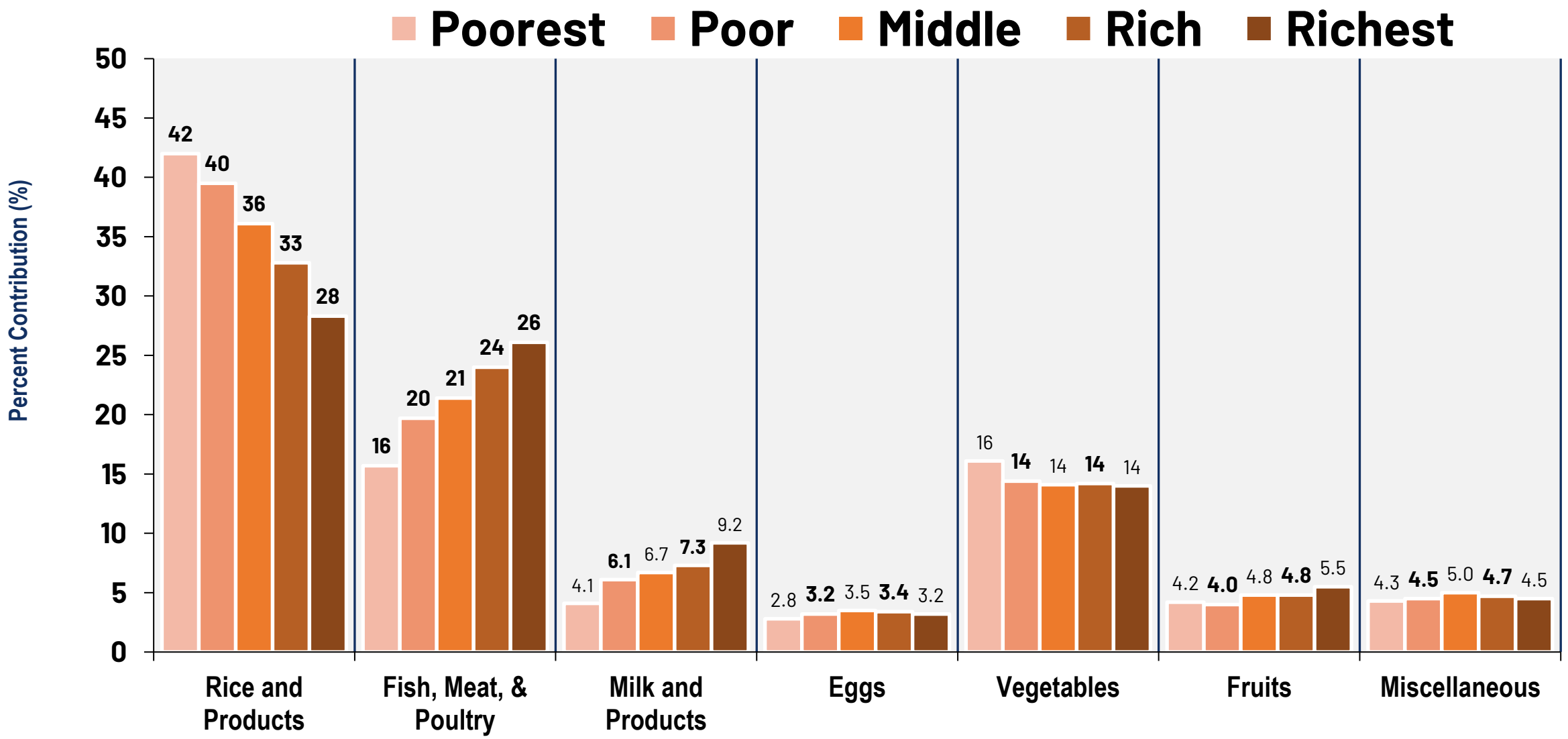
Percent contribution of food groups to total daily household intake, by place of residence: Philippines, 2023



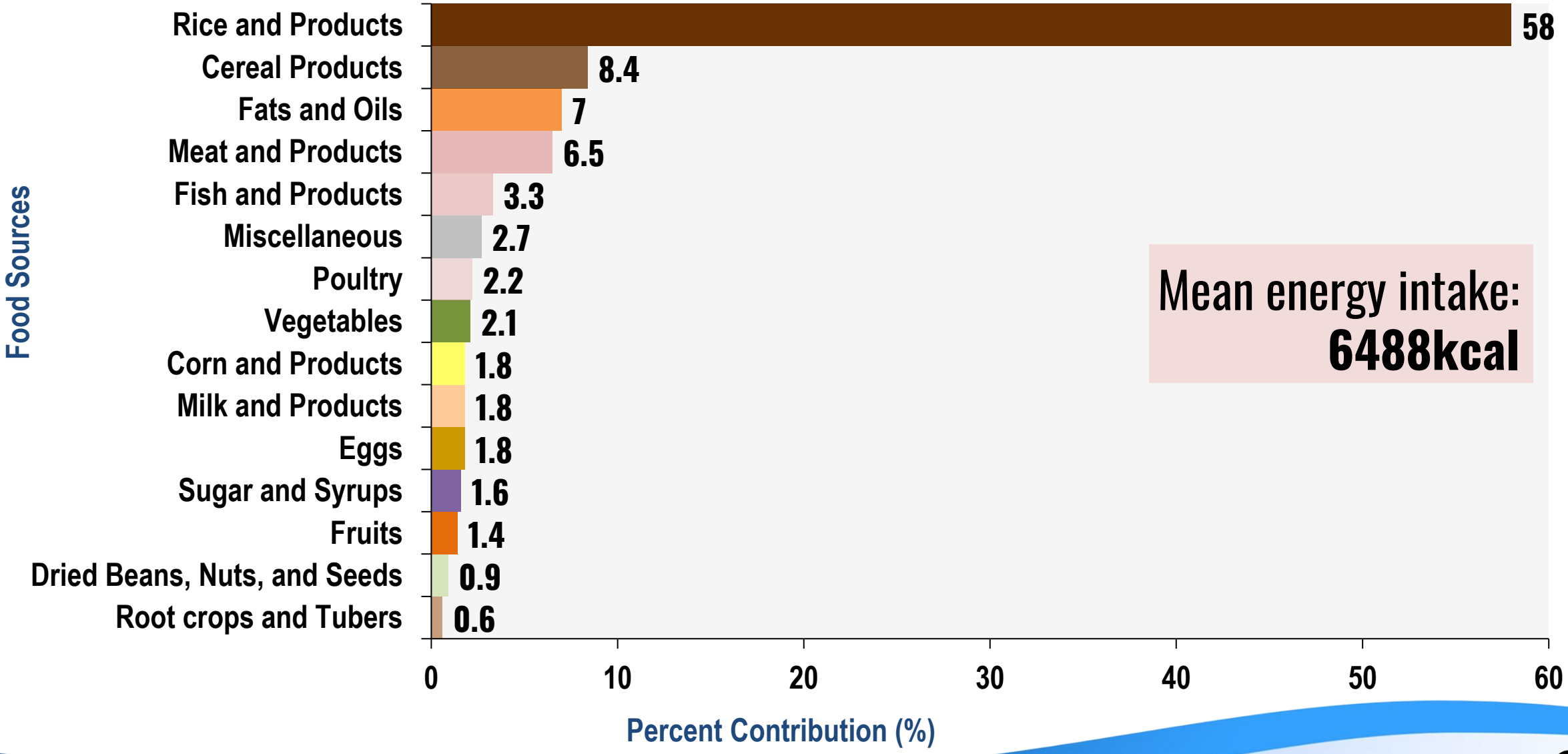


* Significantly different at $p < 0.05$ level of significance

Percent contribution of key food groups to total daily household intake, by wealth quintile: Philippines, 2023



Major food group sources of energy intake among households: Philippines, 2023

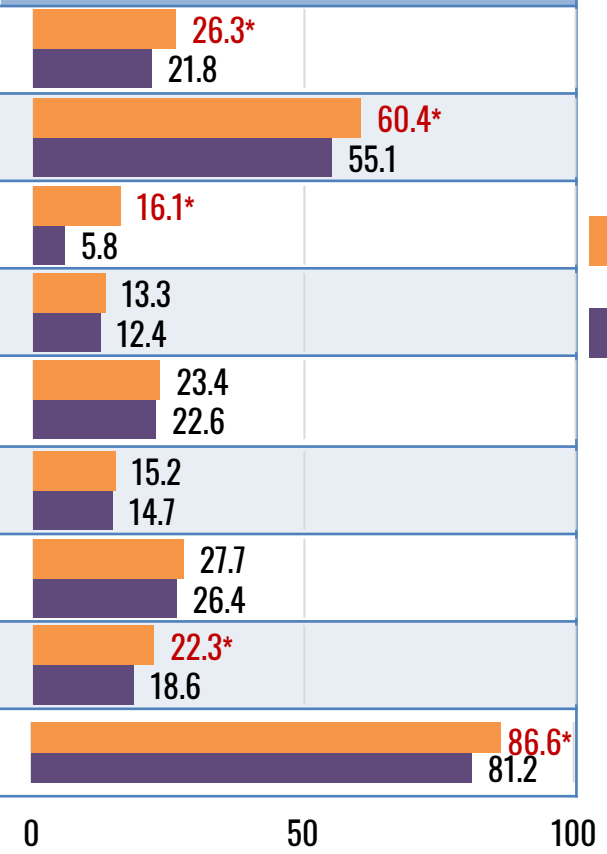


DOST-FNRI

Percentage of households meeting their energy and nutrient requirements: NNS 2023

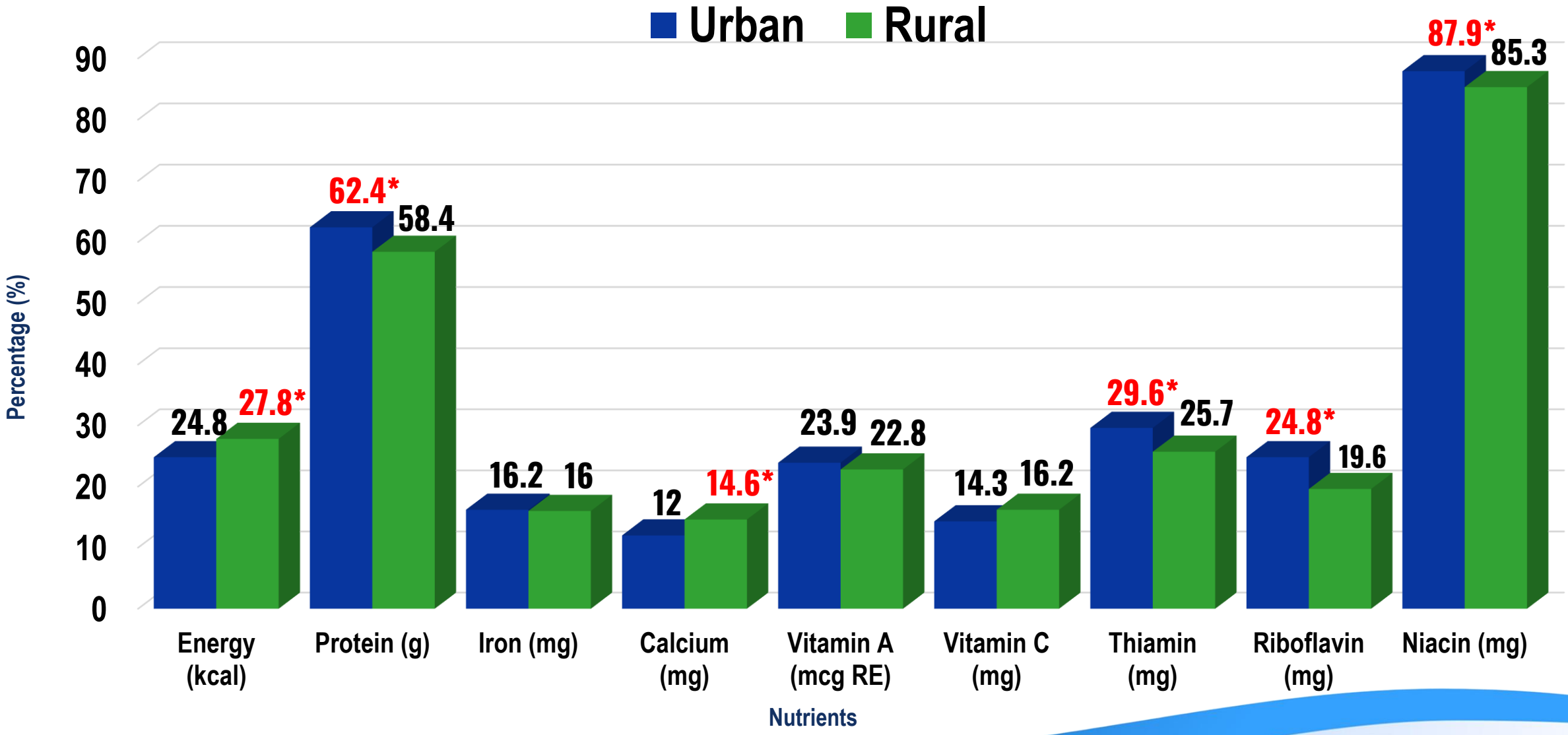
Mean energy and nutrient intake, and percentage of households meeting their energy and nutrient requirements: Philippines, 2023

Energy and Nutrients	2023		2018-2019		Proportion of HH meeting 100% of energy recommendation and EAR for nutrients
	Mean Intake	95% CI	Mean Intake	95% CI	
Energy (kcal)	6487.9	6404.2 – 6571.7	6523.9	6343.9 – 6704.0	26.3* / 21.8
Protein (g)	202.5	199.7 – 205.3	205.9	201.1 – 210.7	60.4* / 55.1
Iron (mg)	32.1	31.7 – 32.6	32.0	31.0 – 33.1	16.1* / 5.8
Calcium (mg)	1340.2	1313.0 – 1367.5	1367.4	1336.5 – 1398.3	13.3 / 12.4
Vitamin A (mcg RE)	1647.1	1547.9 – 1746.3	1734.5	1620.4 – 1848.6	23.4 / 22.6
Vitamin C (mg)	94.9	92.2 – 97.5	98.3	90.5 – 106.1	15.2 / 14.7
Thiamin (mg)	2.8	2.8 – 2.9	2.9	2.8 – 3.0	27.7 / 26.4
Riboflavin (mg)	2.7	2.7 – 2.8	2.7	2.6 – 2.8	22.3* / 18.6
Niacin (mg)	65.3	64.4 – 66.2	66.3	64.7 – 67.9	86.6* / 81.2



* Significantly different at $p < 0.05$ level of significance

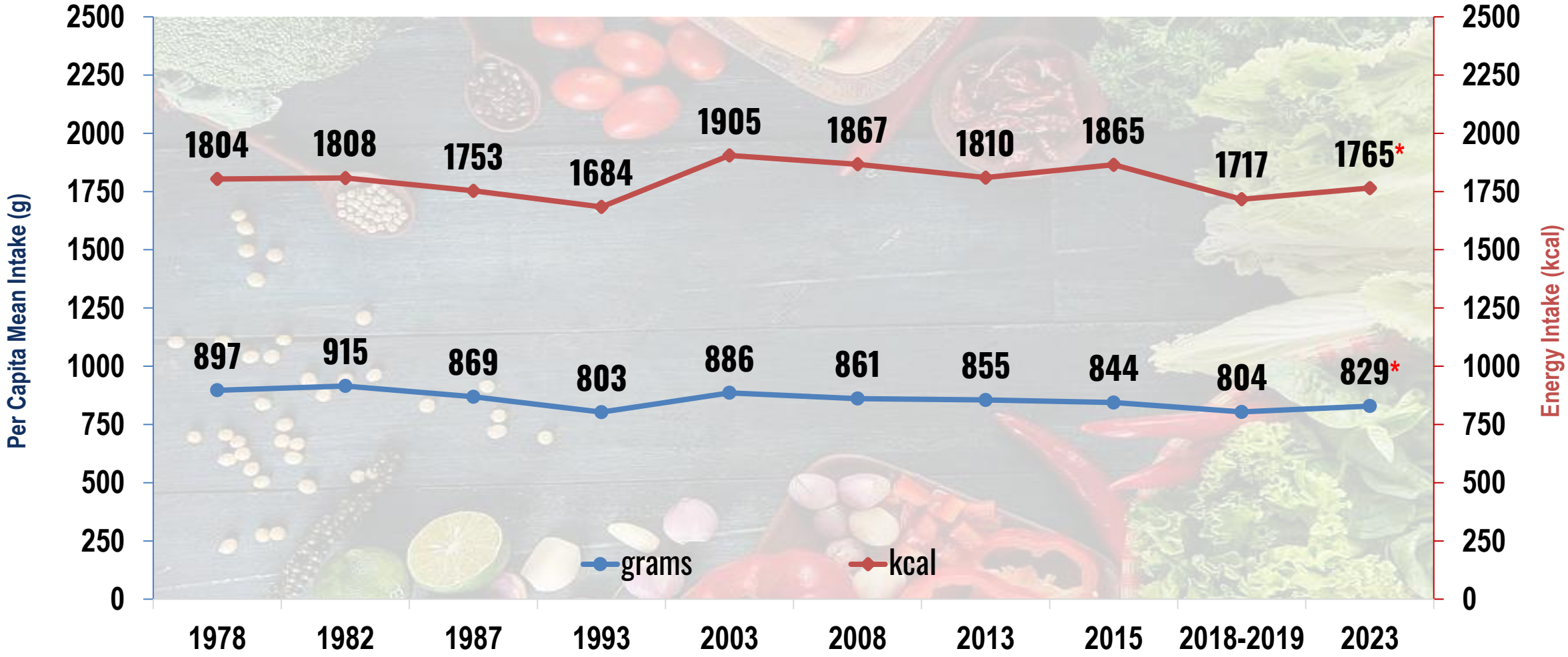
Percentage of households meeting 100% REI and EAR for nutrients of rural and urban areas: Philippines, 2023



* Significantly different at p < 0.05 level of significance

Trends in daily per capita food intake: Philippines, 1978-2023

Trends in the One-Day Per Capita Food and Energy Intake: Philippines, 1978 to 2023

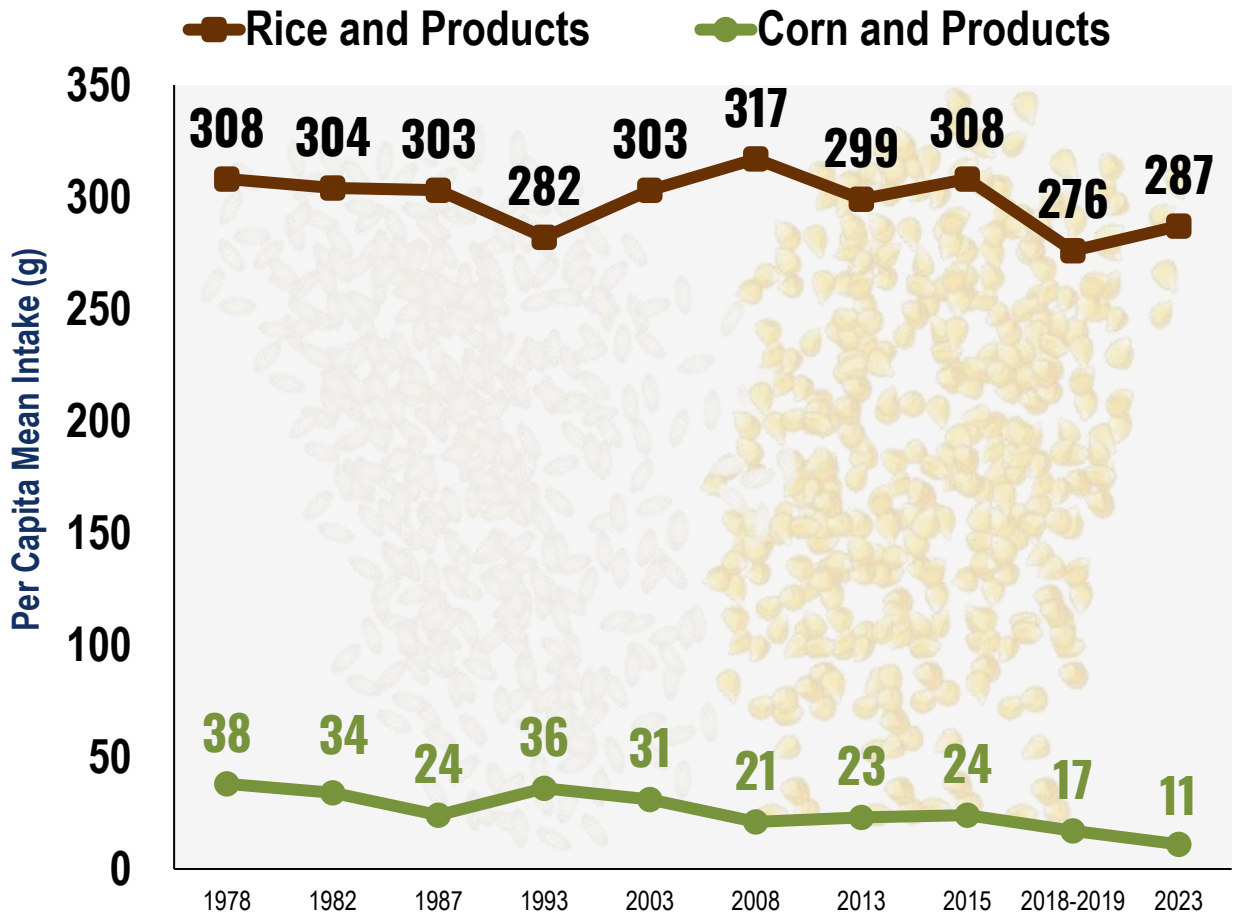


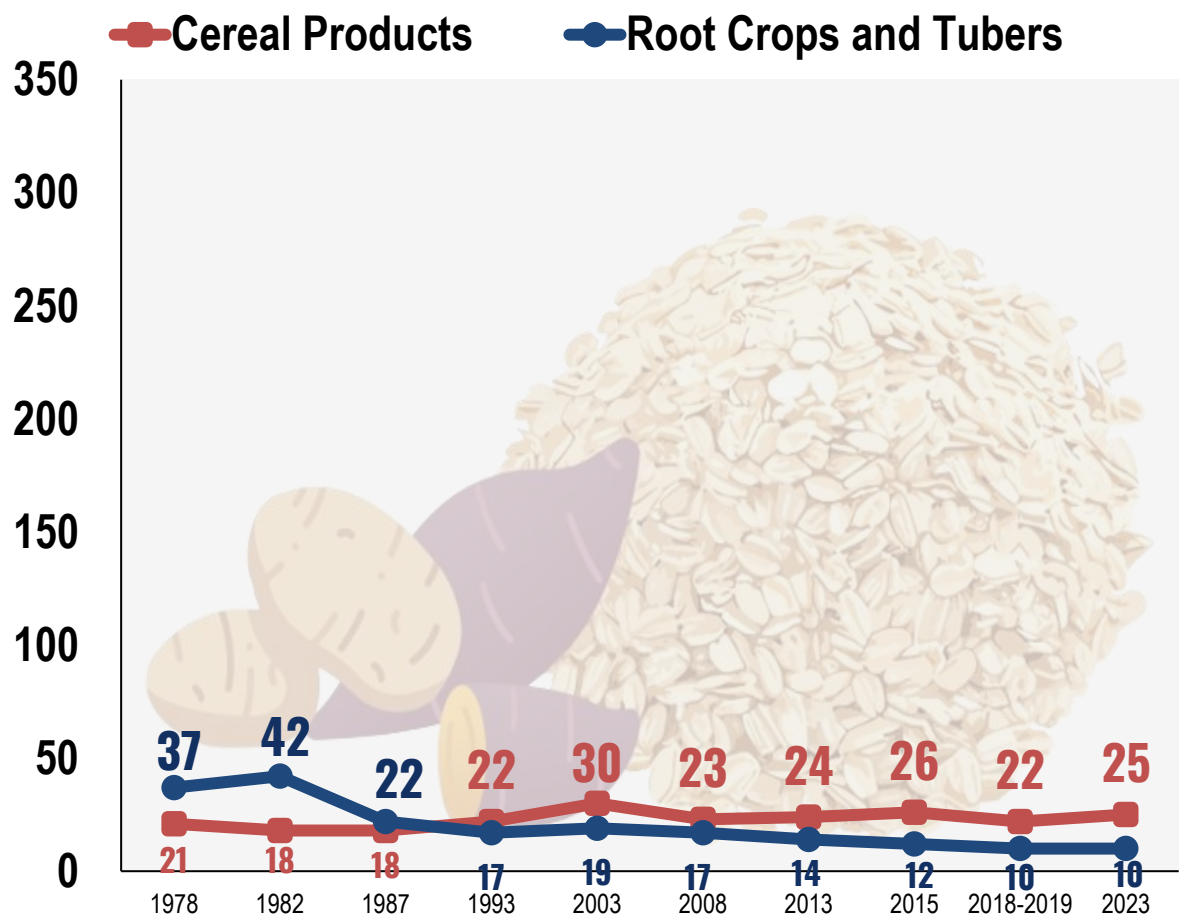
* Significantly different at $p < 0.05$ level of significance

DOST-FNRI

Trends in mean one-day per capita food consumption among Filipino households: Philippines, 1978 to 2023

ENERGY-GIVING FOODS

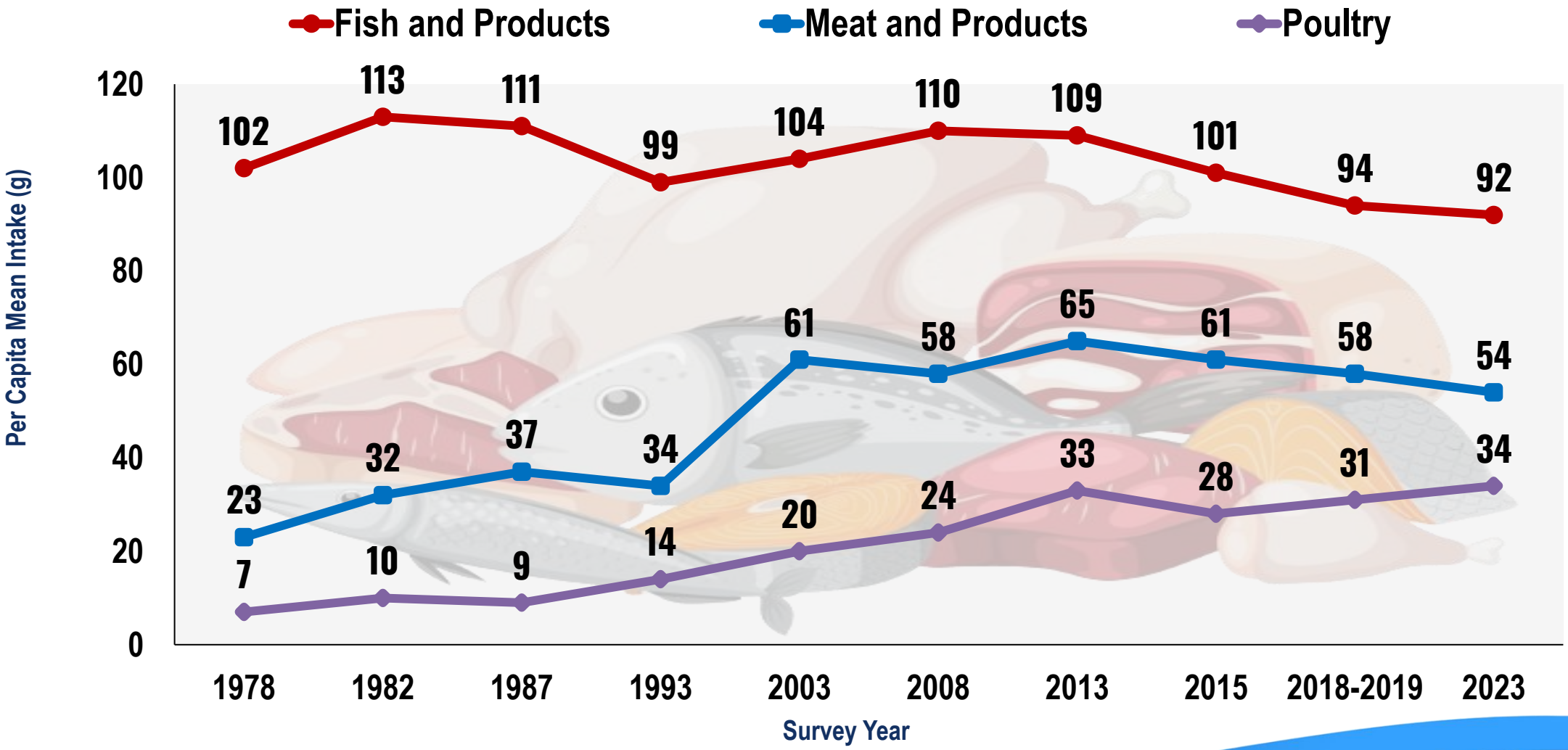




Note: No statistical difference was observed between 2018/19 and 2023

Trends in mean one-day per capita food consumption among Filipino households: Philippines, 1978 to 2023

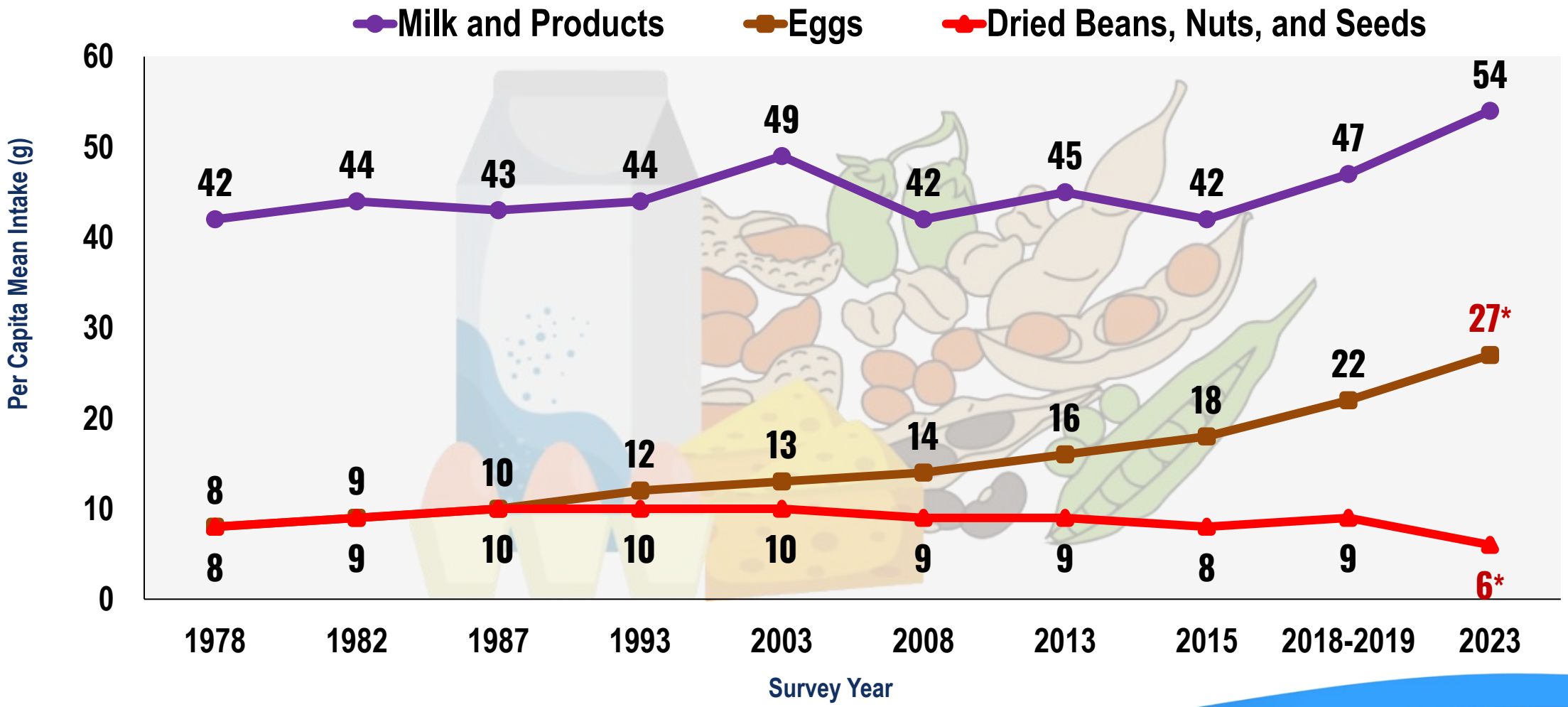
BODY-BUILDING FOODS



Note: No statistical difference was observed between 2018/19 and 2023

Trends in mean one-day per capita food consumption among Filipino households: Philippines, 1978 to 2023

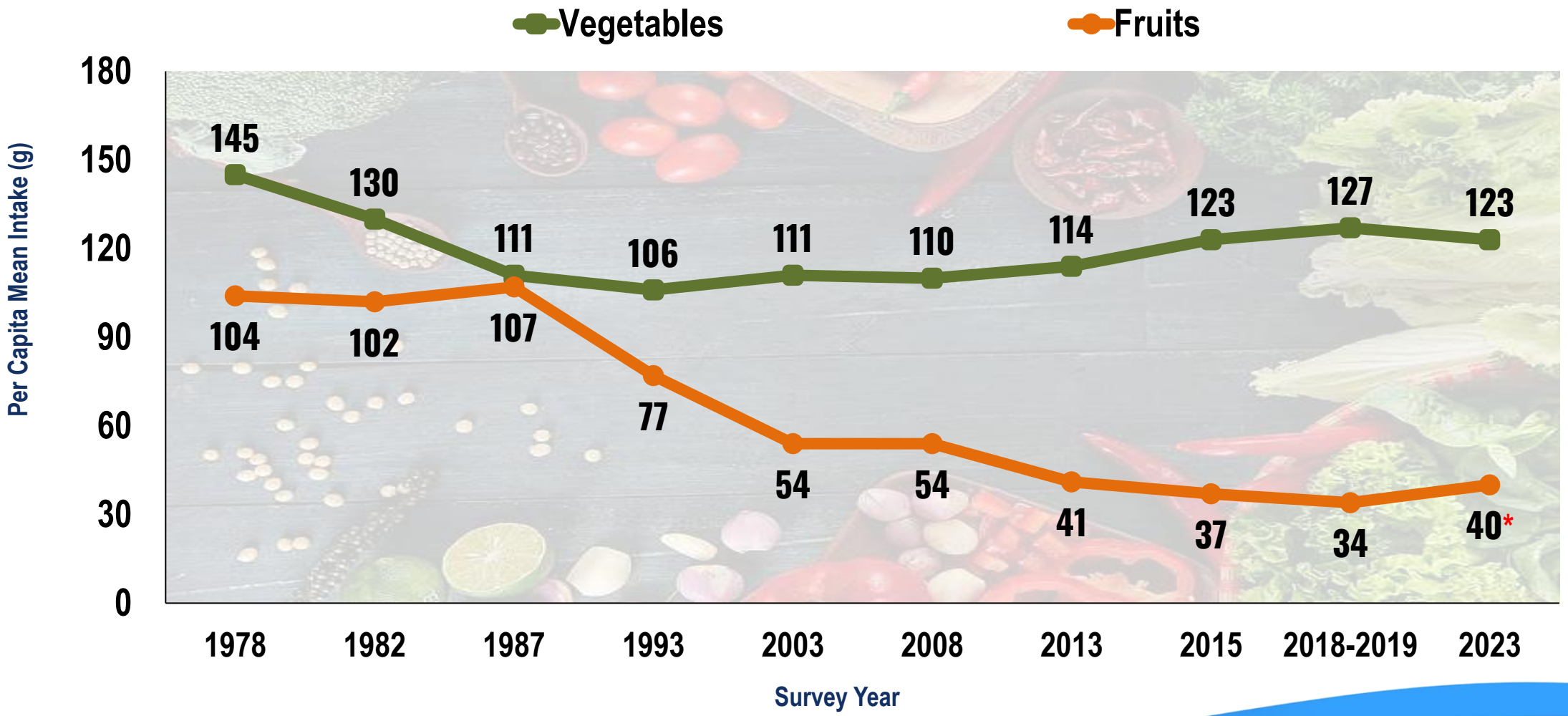
BODY-BUILDING FOODS



* Significantly different at p < 0.05 level of significance

Trends in Mean One-day per Capita Food Consumption among Filipino Households: Philippines, 1978 to 2023

BODY-REGULATING FOODS



* Significantly different at p < 0.05 level of significance

DOST-FNRI

FOOD CONSUMPTION SURVEY





Plate waste

refers to the **edible portions of food which are left on the dining table;** or

on the plates after the family has finished eating and are usually given to pets or discarded.

Household plate waste and percent contribution of the most wasted food groups : Philippines, 2023

Most Wasted Foods:

 Rice

 Vegetables

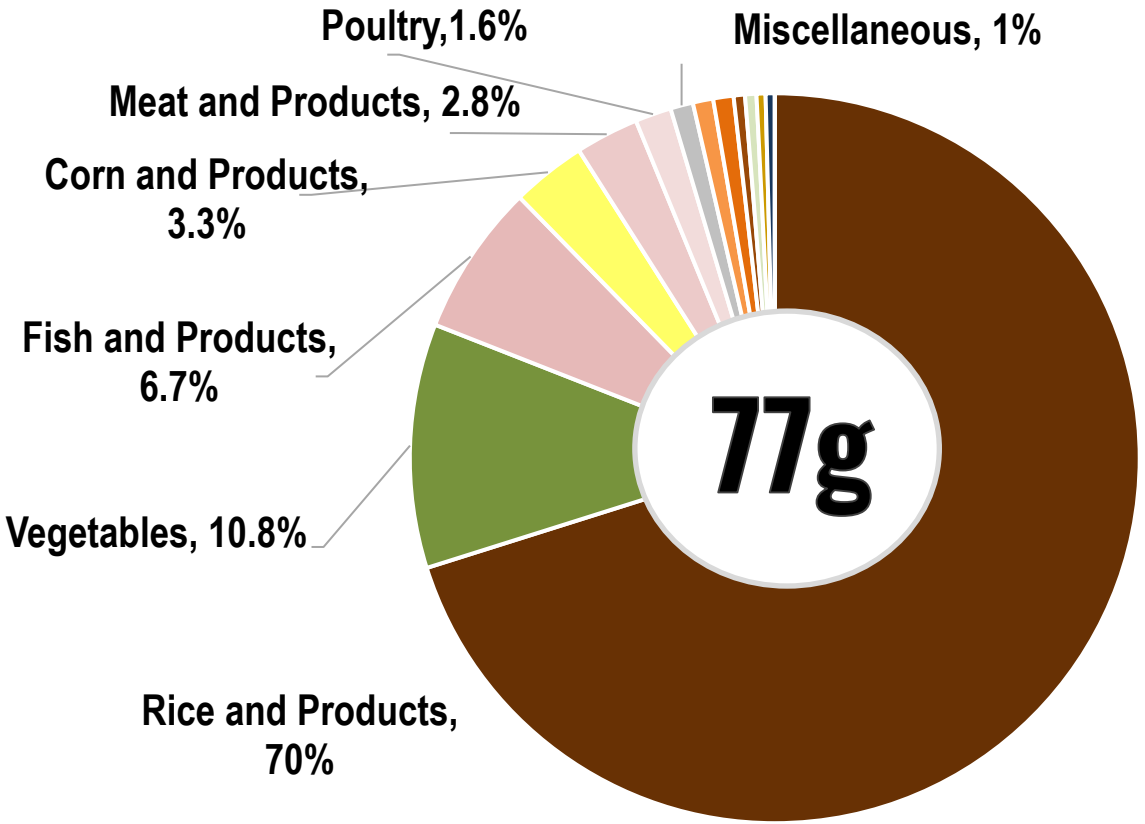
 Fish & Products



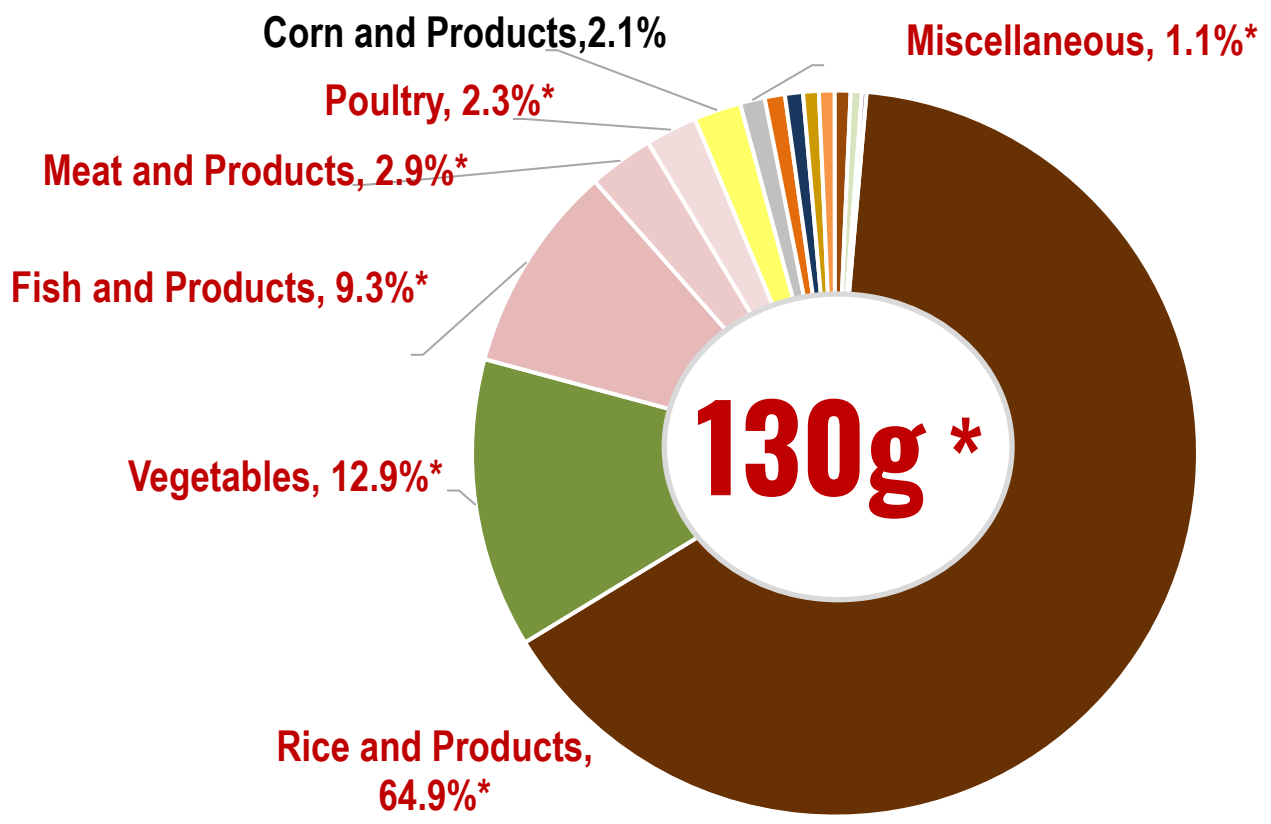
Note: All other Food Groups without values have ≤1.0%

Household plate waste and percent contribution of food groups to total food waste: Philippines, 2018-2019, 2023

2018-2019



2023



* Significantly different at $p < 0.05$ level of significance

Note: All other Food Groups without values have ≤1.0%

Mean one-day household plate waste and percent contribution of the wasted food groups, by place of residence: Philippines, 2023

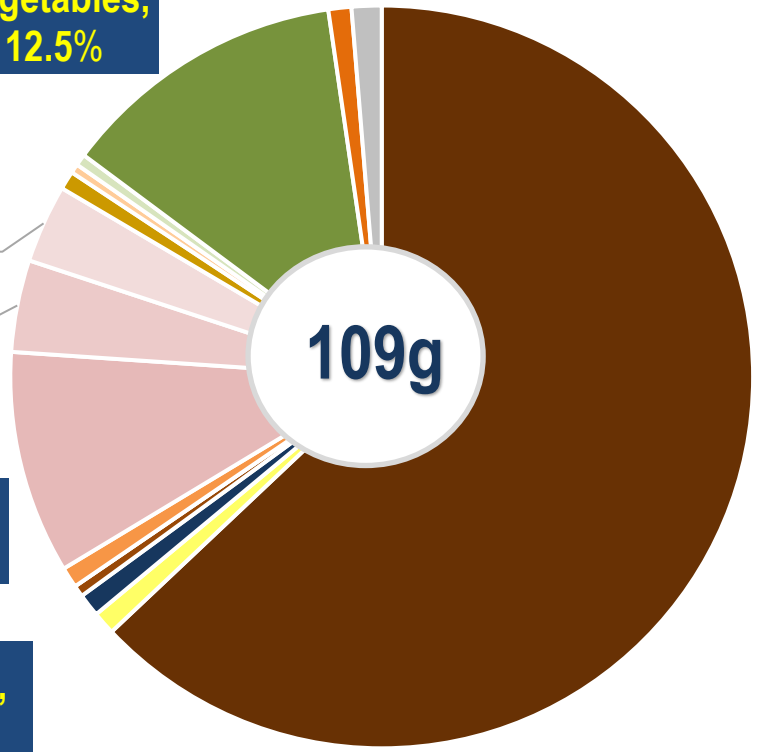
Most Wasted Foods:

 Rice

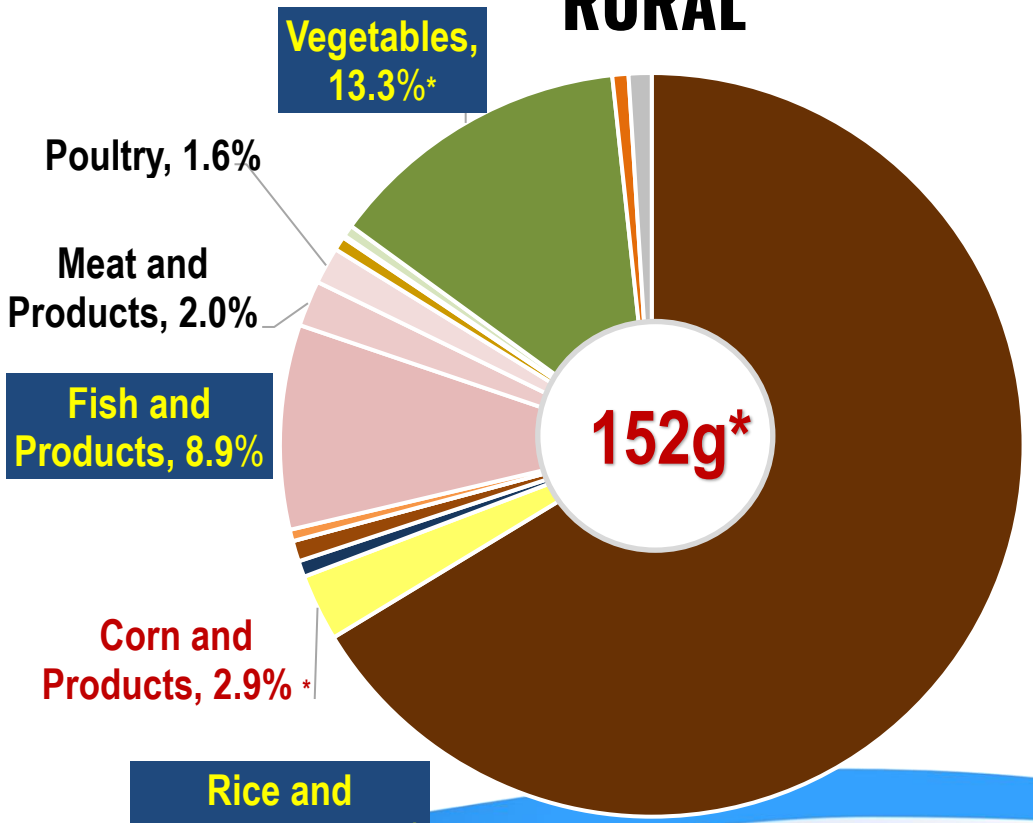
 Vegetables

 Fish & Products

URBAN



RURAL



* Significantly different at p < 0.05 level of significance

Note: All other Food Groups without values have ≤1.0%

Highlights of the 2023 NNS FCS: Household Level

Overall Filipino Diet

- The average daily food intake for Filipinos is **largely composed of rice, vegetables, and fish**, contributing 35.7%, 14.5%, and 10.8% of the daily household intake, respectively.
  - Rice, in particular, is a staple food, providing the majority (58.0%) of energy in the Filipino diet.
- Intake of **fruits**, and other plant source foods namely, **corn, root crops and tubers, and dried beans/nuts/legumes** are still low.

Dietary Adequacy

- The proportion of households meeting their required energy, protein, iron, niacin, and riboflavin **significantly increased** between 2018/19 and 2023.

Highlights of the 2023 NNS FCS: Household Level

By Wealth Quintile

- **Rice and products** play a significant role in the diets of poorer households compared to richer households, suggesting a higher reliance on rice as a staple food among lower income groups.

- **Fish, meat, and poultry** contribute a higher percentage to the daily food intake of richer households compared to poorer households, suggesting that animal-based protein sources are more readily available and consumed by higher socioeconomic groups.

- **Fruit and vegetable intake** generally did not differ significantly across different wealth quintiles, except that the poorest households showed higher vegetable intake compared to other income groups.

Highlights of the 2023 NNS FCS: Household Level

By Place of Residence

- **Urban households** have higher food intake of **cereal products, meat and products, poultry, and milk** than rural households.
- **Rural households** have higher intake of **rice and products, fish and products, vegetables, corn and products, and sugar and sweets,** than urban households.

Plate Waste

- **Rice and products, vegetables, and fish** were among the most wasted foods among households in 2023.
  - A substantial increase in plate waste was noted, with households discarding **53 grams more food on average between 2018-2019 and 2023**.
  - **Rural households** were discarding more foods than urban households, particularly for **rice, corn and products, and vegetables**.

2023 NATIONAL NUTRITION SURVEY (NNS)

FOOD CONSUMPTION SURVEY: INDIVIDUAL LEVEL



INDIVIDUAL FOOD CONSUMPTION SURVEY (IFCS)

The **IFCS** provides data on the intake, food quantities, and nutrient adequacies of the different age or population groups.



Infants and Young Children



School-age Children



Adolescents



Adults



Older persons



Pregnant Women



Lactating Mothers

INDIVIDUAL FOOD CONSUMPTION SURVEY (IFCS)

SIGNIFICANCE

The IFCS provides information vital for:

- **Shaping** food and health **guidelines**
- **Planning** and **evaluating** programs
- **Understanding** food consumption patterns
- **Developing strategies** for health promotion and disease prevention
- **Track the progress** in addressing SDG targets by 2030, particularly Goals 2 and 3











METHOD OF DATA COLLECTION

24-hour Food Recall

- Face-to-face interview
- Two (2) Non-Consecutive Days 24-hr Food Recall







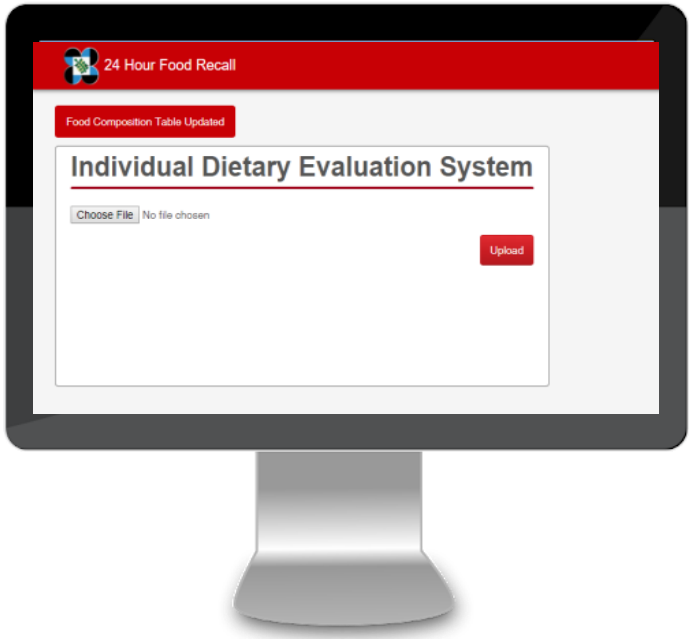


DATA PROCESSING & ANALYSIS

Individual Dietary Evaluation System (IDES)

Developed by FNRI was used to evaluate the energy and nutrient content of foods consumed by each individual.

This system contains the data of the expanded Food Composition Table (FCT).



COVERAGE BY POPULATION GROUP

n=57,272

Infants and Preschool Children, 6 months to 5 years old



6–11 m (n=424)
1–2 y (n=1,725)
3–5 y (n=3,126)

n=5,275

School-age Children, 6 to 12 years old



6–9 y (n=5,189)
10–12 y (n=4,649)

n=9,838

Adolescents, 13 to 18 years old



13–15 y (n=4,313)
16–18 y (n=3,452)

n=7,765

Adults, 19 to 59 years old



19–29 y (n=5,916)
30–49 y (n=11,965)
50–59 y (n=6,409)

n=24,290

Older Persons, ≥60 years old



60–69 y (n=5,247)
≥70 y (n=3,195)

n=8,442

Pregnant Women



n=349

Lactating Mothers



n=1,313

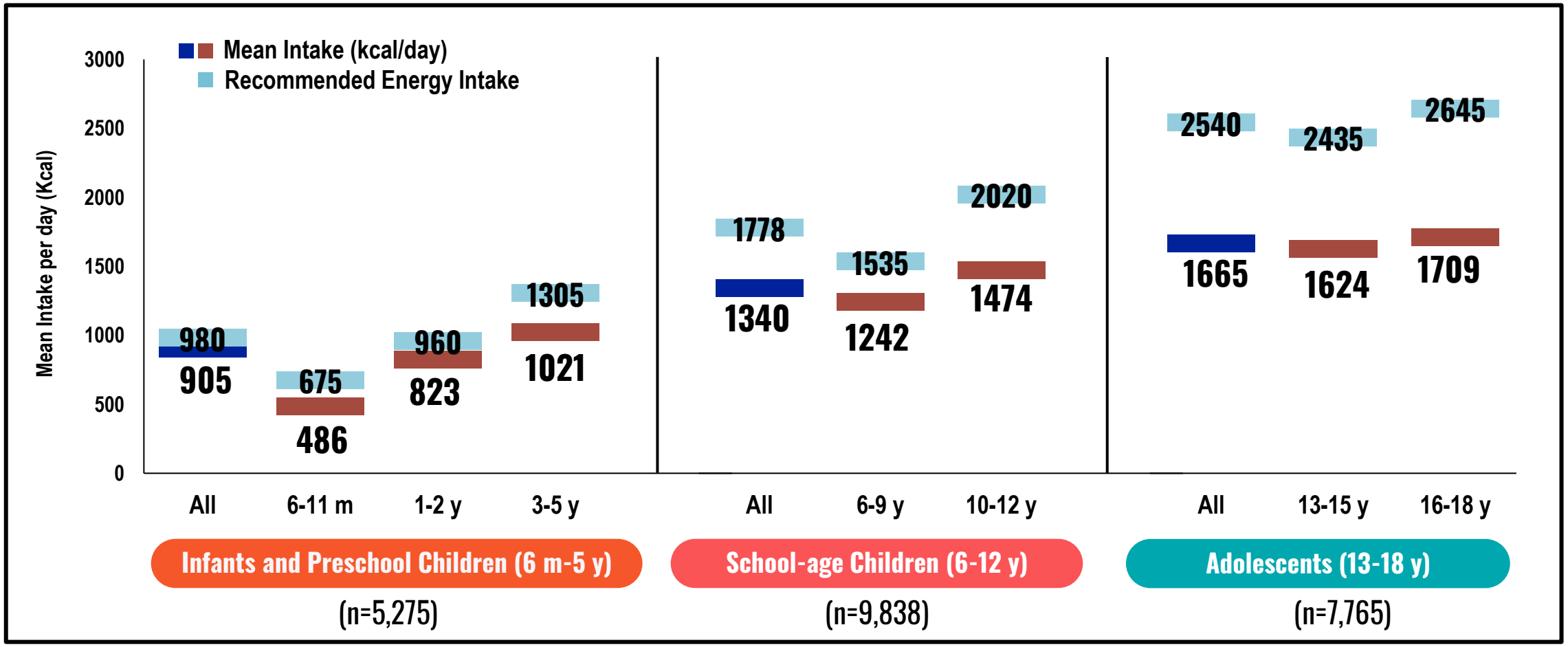
INDIVIDUAL FOOD CONSUMPTION

RESULTS



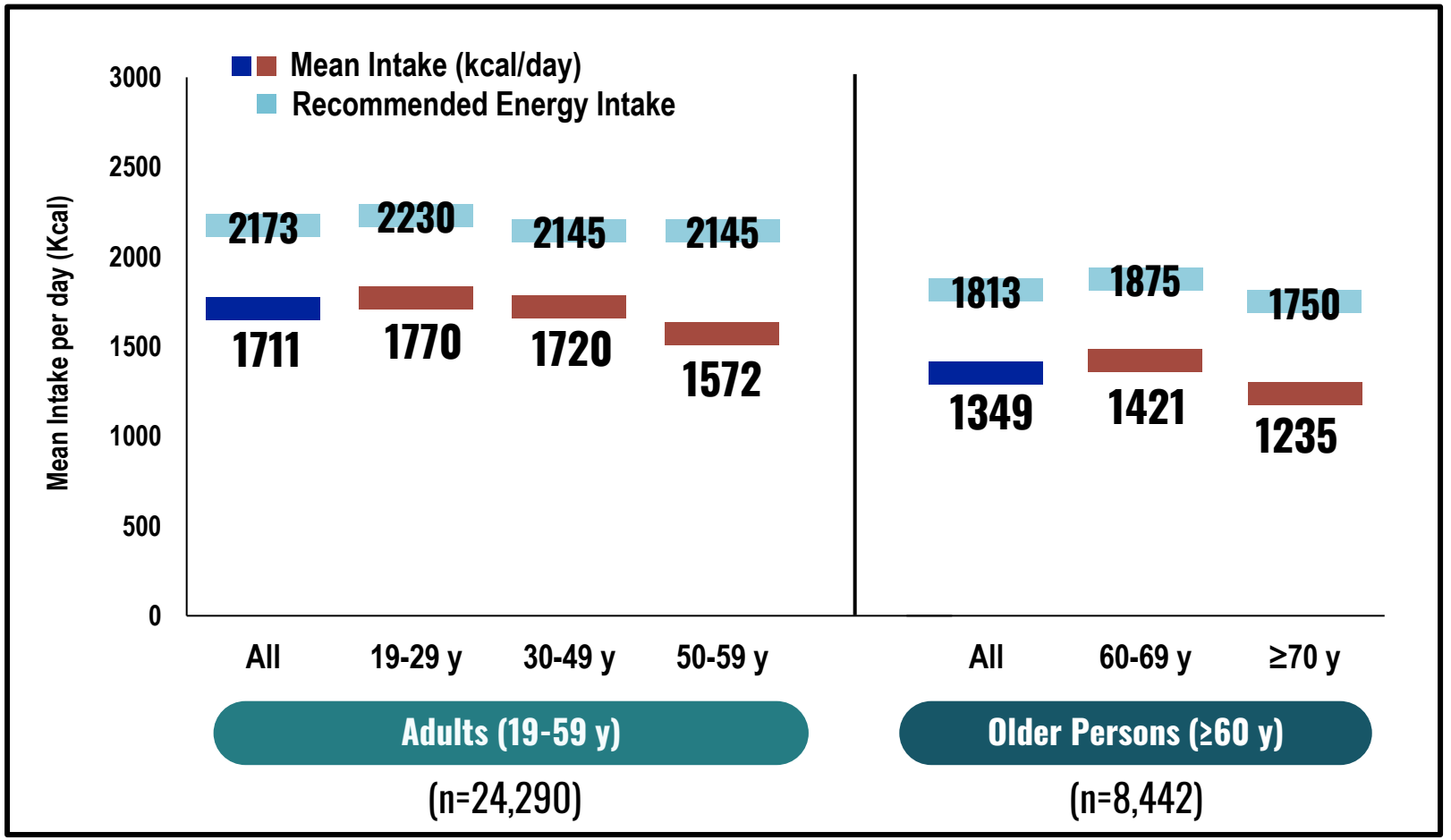
ENERGY AND NUTRIENT INTAKE AND INADEQUACY and MAJOR FOOD SOURCES

Mean ENERGY INTAKE and corresponding energy recommendations* by age and population group: Philippines, 2023



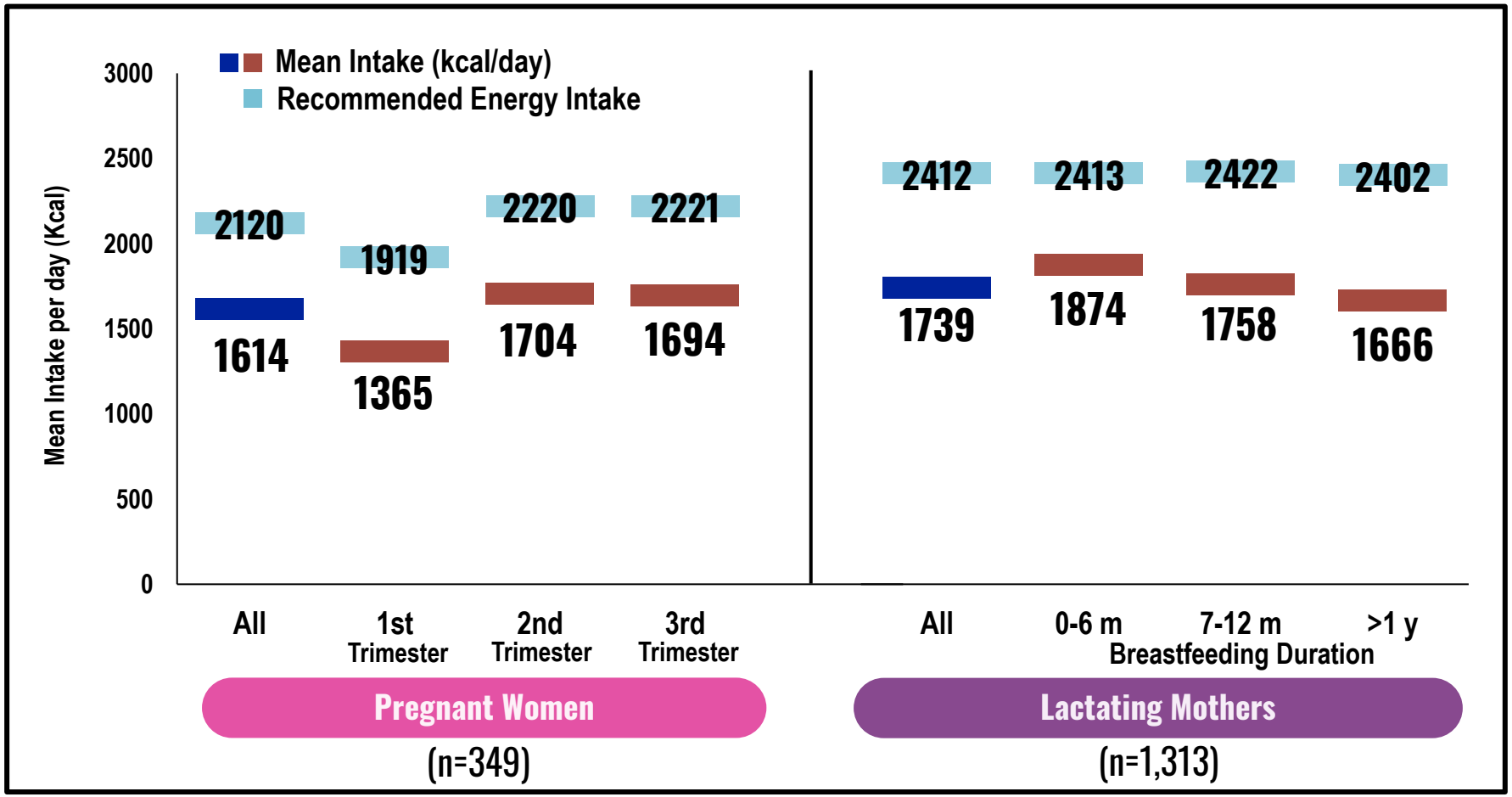
*Based on the Recommended Energy Intake (REI) from the 2015 Philippine Dietary Reference Intakes (PDRI)

Mean ENERGY INTAKE and corresponding energy recommendations* by age and population group: Philippines, 2023



*Based on the Recommended Energy Intake (REI) from the 2015 Philippine Dietary Reference Intakes (PDRI)

Mean ENERGY INTAKE and corresponding energy recommendations* by age and population group: Philippines, 2023



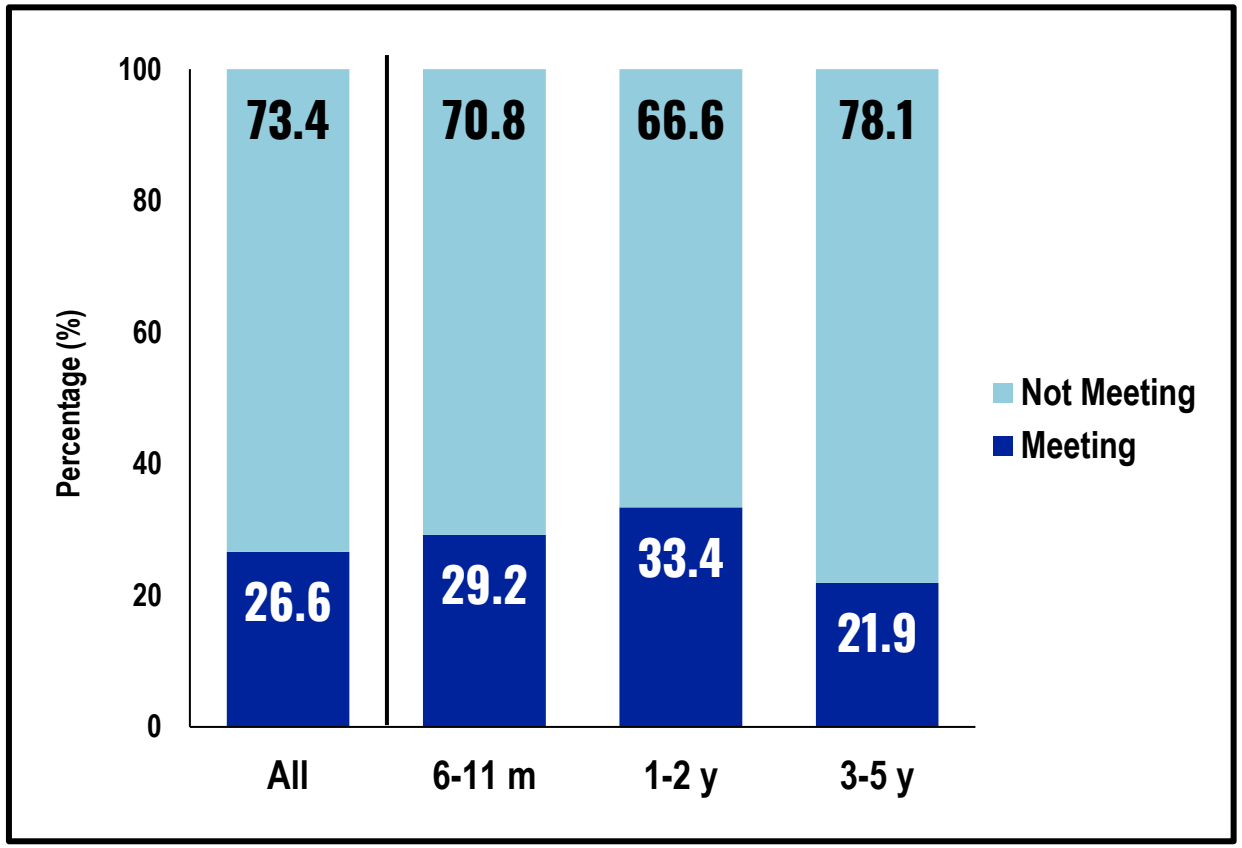
*Based on the Recommended Energy Intake (REI) from the 2015 Philippine Dietary Reference Intakes (PDRI)

INFANTS & PRESCHOOL CHILDREN (6 MONTHS TO 5 YEARS OLD)





Percentage of infants and preschool children, MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=5,275)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

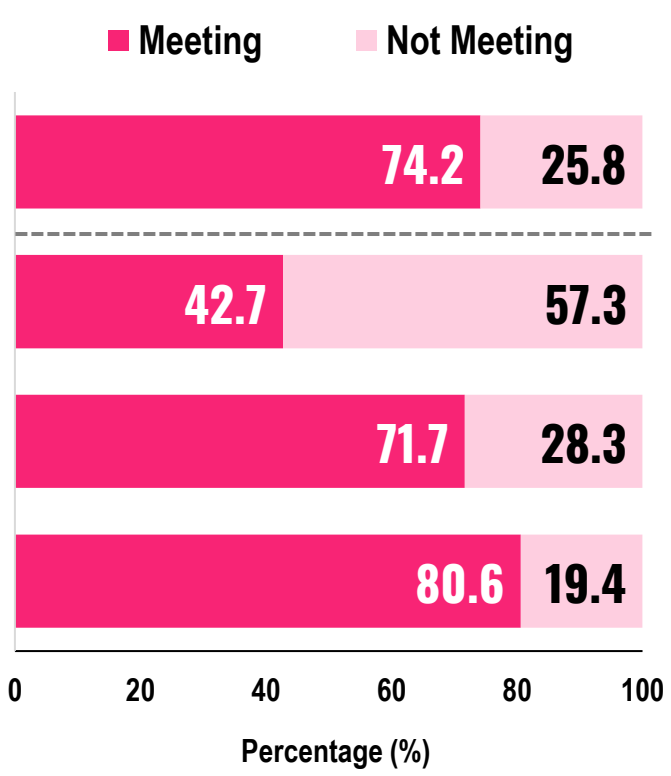
1	Well-milled Rice	29.3%
2	Instant Powdered Filled Milk	13.4%
3	Growing-up Milk Powder (1-3 y)	6.8%
4	Salted Cracker	2.9%
5	Infant Formula Milk[a]	2.4%

°*In raw edible portion (REP) form*
[a]*Major source among 6-11 m (43.5%)*

Mean PROTEIN INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=5,275)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	15	28
6-11 m	14	20
1-2 y	15	27
3-5 y	18	31

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

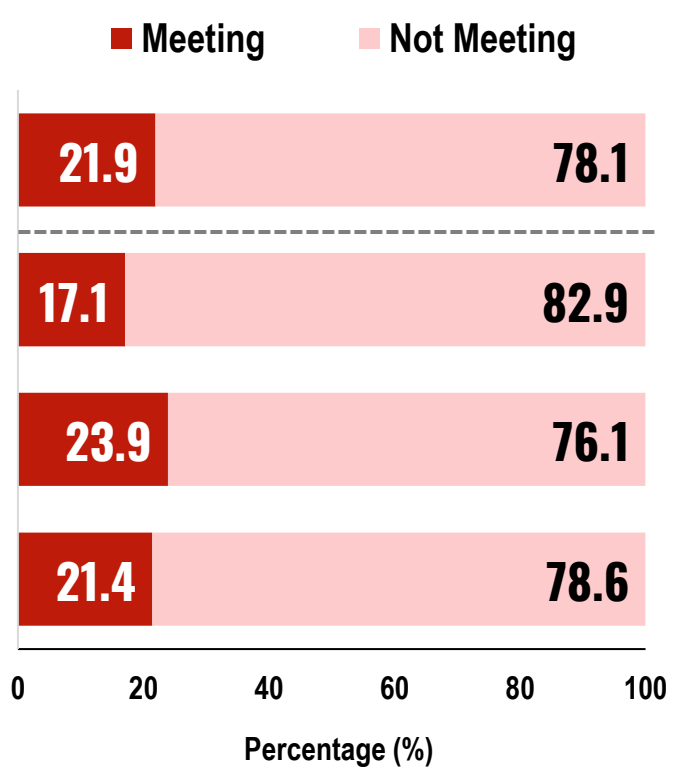
1.  Instant Powdered Filled Milk — 22.1%
2.  Well-milled Rice — 19.4%
3.  Growing-up Milk Powder (1-3 y) — 6.7%
4.  Egg — 4.9%
5.  Infant Formula Milk[a] — 3.6%

°*In raw edible portion (REP) form*
[a] *Major source among 6-11 m (51.0%)*

Mean IRON INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=5,275)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	7.3	5.2
6-11 m	7.8	4.1
1-2 y	6.7	4.9
3-5 y	7.5	5.5

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

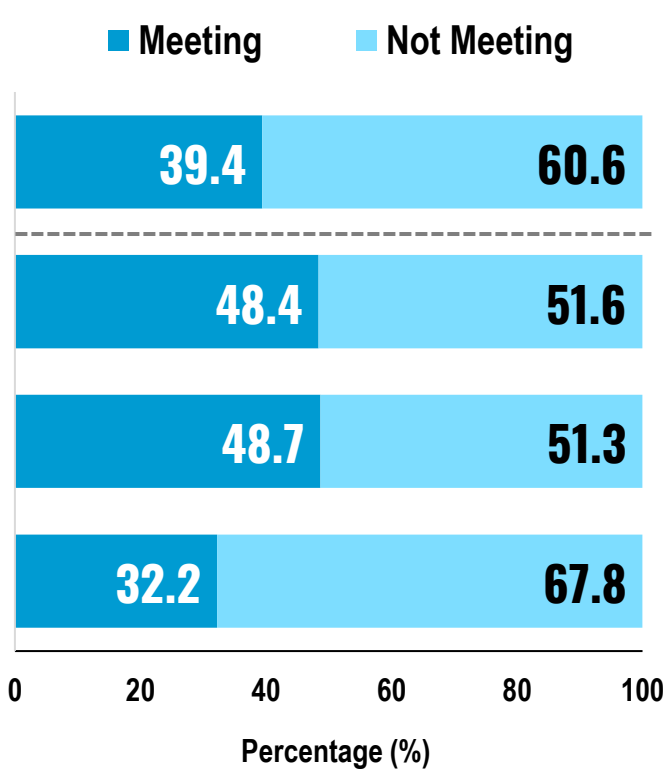
1	Growing-up Milk Powder (1-3 y)	14.8%
2	Well-milled Rice	14.4%
3	Infant Formula Milk[a]	5.2%
4	Chocolate-flavor Powdered Drink	5.0%
5	Hotdog	3.8%

°*In raw edible portion (REP) form*
[a]*Major source among 6-11 m (63.9%)*

Mean CALCIUM INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=5,275)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	0.44	0.49
6-11 m	-	0.62
1-2 y	0.44	0.59
3-5 y	0.44	0.41

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

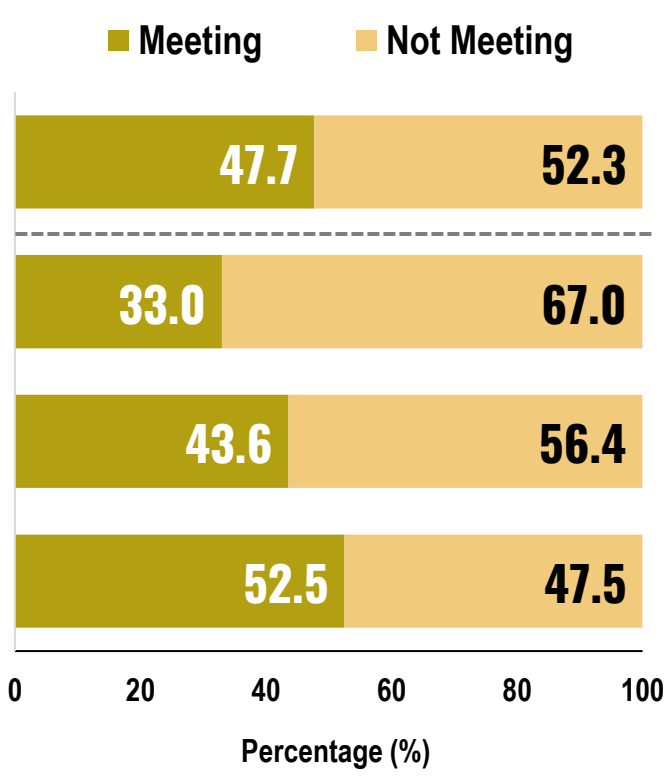
1.  Instant Powdered Filled Milk — 41.5%
2.  Growing-up Milk Powder (1-3 y) — 14.3%
3.  Infant Formula Milk[a] — 7.9%
4.  Growing-up Milk Powder (3-5 y) — 6.0%
5.  Well-milled Rice — 4.1%

°In raw edible portion (REP) form
[a]Major source among 6-11 m (60.8%)

Mean VITAMIN A INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=5,275)

Age	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	203	325
6-11 m	190	258
1-2 y	200	330
3-5 y	220	333

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

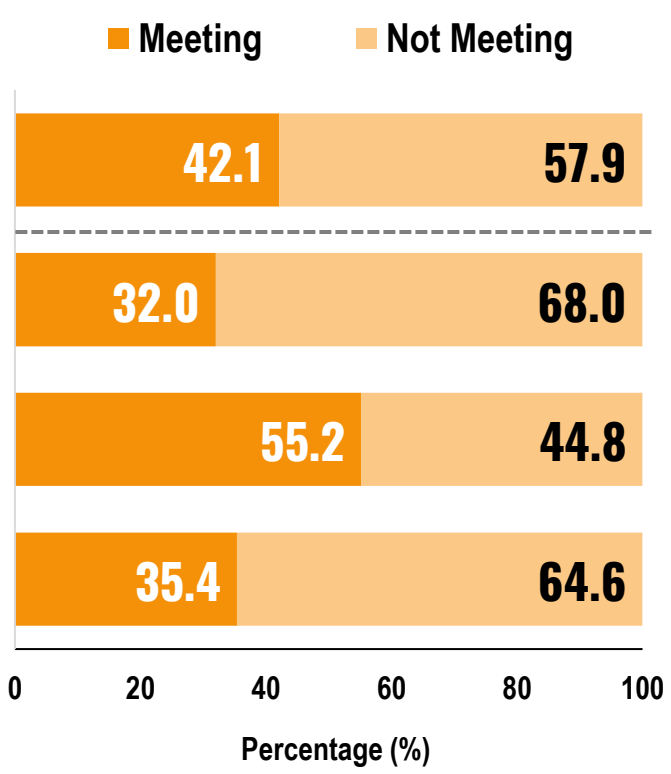
1.  Instant Powdered Filled Milk — 55.4%
2.  Egg — 7.0%
3.  Growing-up Milk Powder (3-5 y) — 4.1%
4.  Chocolate Cake — 3.0%
5.  Infant Formula Milk[a] — 2.7%

°*In raw edible portion (REP) form*
[a] *2nd major source among 6-11 m (32.4%)*

Mean VITAMIN C INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=5,275)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	14	23
6-11 m	-	27
1-2 y	12	31
3-5 y	17	18

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

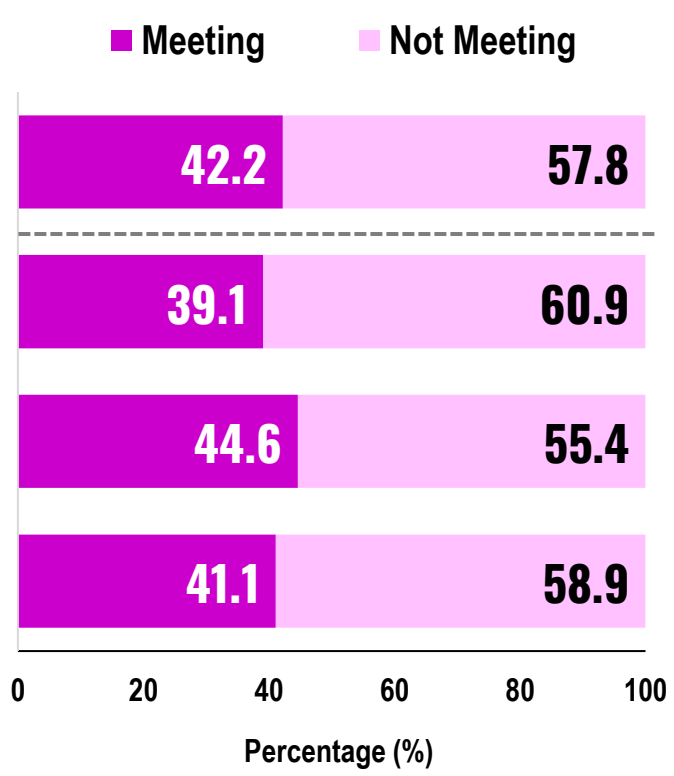
1.  Growing-up Milk Powder (1-3 y) — 31.4%
2.  Instant Powdered Filled Milk — 14.3%
3.  Infant Formula Milk[a] — 7.5%
4.  Growing-up Milk Powder (3-5 y) — 7.1%
5.  Chocolate-flavor Powdered Drink — 4.5%

°*In raw edible portion (REP) form*
[a] *Major source among 6-11 m (61.7%)*

Mean THIAMIN INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	0.4	0.5
6-11 m	0.3	0.4
1-2 y	0.4	0.5
3-5 y	0.5	0.5

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

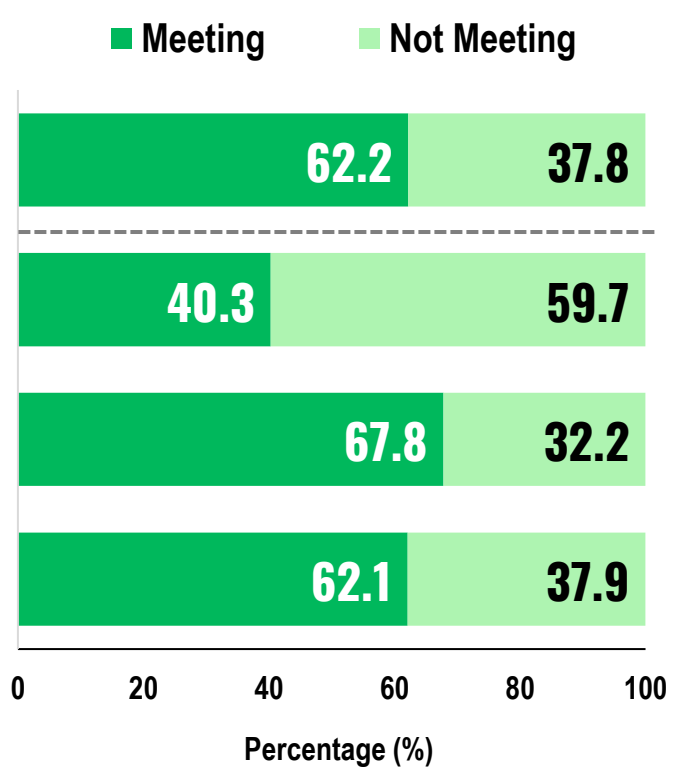
1.  Well-milled Rice — 15.7%
2.  Instant Powdered Filled Milk — 13.1%
3.  Growing-up Milk Powder (1-3 y) — 11.7%
4.  Chocolate-flavor Powdered Drink — 11.4%
5.  Infant Formula Milk[a] — 4.4%

°*In raw edible portion (REP) form*
[a] *Major source among 6-11 m (57.0%)*

Mean RIBOFLAVIN INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	0.4	0.9
6-11 m	0.3	0.6
1-2 y	0.4	1.1
3-5 y	0.5	0.8

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

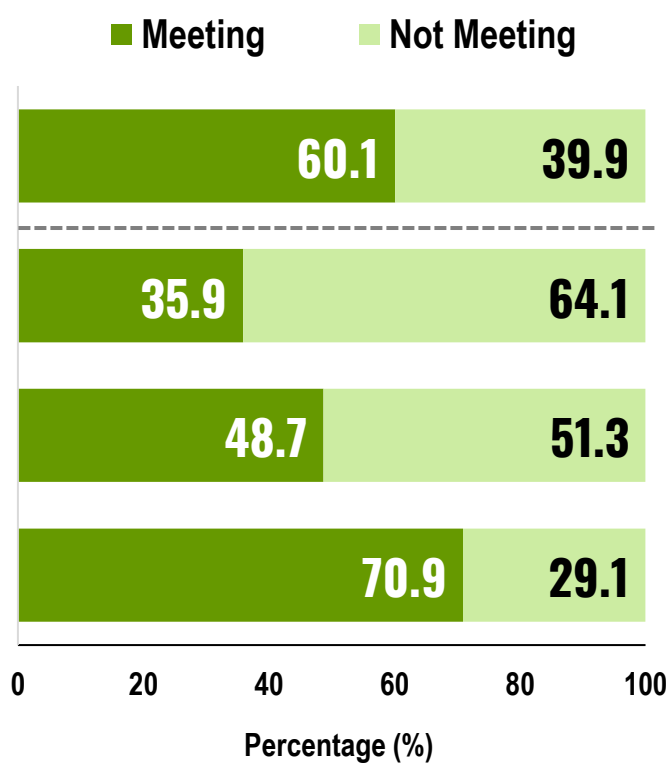
1.  Instant Powdered Filled Milk — 47.2%
2.  Growing-up Milk Powder (1-3 y) — 11.1%
3.  Egg — 5.0%
4.  Well-milled Rice — 4.2%
5.  Chocolate-flavor Powdered Drink — 3.8%

°*In raw edible portion (REP) form*
Note: *Infant formula milk is the 2nd major source among 6-11 m (19.8%)*

Mean NIACIN INTAKE and percentage of infants and preschool children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	4.5	7.0
6-11 m	3.5	5.9
1-2 y	5.0	6.0
3-5 y	5.0	7.8

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

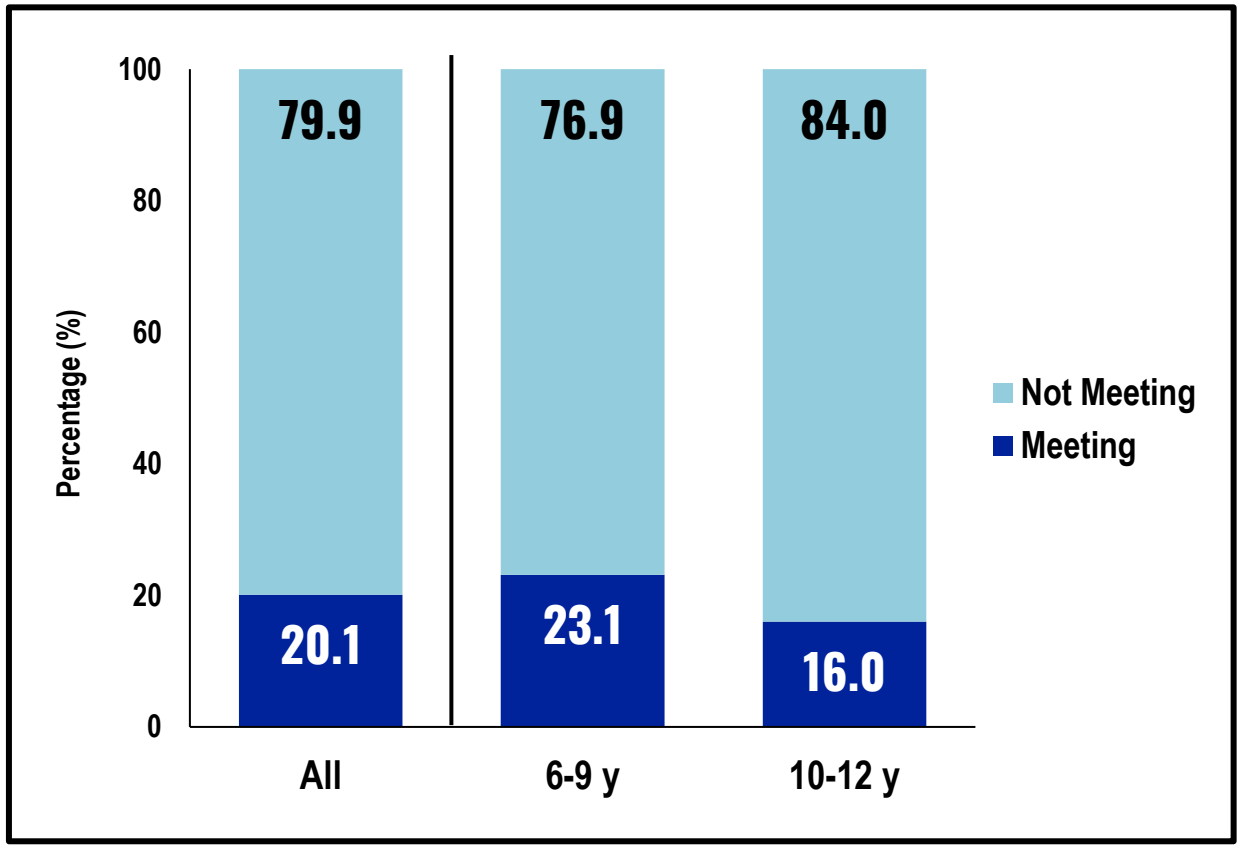
1.  Well-milled Rice — 29.6%
2.  Growing-up Milk Powder (1-3 y) — 10.0%
3.  Infant Formula Milk[a] — 6.5%
4.  Chocolate-flavor Powdered Drink — 4.7%
5.  Chicken Meat — 2.9%

°*In raw edible portion (REP) form*
[a] *Major source among 6-11 m (74.2%)*



SCHOOL-AGE CHILDREN (6 TO 12 YEARS OLD)

Percentage of school-age children MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=9,838)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

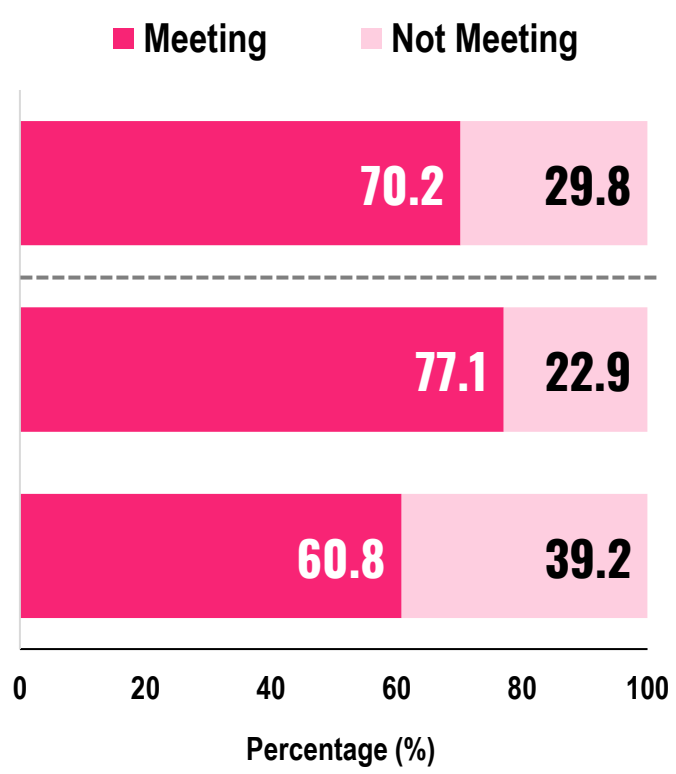
1	Well-milled Rice	49.6%
2	Instant Noodles	2.0%
3	Egg	1.9%
4	Instant Powdered Filled Milk	1.7%
5	*Pandesal*	1.5%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=9,838)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	30	40
6-9 y	24	37
10-12 y	37	45

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

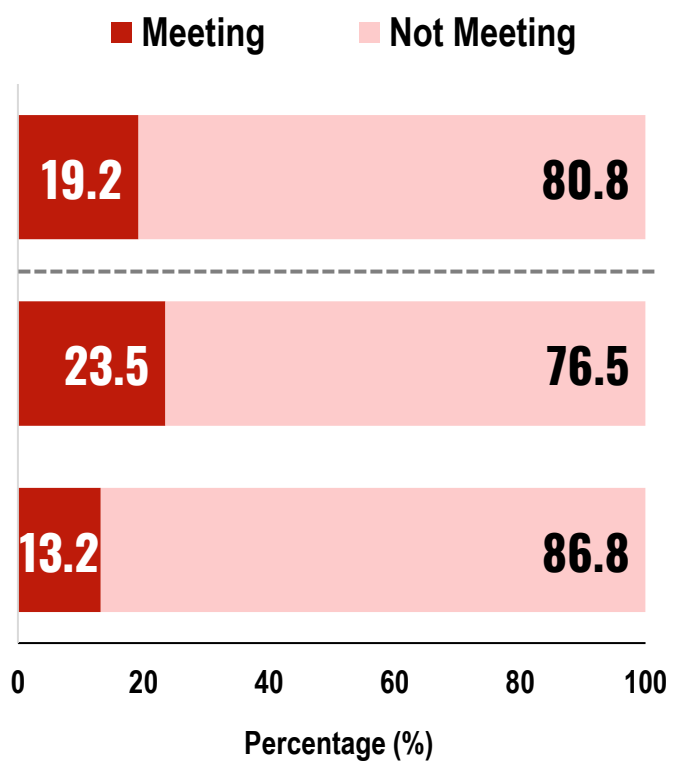
1.  Well-milled Rice — 34.3%
2.  Egg — 5.6%
3.  Chicken Meat — 3.8%
4.  Instant Powdered Filled Milk — 2.9%
5.  Lean Pork Boston Butt (*Kasim*) — 2.6%

°*In raw edible portion (REP) form*

Mean IRON INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=9,838)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	10.8	6.9
6-9 y	8.2	6.4
10-12 y	13.4	7.5

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

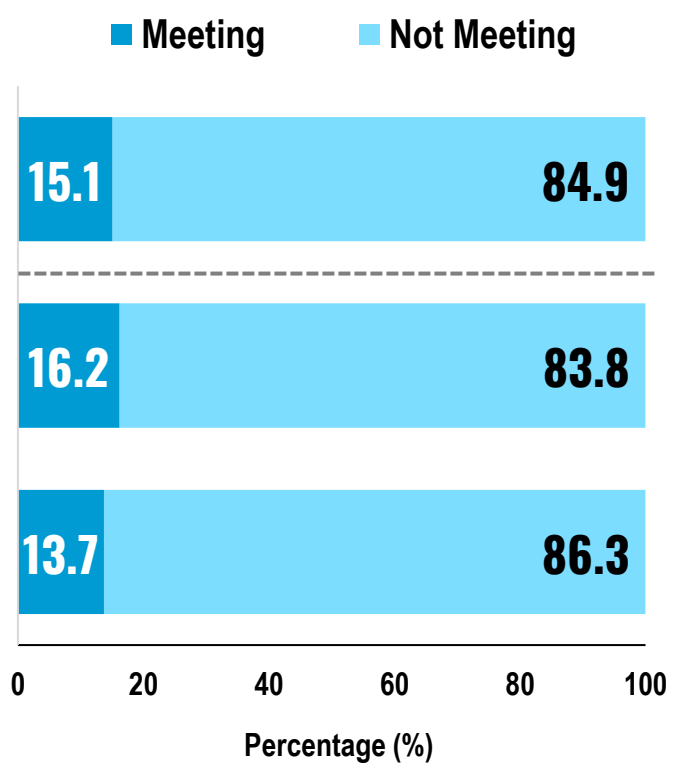
1.  Well-milled Rice — 27.2%
2.  Hotdog — 4.8%
3.  Chocolate-flavor Powdered Drink — 4.7%
4.  Egg — 4.6%
5.  *Pandesal* — 2.6%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=9,838)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	0.44	0.28
6-9 y	0.44	0.28
10-12 y	0.44	0.27

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

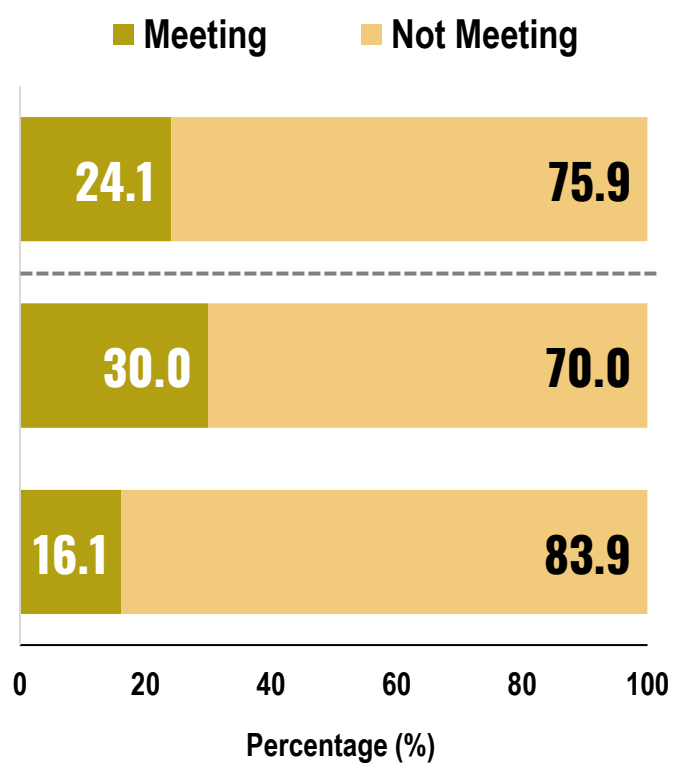
1.  Well-milled Rice — 18.3%
2.  Instant Powdered Filled Milk — 14.0%
3.  Chocolate-flavor Powdered Drink — 2.4%
4.  Chicken Meat — 2.3%
5.  Canned Sardines in Tomato Sauce — 2.2%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=9,838)

Age	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	320	282
6-9 y	271	278
10-12 y	370	288

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

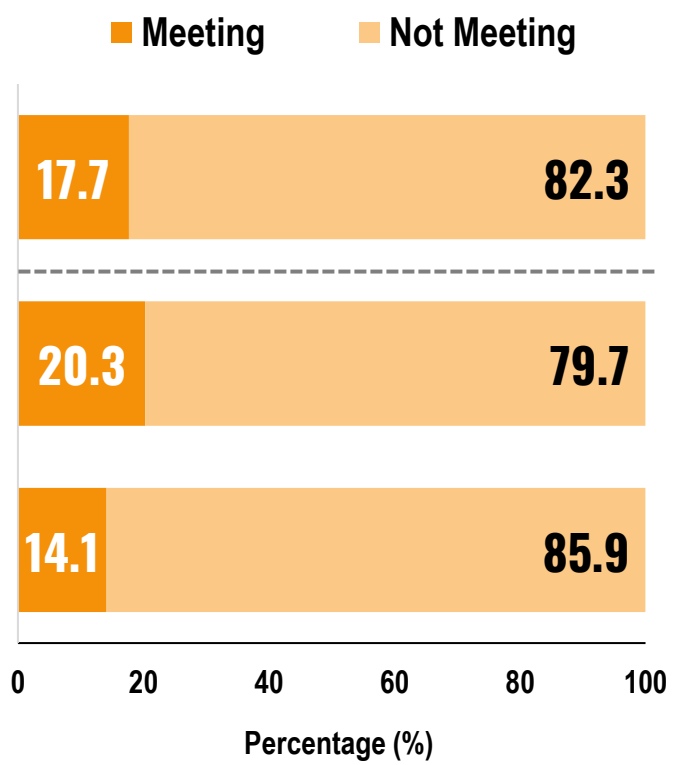
1.  Egg — 13.1%
2.  Instant Powdered Filled Milk — 12.1%
3.  Pork Liver — 11.1%
4.  Chicken Liver — 7.4%
5.  Canned Sardines in Tomato Sauce — 3.8%

°*In raw edible portion (REP) form*

Mean VITAMIN C INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=9,838)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	29	16
6-9 y	23	15
10-12 y	35	18

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

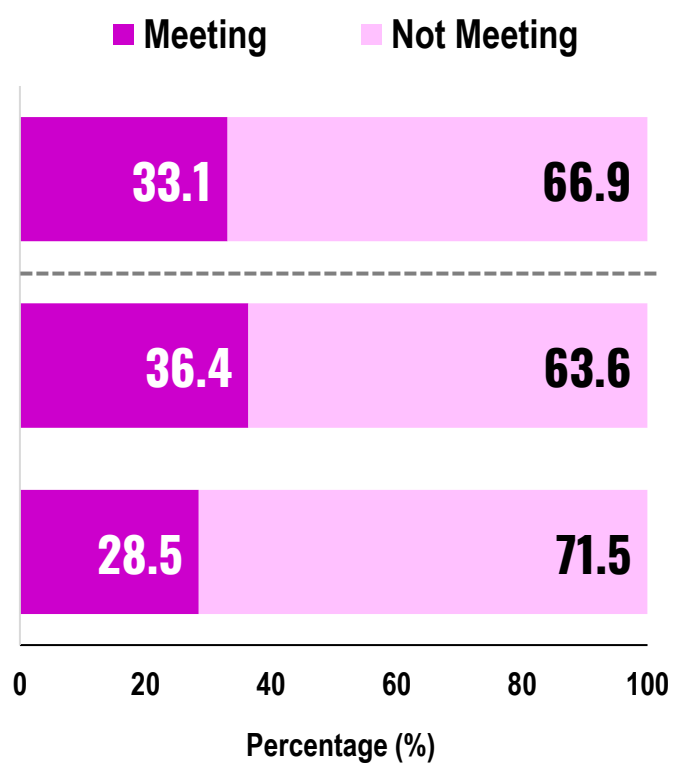
1	Chocolate-flavor Powdered Drink	8.2%
2	Unripe Indian Mango	7.5%
3	*Saba* Banana	5.7%
4	Orange Fruit Juice Drink	5.2%
5	Lemon Tea Drink	4.9%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	0.7	0.6
6-9 y	0.6	0.6
10-12 y	0.8	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

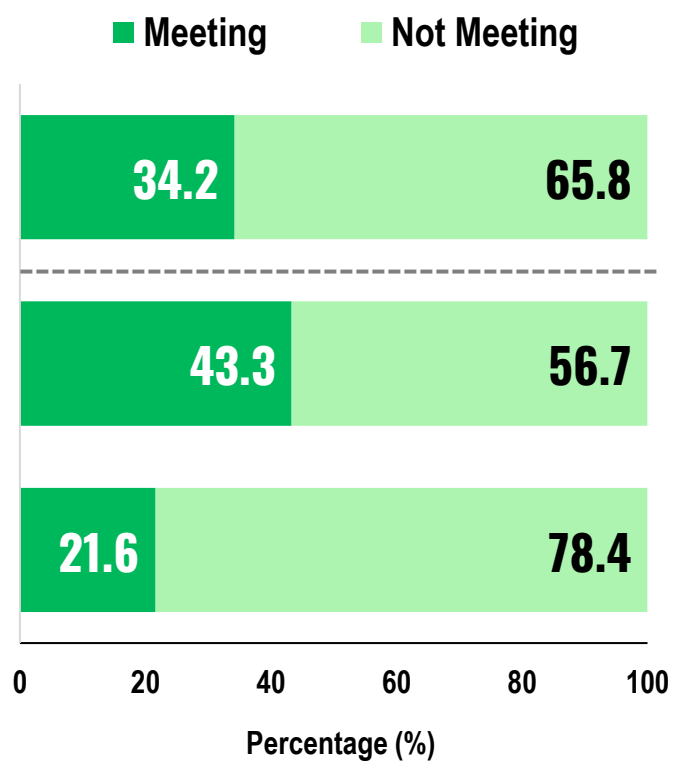
1.  Well-milled Rice — 30.6%
2.  Chocolate-flavor Powdered Drink — 11.3%
3.  Lean Pork Boston Butt (*Kasim*) — 4.4%
4.  Instant Noodles — 4.3%
5.  *Pandesal* — 2.3%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	0.7	0.6
6-9 y	0.6	0.6
10-12 y	0.8	0.6

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

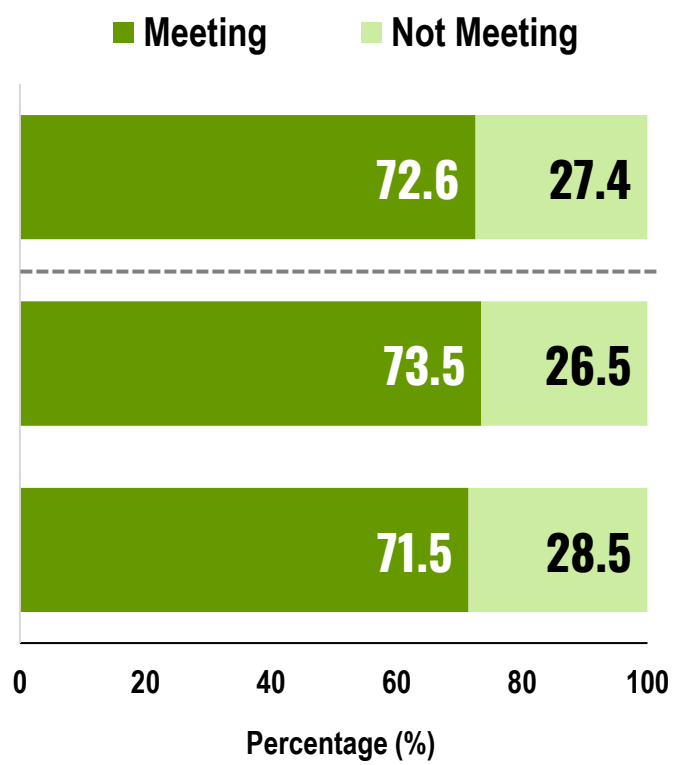
1.  Well-milled Rice — 15.6%
2.  Instant Powdered Filled Milk — 13.1%
3.  Egg — 12.0%
4.  Chocolate-flavor Powdered Drink — 7.1%
5.  Lean Pork Boston Butt (*Kasim*) — 2.7%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of school-age children MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	8.3	12.1
6-9 y	7.0	10.8
10-12 y	10.0	13.9

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

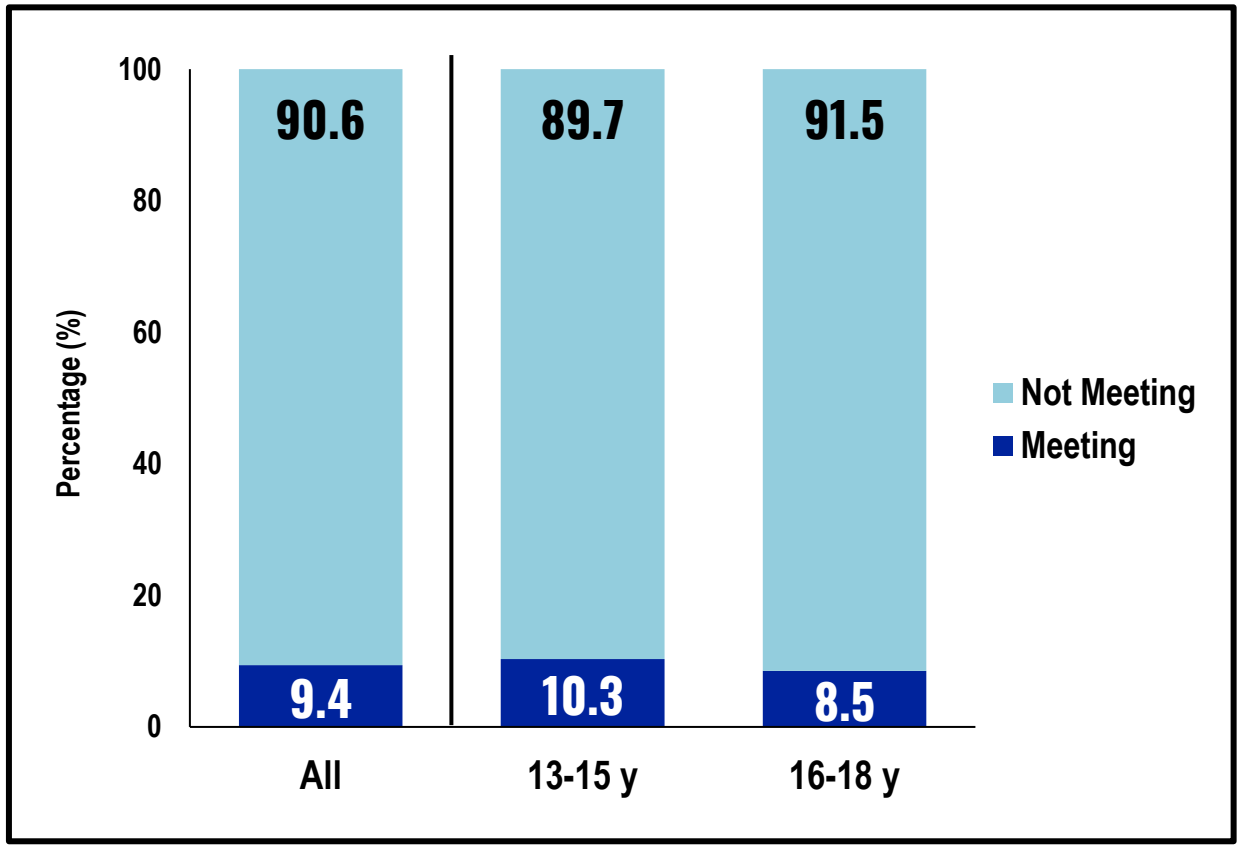
1.  Well-milled Rice — 43.2%
2.  Chicken Meat — 4.1%
3.  Chocolate-flavor Powdered Drink — 3.5%
4.  Chicken Breast — 3.4%
5.  Lean Pork Boston Butt (*Kasim*) — 2.2%

°*In raw edible portion (REP) form*



ADOLESCENTS (13 TO 18 YEARS OLD)

Percentage of adolescents MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=7,765)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

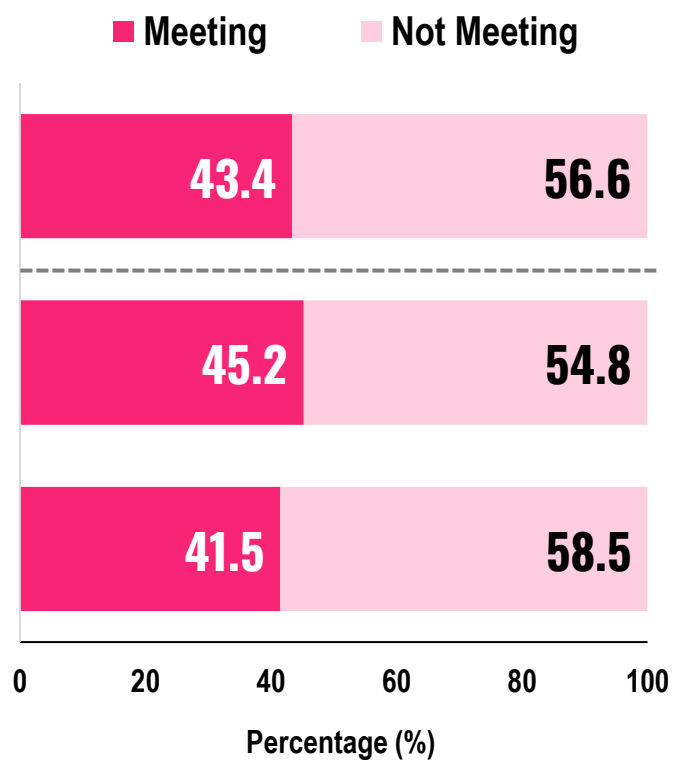
1	Well-milled Rice	55.6%
2	Instant Noodles	2.2%
3	Egg	1.6%
4	Lean Pork Boston Butt (*Kasim*)	1.6%
5	Pork Belly (*Liempo*)	1.6%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=7,765)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	51	51
13-15 y	48	50
16-18 y	54	53

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

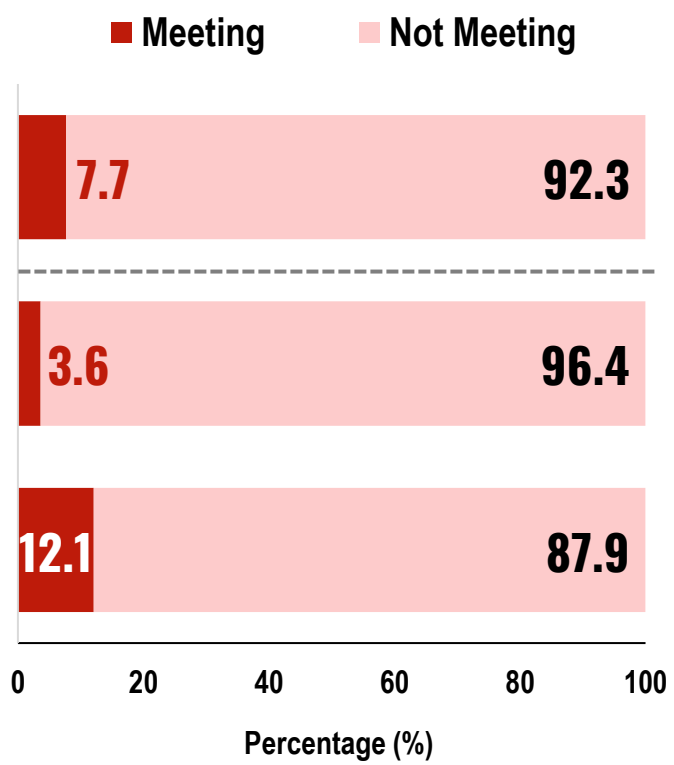
1. Well-milled Rice — 37.4%
2. Egg — 4.6%
3. Chicken Meat — 3.5%
4. Lean Pork Boston Butt (*Kasim*) — 3.2%
5. Chicken Breast — 2.9%

°*In raw edible portion (REP) form*

Mean IRON INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=7,765)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	15.7	8.1
13-15 y	17.3	7.8
16-18 y	14.2	8.3

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

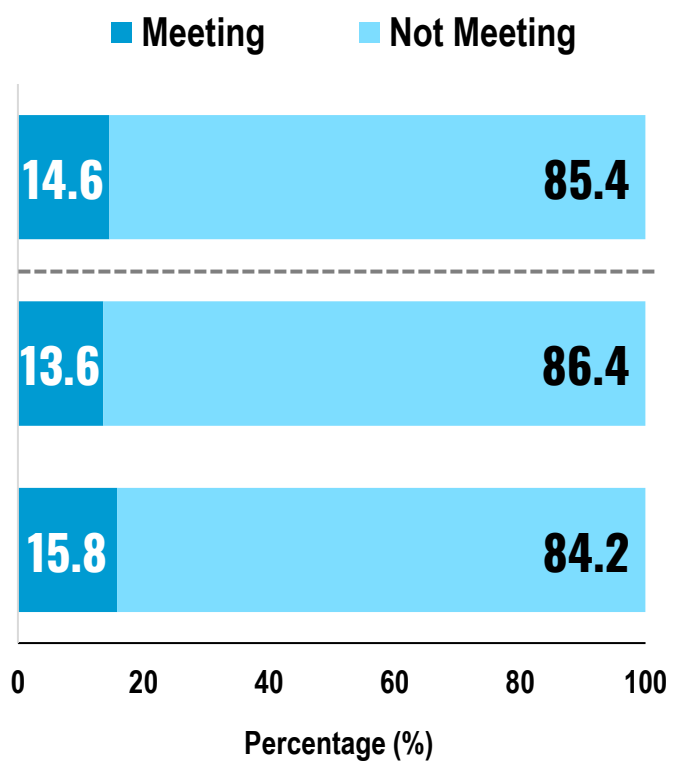
1.  Well-milled Rice — 32.2%
2.  Egg — 4.1%
3.  Hotdog — 3.4%
4.  *Pandesal* — 2.3%
5.  Chocolate-based Cereal Drink — 2.0%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=7,765)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	0.44	0.29
13-15 y	0.44	0.28
16-18 y	0.44	0.29

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

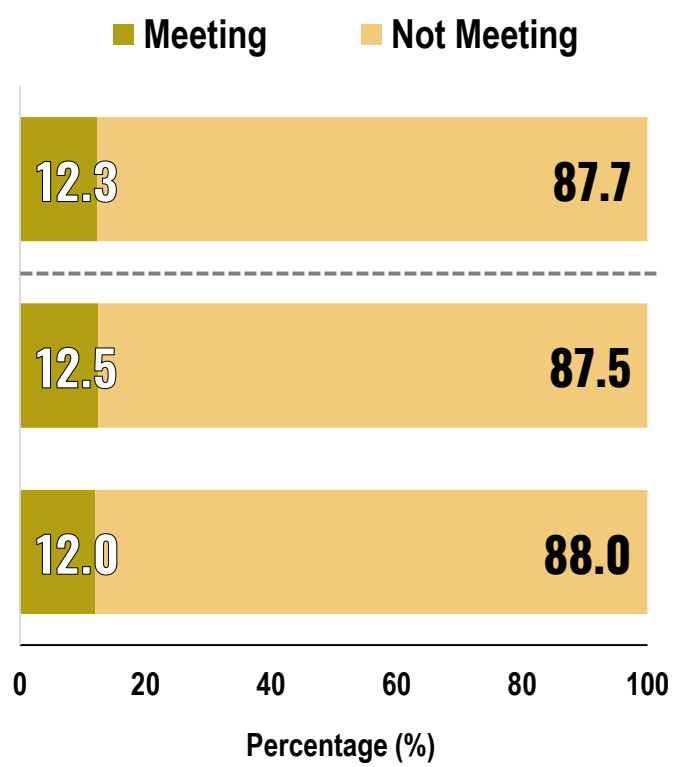
1.  Well-milled Rice — 24.4%
2.  Instant Powdered Filled Milk — 4.9%
3.  Canned Sardines in Tomato Sauce — 3.1%
4.  Chicken Meat — 2.6%
5.  Freshwater Shrimp — 2.4%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=7,765)

Age	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	466	351
13-15 y	438	310
16-18 y	495	394

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

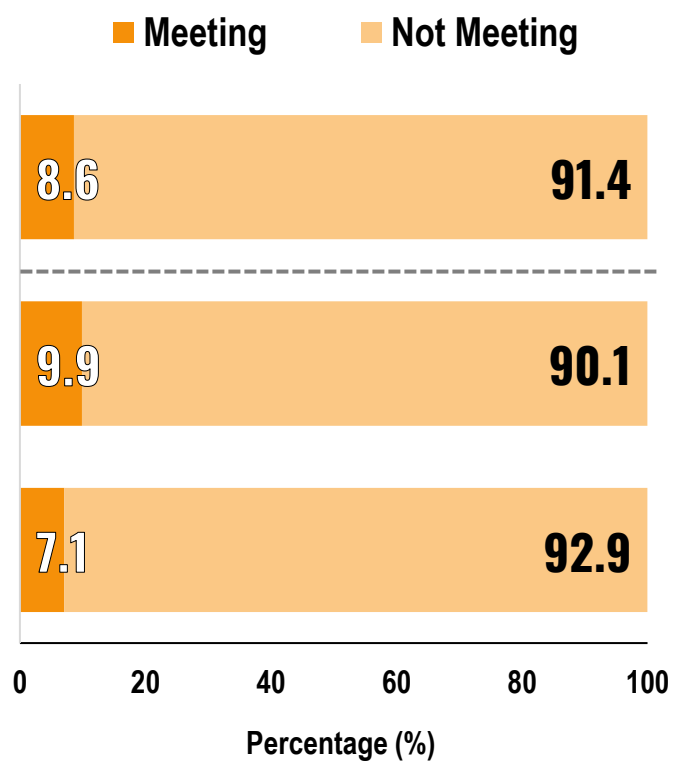
1.  Pork Liver — 25.7%
2.  Egg — 11.0%
3.  Chicken Liver — 8.9%
4.  Canned Sardines in Tomato Sauce — 4.5%
5.  Lean Pork Boston Butt (*Kasim*) — 4.1%

°*In raw edible portion (REP) form*

Mean VITAMIN C INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=7,765)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	51	17
13-15 y	47	18
16-18 y	55	17

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

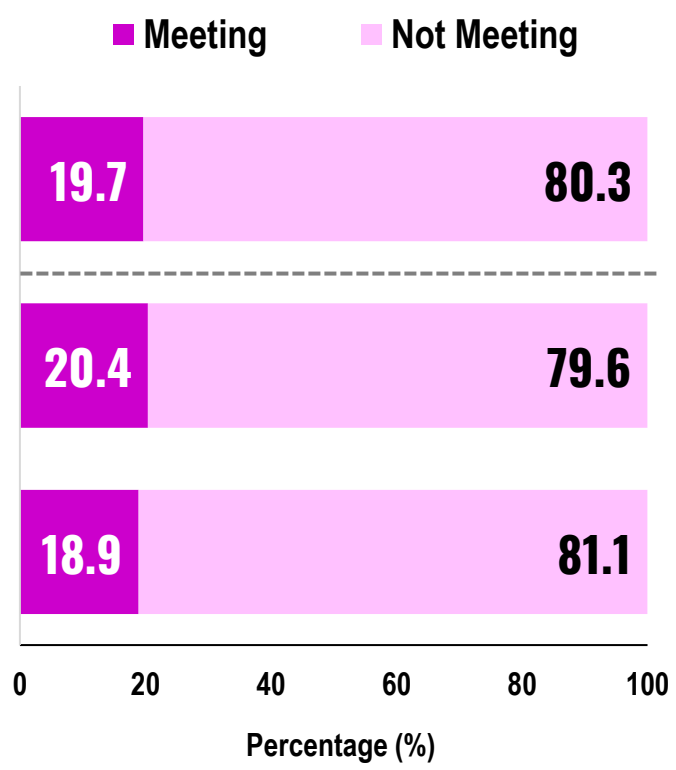
1	Unripe Indian Mango	6.8%
2	*Malunggay*	6.2%
3	*Saba* Banana	5.9%
4	Powdered Iced Tea	4.3%
5	Potato	4.2%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.7
13-15 y	0.9	0.7
16-18 y	1.0	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

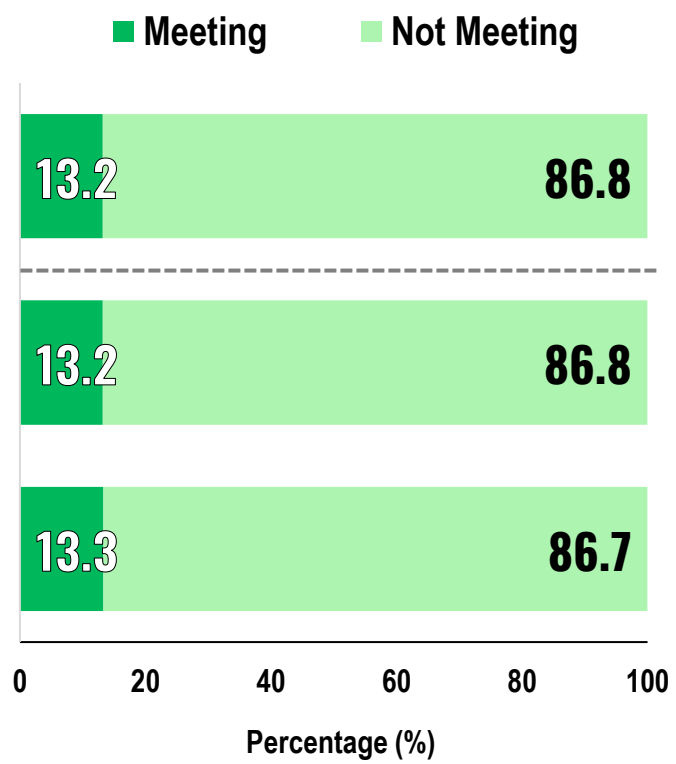
1.  Well-milled Rice — 36.5%
2.  Lean Pork Boston Butt (*Kasim*) — 6.0%
3.  Instant Noodles — 5.2%
4.  Chocolate-flavor Powdered Drink — 4.1%
5.  Pork Belly (*Liempo*) — 3.7%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.6
13-15 y	1.0	0.6
16-18 y	1.1	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

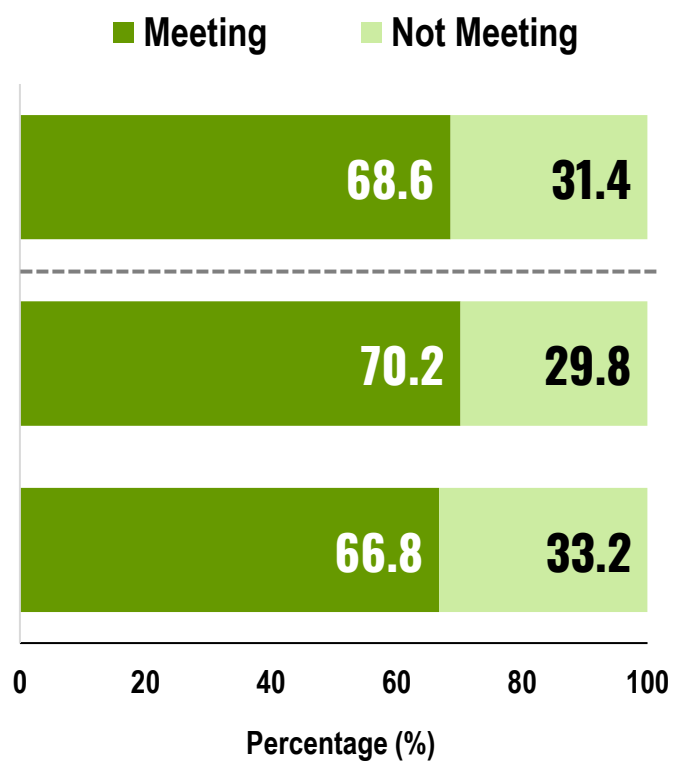
1.  Well-milled Rice — 20.3%
2.  Egg — 11.8%
3.  Instant Powdered Filled Milk — 4.5%
4.  Lean Pork Boston Butt (*Kasim*) — 4.0%
5.  Pork Liver — 3.2%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of adolescents MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	11.8	16.2
13-15 y	11.0	15.6
16-18 y	12.5	16.9

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

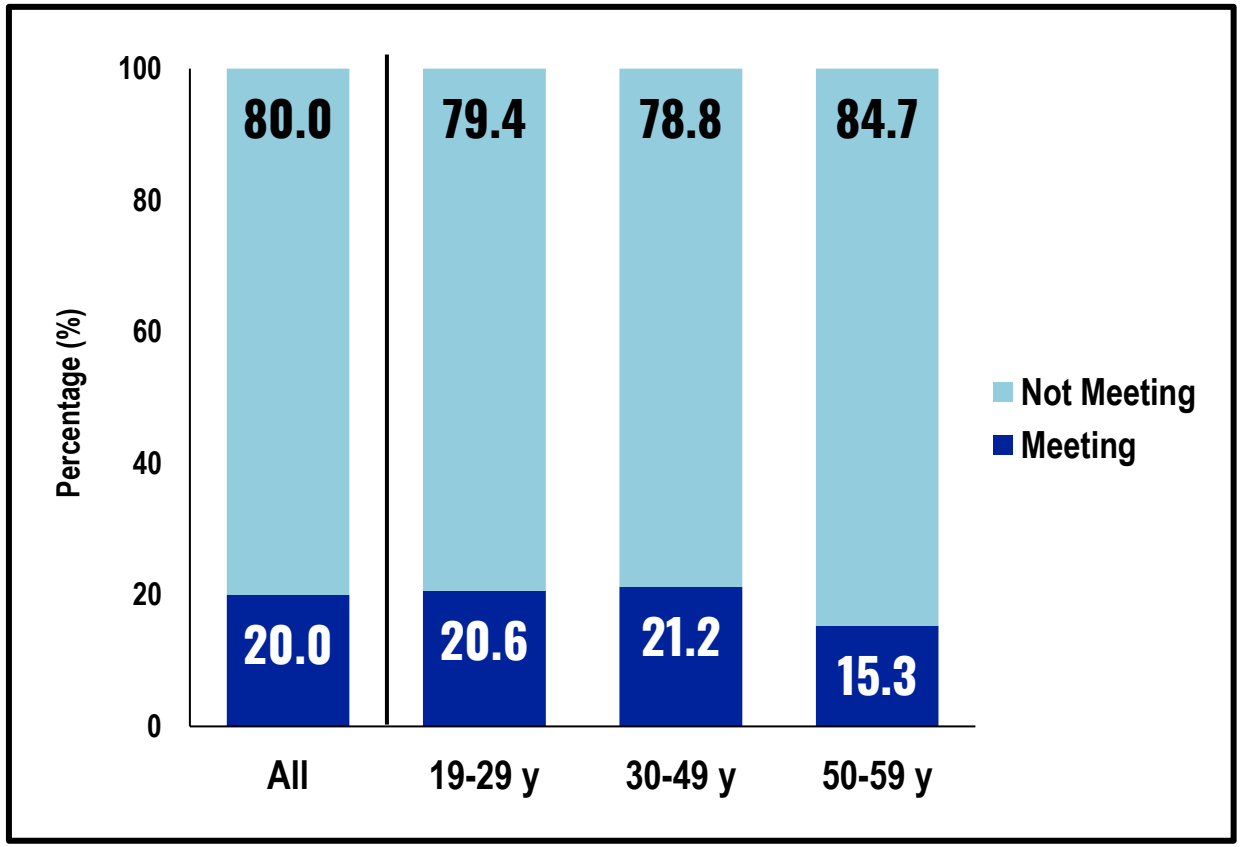
1	Well-milled Rice	44.9%
2	Chicken Breast	4.2%
3	Chicken Meat	3.6%
4	Lean Pork Boston Butt (*Kasim*)	2.6%
5	Frigate Tuna (*Tulingan*)	2.3%

°*In raw edible portion (REP) form*



ADULTS
(19 TO 59 YEARS OLD)

Percentage of adults MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=24,290)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

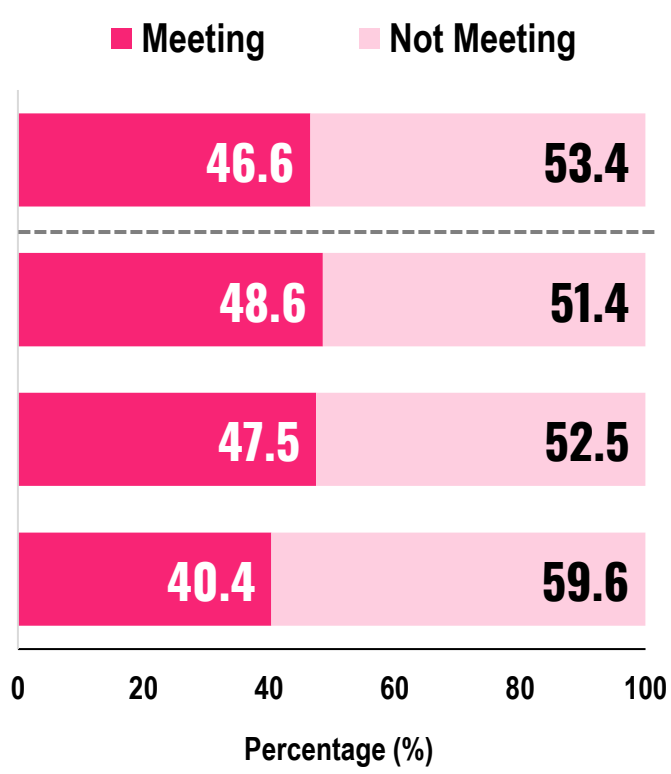
1	Well-milled Rice	54.2%
2	Pork Belly (*Liempo*)	2.0%
3	Lean Pork Boston Butt (*Kasim*)	1.8%
4	3 in 1 Coffee	1.8%
5	*Pandesal*	1.7%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=24,290)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	53	56
19-29 y	53	58
30-49 y	53	56
50-59 y	53	52

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

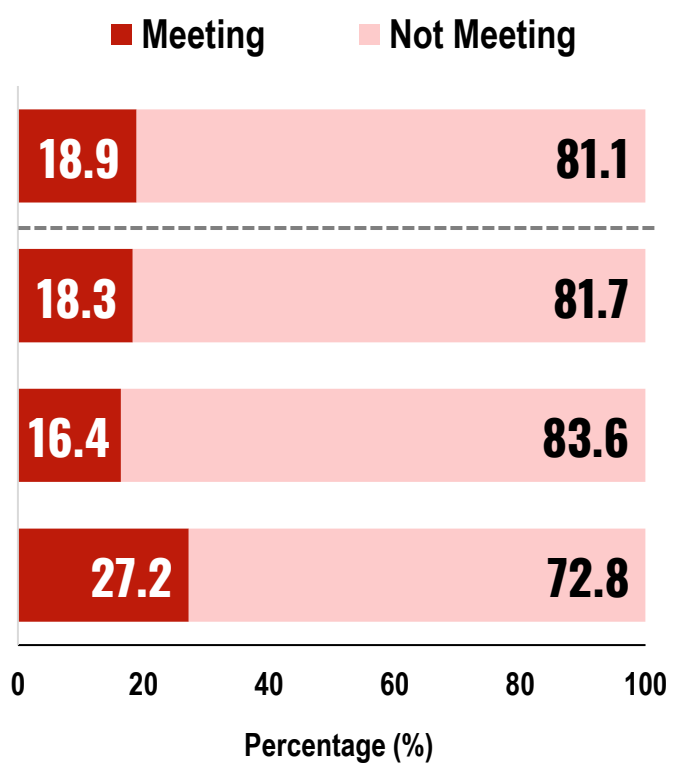
1.  Well-milled Rice — 34.4%
2.  Egg — 3.5%
3.  Lean Pork Boston Butt (*Kasim*) — 3.5%
4.  Chicken Meat — 2.8%
5.  Chicken Breast — 2.5%

°*In raw edible portion (REP) form*

Mean IRON INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=24,290)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	15.4	8.5
19-29 y	18.4	8.8
30-49 y	18.4	8.5
50-59 y	9.5	8.1

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

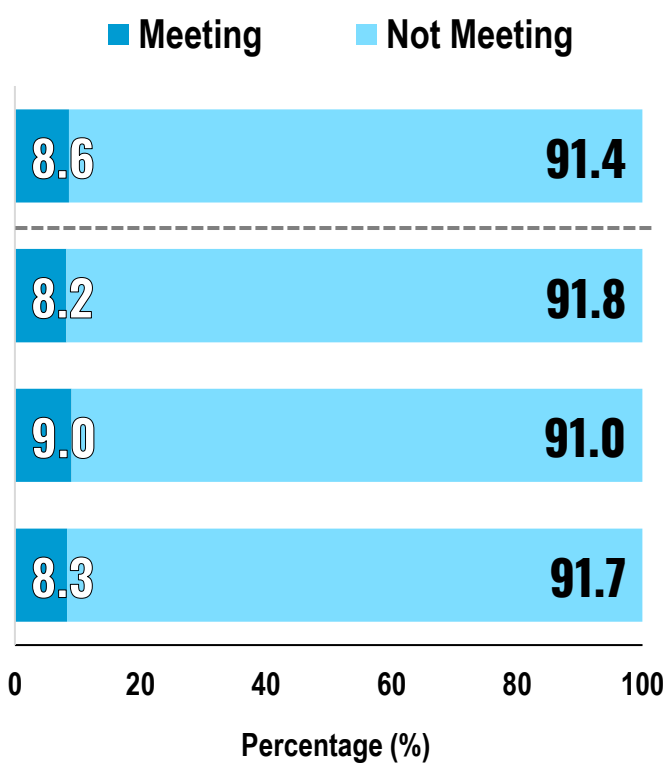
1	Well-milled Rice	30.5%
2	*Pandesal*	3.2%
3	Egg	3.1%
4	Pork Liver	1.7%
5	Lean Pork Boston Butt (*Kasim*)	1.7%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=24,290)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	0.60	0.33
19-29 y	0.60	0.32
30-49 y	0.60	0.34
50-59 y	0.60	0.32

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

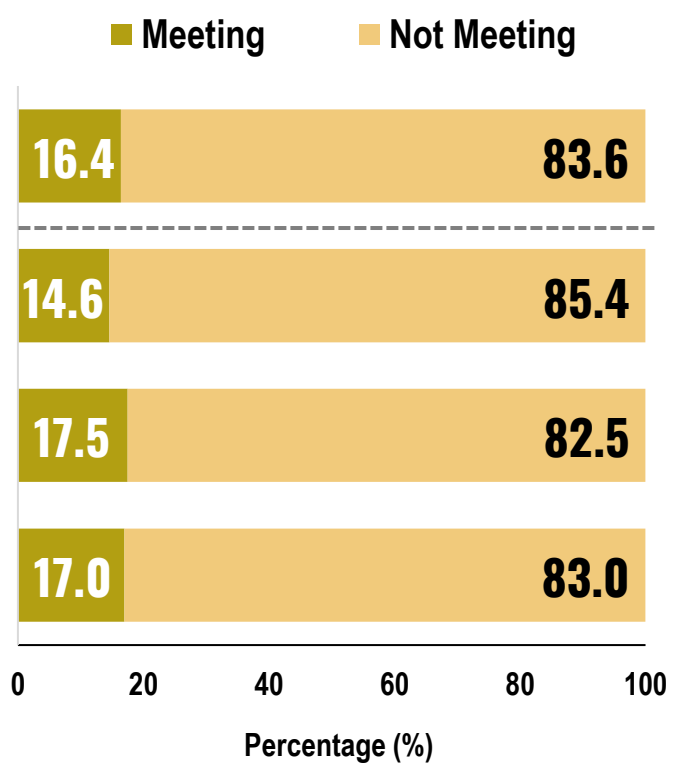
1.  Well-milled Rice — 21.3%
2.  Instant Powdered Filled Milk — 4.3%
3.  Canned Sardines in Tomato Sauce — 2.3%
4.  Freshwater Snail — 2.2%
5.  Chicken Meat — 2.0%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=24,290)

Age	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	466	468
19-29 y	466	493
30-49 y	466	463
50-59 y	466	435

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

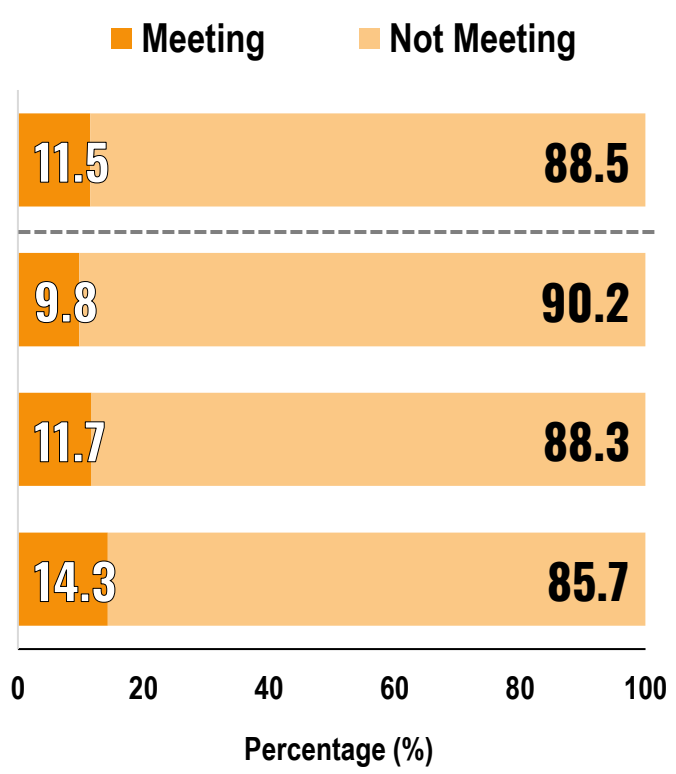
1. Chicken Liver — 20.8%
2. Pork Liver — 20.5%
3. Egg — 6.8%
4. Lean Pork Boston Butt (*Kasim*) — 3.6%
5. Carrot — 2.9%

°*In raw edible portion (REP) form*

Mean VITAMIN C INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=24,290)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	56	24
19-29 y	56	22
30-49 y	56	24
50-59 y	56	29

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

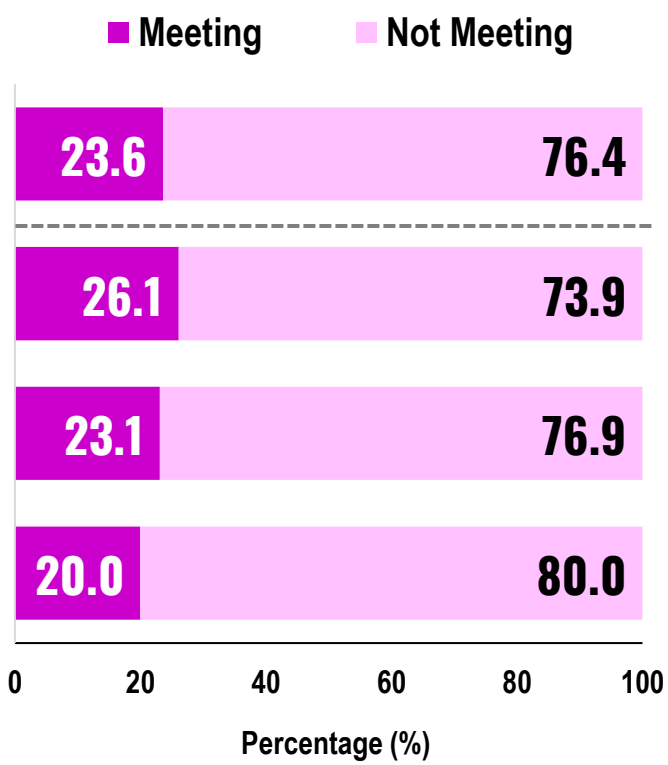
1	*Saba* Banana	7.9%
2	*Malunggay*	7.6%
3	Green Cabbage (*Repolyo*)	4.8%
4	*Pechay* Leaves	4.0%
5	Unripe Indian Mango	3.3%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.8
19-29 y	1.0	0.8
30-49 y	1.0	0.8
50-59 y	1.0	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

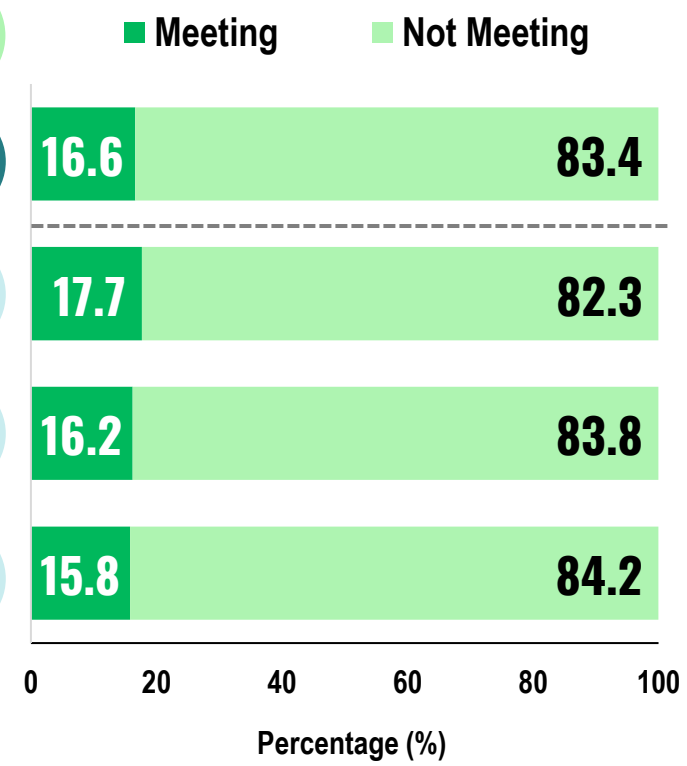
1.  Well-milled Rice — 33.8%
2.  Lean Pork Boston Butt (*Kasim*) — 6.5%
3.  Pork Belly (*Liempo*) — 4.5%
4.  Chocolate-flavor Powdered Drink — 3.8%
5.  Instant Noodles — 2.7%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.7
19-29 y	1.0	0.7
30-49 y	1.0	0.7
50-59 y	1.0	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

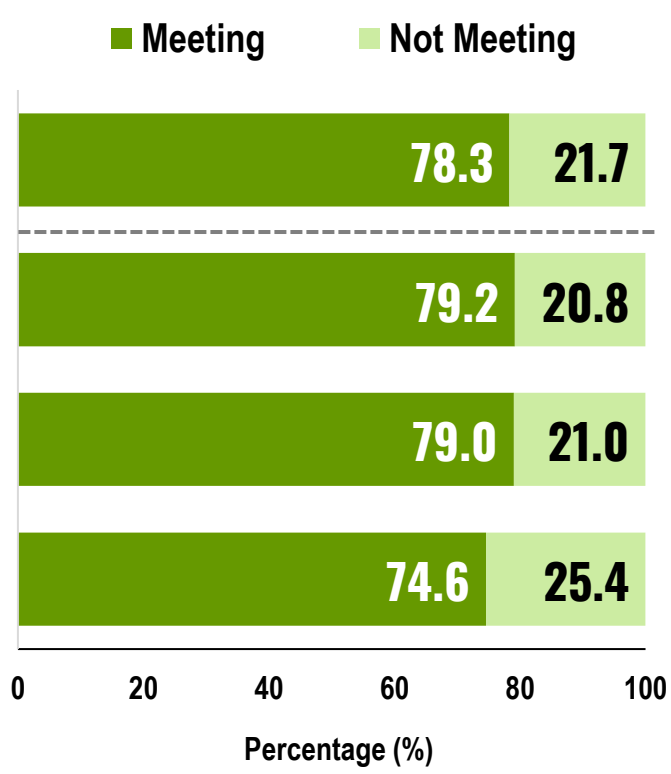
1.  Well-milled Rice — 17.8%
2.  Egg — 8.4%
3.  Chicken Liver — 6.5%
4.  Lean Pork Boston Butt (*Kasim*) — 4.1%
5.  Instant Powdered Filled Milk — 3.9%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of adults MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	11.5	18.4
19-29 y	11.5	18.6
30-49 y	11.5	18.6
50-59 y	11.5	17.2

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

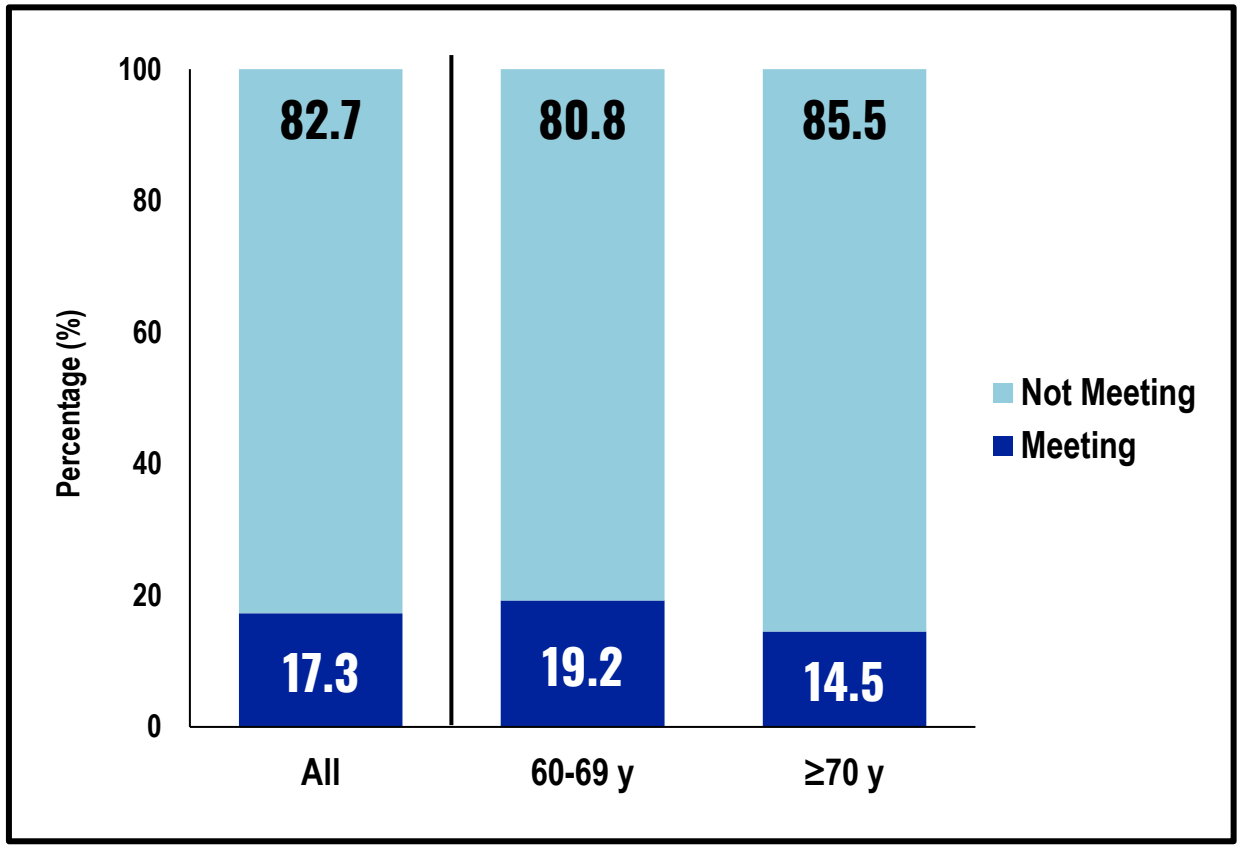
1	Well-milled Rice	39.7%
2	Chicken Breast	3.4%
3	Frigate Tuna (*Tulingan*)	3.1%
4	3 in 1 Coffee	2.9%
5	Chicken Meat	2.8%

°*In raw edible portion (REP) form*



OLDER PERSONS (60 YEARS OLD AND ABOVE)

Percentage of older persons MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=8,442)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

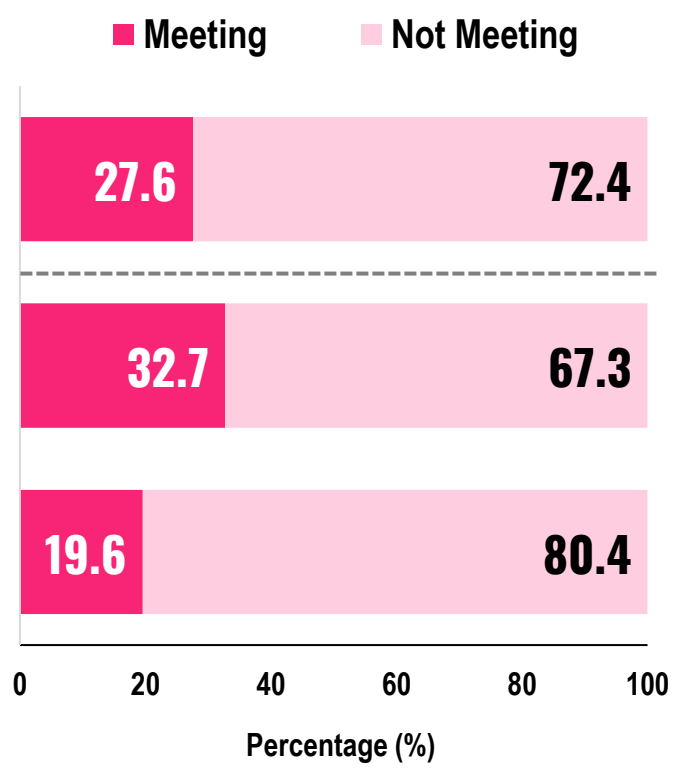
1	Well-milled Rice	50.5%
2	*Pandesal*	3.5%
3	3 in 1 Coffee	2.0%
4	Salted Cracker	1.7%
5	Lean Pork Boston Butt (*Kasim*)	1.5%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=8,442)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	53	45
60-69 y	53	47
≥70 y	53	40

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

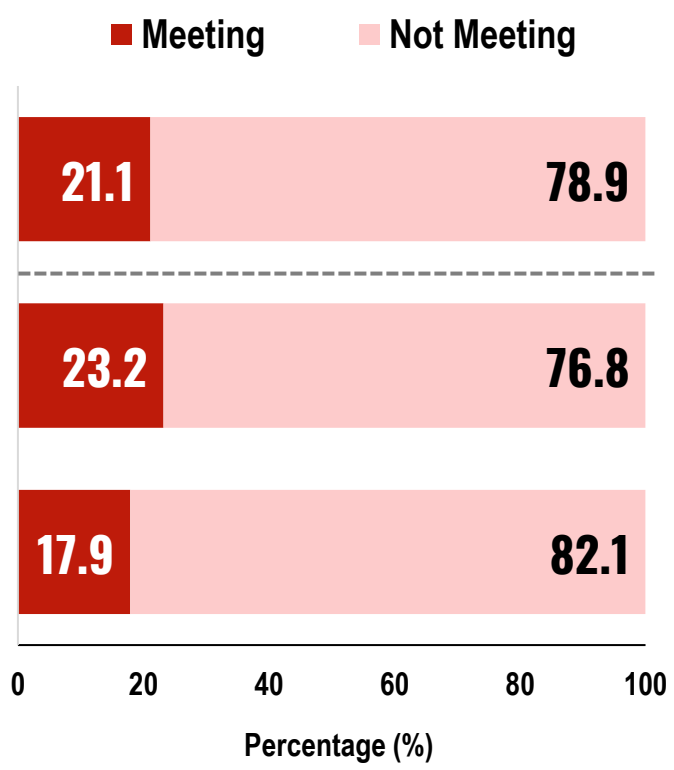
1.  Well-milled Rice — 31.8%
2.  *Pandesal* — 3.2%
3.  Milkfish (*Bangus*) — 3.1%
4.  Egg — 3.0%
5.  Lean Pork Boston Butt (*Kasim*) — 2.9%

°*In raw edible portion (REP) form*

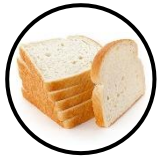
Mean IRON INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=8,442)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	9.5	7.2
60-69 y	9.5	7.5
≥70 y	9.5	6.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

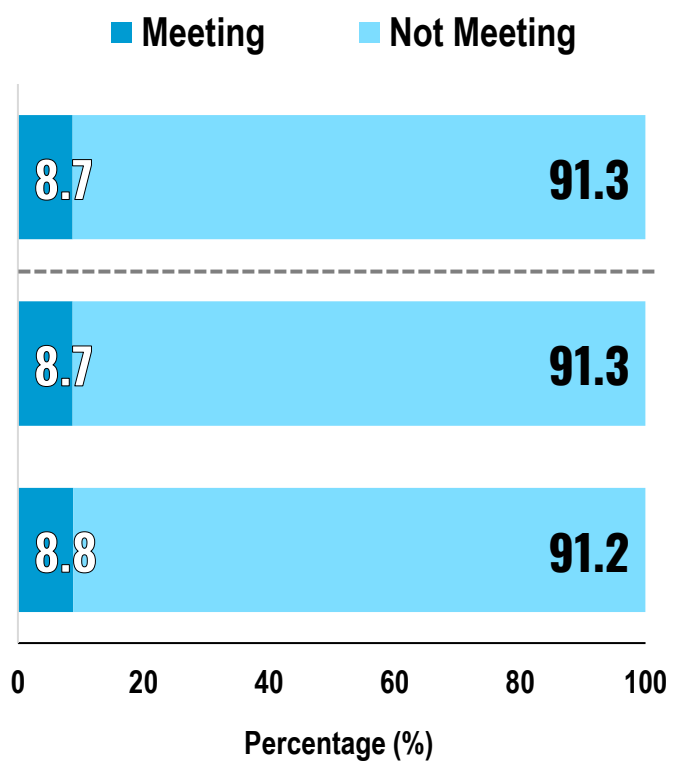
1.  Well-milled Rice — 26.6%
2.  *Pandesal* — 6.0%
3.  White Loaf Bread — 2.7%
4.  Egg — 2.6%
5.  *Pan de Monay* — 2.4%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=8,442)

Age	EAR* (grams/day)	Mean Intake (grams/day)
All	0.60	0.31
60-69 y	0.60	0.32
≥70 y	0.60	0.31

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

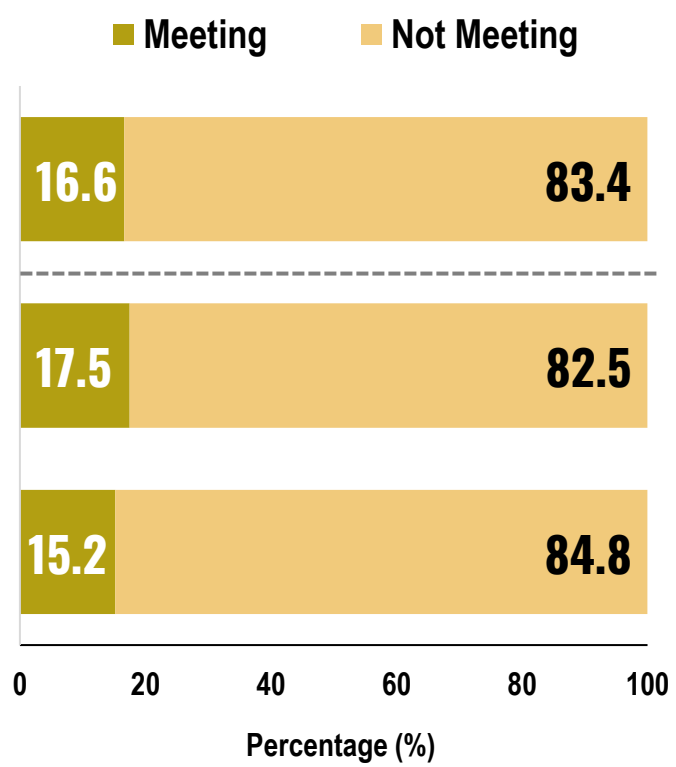
1.  Well-milled Rice — 16.4%
2.  Instant Powdered Filled Milk — 8.0%
3.  Plain Milk Powder — 3.6%
4.  *Malunggay* — 2.4%
5.  Hi-calcium Nonfat/Low-fat Powdered Milk — 1.7%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=8,442)

Age	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	466	383
60-69 y	466	423
≥70 y	466	320

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

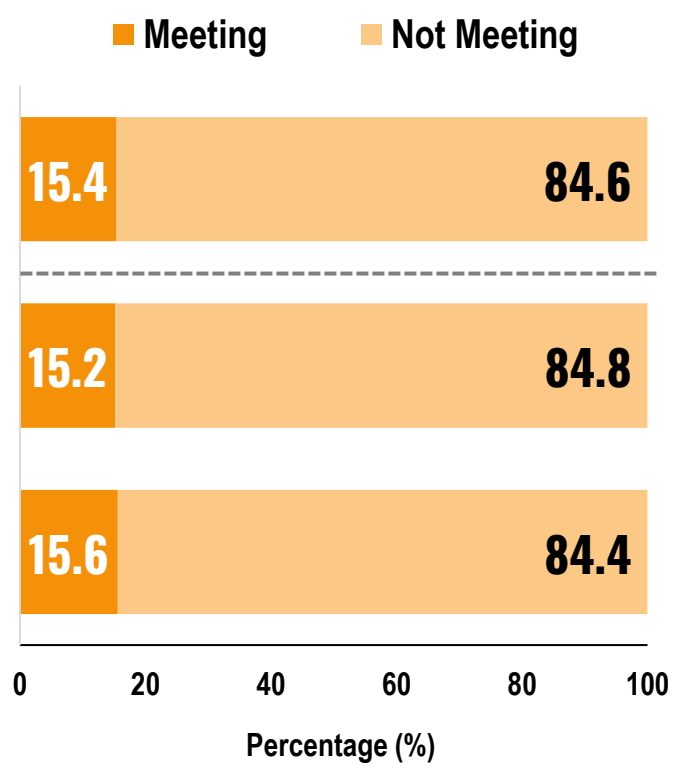
1	Chicken Liver	17.7%
2	Pork Liver	11.2%
3	Instant Powdered Filled Milk	5.8%
4	Egg	5.8%
5	*Malunggay*	5.2%

°*In raw edible portion (REP) form*

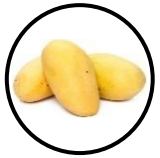
Mean VITAMIN C INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=8,442)

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	56	29
60-69 y	56	30
≥70 y	56	28

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

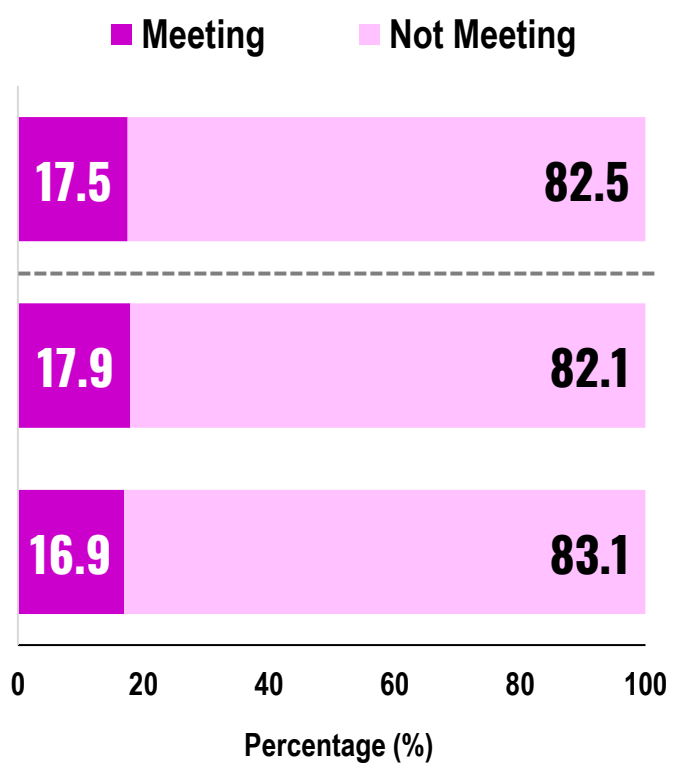
1.  *Malunggay* — 9.7%
2.  *Saba* Banana — 9.2%
3.  Plain Milk Powder — 6.0%
4.  Ripe Manila Super Mango — 3.9%
5.  Ripe Papaya — 3.7%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.7
60-69 y	1.0	0.7
≥70 y	1.0	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

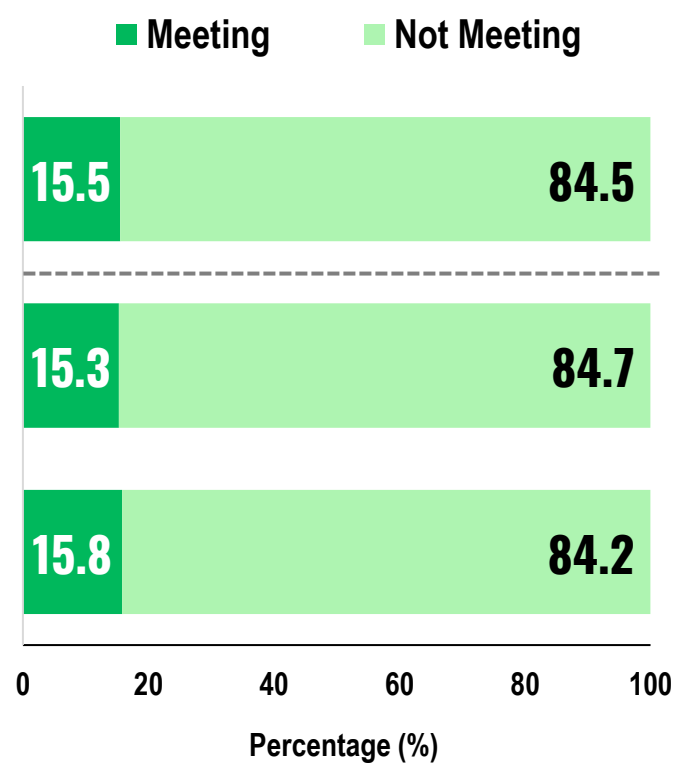
1.  Well-milled Rice — 27.2%
2.  Milk Powder w/ Nano Calcium and Colostrum — 4.9%
3.  Lean Pork Boston Butt (*Kasim*) — 4.8%
4.  *Pandesal* — 4.7%
5.  Plain Milk Powder — 3.9%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.0	0.7
60-69 y	1.0	0.7
≥70 y	1.0	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

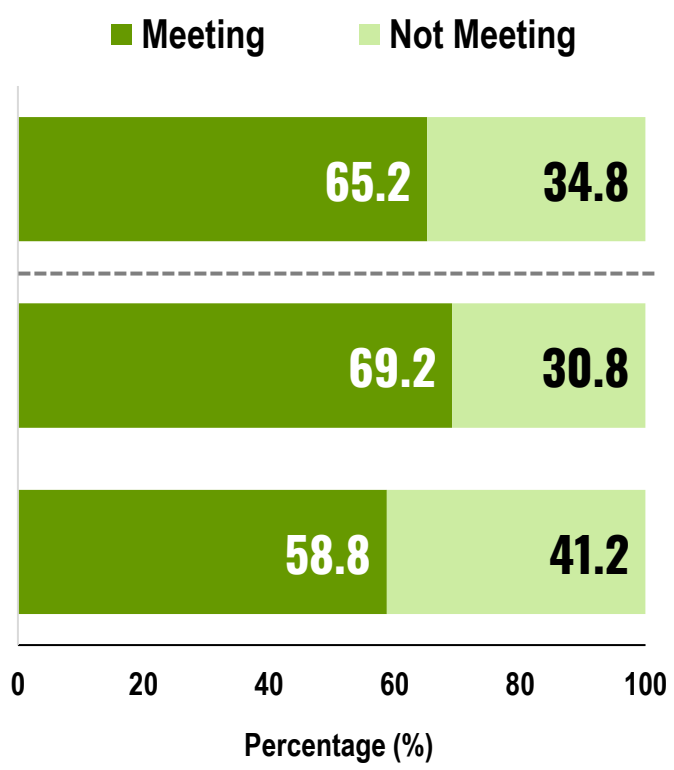
1	Well-milled Rice	14.0%
2	Instant Powdered Filled Milk	7.5%
3	Egg	6.3%
4	Chicken Liver	4.8%
5	Plain Milk Powder	4.4%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of older persons MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Age	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	11.5	15.3
60-69 y	11.5	16.1
≥70 y	11.5	14.0

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

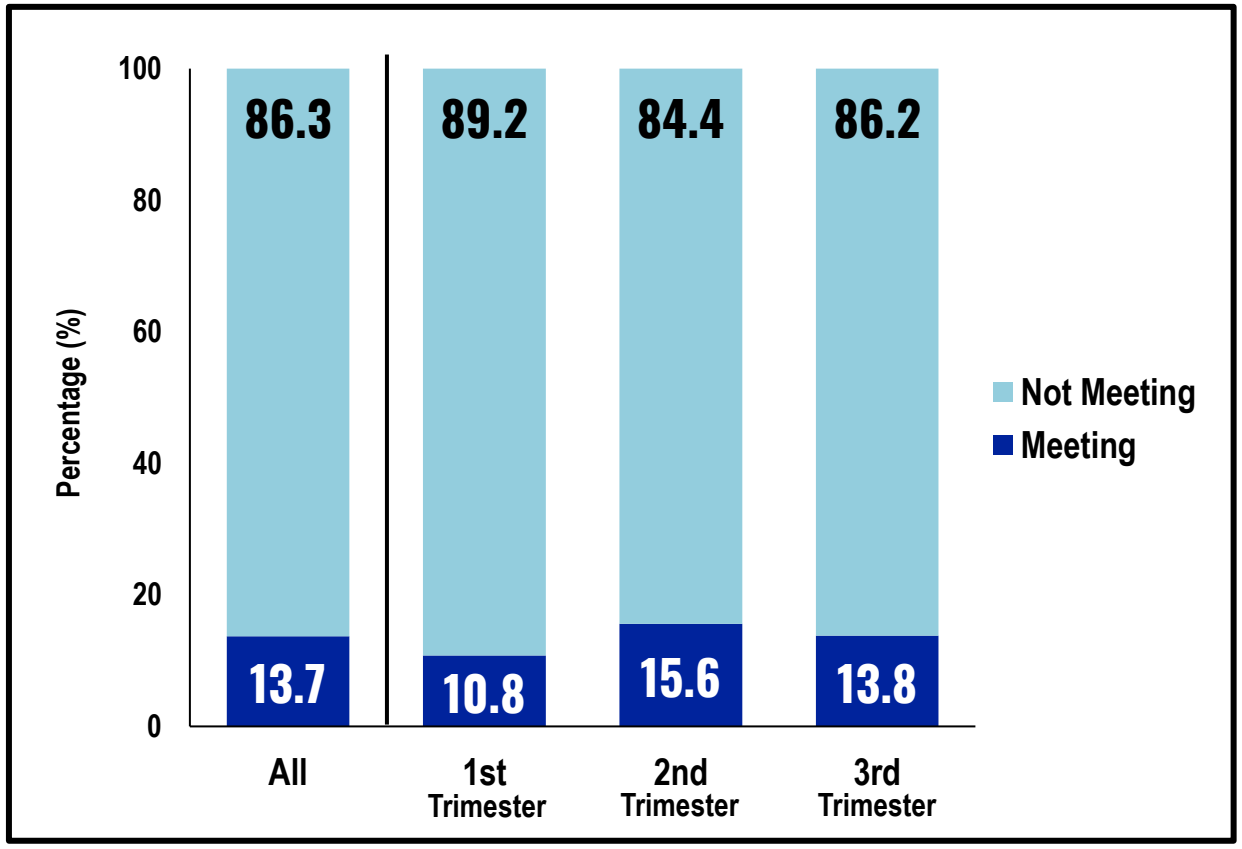
1	Well-milled Rice	35.0%
2	Milkfish (*Bangus*)	3.6%
3	3 in 1 Coffee	3.1%
4	Round Scad (*Galunggong*)	3.0%
5	*Pandesal*	2.8%

°*In raw edible portion (REP) form*



PREGNANT WOMEN

Percentage of pregnant women MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=349)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

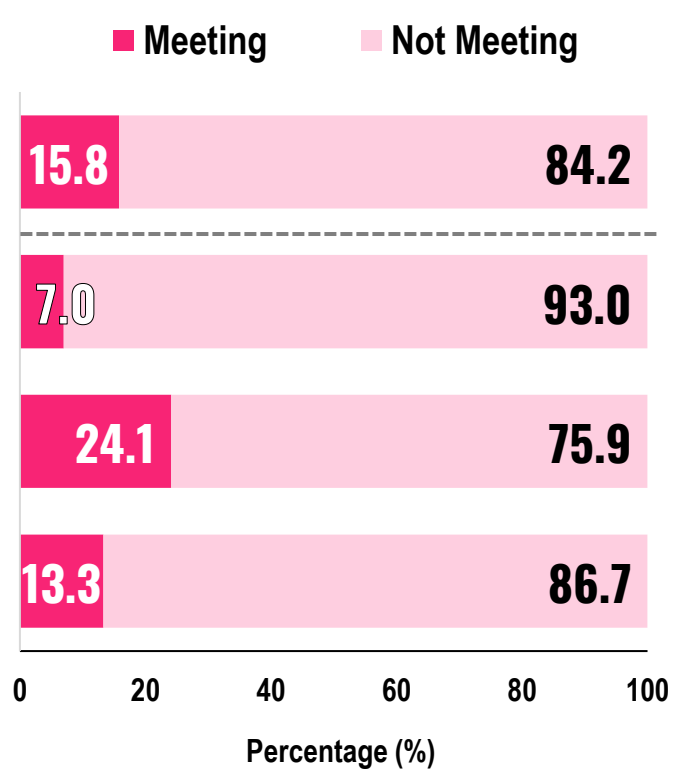
1	Well-milled Rice	47.7%
2	Instant Powdered Filled Milk	2.6%
3	Pork Belly (*Liempo*)	2.4%
4	Instant Noodles	2.0%
5	Salted Cracker	1.9%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=349)

Trimester	EAR* (grams/day)	Mean Intake (grams/day)
All	72	53
1st	72	45
2nd	72	56
3rd	72	55

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

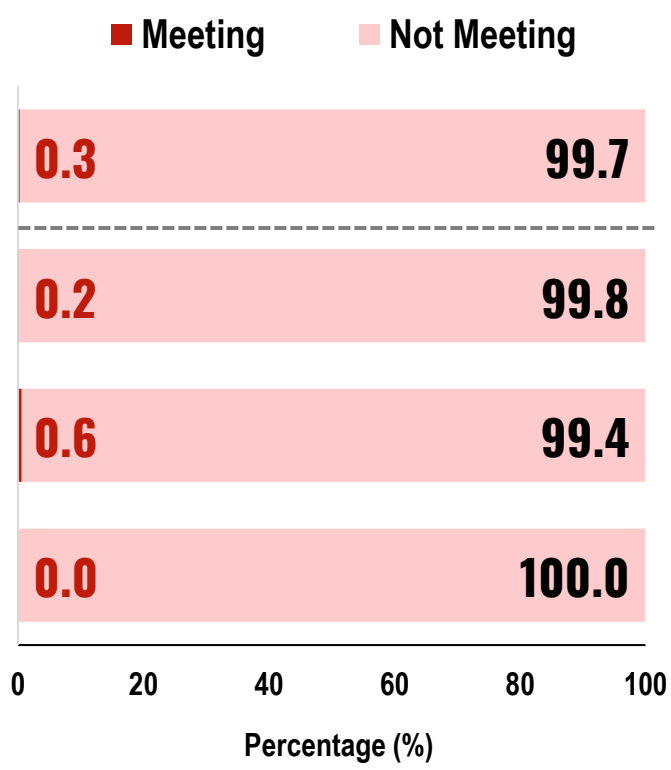
1	Well-milled Rice	30.3%
2	Instant Powdered Filled Milk	4.1%
3	Egg	3.9%
4	Chicken Meat	3.3%
5	Lean Pork Boston Butt (*Kasim*)	2.7%

°*In raw edible portion (REP) form*

Mean IRON INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=349)

Trimester	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	31.7	8.6
1st	31.7	8.0
2nd	31.7	9.1
3rd	31.7	8.3

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

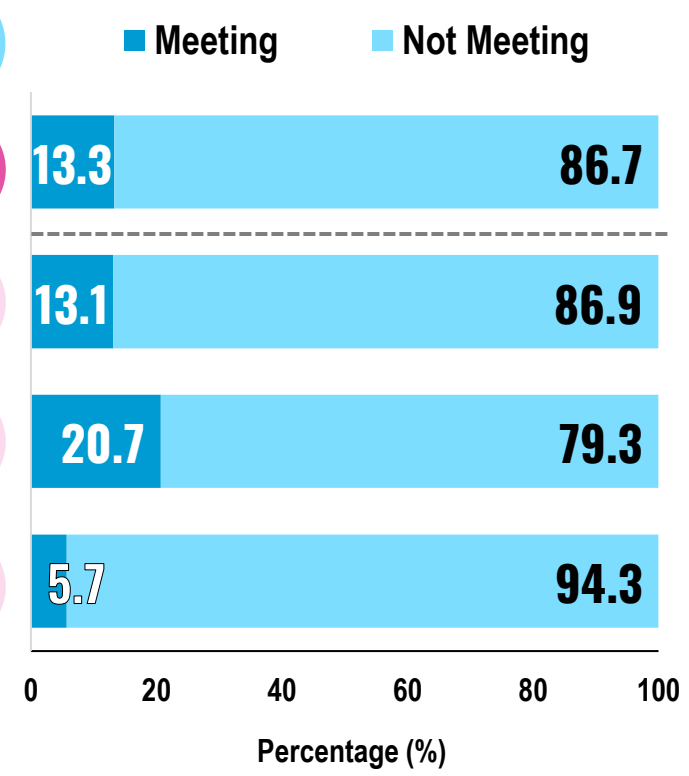
1.  Well-milled Rice — 25.3%
2.  Chocolate-flavor Powdered Drink — 5.1%
3.  Egg — 3.3%
4.  *Pandesal* — 2.9%
5.  Cheeseburger Sandwich — 2.5%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=349)

Trimester	EAR* (grams/day)	Mean Intake (grams/day)
All	0.65	0.38
1st	0.65	0.39
2nd	0.65	0.41
3rd	0.65	0.34

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

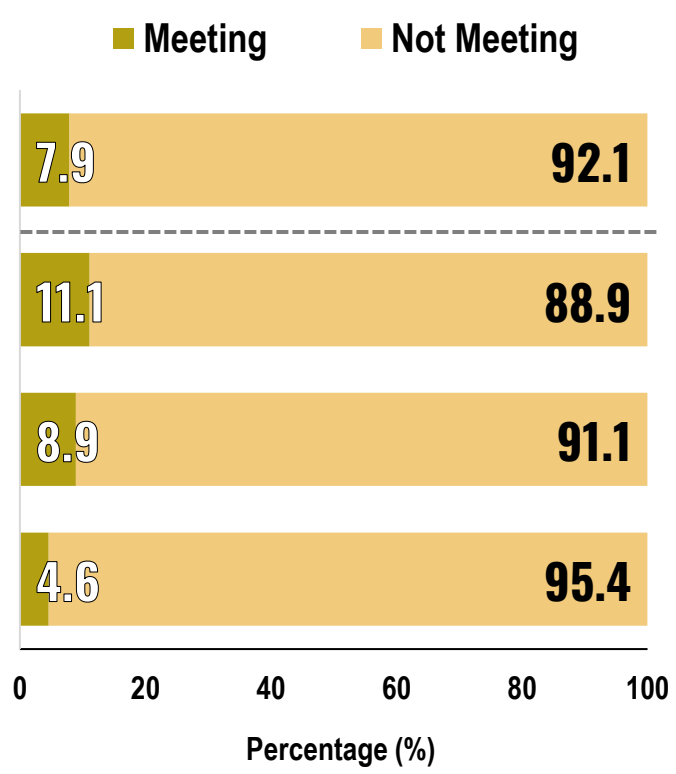
1.  Instant Powdered Filled Milk — 18.7%
2.  Well-milled Rice — 15.4%
3.  Plain Maternal Milk Powder — 3.2%
4.  Canned Sardines in Tomato Sauce — 2.7%
5.  Banana Cue — 2.6%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=349)

Trimester	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	720	452
1st	720	483
2nd	720	585
3rd	720	290

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

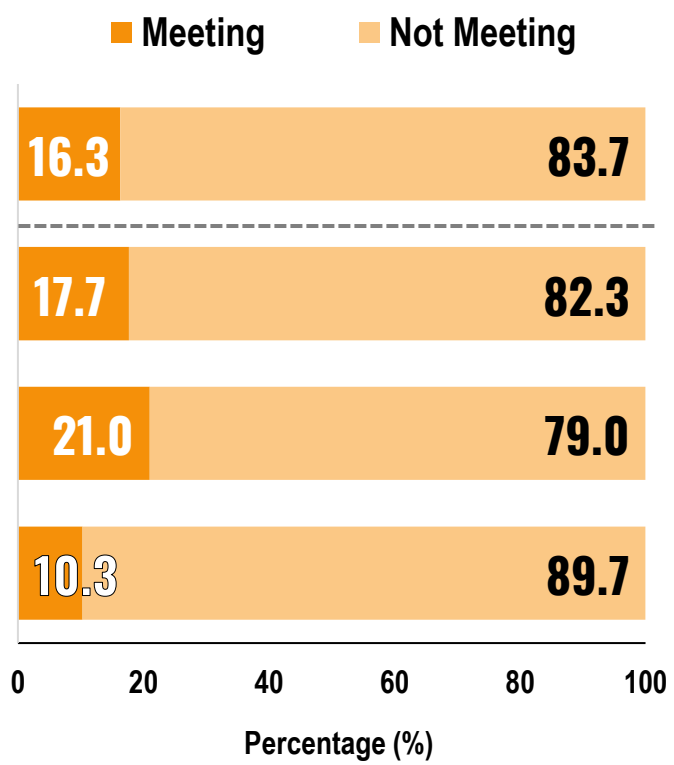
1	Pork Liver	25.1%
2	Instant Powdered Filled Milk	13.8%
3	Egg	7.5%
4	Chicken Liver	4.9%
5	Canned Sardines in Tomato Sauce	4.0%

°*In raw edible portion (REP) form*

Mean VITAMIN C INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=349)

Trimester	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	56	32
1st	56	37
2nd	56	37
3rd	56	25

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

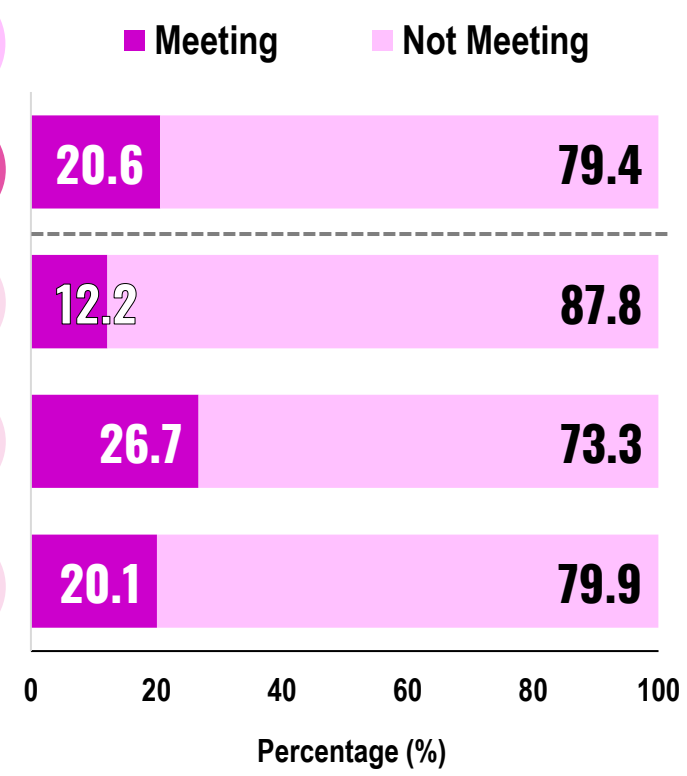
1.  *Saba* Banana — 9.9%
2.  Rambutan — 6.4%
3.  Chocolate-flavor Powdered Drink — 5.3%
4.  *Malunggay* — 5.1%
5.  Cassava Leaves (*Dahon ng Kamoteng Kahoy*) — 4.3%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Trimester	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.2	0.8
1st	1.2	0.7
2nd	1.2	0.9
3rd	1.2	0.8

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

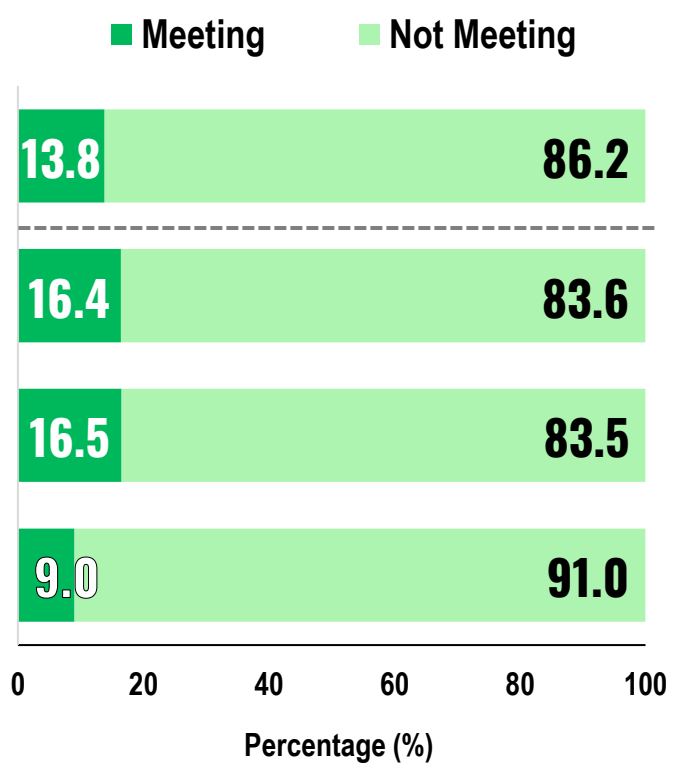
1.  Well-milled Rice — 26.2%
2.  Chocolate-flavor Powdered Drink — 10.9%
3.  Pork Belly (*Liempo*) — 4.7%
4.  Lean Pork Boston Butt (*Kasim*) — 4.5%
5.  Instant Noodles — 3.9%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Trimester	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.4	0.8
1st	1.4	0.8
2nd	1.4	0.9
3rd	1.4	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

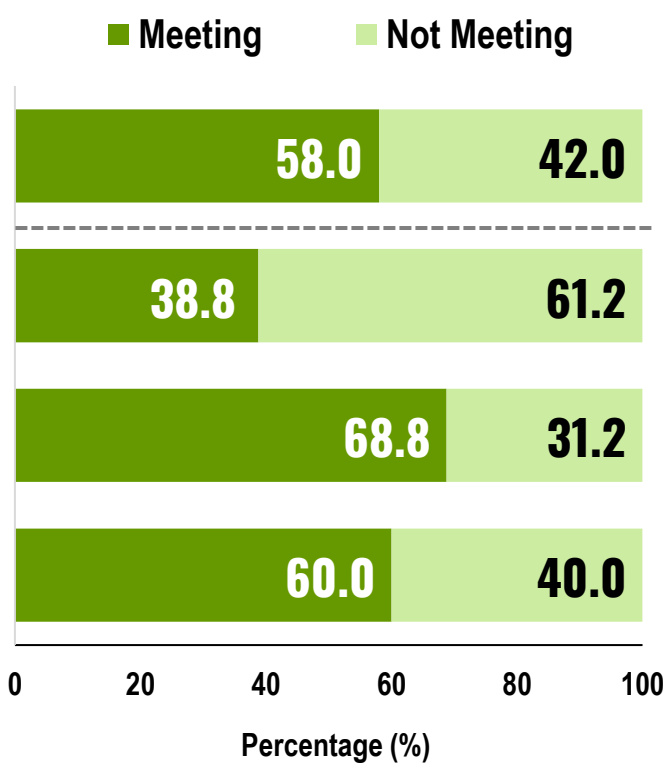
1.  Instant Powdered Filled Milk — 17.5%
2.  Well-milled Rice — 13.1%
3.  Egg — 7.9%
4.  Chocolate-flavor Powdered Drink — 6.7%
5.  Pork Liver — 3.1%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of pregnant women MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Trimester	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	14.0	16.2
1st	14.0	13.4
2nd	14.0	16.9
3rd	14.0	17.4

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

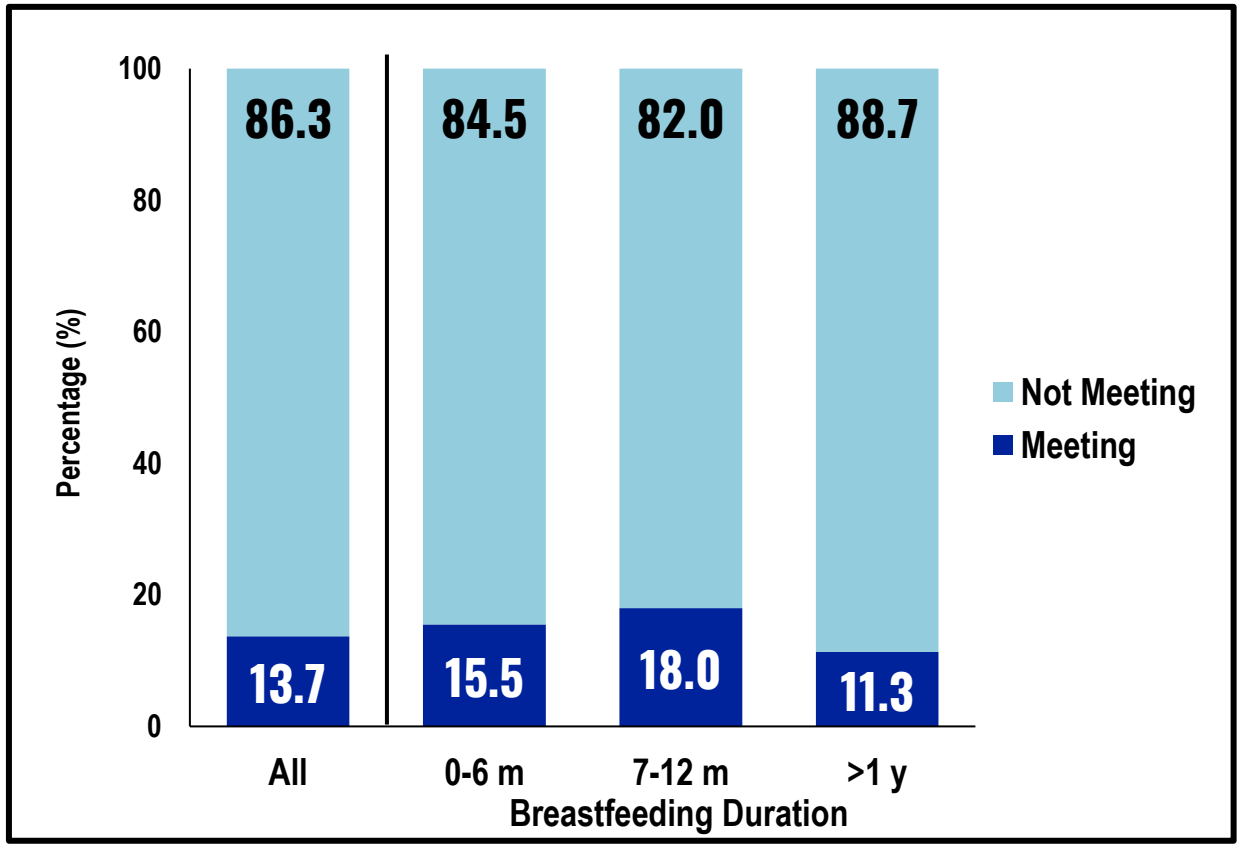
1.  Well-milled Rice — 37.4%
2.  Chicken Meat — 3.5%
3.  Chocolate-flavor Powdered Drink — 3.4%
4.  Chicken Breast — 3.3%
5.  Round Scad (*Galunggong*) — 2.8%

°*In raw edible portion (REP) form*



LACTATING MOTHERS

Percentage of lactating mothers MEETING the Recommended Energy Intake (REI): Philippines, 2023 (n=1,313)



Top 5 food sources° of ENERGY (percent (%) contribution): Philippines, 2023

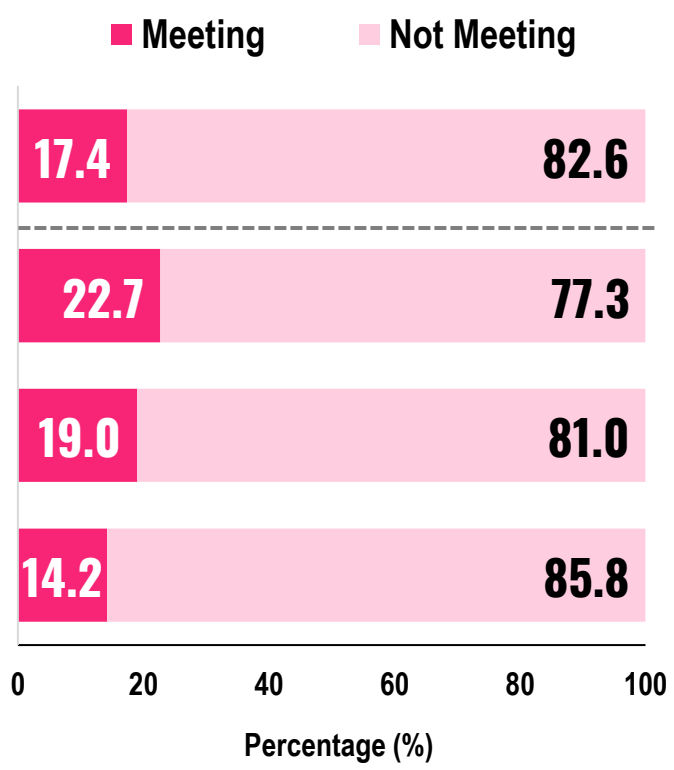
1	Well-milled Rice	55.9%
2	3 in 1 Coffee	2.2%
3	*Pandesal*	2.0%
4	Instant Noodles	1.4%
5	Instant Powdered Filled Milk	1.4%

°*In raw edible portion (REP) form*

Mean PROTEIN INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=1,313)

Breastfeeding Duration	EAR* (grams/day)	Mean Intake (grams/day)
All	72	54
0-6 m	72	60
7-12 m	72	55
>1 y	72	51

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of PROTEIN (% contribution): Philippines, 2023

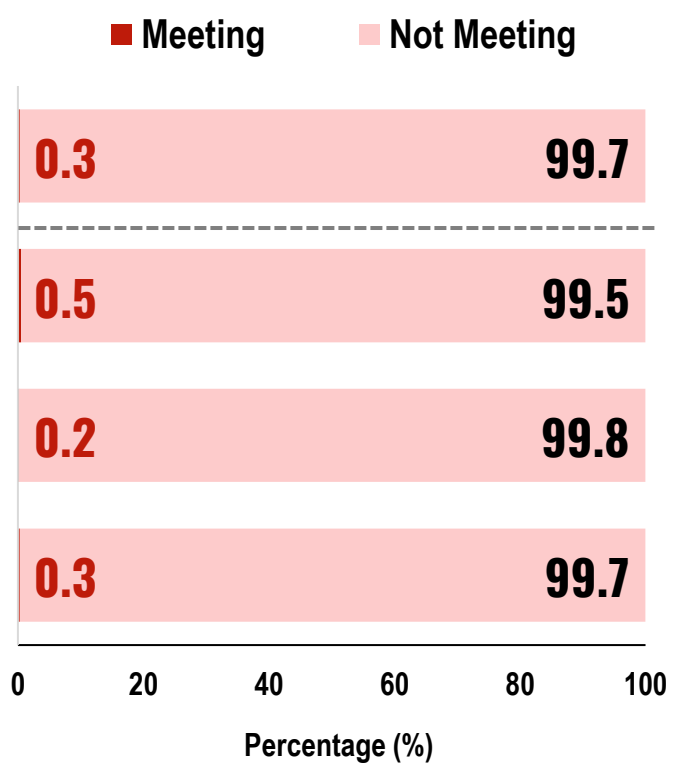
1.  Well-milled Rice — 37.5%
2.  Egg — 3.7%
3.  Chicken Meat — 3.0%
4.  Lean Pork Boston Butt (*Kasim*) — 2.6%
5.  Instant Powdered Filled Milk — 2.3%

°*In raw edible portion (REP) form*

Mean IRON INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=1,313)

Breastfeeding Duration	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	28.2	8.6
0-6 m	28.2	9.7
7-12 m	28.2	8.3
>1 y	28.2	8.2

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of IRON (% contribution): Philippines, 2023

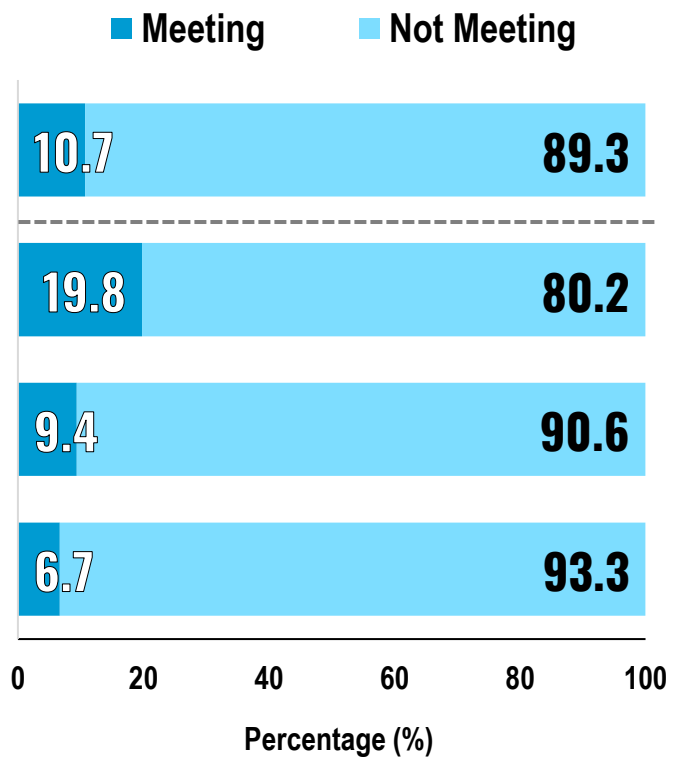
1.  Well-milled Rice — 31.7%
2.  Chocolate-flavor Powdered Drink — 5.7%
3.  *Pandesal* — 3.7%
4.  Egg — 3.2%
5.  Chocolate-based Cereal Drink — 2.3%

°*In raw edible portion (REP) form*

Mean CALCIUM INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=1,313)

Breastfeeding Duration	EAR* (grams/day)	Mean Intake (grams/day)
All	0.61	0.34
0-6 m	0.61	0.42
7-12 m	0.61	0.33
>1 y	0.60	0.31

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of CALCIUM (% contribution): Philippines, 2023

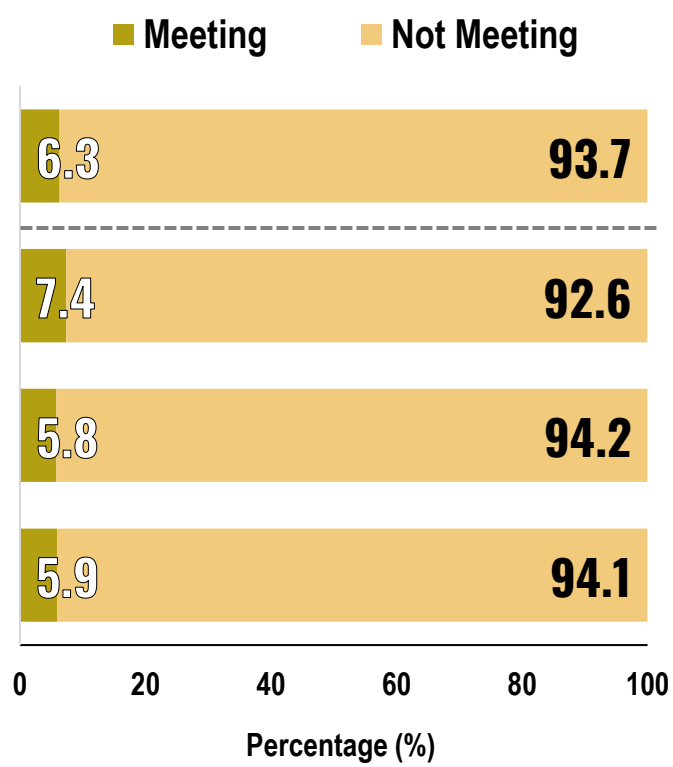
1.  Well-milled Rice — 21.7%
2.  Instant Powdered Filled Milk — 11.9%
3.  Chocolate-flavor Powdered Drink — 2.9%
4.  Chicken Meat — 2.0%
5.  Long-jawed Anchovy (*Dilis*) — 1.8%

°*In raw edible portion (REP) form*

Mean VITAMIN A INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=1,313)

Breastfeeding Duration	EAR* (µg RE/day)	Mean Intake (µg RE/day)
All	800	401
0-6 m	800	415
7-12 m	799	351
>1 y	800	410

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN A (% contribution): Philippines, 2023

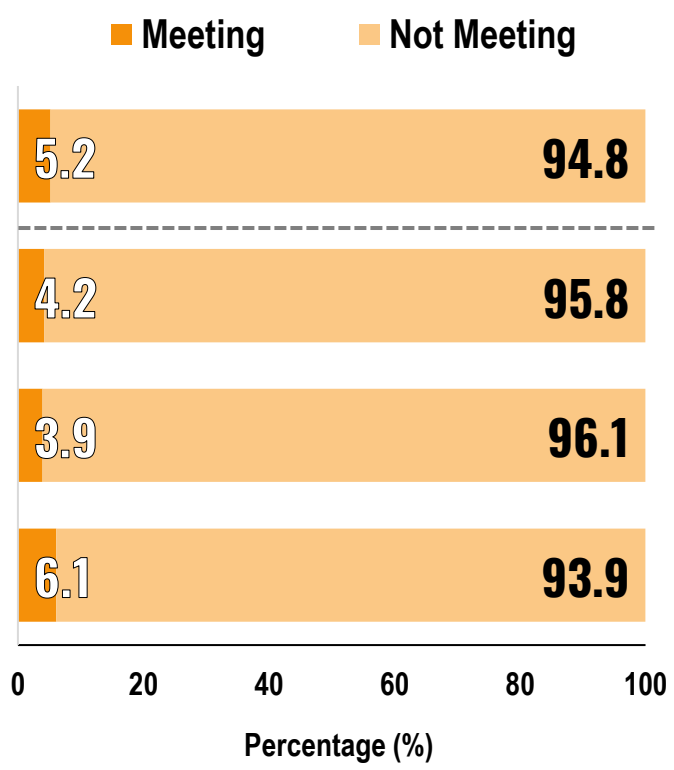
1.  Pork Liver — 17.7%
2.  Chicken Liver — 9.0%
3.  Instant Powdered Filled Milk — 8.9%
4.  Egg — 8.1%
5.  Bun with Custard Filling — 4.0%

°*In raw edible portion (REP) form*

Mean VITAMIN C INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023 (n=1,313)

Breastfeeding Duration	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	76	22
0-6 m	76	23
7-12 m	76	22
>1 y	76	21

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of VITAMIN C (% contribution): Philippines, 2023

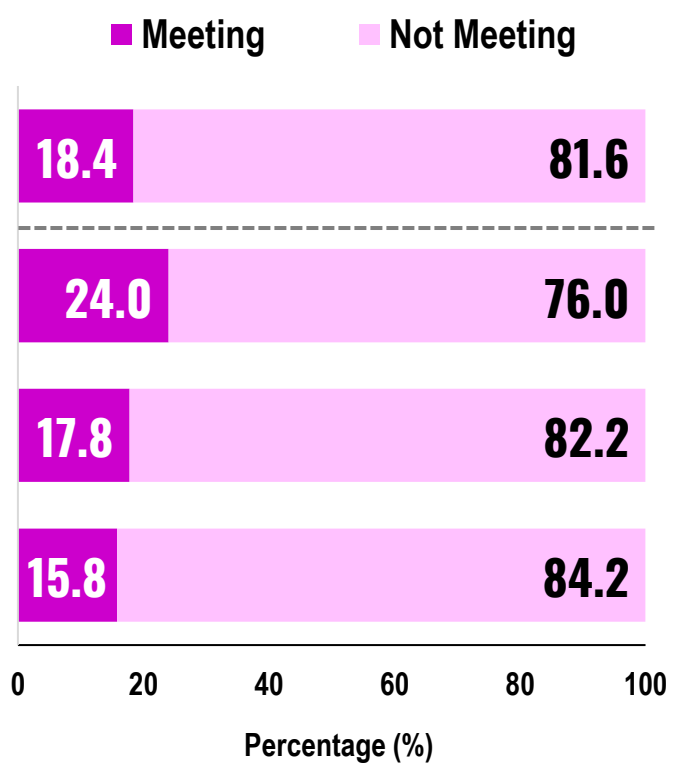
1.  Chocolate-flavor Powdered Drink — 9.3%
2.  *Malunggay* — 9.0%
3.  Taro Leaves (*Dahon ng Gabi*) — 7.5%
4.  *Saba* Banana — 6.3%
5.  Green Cabbage (*Repolyo*) — 3.8%

°*In raw edible portion (REP) form*

Mean THIAMIN INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Breastfeeding Duration	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.1	0.8
0-6 m	1.1	0.9
7-12 m	1.1	0.8
>1 y	1.1	0.8

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of THIAMIN (% contribution): Philippines, 2023

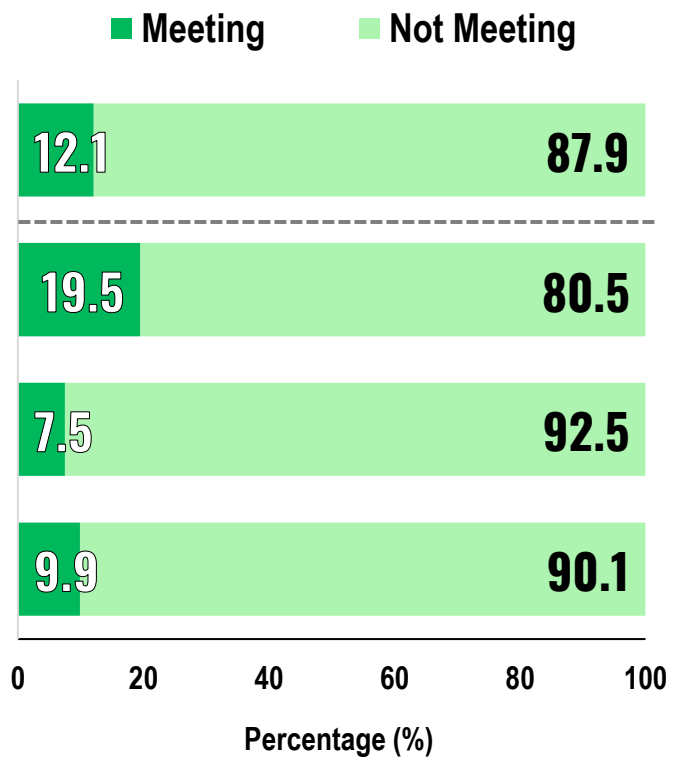
1.  Well-milled Rice — 33.5%
2.  Chocolate-flavor Powdered Drink — 12.8%
3.  Lean Pork Boston Butt (*Kasim*) — 4.5%
4.  Instant Noodles — 3.0%
5.  *Pandesal* — 3.0%

°*In raw edible portion (REP) form*

Mean RIBOFLAVIN INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Breastfeeding Duration	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	1.3	0.8
0-6 m	1.3	0.9
7-12 m	1.3	0.7
>1 y	1.3	0.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of RIBOFLAVIN (% contribution): Philippines, 2023

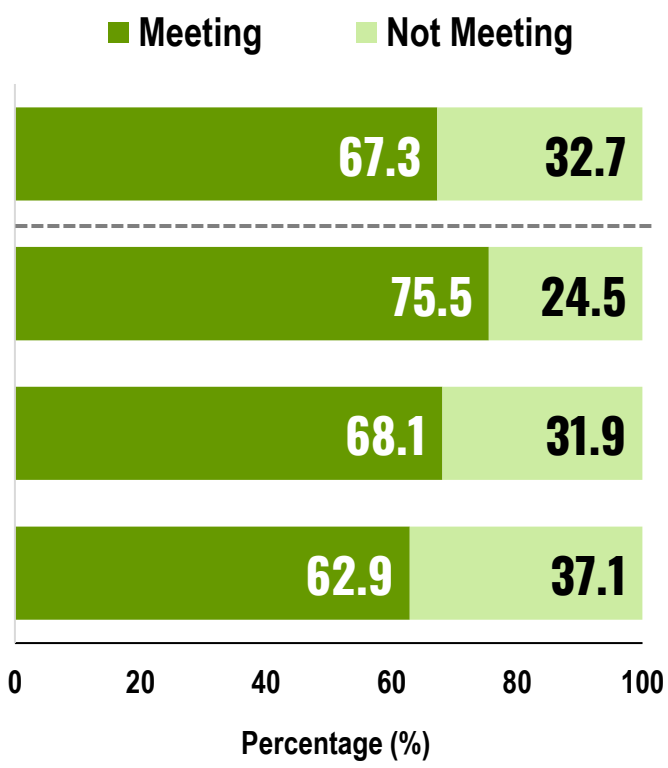
1.  Well-milled Rice — 18.0%
2.  Instant Powdered Filled Milk — 10.9%
3.  Chocolate-flavor Powdered Drink — 8.4%
4.  Egg — 8.3%
5.  3 in 1 Coffee — 3.2%

°*In raw edible portion (REP) form*

Mean NIACIN INTAKE and percentage of lactating mothers MEETING the Estimated Average Requirement (EAR): Philippines, 2023

Breastfeeding Duration	EAR* (milligrams/day)	Mean Intake (milligrams/day)
All	13.4	17.5
0-6 m	13.4	18.5
7-12 m	13.4	18.3
>1 y	13.4	16.7

*Based on the EAR from the 2015 Philippine Dietary Reference Intakes (PDRI)



Top 5 food sources° of NIACIN (% contribution): Philippines, 2023

1.  Well-milled Rice — 43.7%
2.  3 in 1 Coffee — 3.7%
3.  Chocolate-flavor Powdered Drink — 3.7%
4.  Chicken Meat — 3.0%
5.  Chicken Breast — 2.6%

°*In raw edible portion (REP) form*

HIGHLIGHTS

FOOD CONSUMPTION SURVEY: INDIVIDUAL-LEVEL

ENERGY and NUTRIENT INTAKE ACROSS POPULATION GROUPS

- The mean **energy intake** of Filipinos across all age and population groups is **inadequate** based on the **Recommended Energy Intake (REI)**.
- *Across age and population groups:*
  - **Iron** is the **least met micronutrient**, specifically among pregnant women and lactating mothers.
  - The mean intakes of **calcium**, **vitamin A**, **vitamin C**, **thiamin**, and **riboflavin** are **low** across all age and population groups, except among infants and preschool children. Thus, a higher percentage of individuals not meeting their nutrient requirements is observed.
  - **Protein inadequacy** is most notable among pregnant and lactating women.



HIGHLIGHTS

FOOD CONSUMPTION SURVEY: INDIVIDUAL-LEVEL

MAJOR FOOD SOURCES OF ENERGY and NUTRIENTS

❖ *Major food sources of energy and nutrients:*

- ❖ **Well-milled rice** is the **primary source energy and most nutrients,** specifically **protein, iron, calcium**, **thiamin**, **riboflavin**, and **niacin**, across all age and population groups, except **infants and preschool children** where different kinds of **milk** are their top source of nutrients.
- ❖ **Pork liver** and **chicken liver** are included in the **top sources of vitamin A** of all age and population groups, except infants and preschool children.
- ❖ The **top food sources of vitamin C** in all population groups are ***malunggay*** and **banana *(saba)***, excluding young children where their primary source is milk and milk products.





2023 NATIONAL NUTRITION SURVEY

MATERNAL HEALTH AND NUTRITION

Why do we collect Maternal Health and Nutrition data?



A child's health is closely linked to the mother's health from conception and until the child's second birthday **(First 1000 days Window of Opportunity).**



Sustainable Development Goal No. 3 aims to reduce the global maternal mortality ratio to less than 70 per 100, 000 live births.



The country needs evidence-based data on the factors influencing maternal health and nutrition to **formulate or redefine policies and implement programs**

Eligible Respondents & Data Collection Method



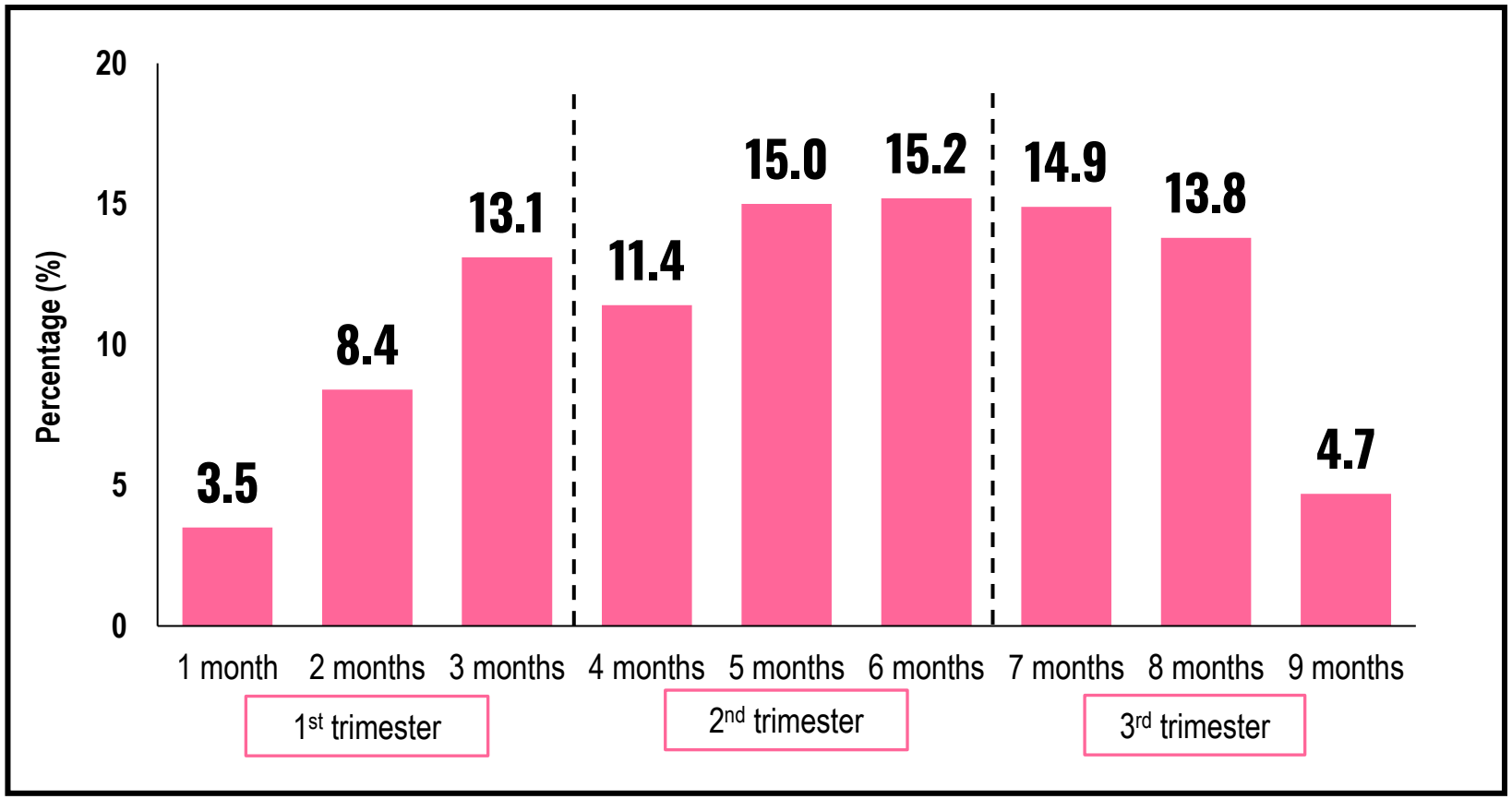
ALL currently pregnant women

Respondents



Face-to-face interview using a structured Questionnaire on Maternal Health seeking behaviors

Percentage of pregnant women by months of pregnancy: Philippines, 2023 (n=682)

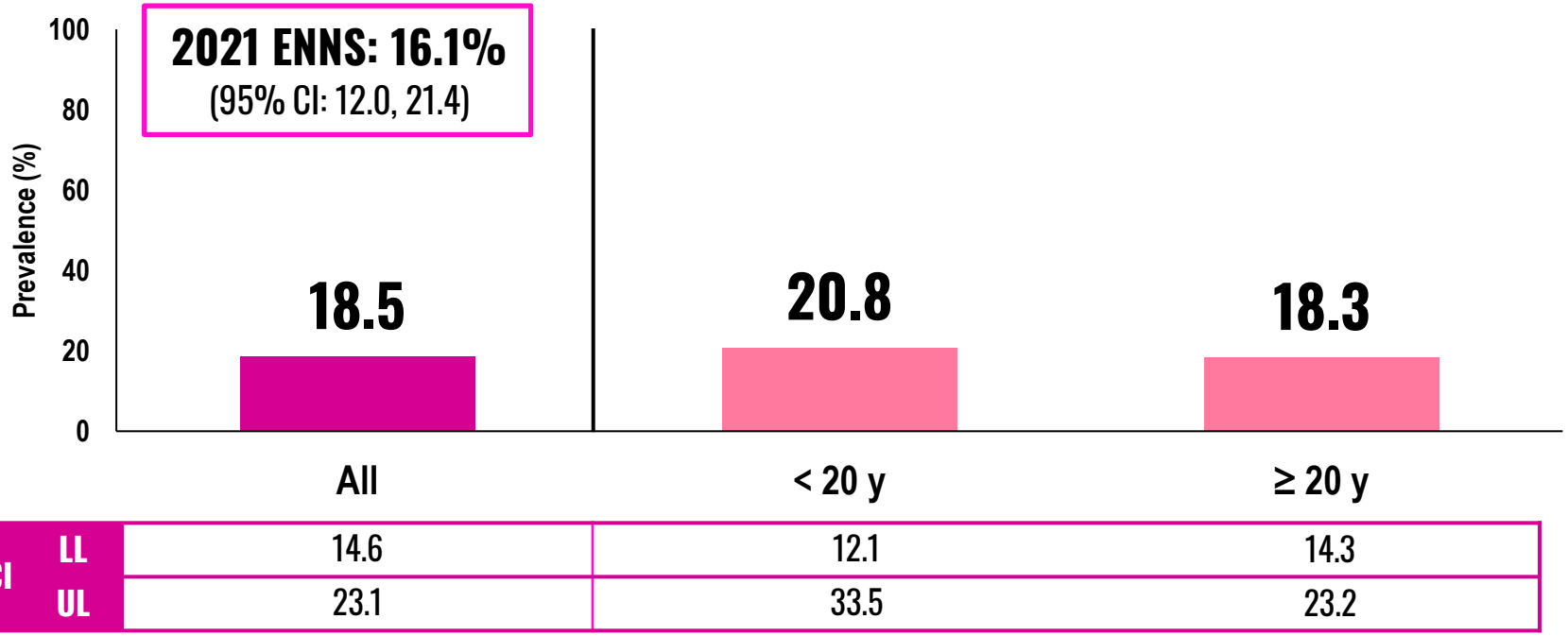


Classification of Nutritionally at-risk Pregnant Women

Cut-off points in classifying the nutritional status of pregnant women based on weight-for-height (Magbitang et al., 1988)

Classification	Percent of Standard
Nutritionally-at-risk	<95%
Not nutritionally-at-risk	≥95%

Prevalence of nutritionally at-risk pregnant women by age group: Philippines, 2023 (n=689)



95% CI	All	< 20 y	≥ 20 y
LL	14.6	12.1	14.3
UL	23.1	33.5	23.2

DOST-FNRI



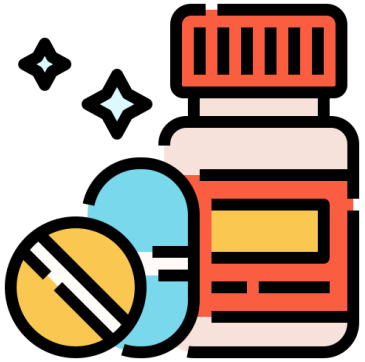
MATERNAL HEALTH AND NUTRITION

Prenatal Care Practices

Prenatal care practices



Timely prenatal visits

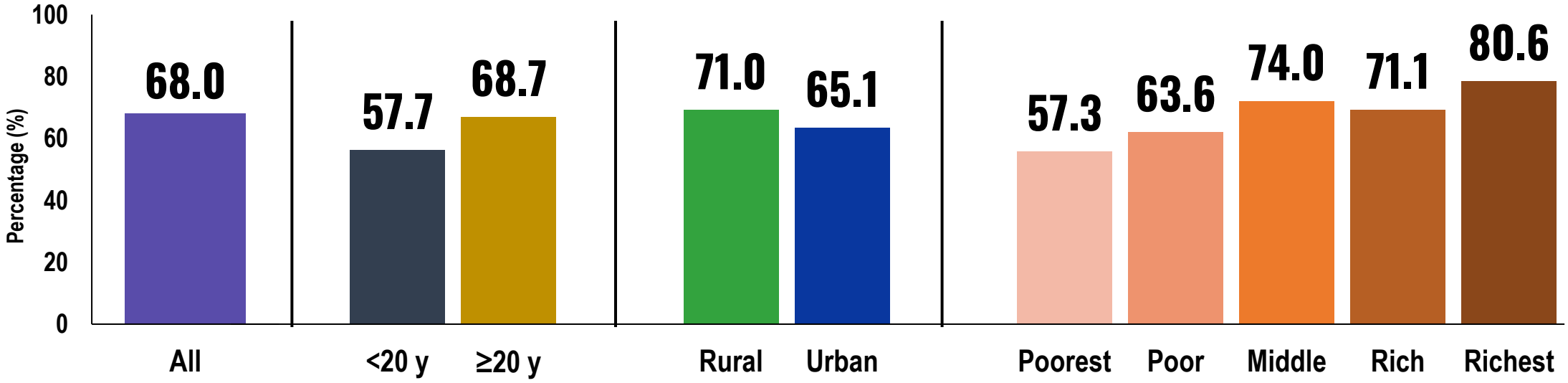


Taking of micronutrient supplements



Prenatal care services availed

Percentage of pregnant women with timely* first prenatal visits by age group, place of residence, and wealth quintile: Philippines, 2023 (n=470)

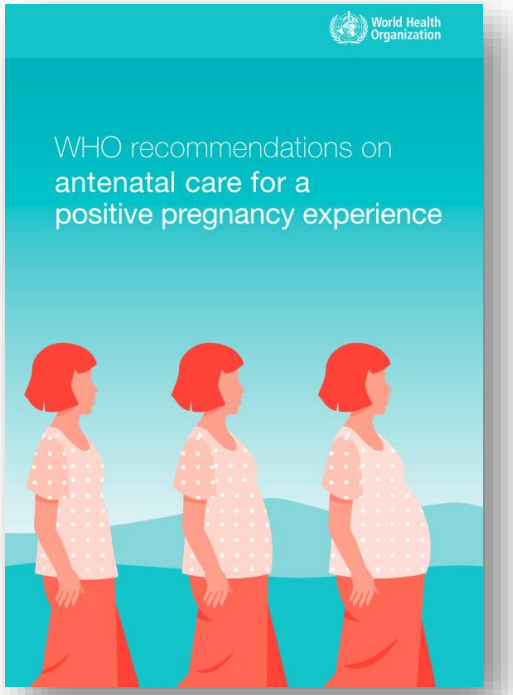


95% CI		Age group		Place of Residence		Wealth Quintile				
LL	59.8	38.8	60.0	62.4	52.8	46.7	50.8	56.4	46.6	64.1
UL	75.3	74.7	76.2	78.3	75.7	67.1	74.7	86.2	87.4	90.6

*within 3 months or first trimester of pregnancy

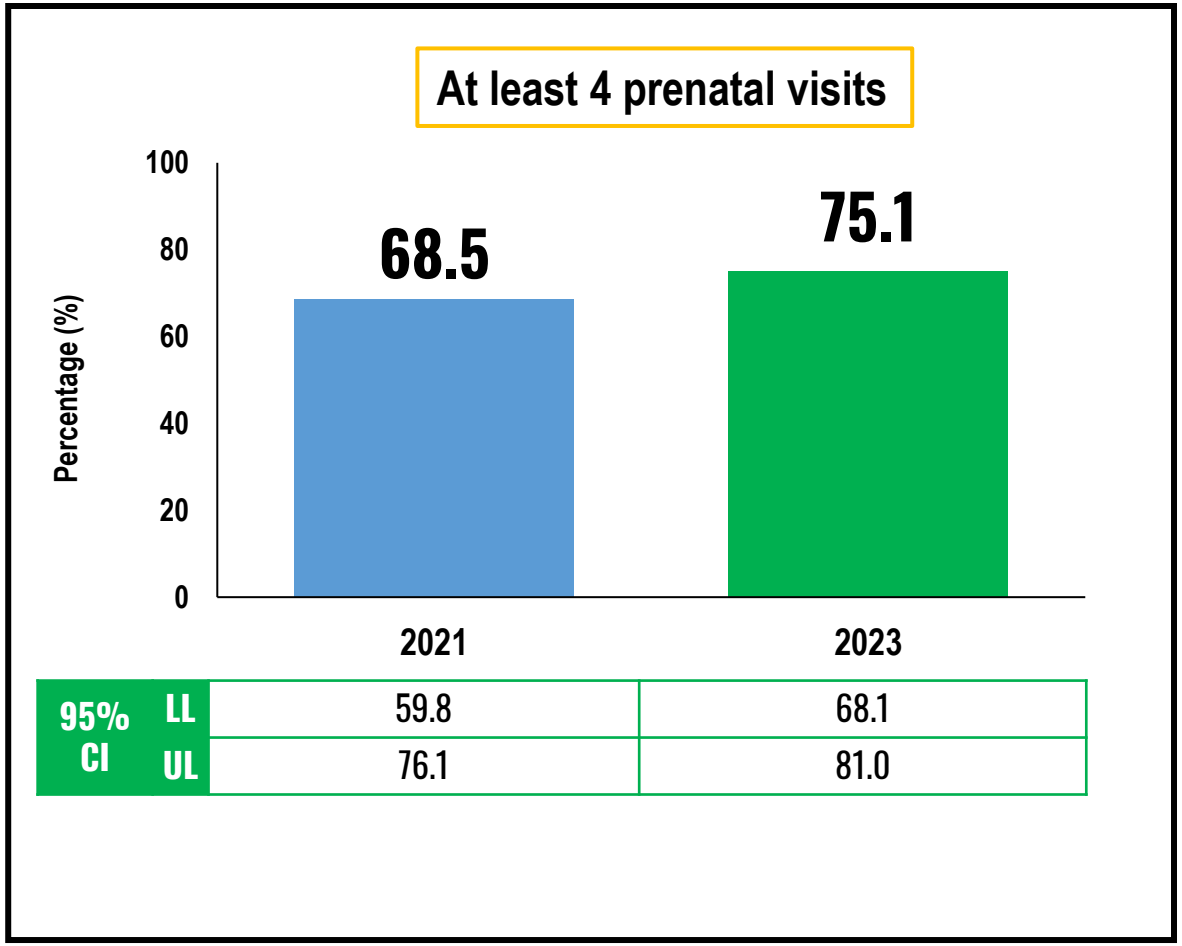
WHO Clinical Guidelines on the Timing of Prenatal Care

WHO FANC model (2001)	WHO ANC – model (2016)
First Trimester	
1st VISIT: 8-12 weeks	Contact 1: Up to 12 weeks (1)
Second Trimester	
2nd VISIT: 24-26 weeks	Contact 2: 20 weeks Contact 3: 26 weeks (2)
Third Trimester	
3rd VISIT: 32 weeks **4TH VISIT: 36-38 weeks**	Contact 4: 30 weeks Contact 5: 34 weeks Contact 6: 36 weeks Contact 7: 38 weeks Contact 8: 40 weeks (5)
Return for delivery at 41 weeks if not given birth yet at 40 weeks.	

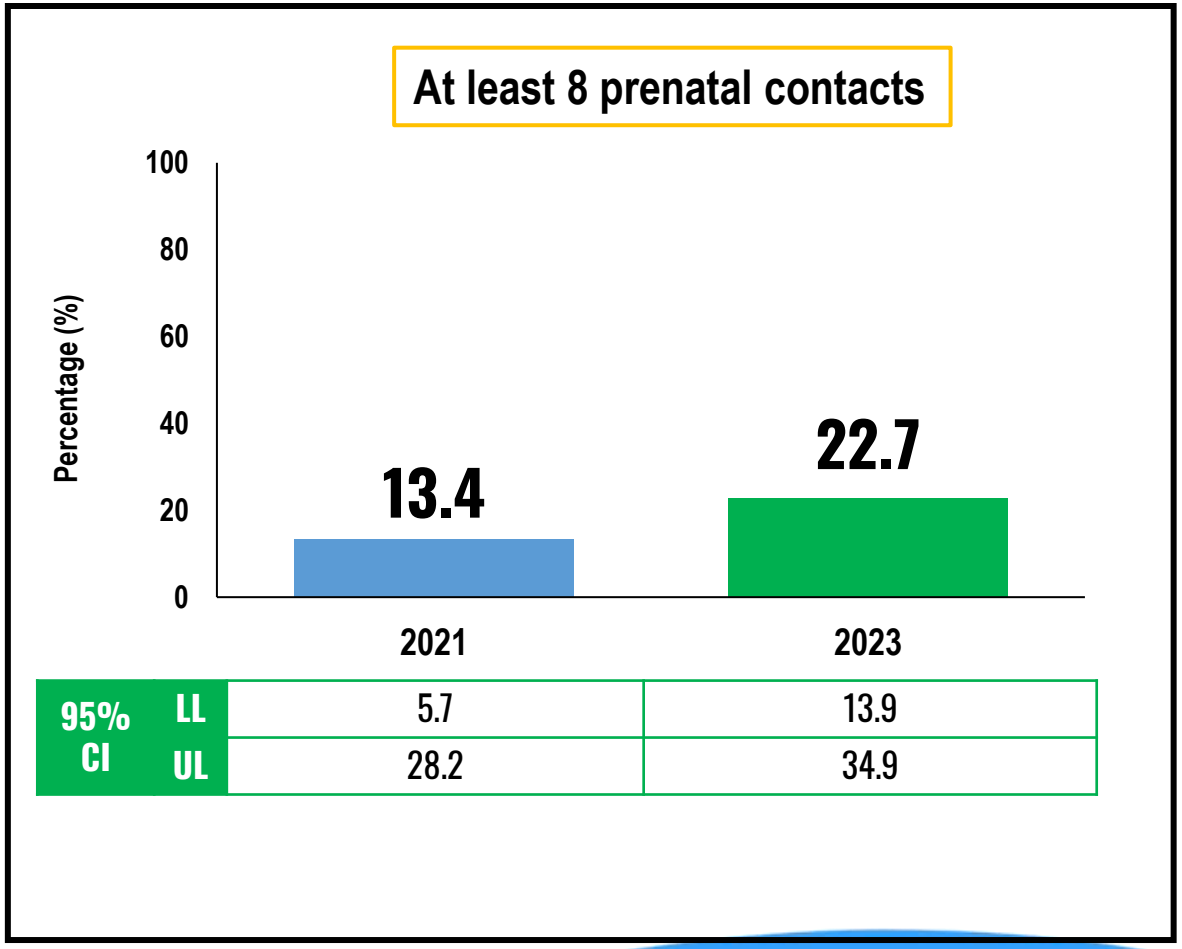


Number of prenatal contacts: at least 8

Percentage of pregnant women in their third trimester with at least four prenatal visits and at least eight prenatal contacts by age group, place of residence, and wealth quintile (n=230)

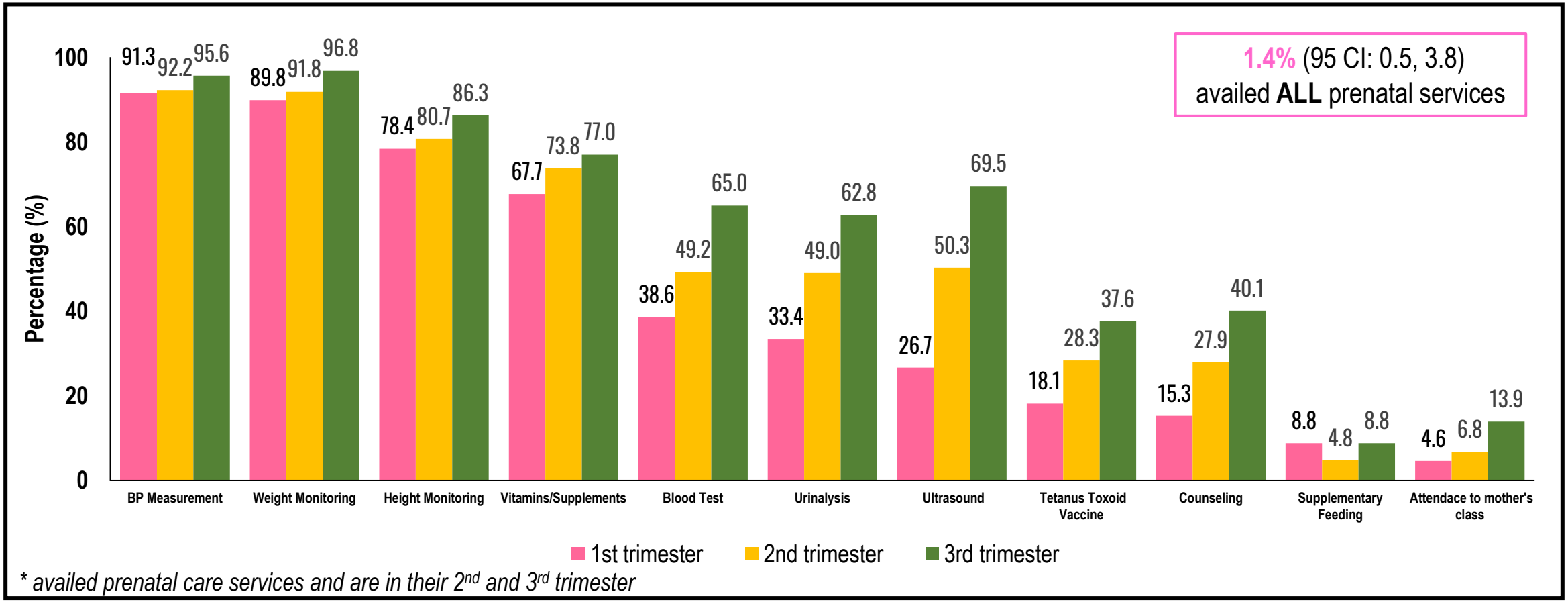


		2021	2023
95% CI	LL	59.8	68.1
	UL	76.1	81.0



		2021	2023
95% CI	LL	5.7	13.9
	UL	28.2	34.9

Percentage of pregnant women* by trimester, by type of prenatal services availed during pregnancy: Philippines, 2023 (n=566)

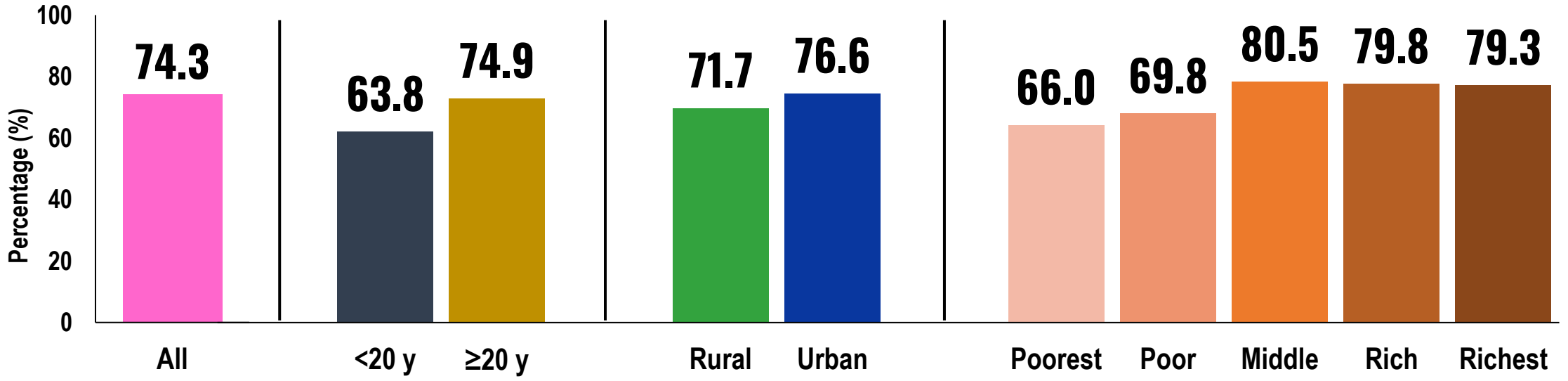


** availed prenatal care services and are in their 2nd and 3rd trimester*

Multiple responses

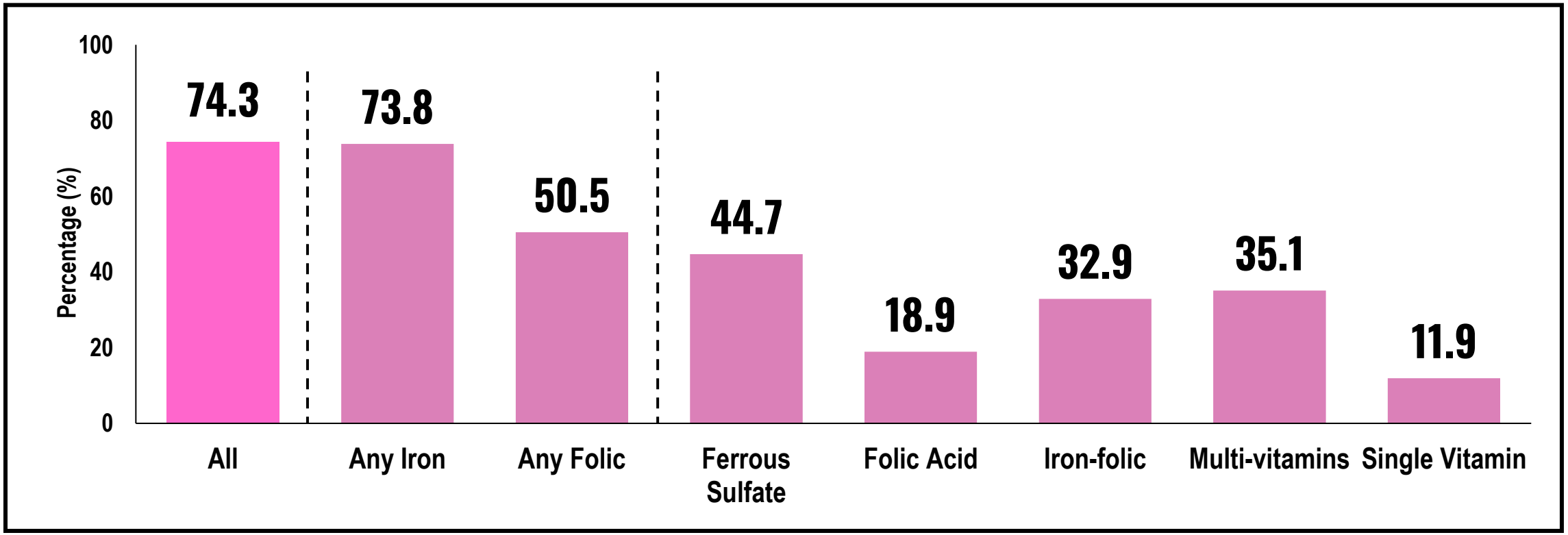
Based on Maternal Newborn Child Health and Nutrition (MNCHN) Strategy

Percentage of pregnant women who took micronutrient supplements by age group, place of residence, and wealth quintile: Philippines, 2023 (n=682)



95% CI		Age group		Place of Residence		Wealth Quintile				
LL	69.0	45.8	69.2	65.1	68.4	58.8	58.0	67.2	65.6	55.3
UL	79.0	78.6	79.8	77.4	83.3	72.6	79.5	89.3	89.1	92.3

Percentage of pregnant women who took micronutrient supplements and type of supplements taken: Philippines, 2023 (n=682)



Multiple responses

HIGHLIGHTS

MATERNAL HEALTH AND NUTRITION SURVEY

NUTRITIONAL STATUS OF PREGNANT WOMEN



2 in 10 (18.5%) pregnant women are **nutritionally-at-risk.**

PRENATAL CARE PRACTICES



7 in 10 (68.0%)
had **timely first prenatal visit**

Among pregnant women in their third trimester:



3 in 4 (75.1%) had **at least 4** prenatal visits



1 in 5 (22.7%) had **at least 8** prenatal contacts

DOST-FNRI

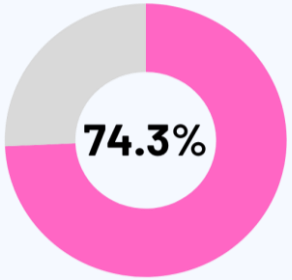
HIGHLIGHTS

MATERNAL HEALTH AND NUTRITION SURVEY

PRENATAL CARE PRACTICES



❖ Majority of the pregnant women availed **BP measurement, weight & height monitoring**, and provision of **micronutrient supplements**.



3 in 4 pregnant women took micronutrient supplements



Ferrous sulfate was the most commonly taken micronutrient supplement



2023 NATIONAL NUTRITION SURVEY

INFANT AND YOUNG CHILD FEEDING PRACTICES

Why do we collect Infant and Young child Feeding Practices data?

- **The First 1000 days period from conception up to 2 years of a child's life is critical because of the** *high nutrient needs to support rapid growth and development, vulnerability to infectious diseases and their reversible form of malnutrition that may occur if a child's needs are not addressed*

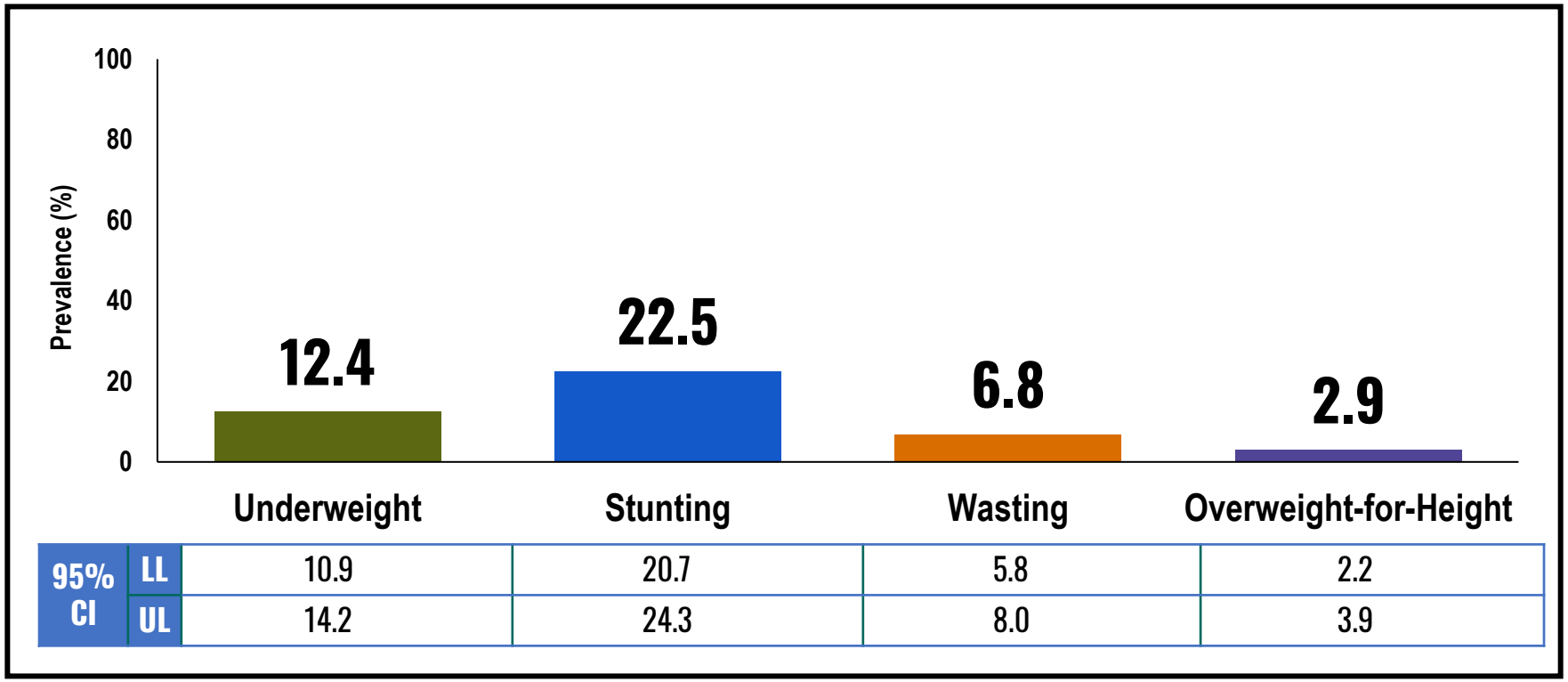


- **Infant and young child feeding (IYCF) practices** directly affect the health, development and nutritional status of children less than 2 years of age

- **Vital inputs to craft and formulate laws and programs to improve IYCF practices and address childhood malnutrition**



Prevalence of malnutrition among infants and young children (0 to 23 months old): Philippines, 2023



		Underweight	Stunting	Wasting	Overweight-for-Height
95% CI	LL	10.9	20.7	5.8	2.2
	UL	14.2	24.3	8.0	3.9

ELIGIBLE SUBJECTS AND RESPONDENTS





Eligible Subjects

❖ Infants and young children 0-23 months

*Who is considered an **infant**?*

✓ *A child 0-12 months of age*

*Who is considered a **young child**?*

✓ *A child from the age 12 months and one day up to 23 months*

Respondents

❖ Mothers or primary caregivers

Data Collection Method

- Face-to-face interview using a structured IYCF Questionnaire
- One-day 24-Hour Food Recall
- Checklist of Food and liquid intake of children 0-23 months

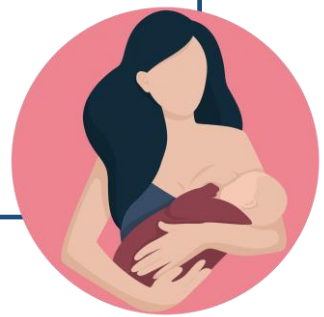


eDCS ELECTRONIC DATA COLLECTION SYSTEM

IYCF INDICATORS

BREASTFEEDING PRACTICES

- Timely breastfeeding initiation
- Exclusive breastfeeding
- Continued breastfeeding up to 1 year
- Continued breastfeeding up to 2 years



COMPLEMENTARY FEEDING PRACTICES

- Timely introduction to solid, semi-solid and soft foods
- Minimum Dietary Diversity
- Minimum Meal Frequency
- Minimum Acceptable Diet



IYCF DEFINITIONS/CONCEPTS



Current feeding practice - the type of feeding based on the foods eaten by the child the previous day using 24-hr food recall method.



Exclusive Breastfeeding - infant only receives solely breast milk for the first six months of life



Predominant breastfeeding - when breast milk is the main source of nutrition for an infant, but they may also consume other liquids



Prelacteal Feed - any liquids or drinks given to newborn temporarily within the first 3 days of life before the mother's milk begin to flow regularly.

DIFFERENT CURRENT FEEDING PRACTICES



(01) Exclusive breastfeeding

(02) Predominant breastfeeding



(03) Breastfeeding + breastmilk substitute (BMS) + complementary foods (CF)

(04) Pure BMS

(05) BMS with CF

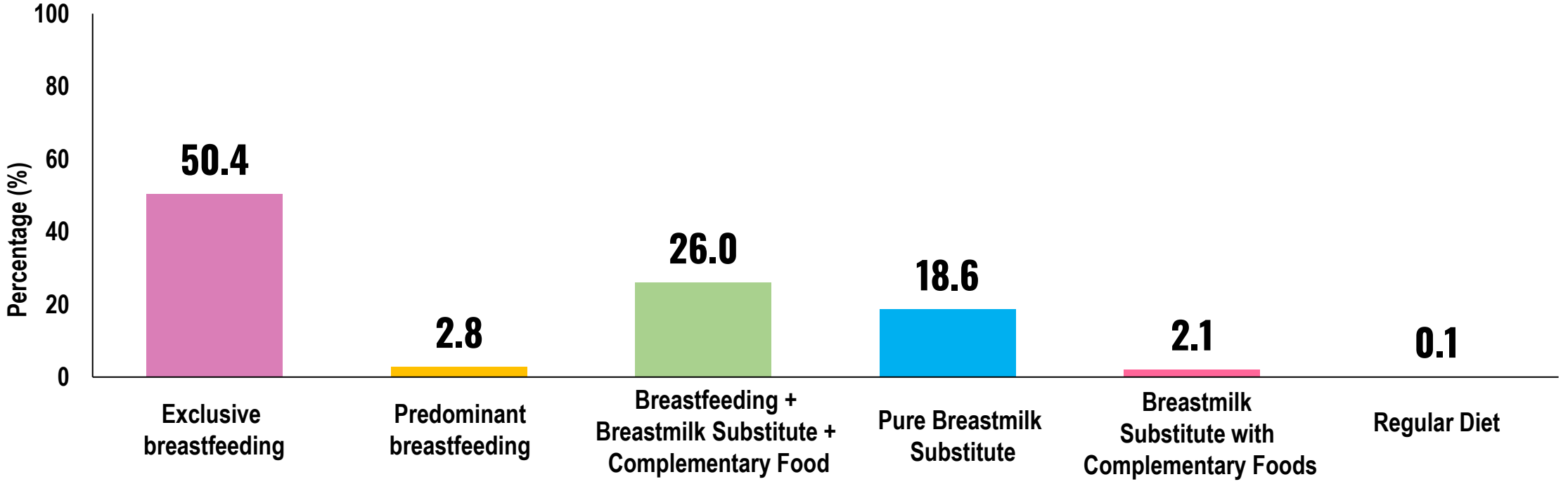
(06) Regular Diet





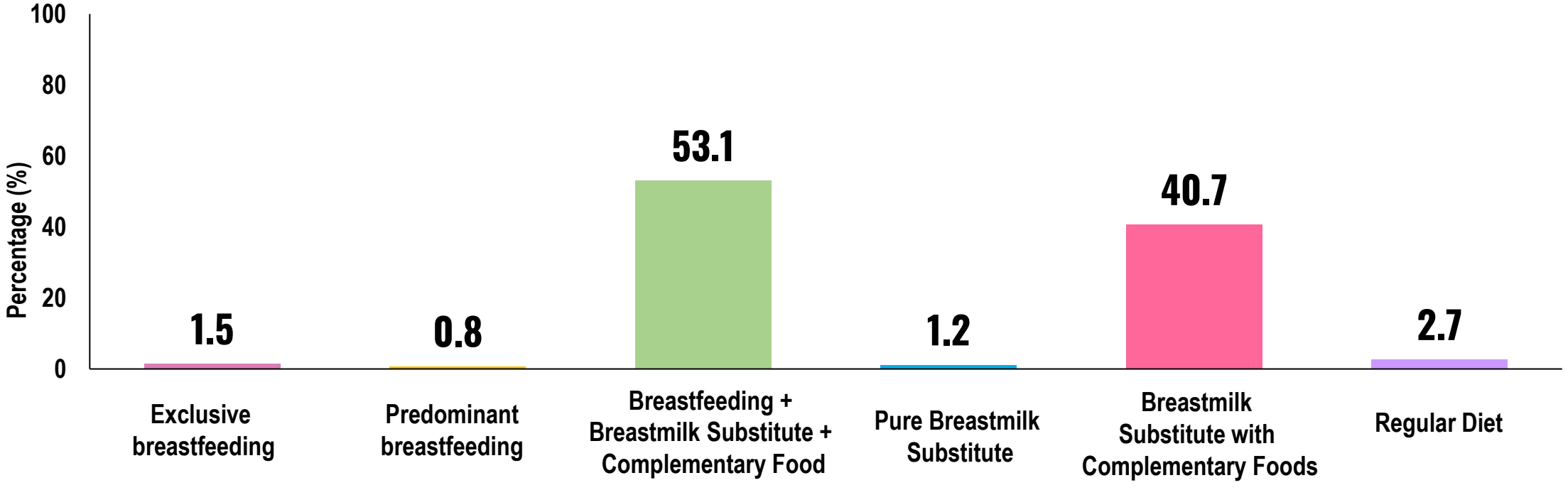


Percentage of infants and young children, 0-5.9 months by current feeding practices: Philippines, 2023 (n=737)



95% CI	Exclusive breastfeeding	Predominant breastfeeding	Breastfeeding + Breastmilk Substitute + Complementary Food	Pure Breastmilk Substitute	Breastmilk Substitute with Complementary Foods	Regular Diet
LL	44.9	1.7	21.7	14.8	1.1	0.0
UL	55.9	4.6	30.8	23.1	3.9	0.5

Percentage of infants and young children, 6-23.9 months by current feeding practices: Philippines, 2023 (n=2,552)



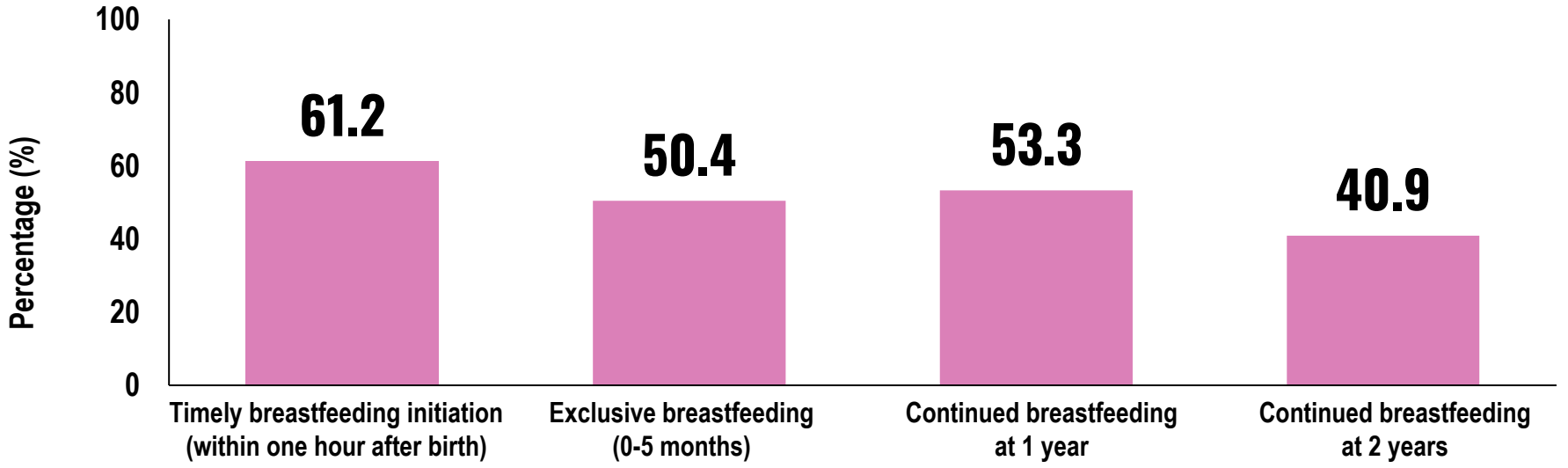
95% CI	LL	1.0	0.5	50.6	0.7	38.3	2.1
	UL	2.2	1.3	55.5	2.1	43.2	3.4



BREASTFEEDING PRACTICES

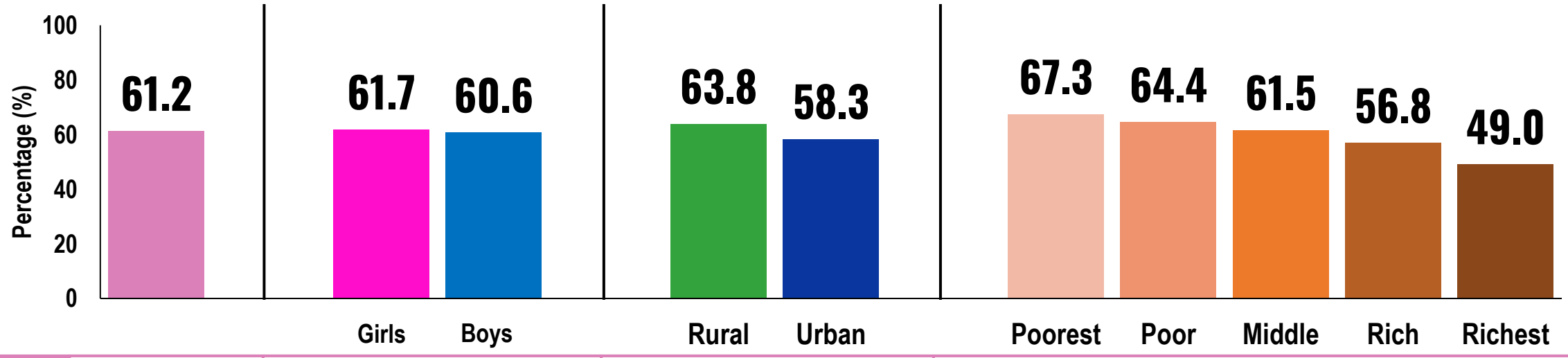
1. **Early initiation of breastfeeding** within one hour after birth
2. **Exclusive breastfeeding** from birth up to 5.9 months
3. **Continued breastfeeding** up to 1 year (12-15 months)
4. **Continued breastfeeding** up to 2 years and beyond (20-23 months)

Percentage of infants and young children (0 to 23 months old) by breastfeeding practices: Philippines, 2023



95% CI	Timely breastfeeding initiation (within one hour after birth)	Exclusive breastfeeding (0-5 months)	Continued breastfeeding at 1 year	Continued breastfeeding at 2 years
LL	58.7	44.9	48.6	35.1
UL	63.5	55.9	58.0	47.0

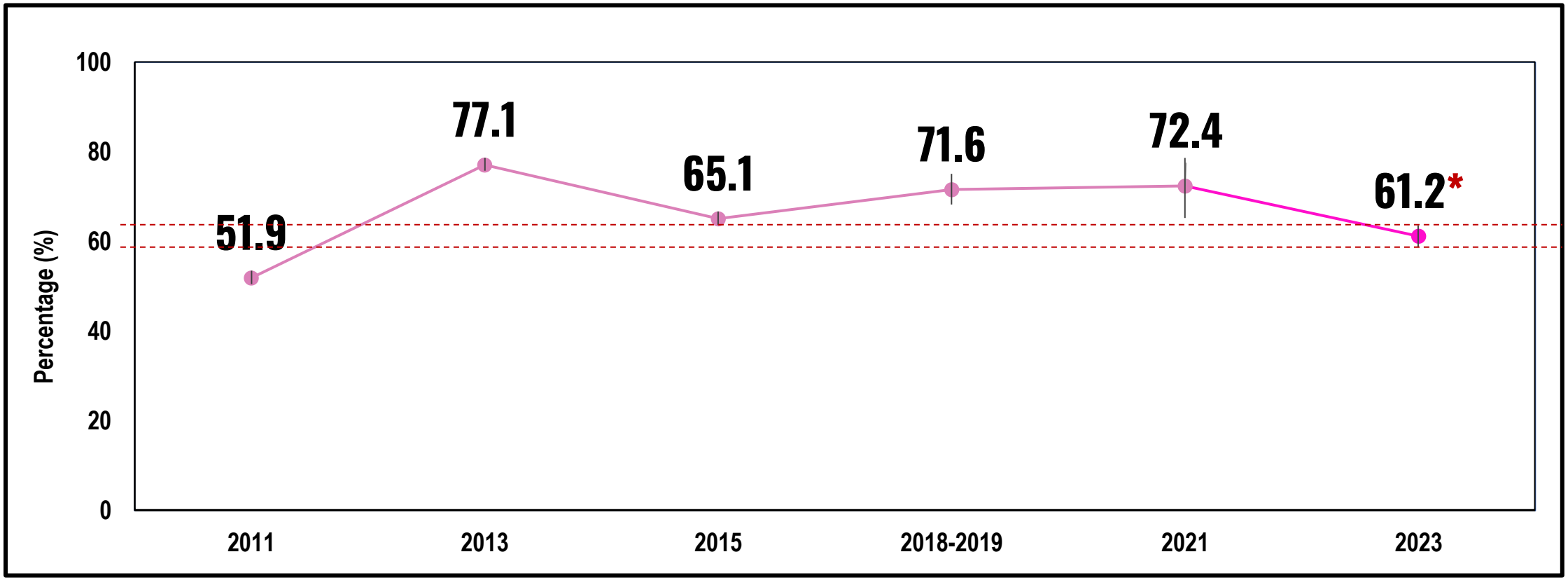
Percentage of children 0-23 months with timely breastfeeding initiation* by sex, place of residence, and wealth quintile: Philippines, 2023 (n=3,289)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	58.7	58.0	57.1	60.6	54.8	63.0	59.0	55.7	50.2	40.8
UL	63.5	65.2	64.1	66.9	61.8	71.3	69.5	66.9	63.1	57.3

*Timely breastfeeding initiation- early initiation of breastfeeding within one hour after birth

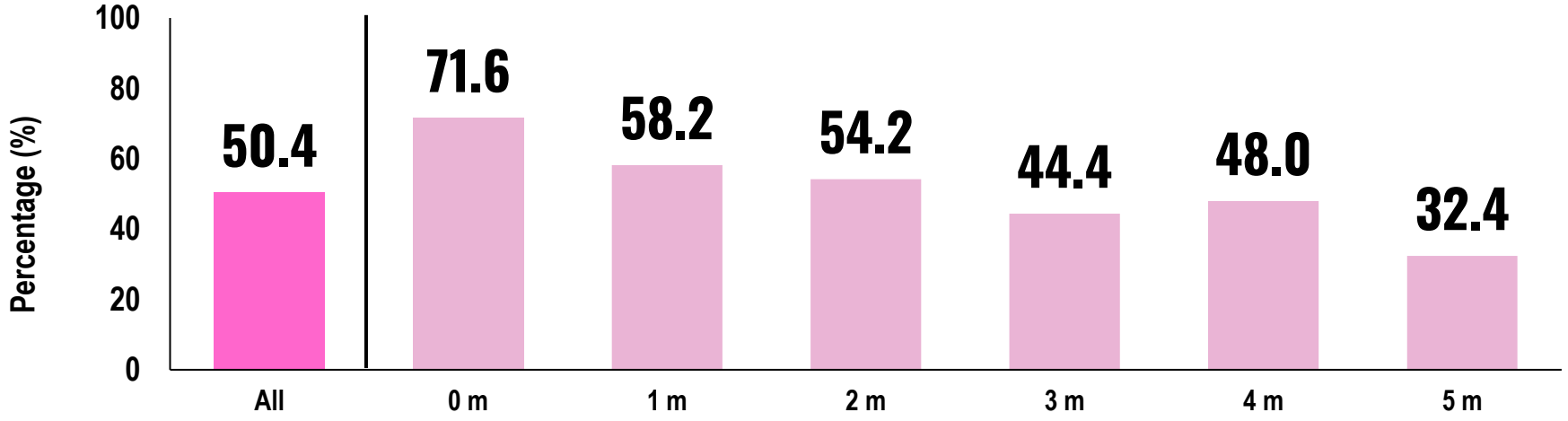
Trends in the percentage of children 0-23 months with timely breastfeeding initiation**: Philippines, 2011-2023



*significant decrease from 2021 ($p < 0.05$)

**within 1 hour after delivery among children 0-23 months

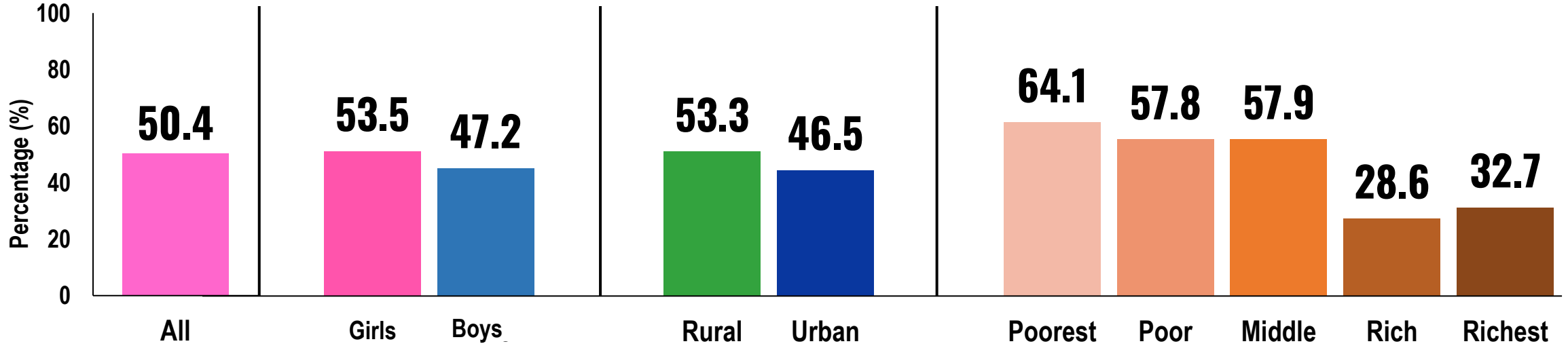
Percentage of exclusively breastfed* infants 0-5 months by single age in months: Philippines, 2023 (n=737)



95% CI		All	0 m	1 m	2 m	3 m	4 m	5 m
	LL	44.9	54.9	45.3	41.5	33.2	35.8	22.5
	UL	55.9	83.9	70.1	66.4	56.2	60.4	44.1

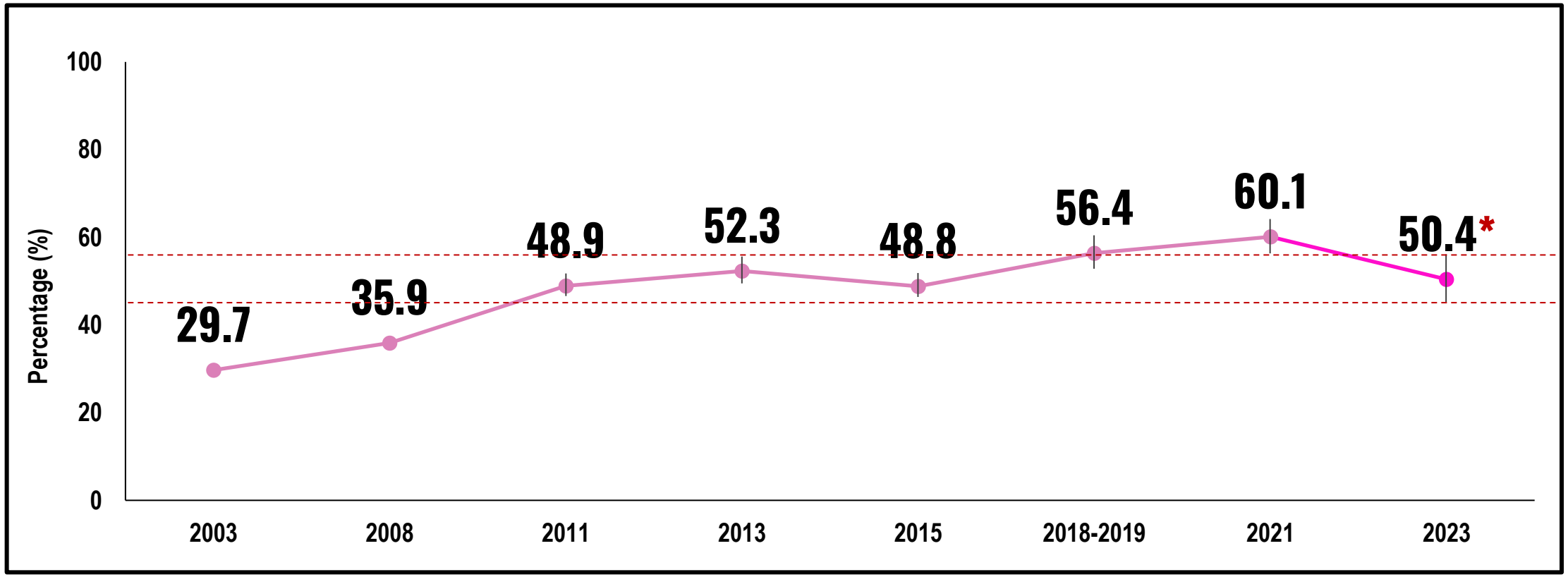
**Exclusively breastfed-* *an infant was fed exclusively with breast milk during the previous day*

Percentage of exclusively breastfed infants 0-5 months by sex, place of residence, and wealth quintile: Philippines, 2023 (n=737)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	44.9	45.9	40.3	46.5	38.3	54.7	47.2	46.1	18.3	20.6
UL	55.9	61.0	54.3	60.0	54.9	72.6	67.7	68.8	41.9	47.7

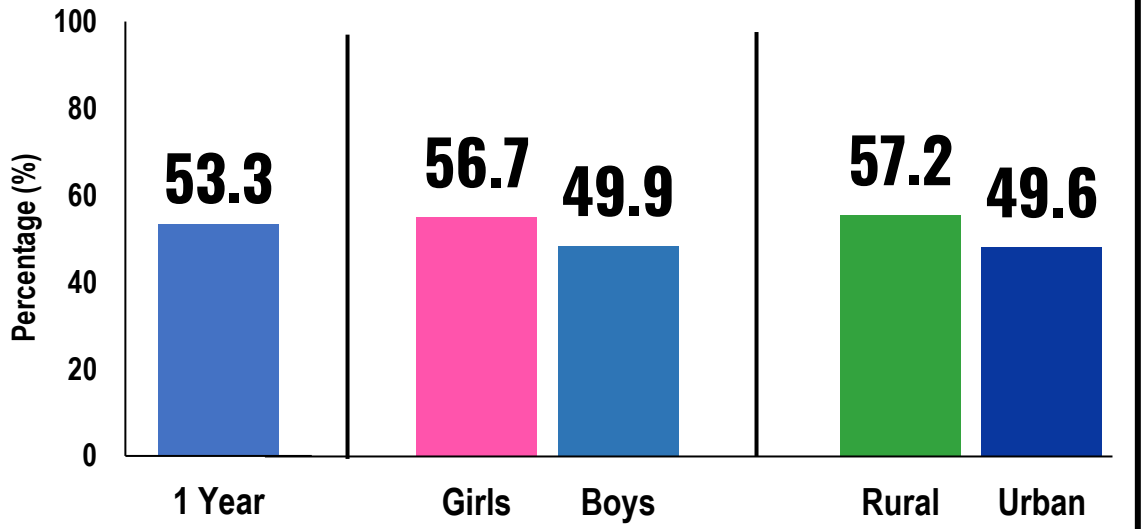
Trends in the percentage of exclusively breastfeeding** among infants 0-5 months old: Philippines, 2003-2023



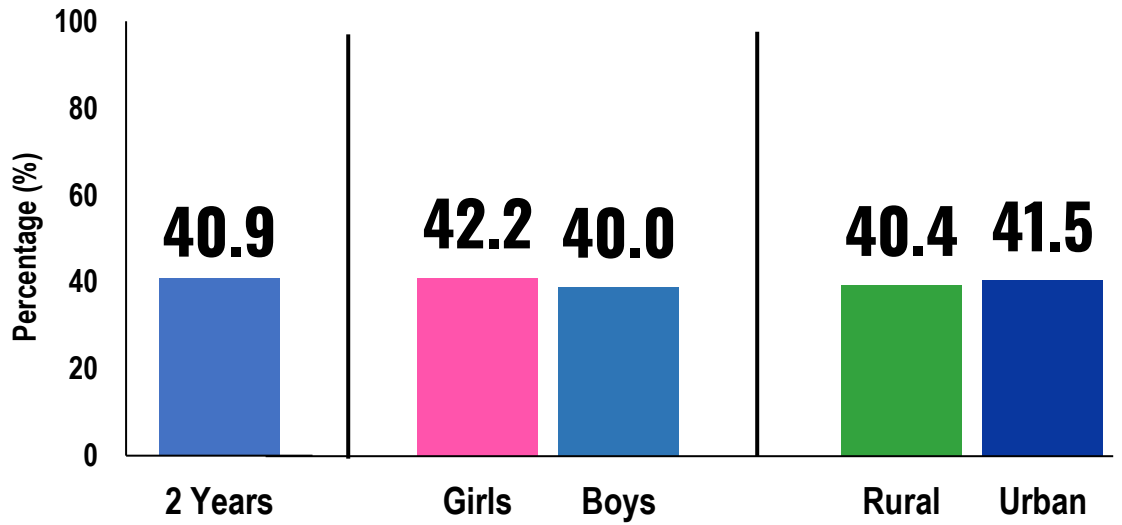
*significant decrease from 2021 ($p < 0.05$)

**receives only breastmilk with no other solids, liquids, or water given except for vitamins, minerals, or medicine in the form of drops

Percentage of children with continued breastfeeding* at 1 year and 2 years by sex and place of residence: Philippines, 2023



95% CI		Sex		Place of Residence	
LL	48.6	49.2	42.9	50.9	42.3
UL	58.0	63.9	56.9	63.3	56.9

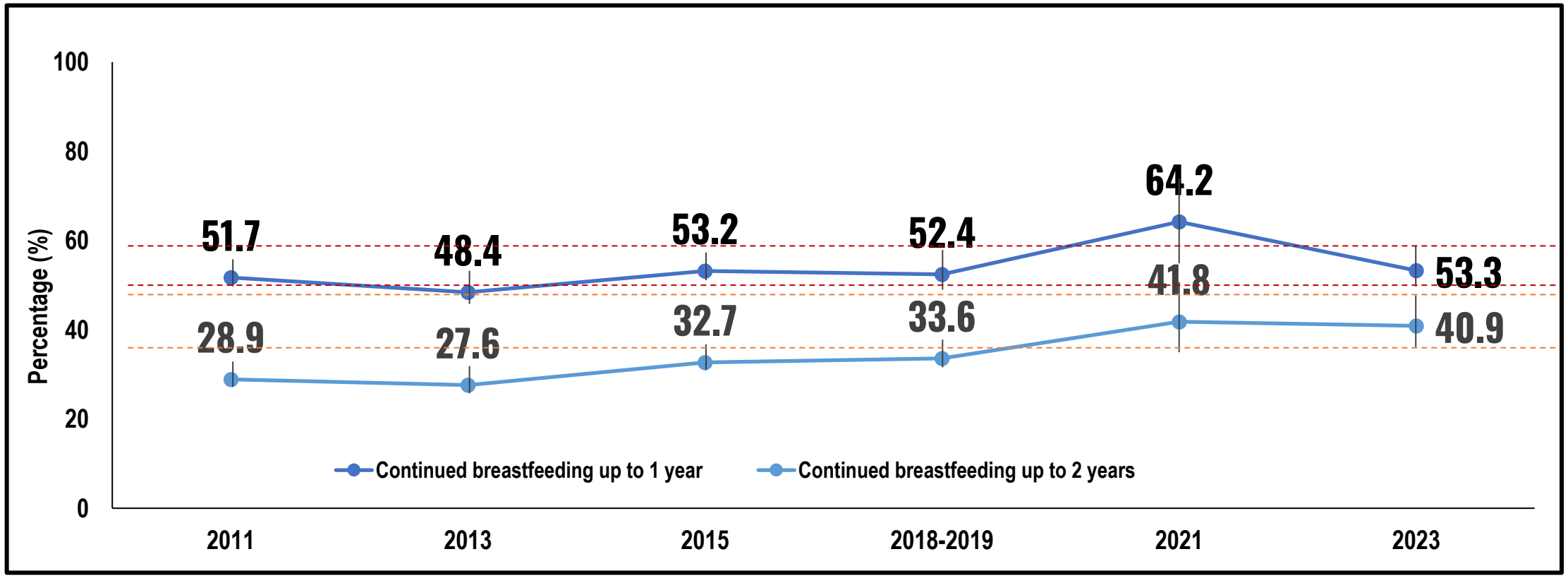


95% CI		Sex		Place of Residence	
LL	35.1	34.1	32.6	33.9	31.8
UL	47.0	50.7	47.8	47.3	51.8

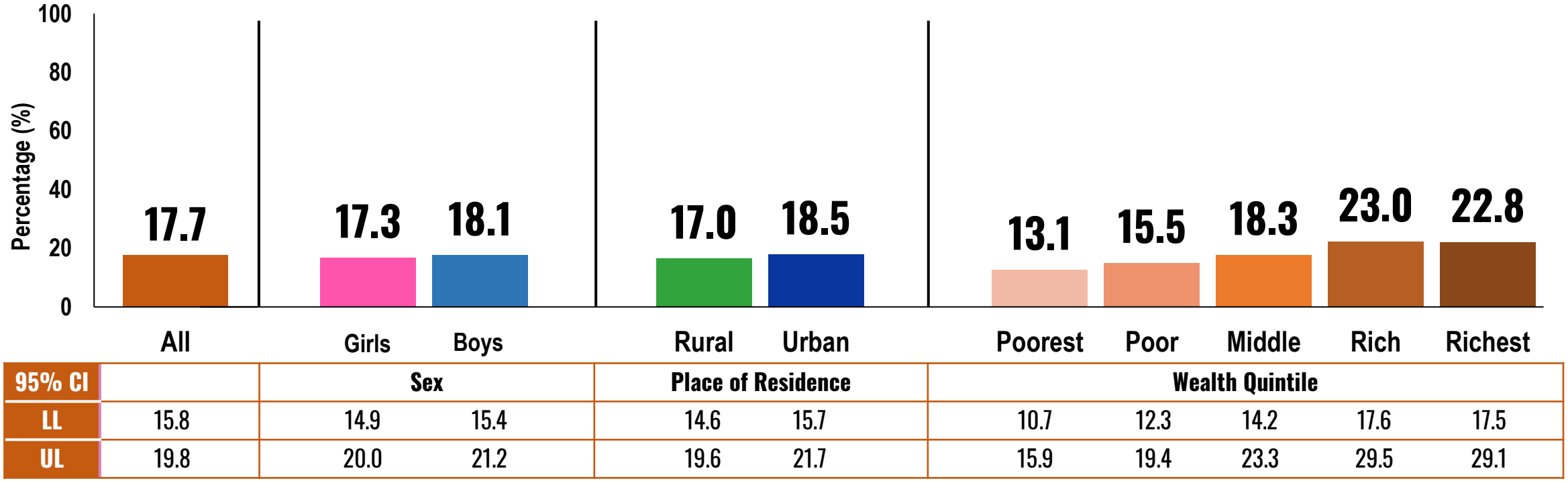
**Continued breastfeeding at 1 year -* *a child between the age of 12-15 months was fed breastmilk during the previous day (n=578)*

**Continued breastfeeding at 2 years -* *a child between the age of 20-23 months was fed breastmilk during the previous day (n=530)*

Trends in the percentage of young children who were breastfed until 1 year and 2 years of age: Philippines, 2011 to 2023



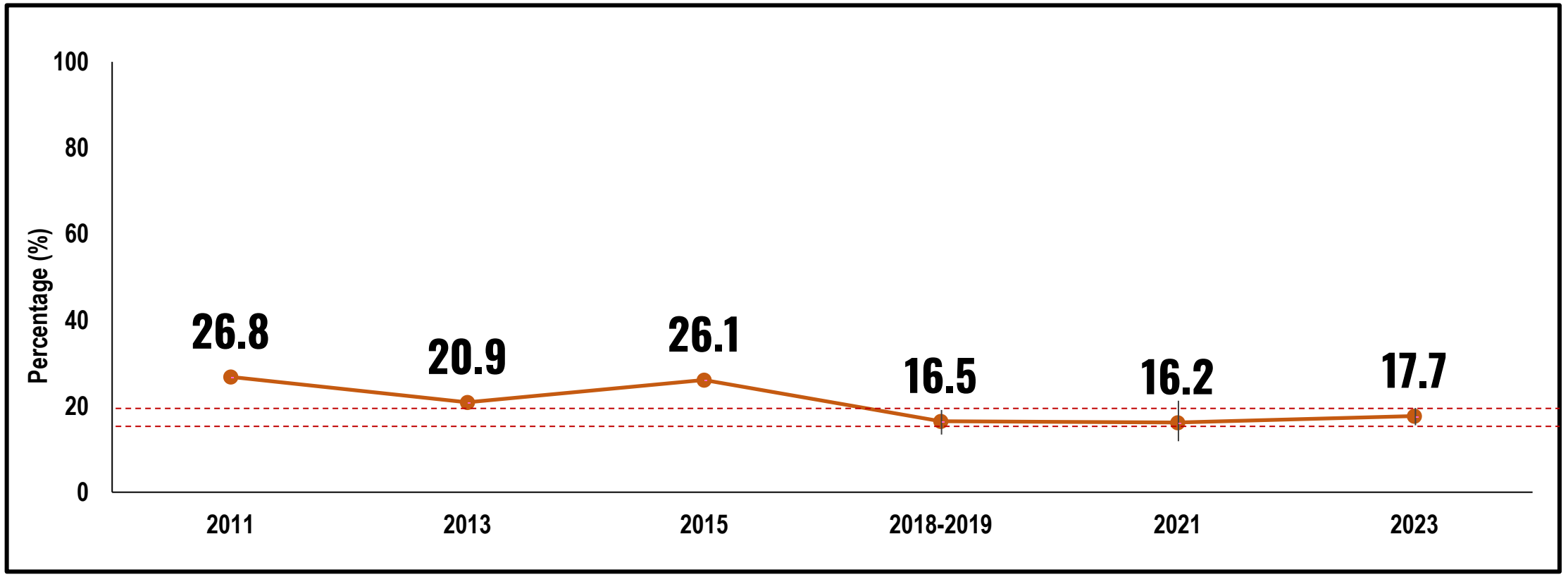
Percentage of children 0 to 23 months old who were given pre-lacteal feeds* by sex, place of residence, and wealth quintile: Philippines, 2023 (n=3,248)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	15.8	14.9	15.4	14.6	15.7	10.7	12.3	14.2	17.6	17.5
UL	19.8	20.0	21.2	19.6	21.7	15.9	19.4	23.3	29.5	29.1

any liquids or drinks given to newborn temporarily within the first 3 days of life before the mother's milk begin to flow regularly.

Trends in the percentage of children 0 to 23 months old who were given pre-lacteal feeds*: Philippines, 2023



*any liquids or drinks given to newborn temporarily within the first 3 days of life before the mother's milk begin to flow regularly.

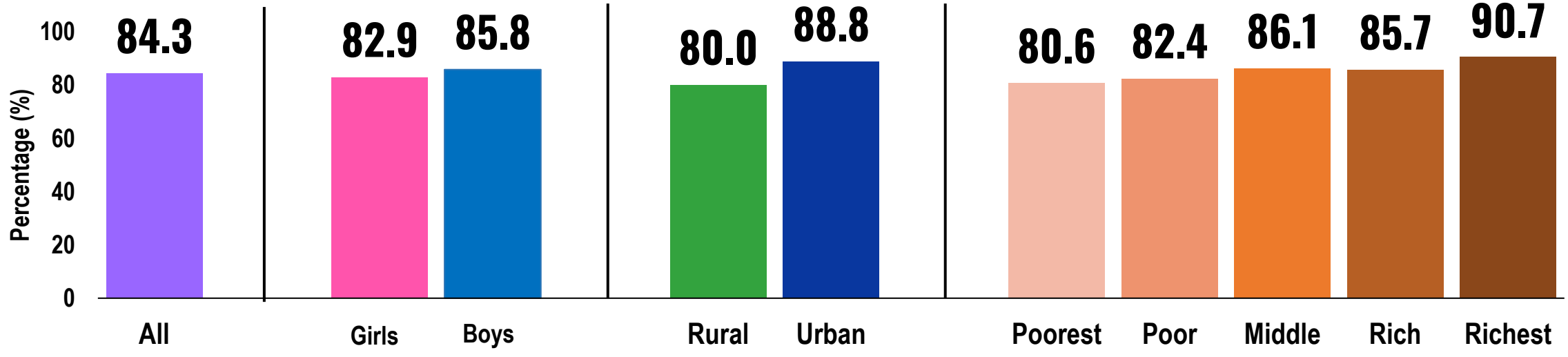


COMPLEMENTARY FEEDING PRACTICES

Starting at 6 months of age, infants should be fed safe, adequate, and nutrient rich foods throughout the day while continuing to receive breastmilk.

1. Timely introduction to solid, semi-solid and soft foods
2. Minimum Dietary Diversity
3. Minimum Meal Frequency
4. Minimum Acceptable Diet

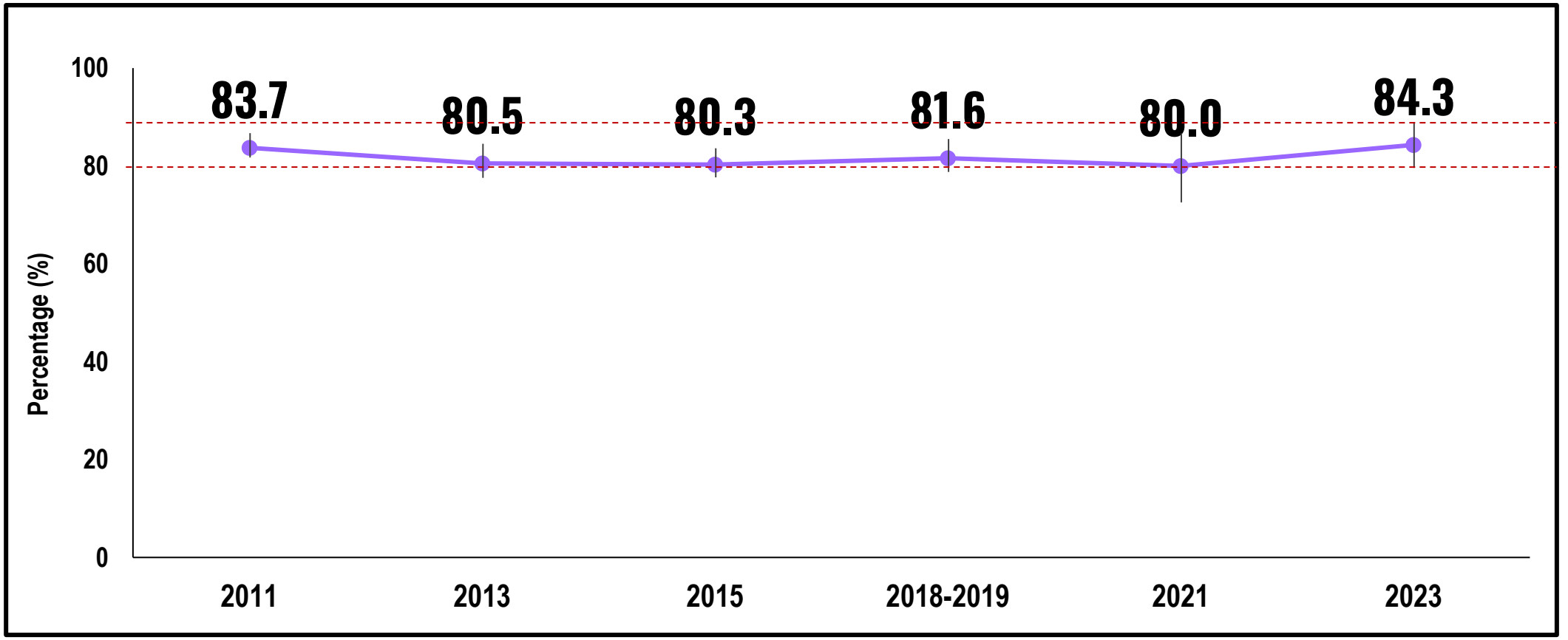
Percentage of infants 6-8 months old who were given timely complementary foods* by sex, place of residence, and wealth quintile: Philippines, 2023 (n=422)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	79.1	74.3	78.5	72.4	80.2	70.7	71.8	69.8	67.7	65.7
UL	88.4	89.1	90.9	85.9	93.9	87.7	89.6	94.3	94.5	98.0

Timely complementary foods- infant aged between 6 to 8 months consumed solid, semi-solid or soft foods during the previous day

Trends in the percentage of infants 6-8 months old who received timely complementary feeding: Philippines, 2011-2023



**Timely complementary foods-* *infant aged between 6 to 8 months consumed solid, semi-solid or soft foods during the previous day*

Minimum Diet Diversity (MDD)

Children 6-23 months of age who consumed foods and beverages from at least 5 or more food groups (DDS ≥ 5) of the following 8 food groups

1. Breastmilk



2. Grains, roots and tubers



3. Dairy Products



4. Legumes and Nuts



5. Flesh Meat



6. Eggs



7. Vitamin A rich fruits & vegetables

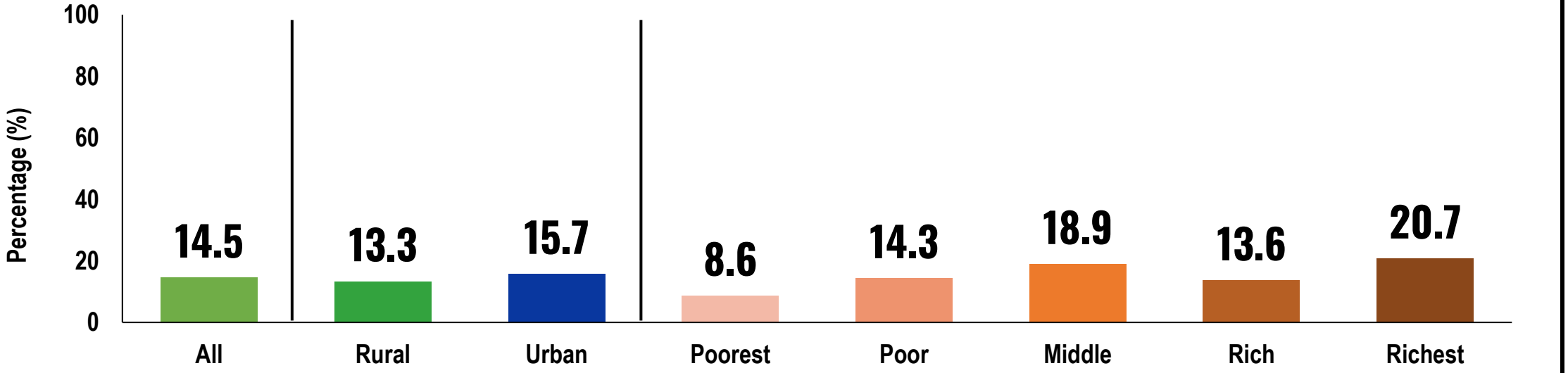


8. Other fruits & vegetables



Reference: *2021 World Health Organization and the United Nations Children's Fund (UNICEF) - Indicators for Assessing Infant and Young Child Feeding Practices: Definitions and Measurement Methods.*

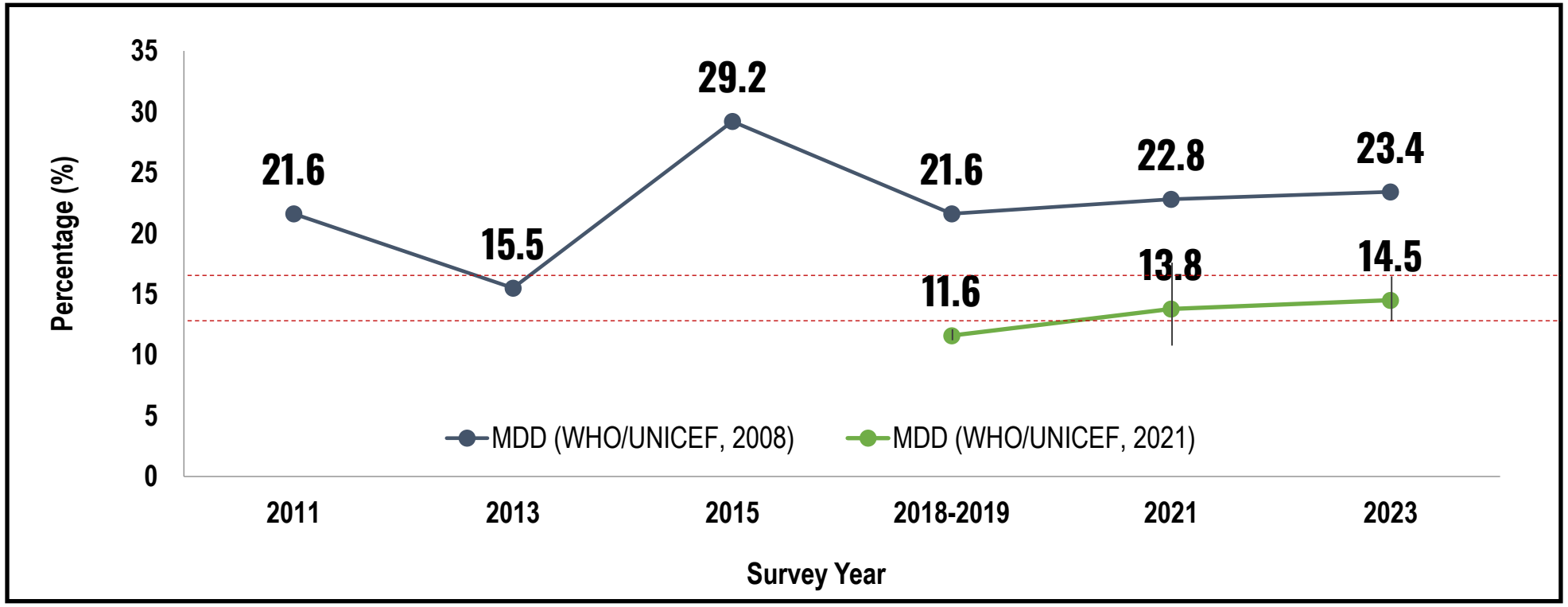
Percentage of children 6-23 months old who met the Minimum Dietary Diversity * (MDD) by place of residence and wealth quintile: Philippines, 2023 (n=2,552)



95% CI		Place of Residence		Wealth Quintile				
LL	12.7	11.4	13.0	6.5	11.2	14.5	9.5	14.9
UL	16.4	15.6	18.8	11.3	18.1	24.1	18.9	27.9

consumed at least five (5) out of eight (8) food groups during the previous day (WHO/UNICEF, 2021)

Trends in the percentage of children 6-23 months meeting the Minimum Dietary Diversity* (MDD): Philippines, 2011-2023



*consumed at least five (5) out of eight (8) food groups during the previous day (WHO/UNICEF, 2021)

Minimum Meal Frequency (MMF)

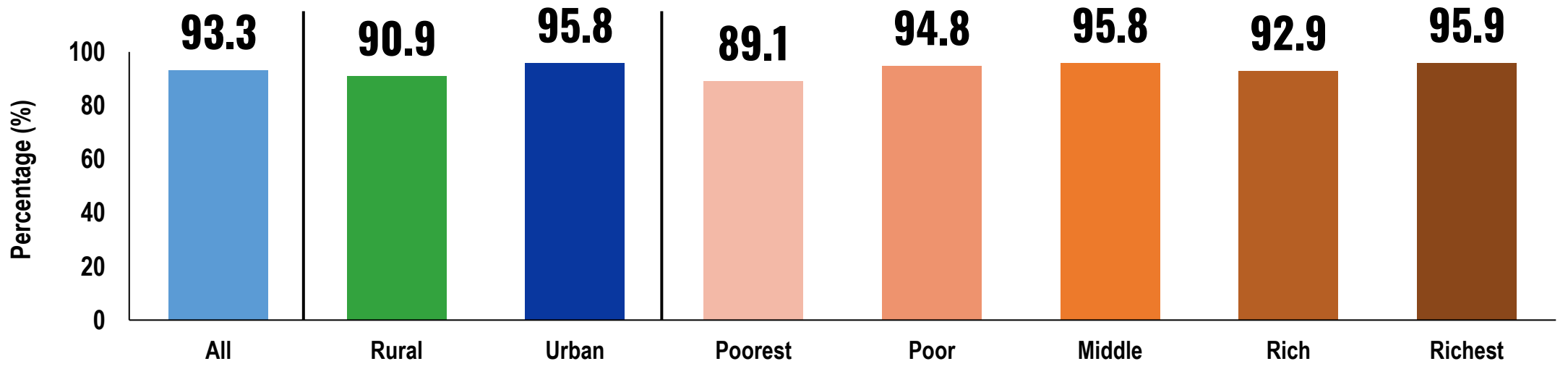
Breastfed and non-breastfed children 6-23 months of age received solid, semi-solid or soft foods (but also including milk feeds for non-breastfed children) the minimum number of times or more



Minimum meal is defined as:

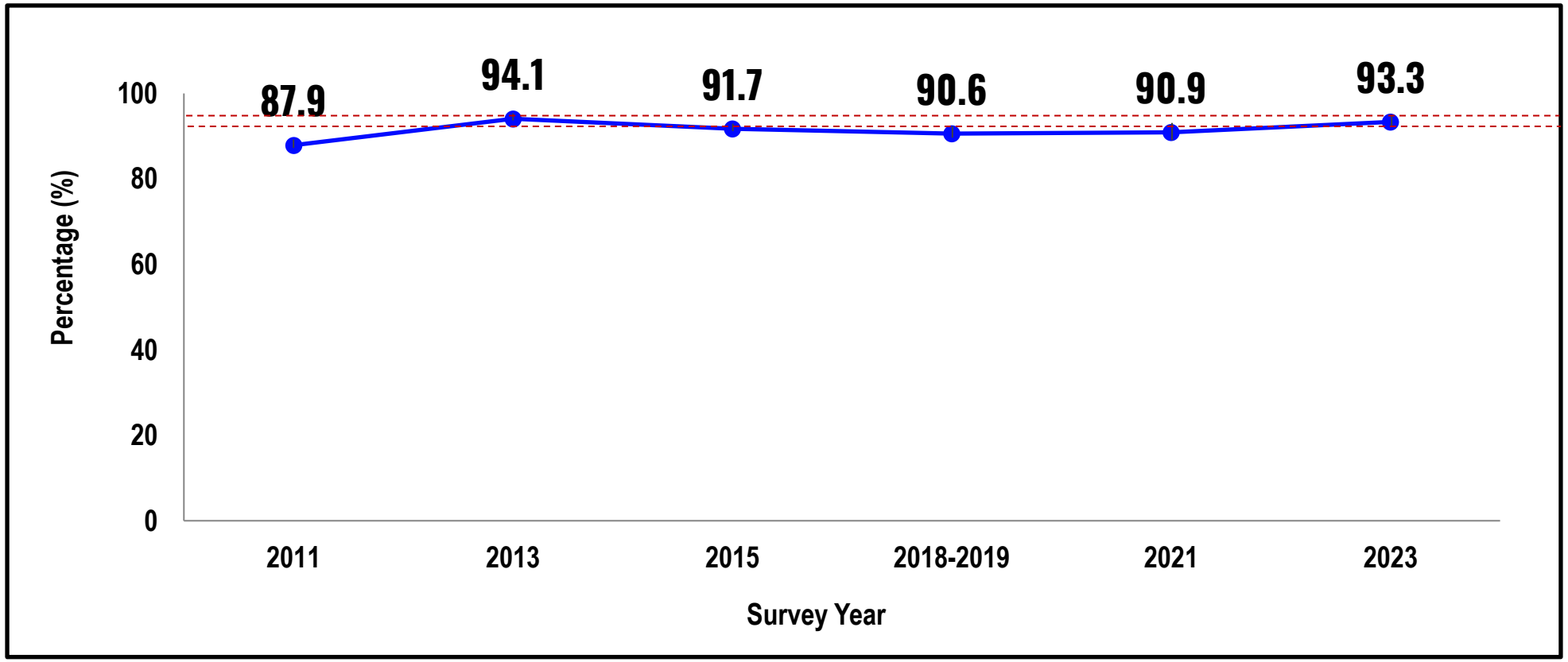
- 2 times for breastfed infants 6-8 months
- 3 times for breastfed children 9-23 months
- 4 times for non-breastfed children 6-23 months

Percentage of children 6-23 months old who met the Minimum Meal Frequency (MMF) by place of residence and wealth quintile: Philippines, 2023 (n=2,552)



			Place of Residence		Wealth Quintile			
95% CI	LL	91.9	88.4	94.2	85.9	92.6	93.1	87.6	91.4
	UL	94.5	92.9	97.0	91.7	96.3	97.5	96.0	98.1

Trends in the percentage of children 6-23 months old who met the Minimum Meal Frequency (MMF): Philippines, 2011-2023

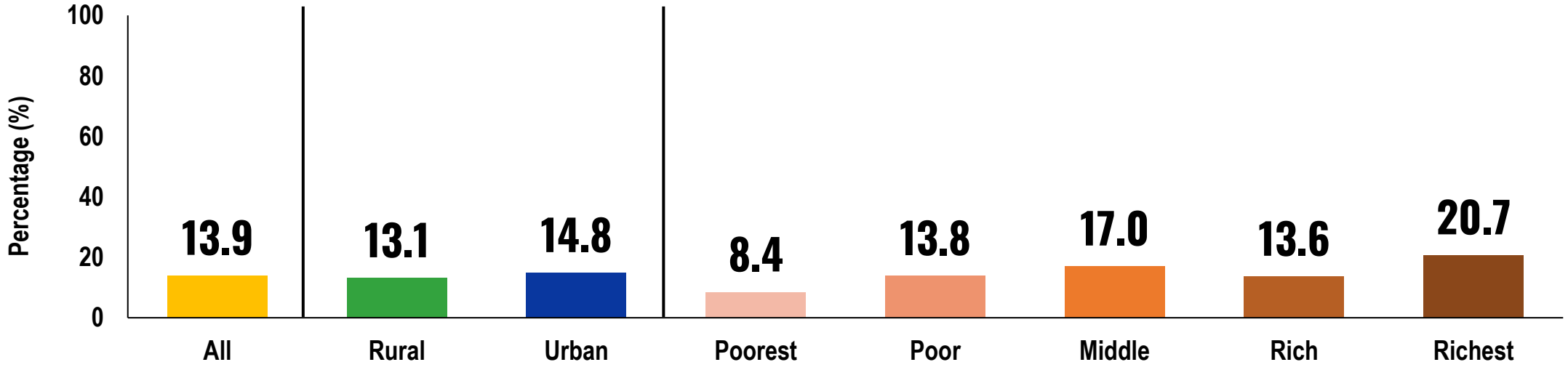


Minimum Acceptable Diet (MAD)

Children 6-23 months meeting both the minimum dietary diversity and the minimum meal frequency to ensure both dietary and nutrient adequacy (WHO, 2021)

Reference: *2021 World Health Organization and the United Nations Children's Fund (UNICEF) - Indicators for Assessing Infant and Young Child Feeding Practices: Definitions and Measurement Methods.*

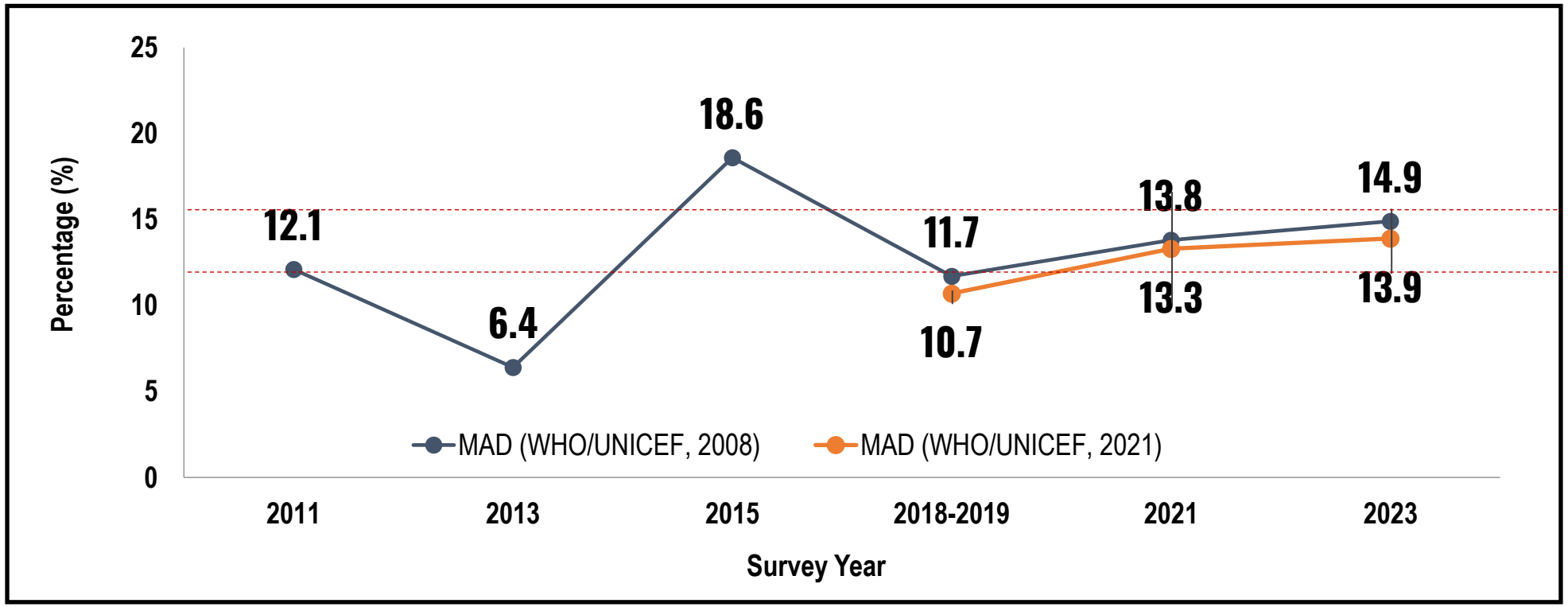
Percentage of children 6-23 months old who met the Minimum Acceptable Diet (MAD)* by place of residence and wealth quintile: Philippines, 2023 (n=2,552)



95% CI		Place of Residence		Wealth Quintile				
LL	12.1	11.1	11.9	6.3	10.7	12.6	9.5	14.9
UL	16.0	15.3	18.2	11.1	17.6	22.5	18.9	27.9

** for breastfed infants: receiving at least the minimum dietary diversity (MDD) and minimum meal frequency (MMF) during the previous day for non- breastfed infants: receiving at least the minimum dietary diversity (MDD) and minimum meal frequency (MMF) during the previous day as well as at least two milk feeds (WHO/UNICEF, 2021)*

Trends in the percentage of children 6-23 months meeting the Minimum Acceptable Diet* (MAD): Philippines, 2011-2023



** for breastfed infants: receiving at least the minimum dietary diversity (MDD) and minimum meal frequency (MMF) during the previous day for non- breastfed infants: receiving at least the minimum dietary diversity (MDD) and minimum meal frequency (MMF) during the previous day as well as at least two milk feeds (WHO/UNICEF, 2021)*

HIGHLIGHTS

INFANT AND YOUNG CHILD FEEDING SURVEY

BREASTFEEDING PRACTICES

- **6 in 10** (61.2%) infants and young children were **early initiated to breastfeeding**.
- **5 in 10** (50.4%) infants below 6 months of age were **exclusively breastfed**.
- Early initiation to breastfeeding and Exclusive breastfeeding **significantly decreased in 2023**.



- **5 in 10** (53.3%) young children were **continued to be breastfed for one year**
- **4 in 10** (40.9%) young children were **continued to be breastfed for two years**
- **2 in 10** (17.7%) infants were **given pre-lacteal feeds.**

DOST-FNRI

HIGHLIGHTS

INFANT AND YOUNG CHILD FEEDING SURVEY

COMPLEMENTARY FEEDING PRACTICES



8 in 10 (84.3%)
children were **timely introduced to complementary food**



1 in 7 (14.5%) met the minimum dietary diversity **(MDD)**



9 in 10 (93.3%) met the minimum meal frequency **(MMF)**



1 in 7 (13.9%) met the minimum acceptable diet **(MAD)**

❖ Complementary feeding practices **improved** in 2023 for all indicators

BIOCHEMICAL SURVEY

Data on micronutrient status of different population groups

- ❑ *Vitamin A*
- ❑ *Iodine*
- ❑ *Anemia*

Data on the use of iodized salt among Filipino households



SIGNIFICANCE

- Formulating policies (i.e., fortification, supplementation)
- Guide in the selection of appropriate interventions
- Monitoring and evaluating the nutrition intervention programs
- Tracking progress in addressing SDG target by 2030 (Goal 2 - Zero Hunger)

METHODOLOGY

METHOD OF DATA COLLECTION



- Collection of blood samples by Registered Medical Technologists **among participants, 6 months old and above, including pregnant and lactating women**
  - ✓ 6-71 months old via Finger Prick Method
  - ✓ 6 years old & above via Venipuncture Method
- Collection of urine samples (5 ml of mid-stream urine) **among participants, 6 years old and above including pregnant and lactating women**
- Analysis at the DOST-FNRI Laboratory

Vitamin A Analysis

01 | Serum samples and IHRM

Aliquot 100 µL serum blood sample into a test tube.



02 | Internal Standard

Add 100 µL Retinyl Acetate to the sample then vortex to mix.



03 |

Add 500 µL hexane and vortex for 2 minutes.



04 | Separation

Centrifuge for 20 minutes at 3000 rpm and 25°C.



05 |

Transfer the supernatant portion of centrifuged sample into another test tube.



06 | Repeat Extraction

Repeat steps 3 - 5, for the second extraction.



07 | Drying

Dry the separated supernatant with nitrogen gas.



08 |

Dissolve the dried sample with 100 µL of 4:1 methanol and dichloromethane and mix.



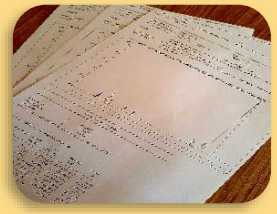
09 |

Transfer the dissolved sample into a vial then subject to HPLC unit.



10 |

Obtain the chromatogram and calculate the result.



Iodine Analysis

01 | Preparation of Standards and Reagents

- 1 µg/mL Iodine Stock Solution
- 1M Ammonium Persulfate Solution
- Arsenous Acid Solution
- Ceric Ammonium Sulfate Solution



02 | Urine samples and QCTM

- Aliquot 250 µL into separate glass test tubes.
- For recovery test, add 10 µL stock solution to 250 µL IHRM.



03 |

- Add 1 mL of 1M Ammonium Persulfate to each tube.
- Mix contents for 10 – 15 seconds in a vortex mixer.



04 | Digestion

- Place the tubes in the heating block with 100°C temperature (95 °C for microplate method).
- Heat for 60 minutes (start the time when the heating block reached required temperature.
- Cool test tube to room temperature.



05 |

Add 2.5 mL Arsenous acid solution to each tube and mix in a vortex mixer.

Stand for 15 minutes.



06 |

Add 300 µL Ceric ammonium sulfate solution, keeping a constant interval of 15 seconds between additions to successive tubes.



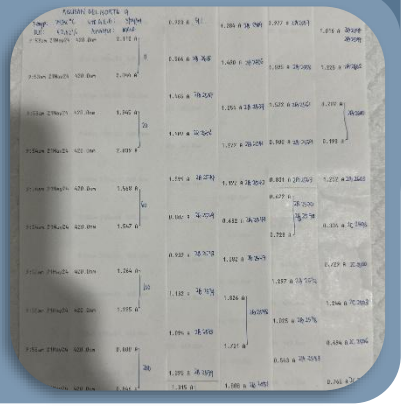
07 |

Exactly 30 minutes after addition of ceric ammonium sulfate solution to the first tube, read the absorbance at 420 nm in a UV-Vis spectrophotometer with a 15 second interval between readings.



08 | Result

Print and calculate result.



RESULTS

VITAMIN A STATUS OF DIFFERENT POPULATION GROUPS IN THE PHILIPPINES: 2023 NNS



GUIDELINES USED FOR THE INTERPRETATION OF VITAMIN A DATA

(WHO/USAID, 1976; WHO/UNICEF/HKI/IVACG, 1982; WHO, 2011)

Vitamin A Level	Serum Retinol (µg/dL)
High	≥ 50
Acceptable	20 - 49
Low	10 - 19
Deficient	< 10

PREVALENCE CUT-OFFS TO DEFINE VITAMIN A DEFICIENCY IN A POPULATION AND ITS LEVEL OF PUBLIC HEALTH SIGNIFICANCE

(WHO, 1996; WHO, 2011)

Public Health Importance Degree of Severity	Prevalence of population with low to deficient SR level (%)
Mild	2 - <10
Moderate	10 - <20
Severe	≥ 20

Total Number of Responses for Vitamin A Data: Philippines, 2023 (n=88,866)





PRESCHOOL CHILDREN (6 MONTHS-5 YEARS OLD)
n = 3,677



SCHOOL-AGE CHILDREN (6-12 YEARS OLD)
n = 13,183



ADOLESCENTS (13-19 YEARS OLD)
n = 13,594



ADULTS (20.0-59.9 YEARS OLD)
n = 40,977



OLDER PERSONS (60 YEARS OLD AND ABOVE)
n = 14,561

WOMEN OF REPRODUCTIVE AGE
(15-49 YEARS OLD)



NON-PREGNANT/NON-LACTATING WOMEN
n = 19,896



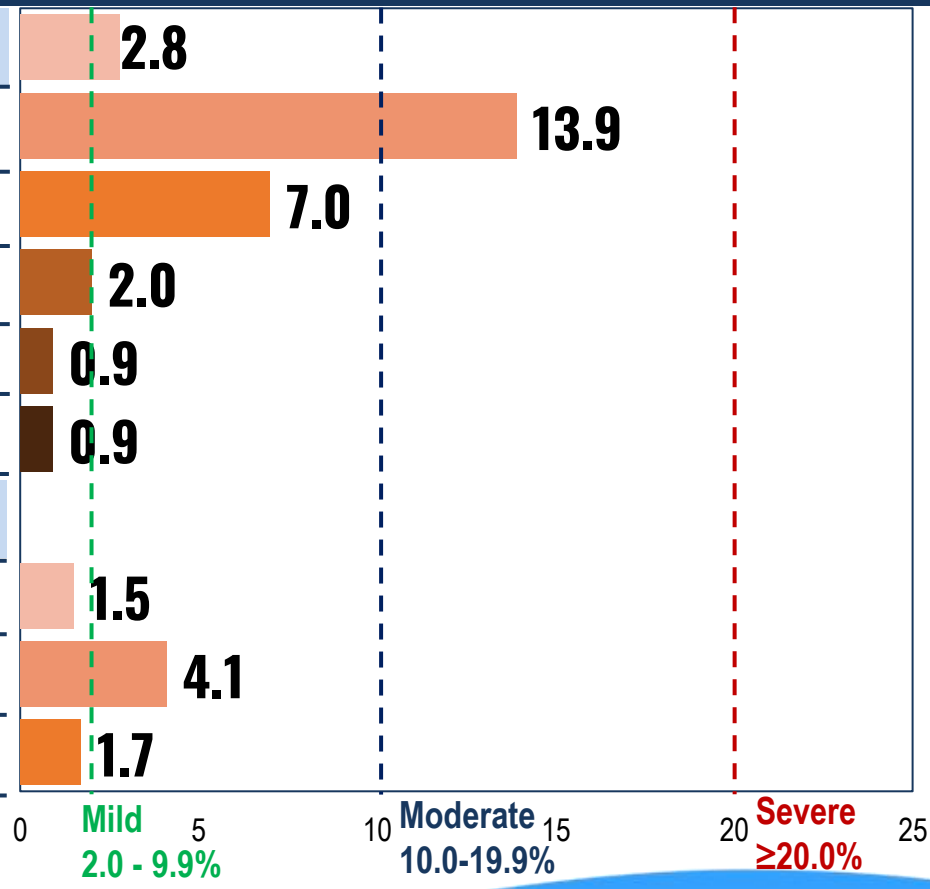
PREGNANT WOMEN
n = 584



LACTATING MOTHERS
n = 2,282

Prevalence of vitamin A deficiency (VAD) by age/physiologic group: Philippines 2023

Age/Physiologic Group	No. of Participants	Prevalence (%)
ALL	**88,866**	**2.8**
Preschool Children, 6 months-5 years old (6-71 months)	3,677	13.9
School-age Children, 6 - 12 years old	13,183	7.0
Adolescents, 13 - 19 years old	13,594	2.0
Adults, 20.0 - 59.9 years old	40,977	0.9
Older Persons, 60 years old and above	14,561	0.9
Women of Reproductive Age, 15 - 49 years old		
Non-pregnant/Non-lactating Women	19,896	1.5
Pregnant Women	584	4.1
Lactating Mothers	2,282	1.7



Preschool Children, 6 months to 5 years old (6 – 71 months)





Mean serum retinol and prevalence of vitamin A deficiency among preschool children, 6 months to 5 years old (6 – 71 months): Philippines, 2023 (n=3,677)



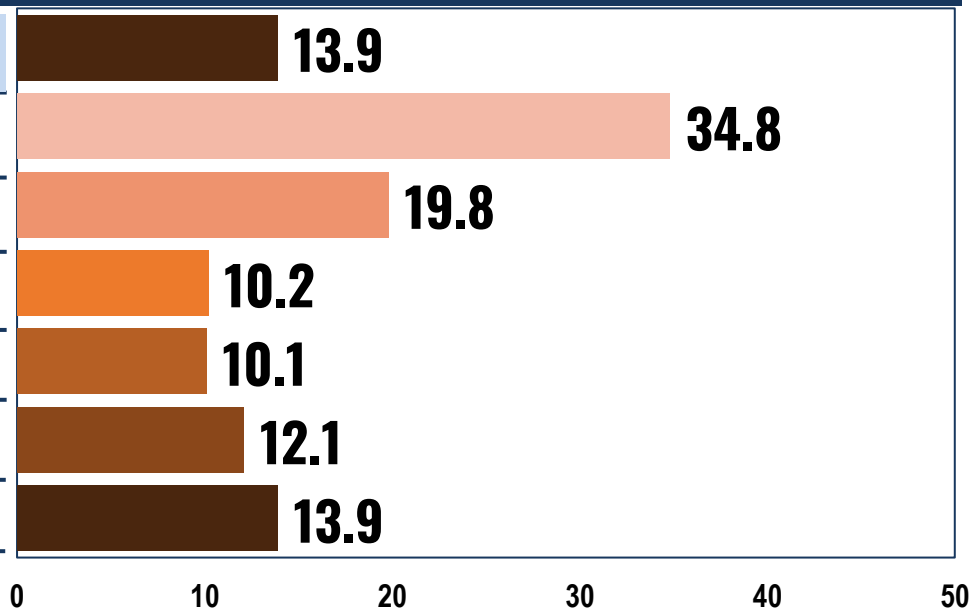


Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
28.9	13.9 95% CI [12.1, 16.0]

*Acceptable VA level - SR of 20-49 µg/dL
**Moderate public health significance

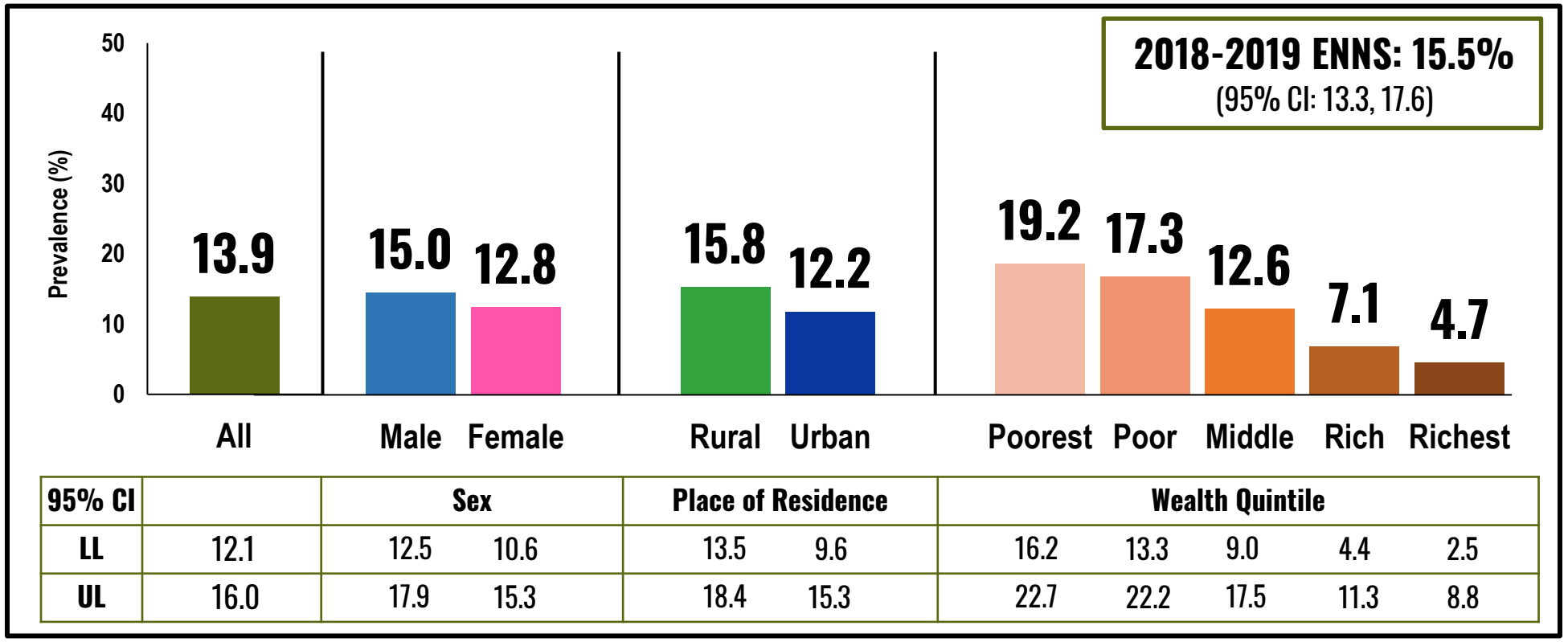
Prevalence of vitamin A deficiency among preschool children (6 to 71 months old) by single age group: Philippines, 2023

Age Group	No. of Participants	95% CI [LL/UL]	Prevalence* (%)
6 months - 5 years old	**3,677**	**[12.1, 16.0]**	**13.9**
6 to 11 months old	167	[23.3, 48.5]	34.8
1 year old	474	[14.7, 26.2]	19.8
2 years old	512	[7.3, 14.1]	10.2
3 years old	688	[7.2, 14.0]	10.1
4 years old	870	[8.9, 16.2]	12.1
5 years old	966	[10.6, 18.1]	13.9



*Mild (2.0-9.9%); Moderate (10.0-19.9%); Severe (≥20%)

Prevalence of vitamin A deficiency among preschool children (6 to 71 months old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=3,677)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	12.1	12.5	10.6	13.5	9.6	16.2	13.3	9.0	4.4	2.5
UL	16.0	17.9	15.3	18.4	15.3	22.7	22.2	17.5	11.3	8.8

Mean serum retinol and prevalence of vitamin A deficiency among preschool children, 6 months to 5 years old (6 – 71 months): Philippines, 2023 (n=3,441)



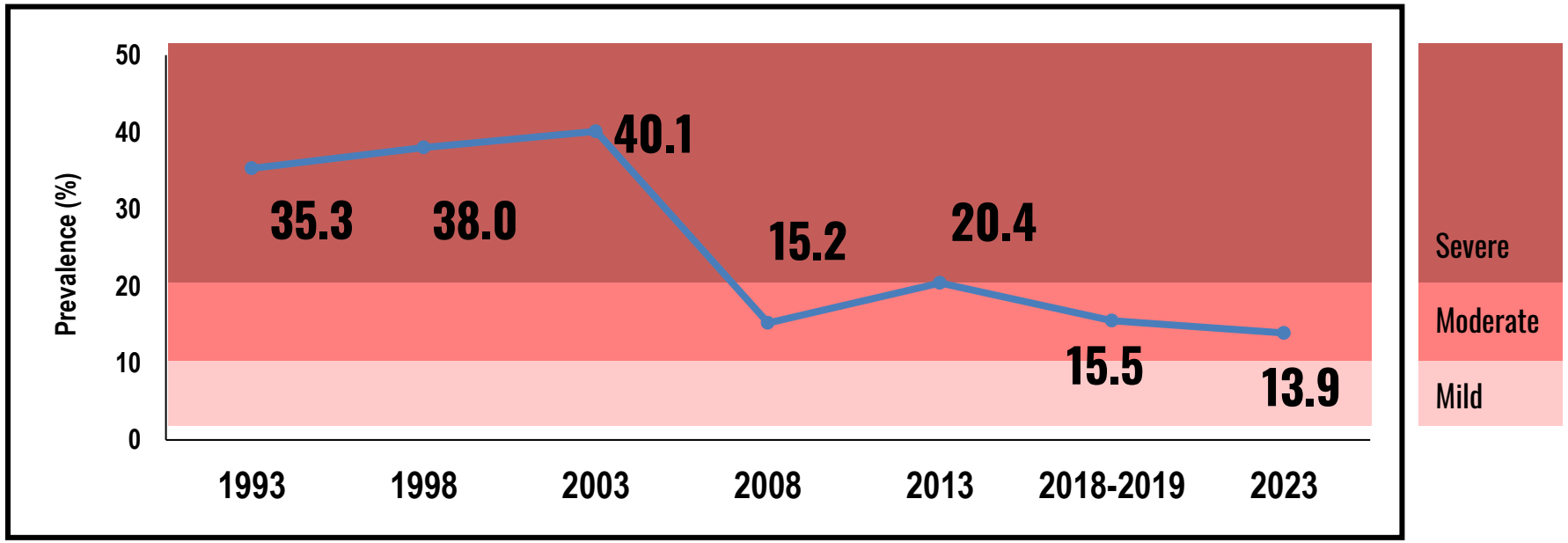


Participation in the Vitamin A Supplementation Program

Response	Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)
Yes	29.0	13.8 95% CI [11.7-16.1]
No	29.3	13.7 95% CI [10.0-18.4]
Do not know	27.5	12.9 95% CI [2.2-49.4]

*Acceptable VA level - SR of 20-49 µg/dL

Trend in the prevalence of vitamin A deficiency among preschool children, 6 months to 5 years old: Philippines, 1993 to 2023



School-age Children, 6 to 12 years old



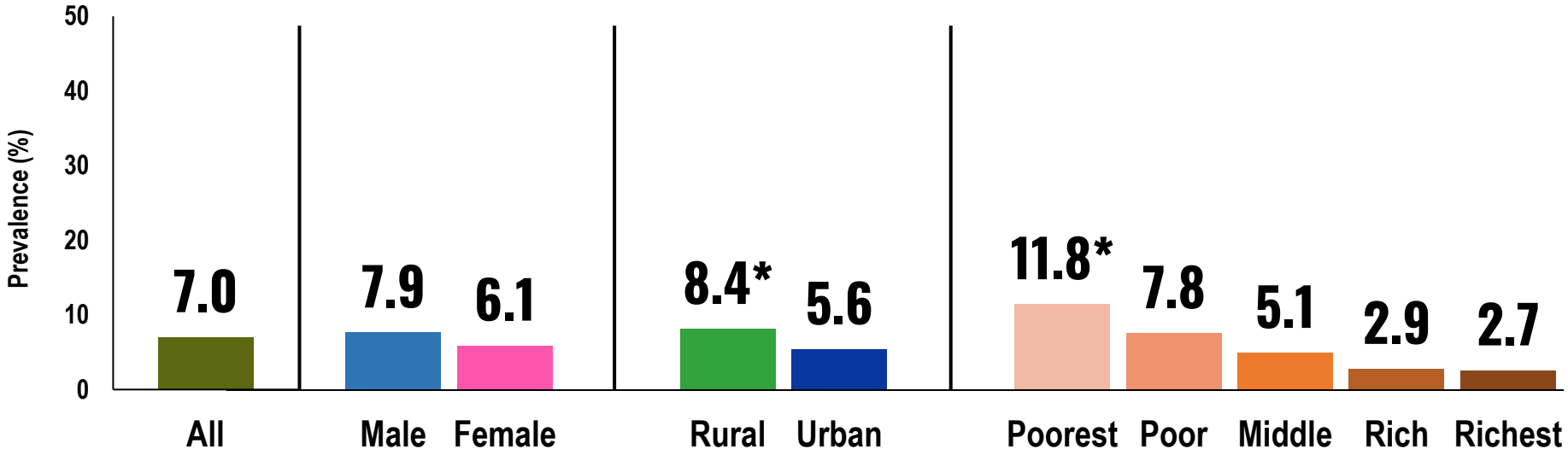
Mean serum retinol and prevalence of vitamin A deficiency among school-age children, 6 to 12 years old: Philippines, 2023 (n=13,183)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
31.6	7.0 95% CI [6.4, 7.8]

*Acceptable VA level - SR of 20-49 µg/dL
**Mild public health significance

Prevalence of vitamin A deficiency among school-age children (6 to 12 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,183)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	6.4	7.0	5.3	7.5	4.8	10.4	6.7	4.0	2.0	1.7
UL	7.8	9.0	7.0	9.5	6.5	13.3	9.2	6.6	4.1	4.0

*significantly different at 5% level of significance

Adolescents, 13 to 19 years old



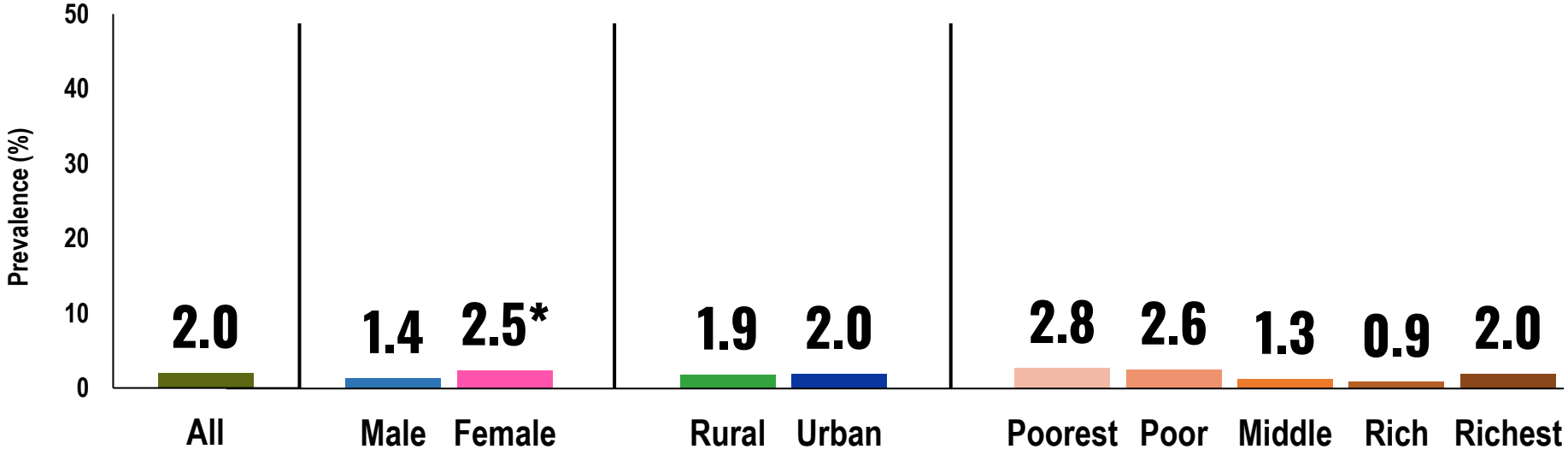
Mean serum retinol and prevalence of vitamin A deficiency among adolescents, 13 to 19 years old: Philippines, 2023 (n=13,594)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
39.2	2.0 95% CI [1.6, 2.4]

*Acceptable VA level - SR of 20-49 µg/dL
**Low to Mild public health significance

Prevalence of vitamin A deficiency among adolescents (13 to 19 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,594)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	1.6	1.1	2.0	1.5	1.5	2.1	1.9	0.9	0.4	1.1
UL	2.4	1.9	3.1	2.4	2.7	3.7	3.6	1.9	1.7	3.9

*significantly different at 5% level of significance

Adults, 20.0 to 59.9 years old



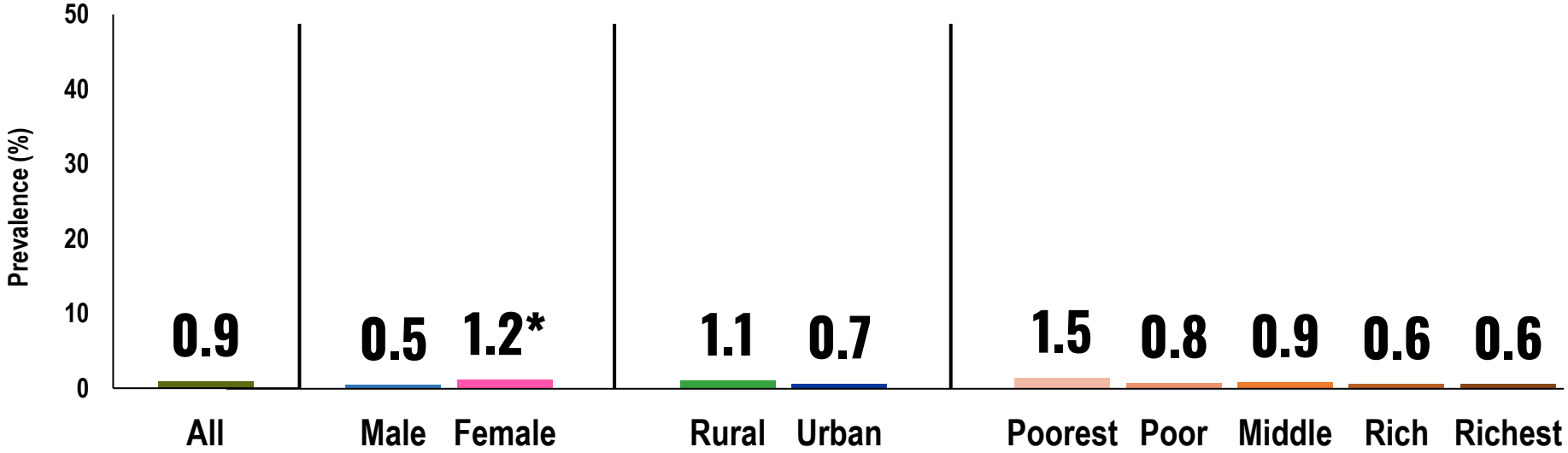
Mean serum retinol and prevalence of vitamin A deficiency among adults, 20.0 to 59.9 years old: Philippines, 2023 (n=40,977)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
51.4	0.9 95% CI [0.8, 1.0]

*Acceptable VA level - SR of 20-49 µg/dL
**Low public health significance

Prevalence of vitamin A deficiency among adults (20.0 to 59.9 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=40,977)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	0.8	0.4	1.0	0.9	0.5	1.2	0.6	0.6	0.4	0.4
UL	1.0	0.7	1.4	1.3	0.9	1.9	1.1	1.3	0.9	0.9

*significantly different at 5% level of significance

Older Persons, 60 years old and above



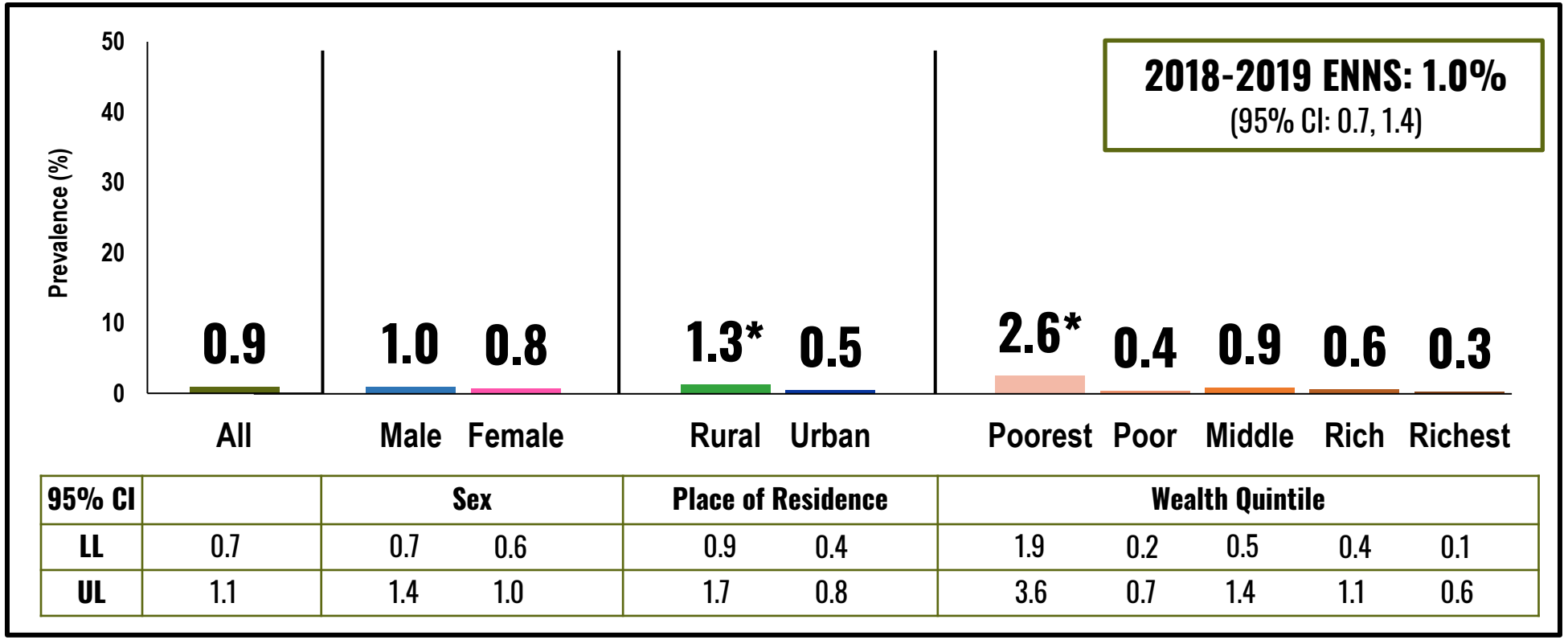
Mean serum retinol and prevalence of vitamin A deficiency among older persons, 60 years old and above: Philippines, 2023 (n=14,561)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
54.8	0.9 95% CI [0.7, 1.1]

*Acceptable VA level - SR of 20-49 µg/dL
**Low public health significance

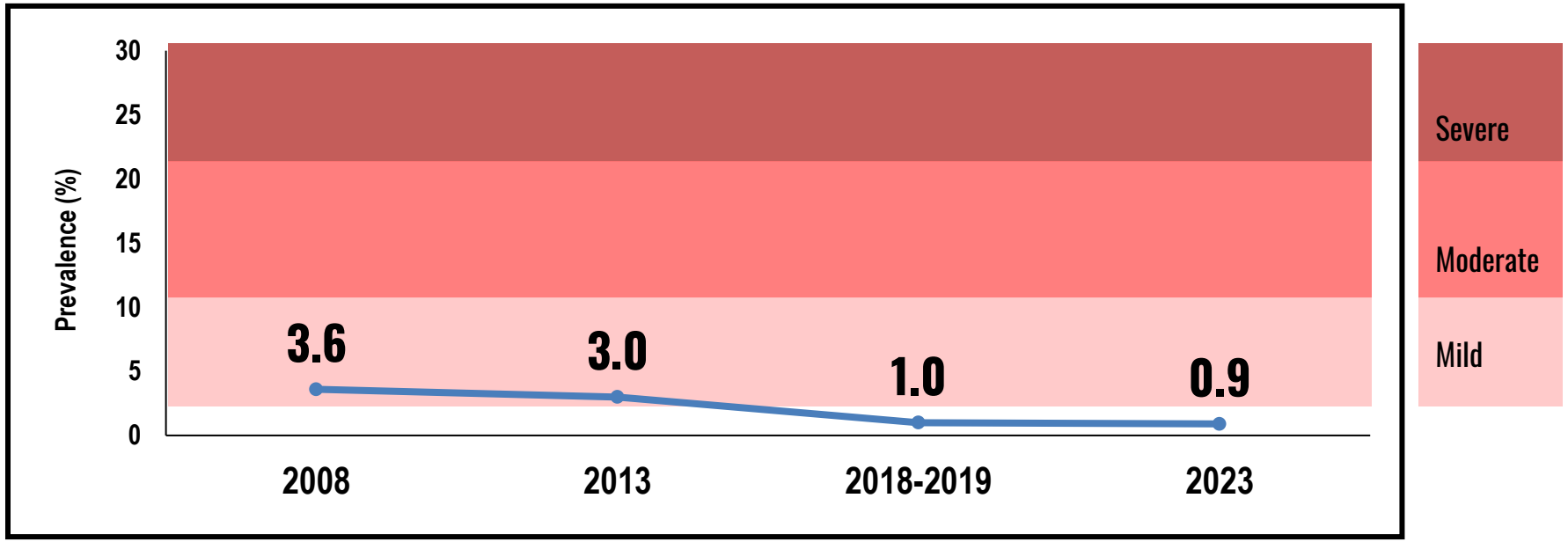
Prevalence of vitamin A deficiency among older persons (60 years old and above) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=14,561)



95% CI		Sex		Place of Residence		Wealth Quintile				
LL	0.7	0.7	0.6	0.9	0.4	1.9	0.2	0.5	0.4	0.1
UL	1.1	1.4	1.0	1.7	0.8	3.6	0.7	1.4	1.1	0.6

*significantly different at 5% level of significance

Trend in the prevalence of vitamin A deficiency among older persons, 60 years old and above: Philippines, 2008 to 2023



Women of Reproductive Age, 15 to 49 years old



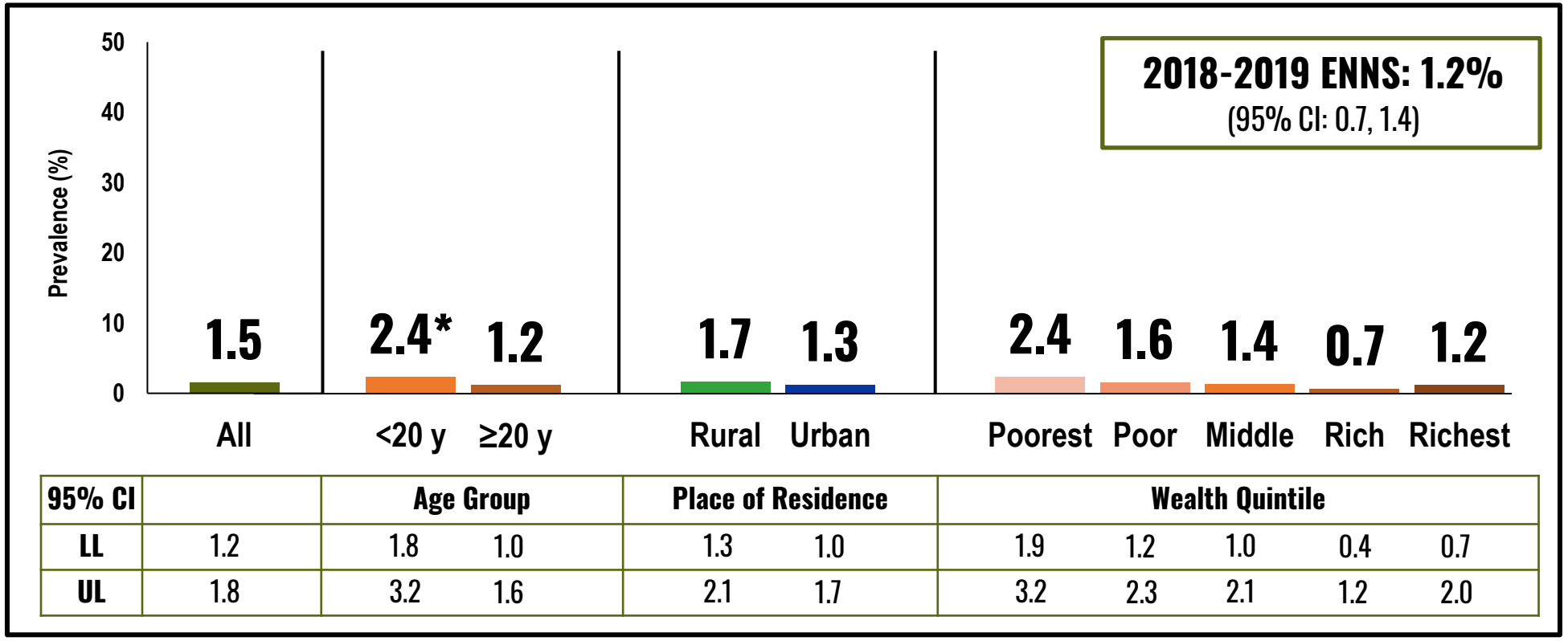
Mean serum retinol and prevalence of vitamin A deficiency among non-pregnant/non-lactating women, 15 to 49 years old: Philippines, 2023 (n=19,896)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
43.3	1.5 95% CI [1.2, 1.8]

*Acceptable VA level - SR of 20-49 µg/dL
**Low public health significance

Prevalence of vitamin A deficiency among non-pregnant/non-lactating women (15 to 49 years old) by age group, place of residence, and wealth quintile: Philippines, 2023 (n=19,896)



95% CI		Age Group		Place of Residence		Wealth Quintile				
LL	1.2	1.8	1.0	1.3	1.0	1.9	1.2	1.0	0.4	0.7
UL	1.8	3.2	1.6	2.1	1.7	3.2	2.3	2.1	1.2	2.0

*significantly different at 5% level of significance

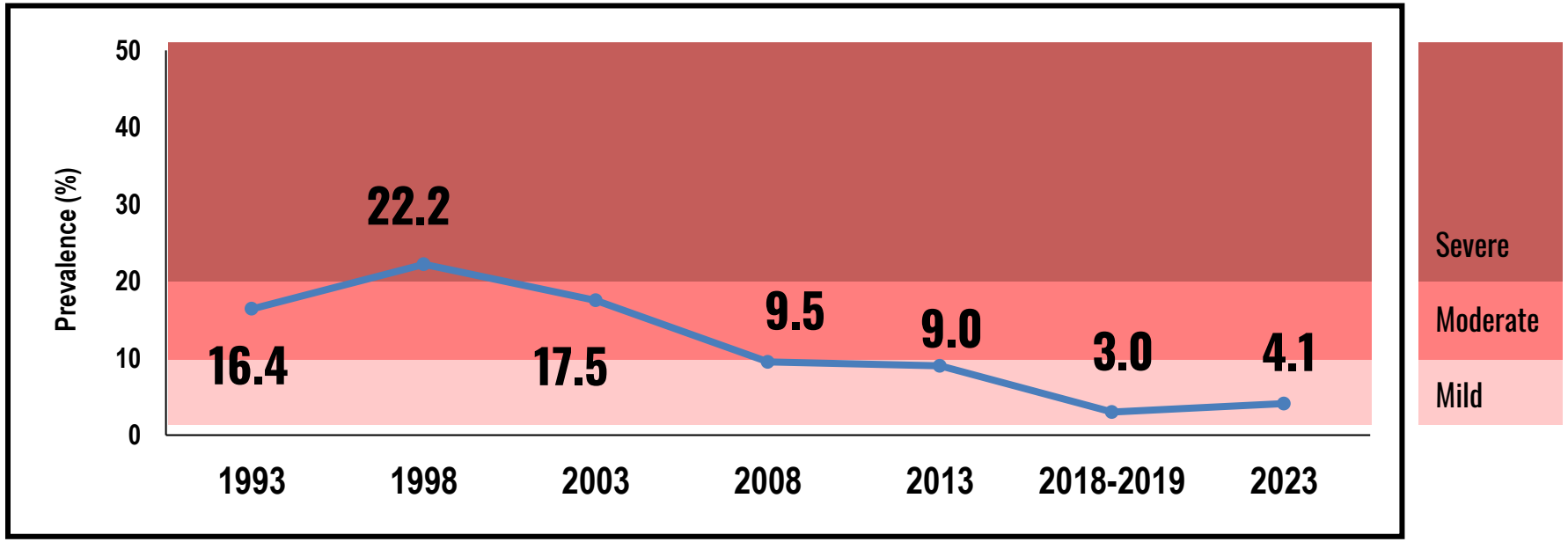
Mean serum retinol and prevalence of vitamin A deficiency among pregnant women, 15 to 49 years old: Philippines, 2023 (n=584)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
37.0	4.1 95% CI [2.5, 6.5]

*Acceptable VA level - SR of 20-49 µg/dL
**Mild public health significance

Trend in the prevalence of vitamin A deficiency among pregnant women, 15 to 49 years old: Philippines, 1993 to 2023



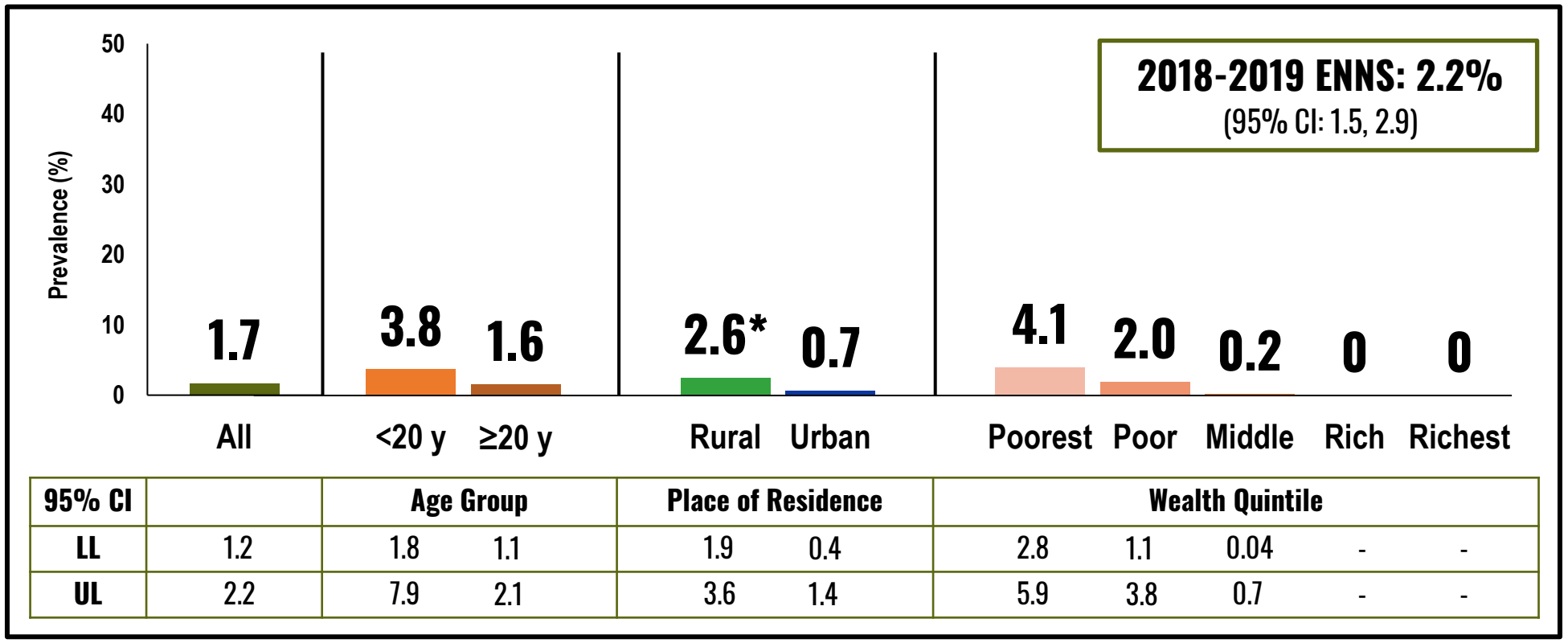
Mean serum retinol and prevalence of vitamin A deficiency among lactating mothers, 15 to 49 years old: Philippines, 2023 (n=2,282)



Mean Serum Retinol (SR) Level (µg/dL)*	Prevalence of low to deficient SR level (%)**
42.6	1.7 95% CI [1.2, 2.2]

*Acceptable VA level - SR of 20-49 µg/dL
**Low to Mild public health significance

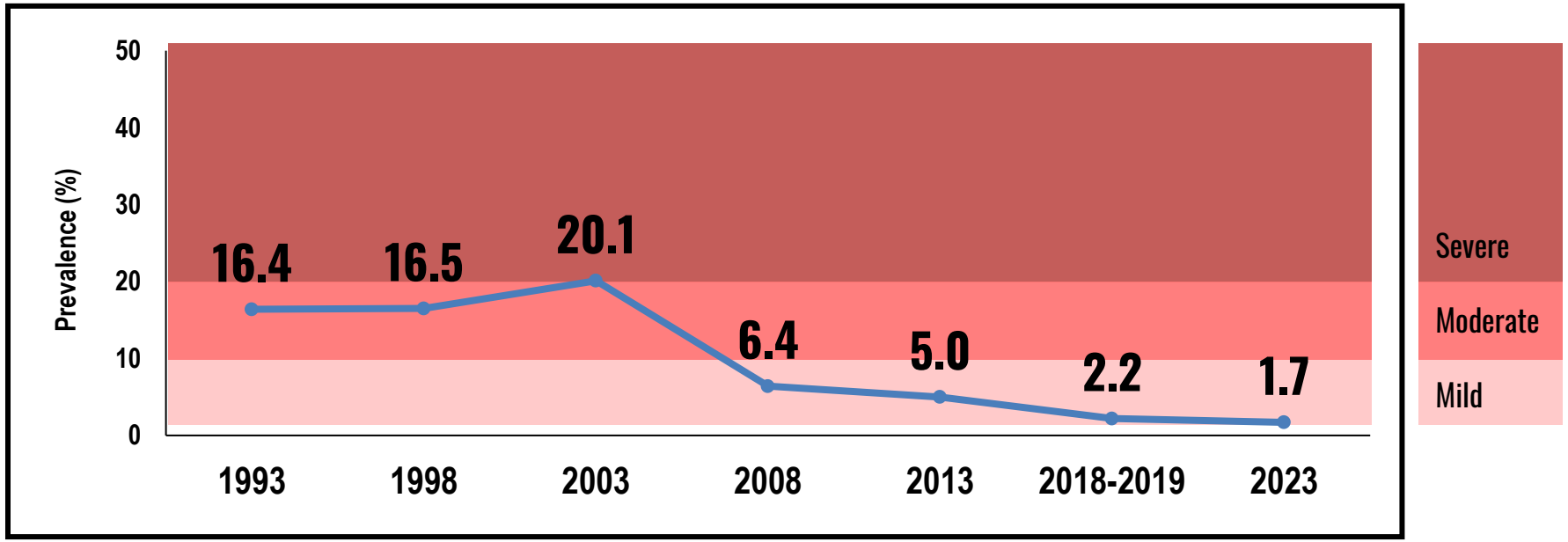
Prevalence of vitamin A deficiency among lactating mothers (15 to 49 years old) by age group, place of residence, and wealth quintile: Philippines, 2023 (n=2,282)



95% CI		Age Group		Place of Residence		Wealth Quintile				
LL	1.2	1.8	1.1	1.9	0.4	2.8	1.1	0.04	-	-
UL	2.2	7.9	2.1	3.6	1.4	5.9	3.8	0.7	-	-

*significantly different at 5% level of significance

Trend in the prevalence of vitamin A deficiency among lactating mothers, 15 to 49 years old: Philippines, 1993 to 2023



HIGHLIGHTS
VITAMIN A STATUS

Vitamin A Deficiency in the Philippines



- Highest VAD prevalence is noted among **preschool children** and considered of moderate public health significance
- Mild public health significance among **school-age children**, and **pregnant women**
- Low to mild public health significance among **adolescents** and **lactating mothers**
- The level of VAD among **adults, older persons** and **non-pregnant/non-lactating women** indicates low public health significance

HIGHLIGHTS

VITAMIN A STATUS

Vitamin A Deficiency in the Philippines



- Across all age groups, VAD is more prevalent among the poorest wealth quintile households.
- Significantly higher among rural dwellers (**school-age children**, **older persons**, and **lactating mothers**)
- The level of VAD is significantly higher among females (**adolescents** and **adults**)

IODINE STATUS OF DIFFERENT POPULATION GROUPS IN THE PHILIPPINES: 2023 NNS



RESULTS

EPIDEMIOLOGICAL CRITERIA FOR ASSESSMENT OF IODINE NUTRITION IN A POPULATION BASED ON MEDIAN OR RANGE OF URINARY IODINE CONCENTRATION IN SCHOOL-AGE CHILDREN*[1]

Urinary Iodine Concentration (UIC), µg/L	Iodine Intake	Iodine Status
≥ 300	Excessive**	Risk of adverse health consequences (iodine-induced hyperthyroidism, autoimmune thyroid disease
100 – 299	**Adequate**	**Adequate iodine nutrition**
50 – 99	Insufficient	Mild iodine deficiency
20 – 49	Insufficient	Moderate iodine deficiency
<20	Insufficient	Severe iodine deficiency

*Applies to adults, but not to pregnant and lactating women

**The term "excessive" means in excess of the amount required to prevent and control ID.

[1] **WHO, 2013**

EPIDEMIOLOGICAL CRITERIA FOR ASSESSING IODINE NUTRITION BASED ON URINARY IODINE CONCENTRATIONS OF LACTATING WOMEN AND PREGNANT WOMEN[1]

Median Value µg/L	Iodine Intake
LACTATING WOMEN	
≥100	Adequate
<100	Insufficient
PREGNANT WOMEN	
≥500	Excessive**
250 – 499	Above requirements
150 – 249	**Adequate**
<150	Insufficient

[1] WHO, 2013

Total Number of Responses for Iodine Data: Philippines, 2023 (n=86,614)



SCHOOL-AGE CHILDREN (6-12 YEARS OLD)
n = 13,768



ADOLESCENTS (13-19 YEARS OLD)
n = 13,774



ADULTS (20.0-59.9 YEARS OLD)
n = 41,374



OLDER PERSONS (60 YEARS OLD AND ABOVE)
n = 14,804

WOMEN OF REPRODUCTIVE AGE
(15-49 YEARS OLD)



NON-PREGNANT/NON-LACTATING WOMEN
n = 20,003



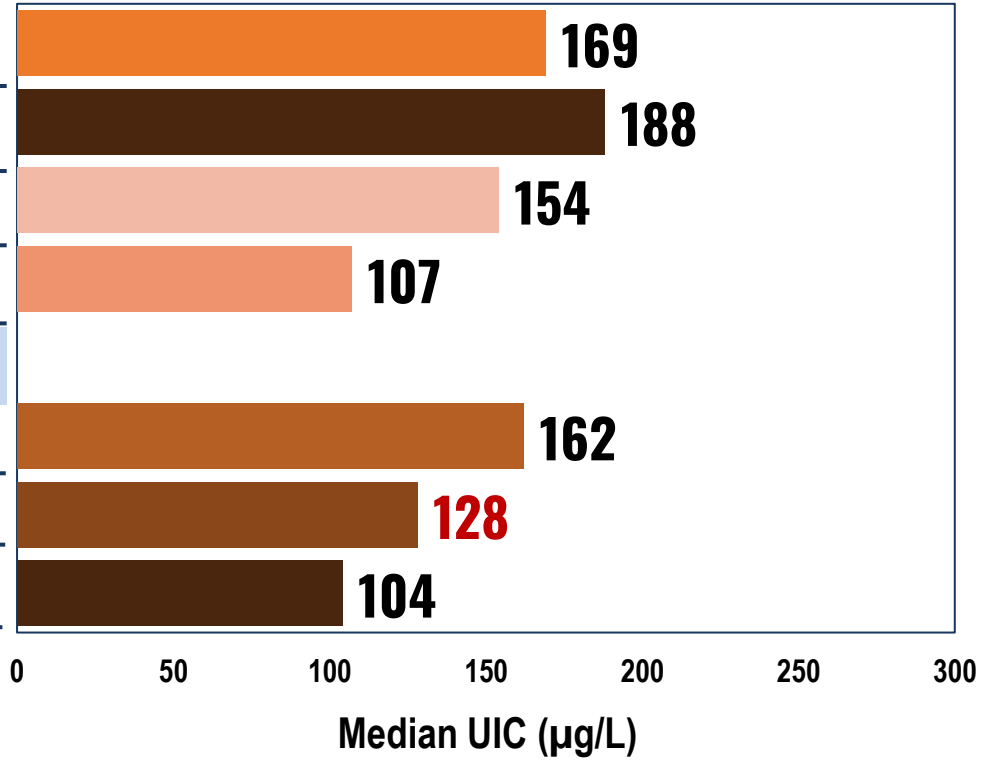
PREGNANT WOMEN
n = 590



LACTATING MOTHERS
n = 2,296

Median urinary iodine concentration (UIC) by age/physiologic group: Philippines, 2023

Age/Physiologic Group	No. of Participants	Median UIC (µg/L)*
School-age Children, 6 - 12 years old	13,768	169
Adolescents, 13 - 19 years old	13,774	188
Adults, 20.0 - 59.9 years old	41,374	154
Older Persons, 60 years old and above	14,804	107
Women of Reproductive Age, 15 - 49 years old		
Non-pregnant/Non-lactating Women	20,003	162
Pregnant Women	590	128
Lactating Mothers	2,296	104



*Adequate iodine intake for school-age children and other age/physiologic groups except Pregnant Women (≥100 µg/L); while (150-249 µg/L) for Pregnant Women

Values in red font indicate Iodine Deficiency

School-age Children, 6 to 12 years old



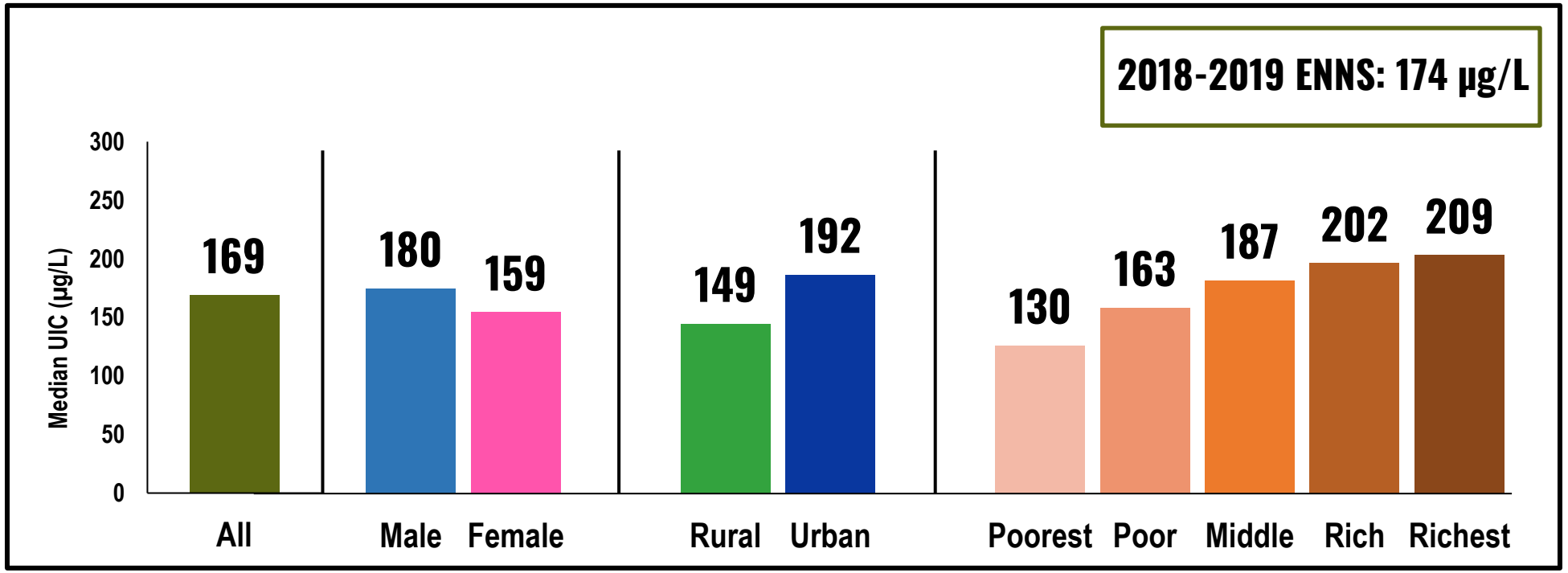
Median UIC and percent UI level of < 50 µg/L among school-age children, 6 to 12 years old: Philippines, 2023 (n=13,768)



Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of School-age Children with Urinary Iodine level of <50 µg/L (%)**
169	11.6

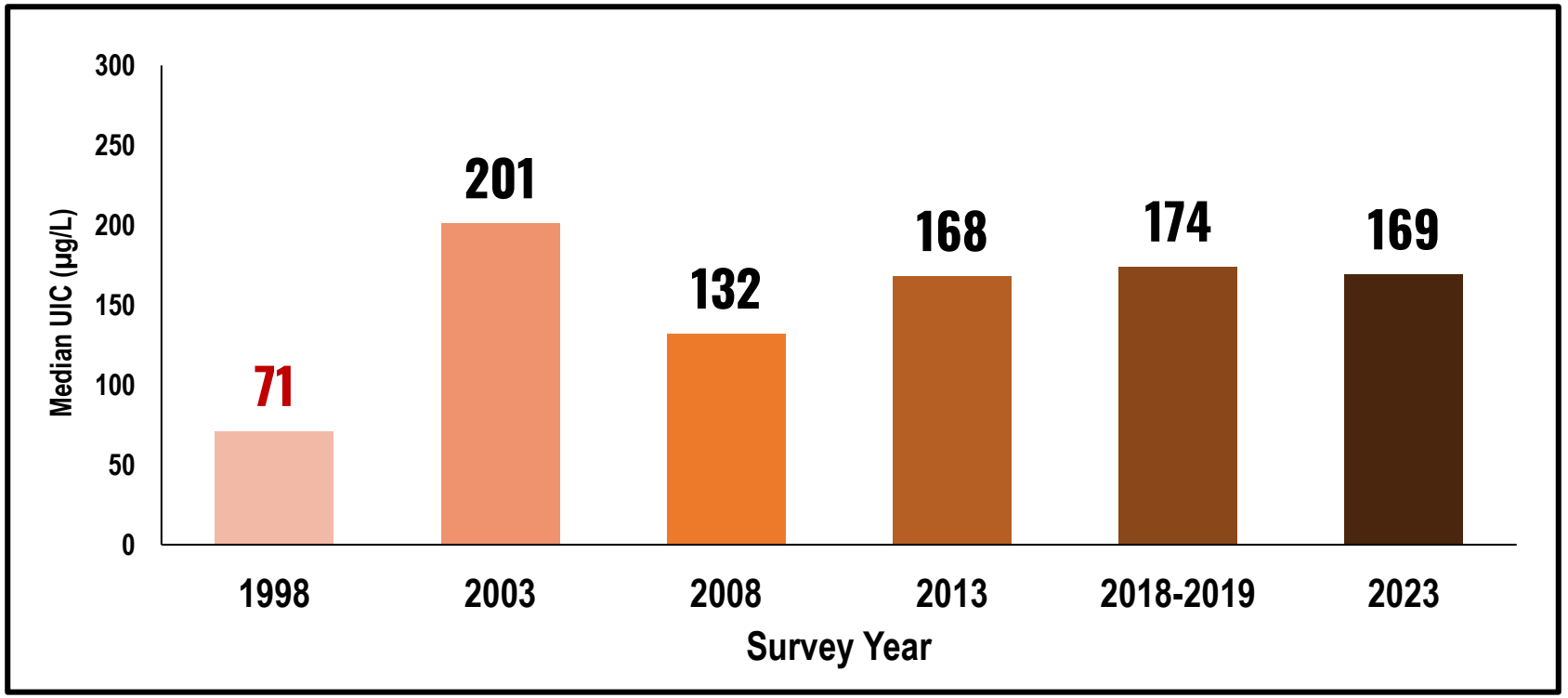
*Adequate iodine intake (UIC: 100-299 µg/L)
**Should be <20%

Median UIC among school-age children (6 to 12 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,768)



Adequate iodine intake (UIC: 100-299 µg/L)

Trend in the median UIC among school-age children, 6 to 12 years old: Philippines, 1998 to 2023



Adequate iodine intake (UIC: 100-299 µg/L)
Values in red font indicate presence of Iodine Deficiency

Adolescents, 13 to 19 years old



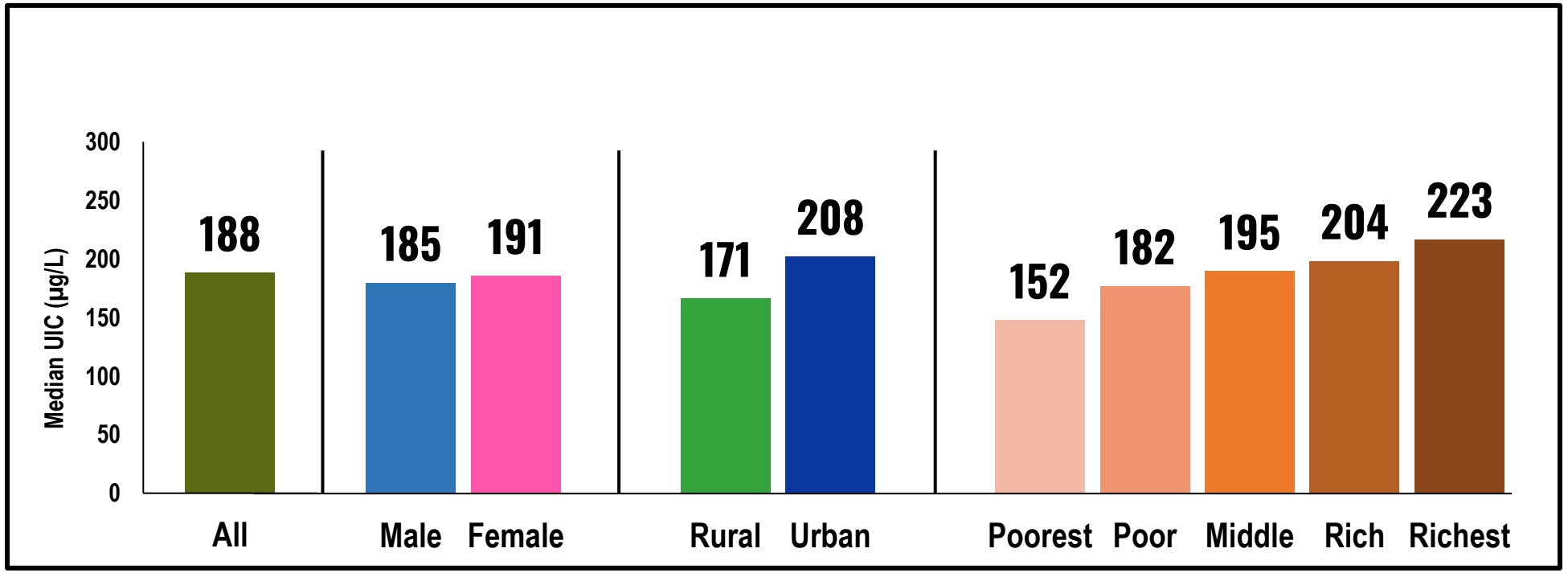
Median UIC and percent UI level of < 50 µg/L among adolescents, 13 to 19 years old: Philippines, 2023 (n=13,774)

Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Adolescents with Urinary Iodine level of <50 µg/L (%)**
188	9.0



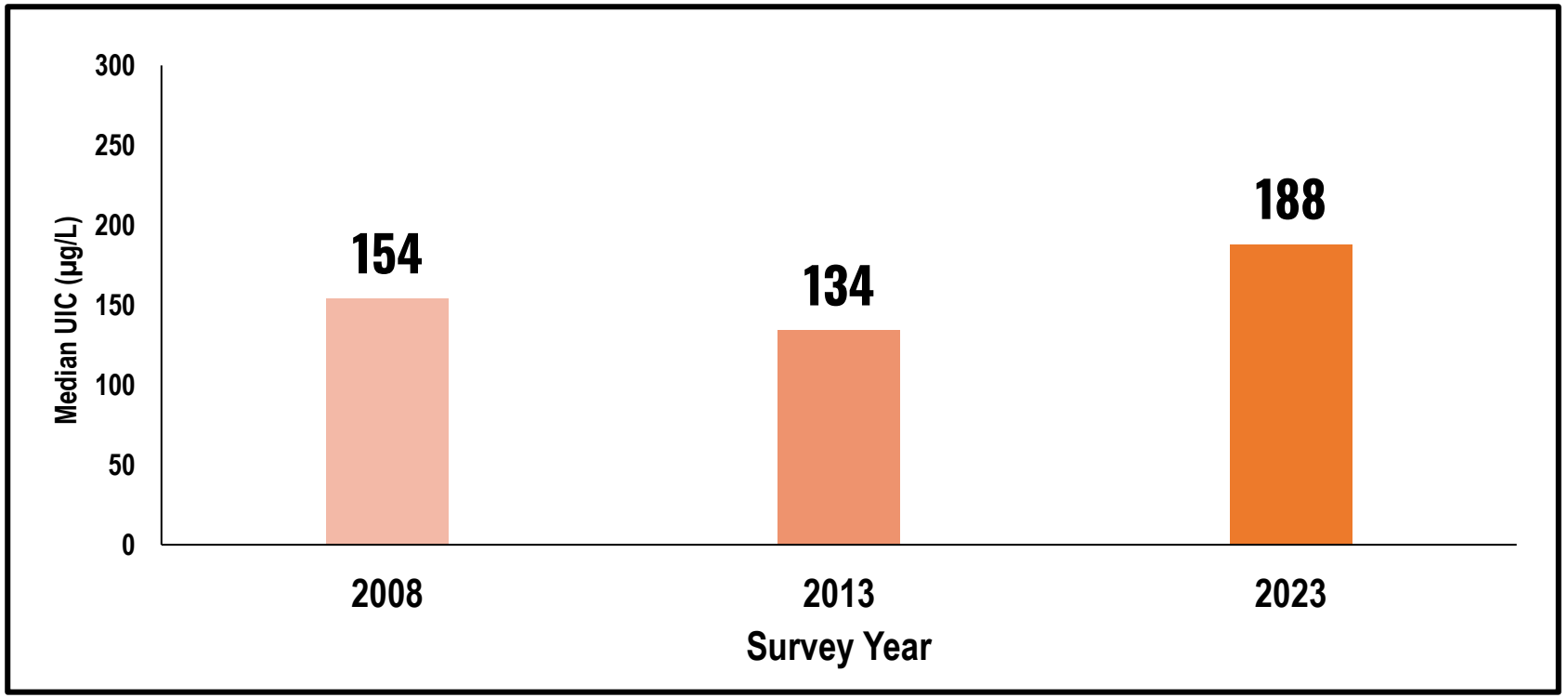
*Adequate iodine intake (UIC: 100-299 µg/L)
**Should be <20%

Median UIC among adolescents (13 to 19 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,774)



Adequate iodine intake (UIC: 100-299 µg/L)

Trend in the median UIC among adolescents, 13 to 19 years old: Philippines, 2008 to 2023



Adequate iodine intake (UIC: 100-299 µg/L
No data in 2018, 2019, and 2021

Adults, 20.0 to 59.9 years old



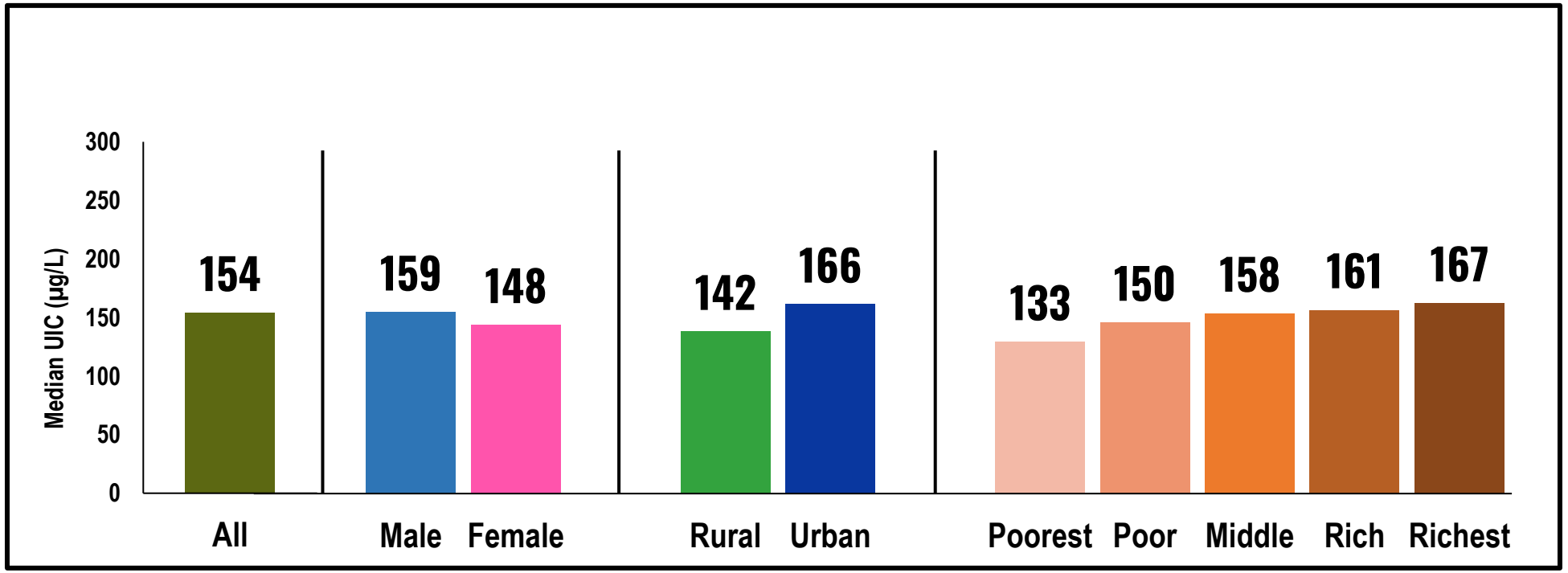
Median UIC and percent UI level of < 50 µg/L among adults, 20.0 to 59.9 years old: Philippines, 2023 (n=41,374)



Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Adults with Urinary Iodine level of <50 µg/L (%)**
154	12.8

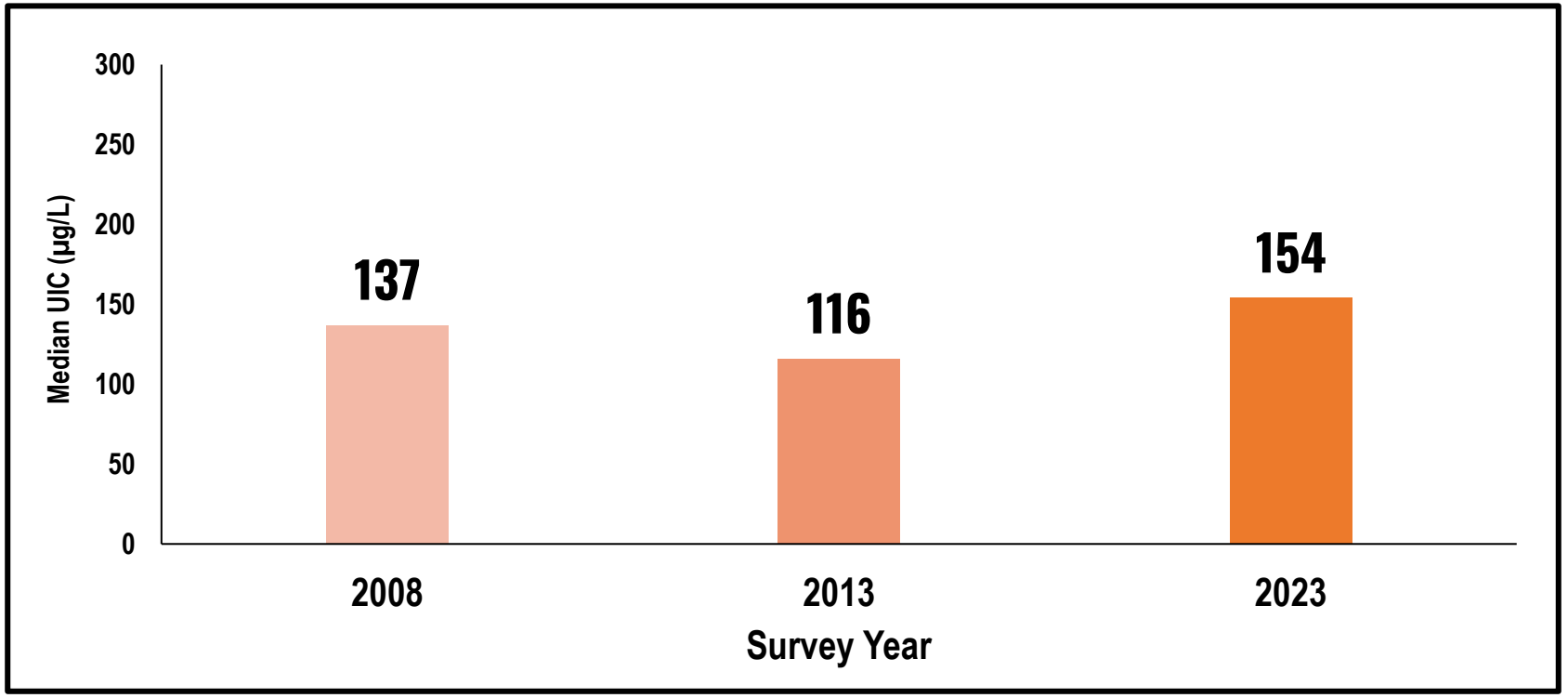
*Adequate iodine intake (UIC: 100-299 µg/L)
**Should be <20%

Median UIC among adults (20.0 to 59.9 years old) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=41,374)



Adequate iodine intake (UIC: 100-299 µg/L)

Trend in the median UIC among adults, 20.0 to 59.9 years old: Philippines, 2008 to 2023



Adequate iodine intake (UIC: 100-299 µg/L)
No data in 2018, 2019, and 2021

Older Persons, 60 years old and above



Median UIC and percent UI level of < 50 µg/L among older persons, 60 years old and above: Philippines, 2023 (n=14,804)

Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Older Persons with Urinary Iodine level of <50 µg/L (%)**
107	21.3

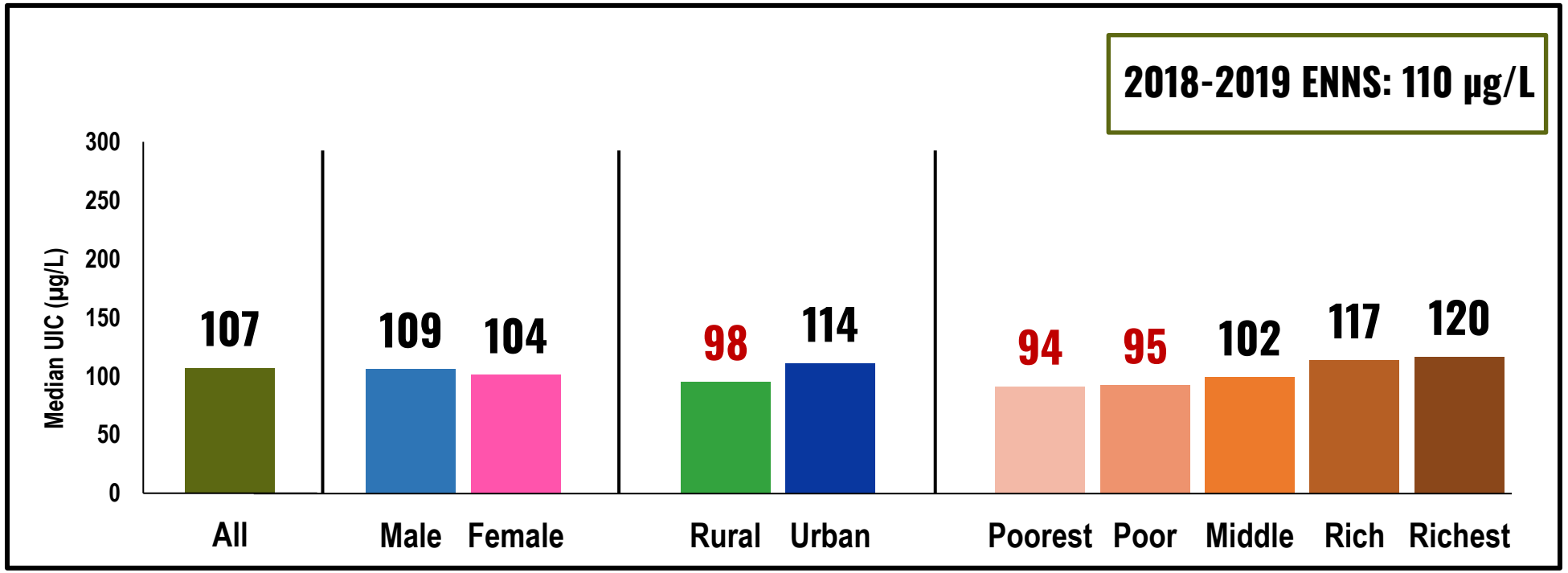


Values in red font indicate presence of Iodine Deficiency

*Adequate iodine intake (UIC: 100-299 µg/L)

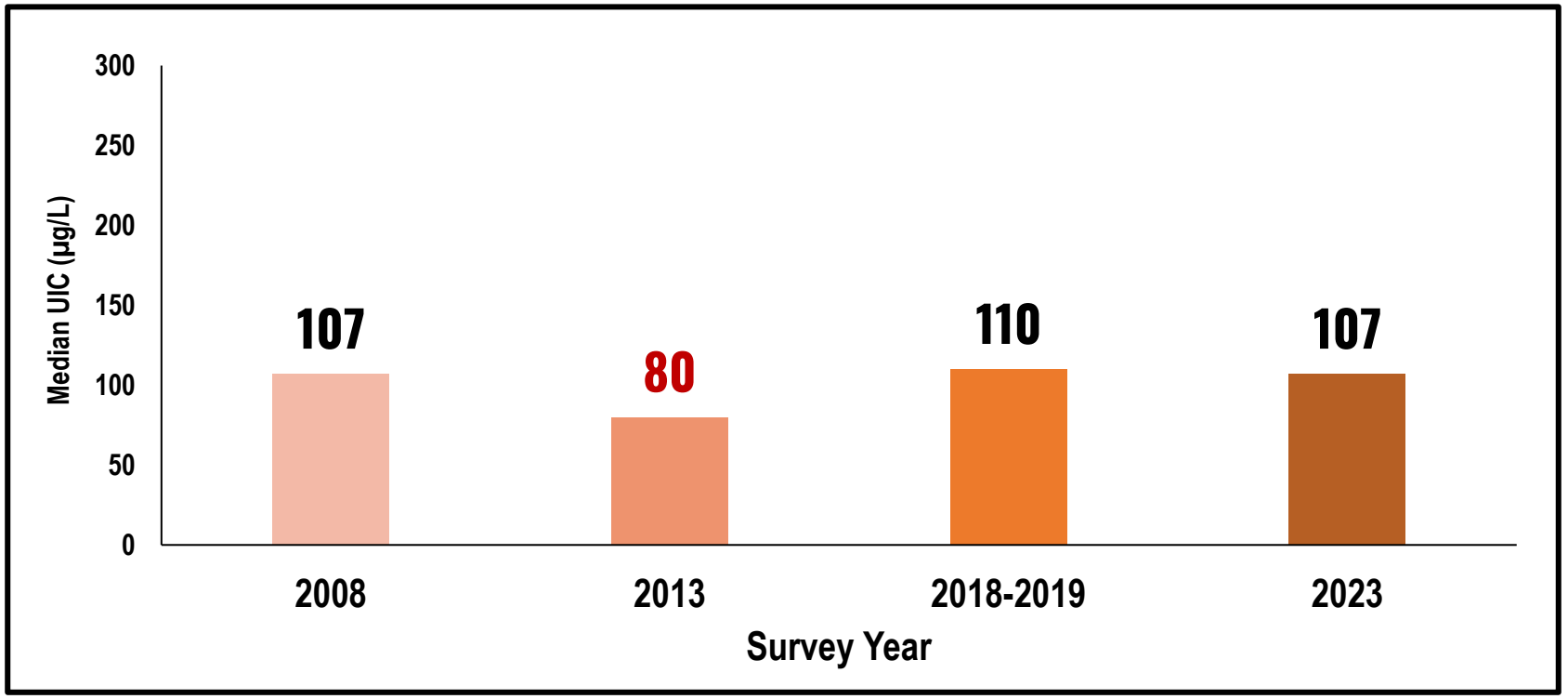
**Should be <20%

Median UIC among older persons (60 years old and above) by sex, place of residence, and wealth quintile: Philippines, 2023 (n=14,804)



Adequate iodine intake (UIC: 100-299 µg/L)
Values in red font indicate presence of Iodine Deficiency

Trend in the median UIC among older persons, 60 years old and above: Philippines, 2008 to 2023



Adequate iodine intake (UIC: 100-299 µg/L)
Values in red font indicate presence of Iodine Deficiency
Pockets of IODINE DEFICIENCY exist in 2018-2019 and 2023
No data in 2021

Women of Reproductive Age, 15 to 49 years old



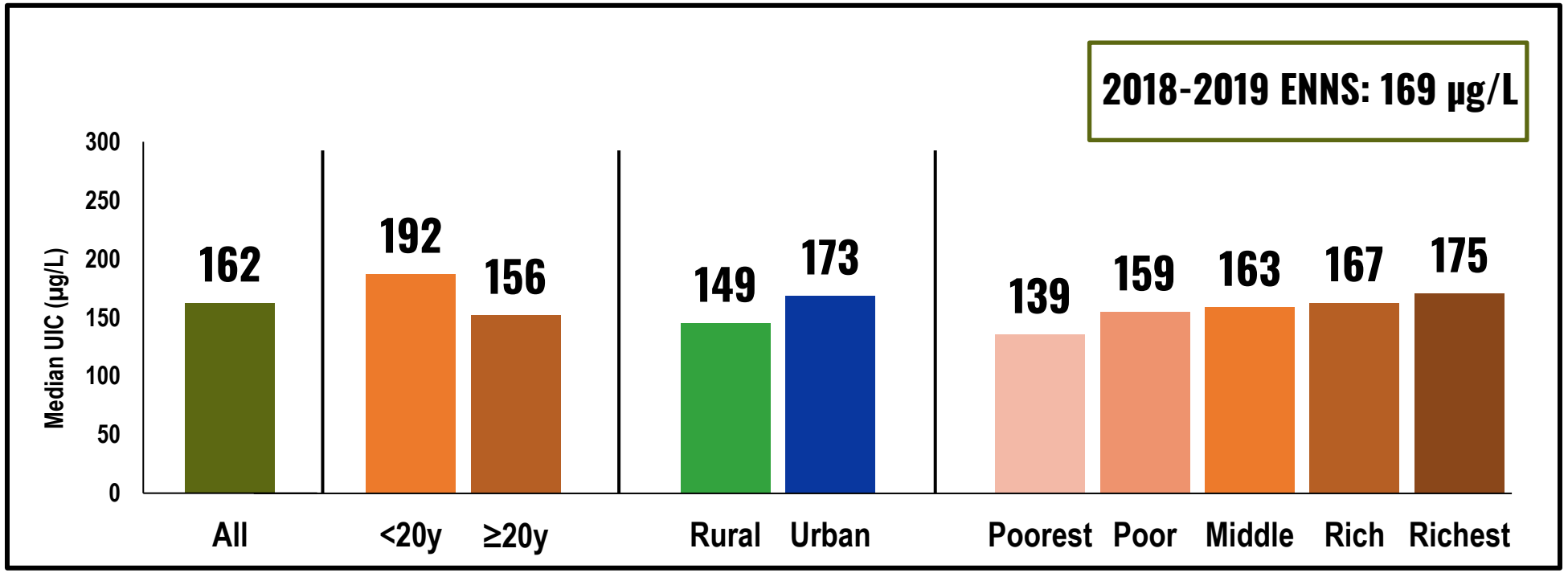
Median UIC and percent UI level of < 50 µg/L among non-pregnant/non-lactating women, 15 to 49 years old: Philippines, 2023 (n=20,003)



Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Non-pregnant/Non-lactating Women with Urinary Iodine level of <50 µg/L (%)**
162	12.3

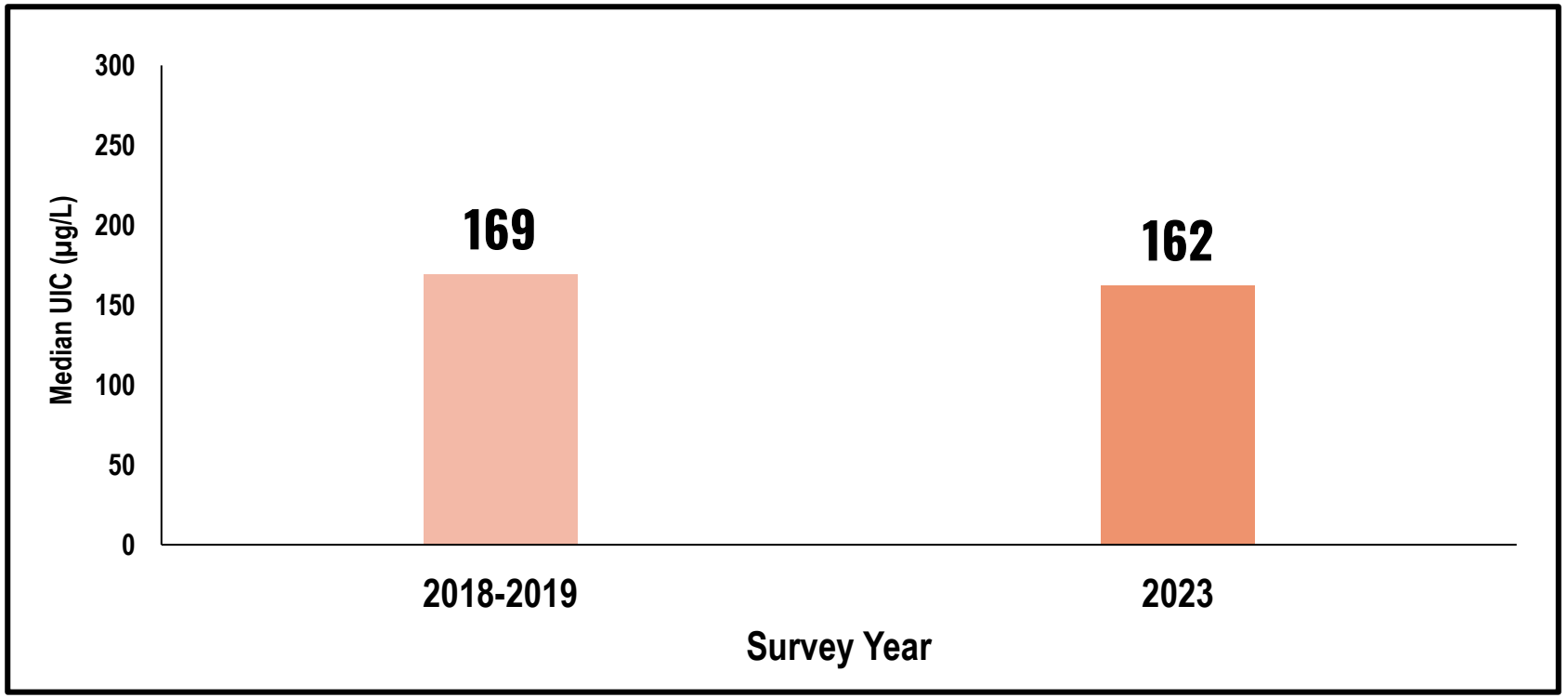
*Adequate iodine intake (UIC: 100-299 µg/L)
**Should be <20%

Median UIC among non-pregnant/non-lactating women (15 to 49 years old) by age group, place of residence, and wealth quintile: Philippines, 2023 (n=20,003)



Adequate iodine intake (UIC: 100-299 µg/L)

Trend in the median UIC among non-pregnant/non-lactating women, 15 to 49 years old: Philippines, 2018-2019 to 2023



Adequate iodine intake (UIC: 100-299 µg/L)
No data in 2021

Median UIC and percent UI level of < 50 µg/L among pregnant women, 15 to 49 years old: Philippines, 2023 (n=590)

Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Pregnant Women with Urinary Iodine level of <50 µg/L (%)**
128	21.8

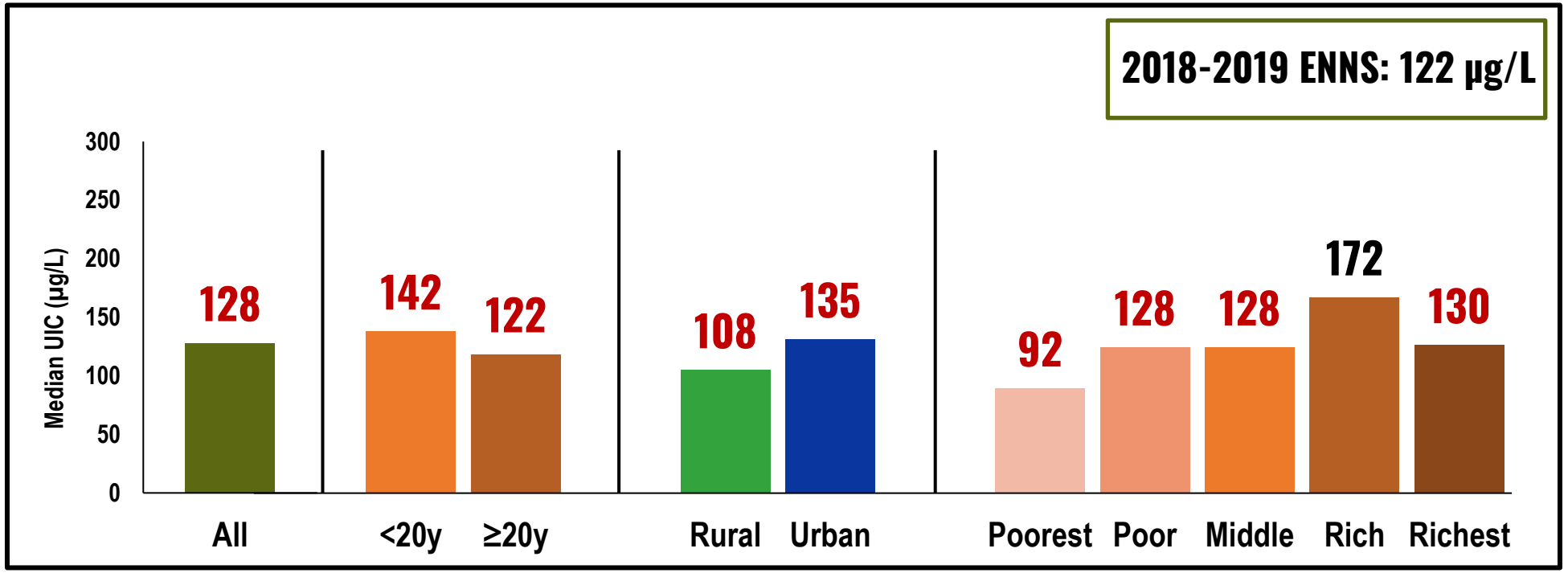


Values in red font indicate presence of Iodine Deficiency

*Adequate iodine intake (UIC: 150-249 µg/L)

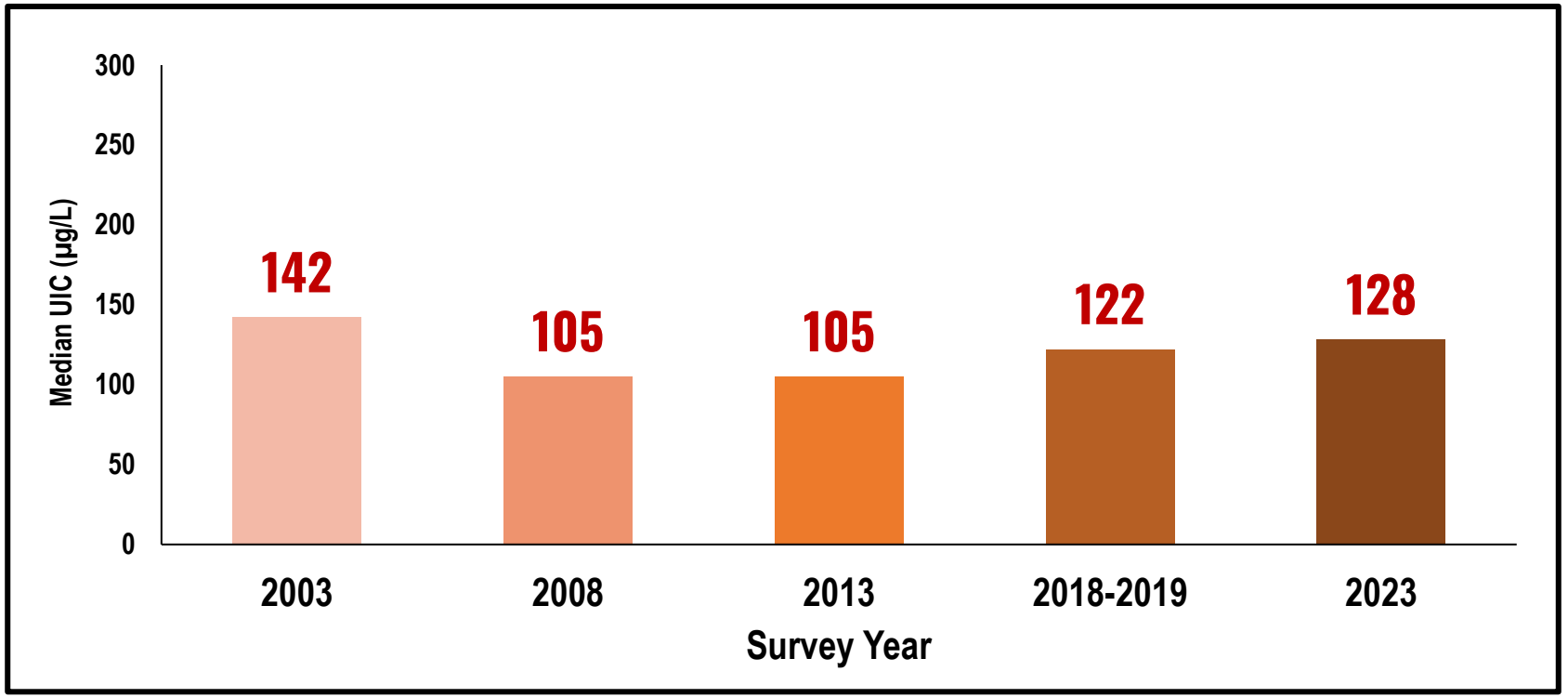
**Should be <20%

Median UIC among pregnant women (15 to 49 years old) by age group, place of residence, and wealth quintile: Philippines, 2023 (n=590)



Adequate iodine intake (UIC: 150-249 µg/L)
Values in red font indicate presence of Iodine Deficiency

Trend in the median UIC among pregnant women, 15 to 49 years old: Philippines, 2003 to 2023



Adequate iodine intake (UIC: 150-249 µg/L)
Values in red font indicate presence of Iodine Deficiency
No data in 2021

Median UIC and percent UI level of < 50 µg/L among lactating mothers, 15 to 49 years old: Philippines, 2023 (n=2,296)



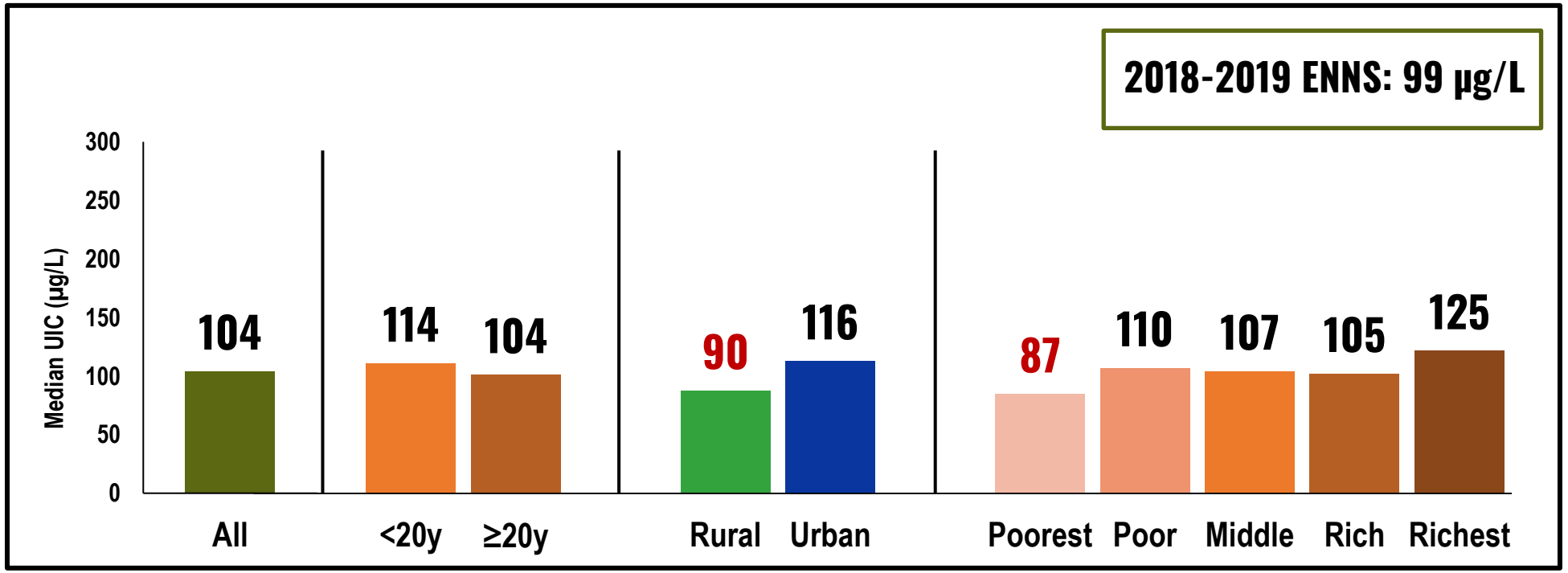
Median Urinary Iodine Concentration (UIC) (µg/L)*	Percentage of Lactating Mothers with Urinary Iodine level of <50 µg/L (%)**
104	20.1

Values in red font indicate presence of Iodine Deficiency

*Adequate iodine intake (UIC: ≥ 100 µg/L)

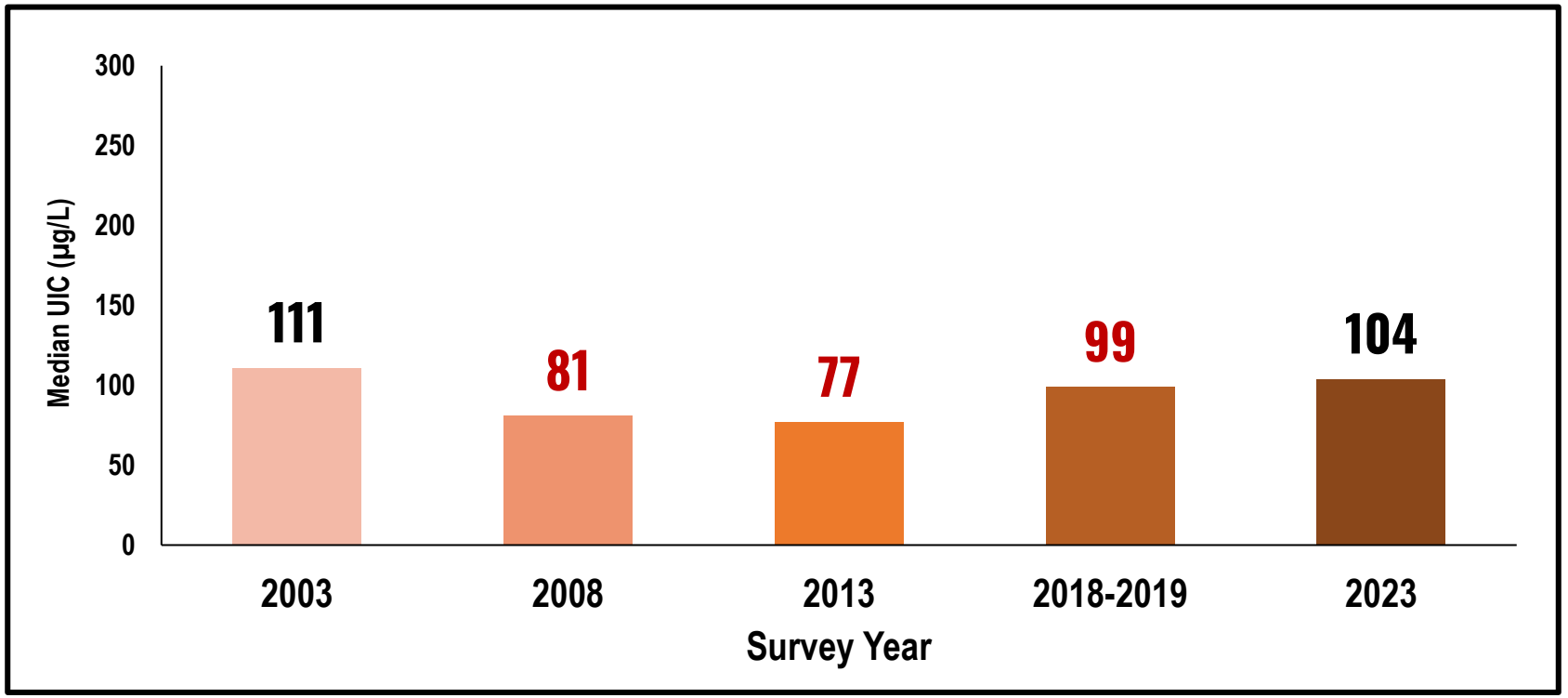
**Should be <20%

Median UIC among lactating mothers (15 to 49 years old) by age group, place of residence, and wealth quintile: Philippines, 2023 (n=2,296)



Adequate iodine intake (UIC: 100-299 µg/L)
Values in red font indicate presence of Iodine Deficiency

Trend in the median UIC among lactating mothers, 15 to 49 years old: Philippines, 2003 to 2023

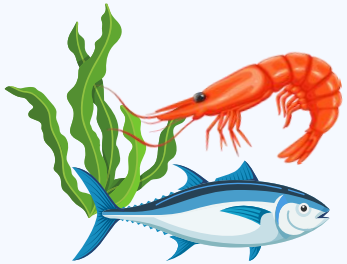


Adequate iodine intake (UIC: ≥100 µg/L)
Values in red font indicate presence of Iodine Deficiency
Pockets of IODINE DEFICIENCY exist in 2023
No data in 2021

HIGHLIGHTS

IODINE

Iodine Status in the Philippines





- The median UIC reflects **adequate iodine nutrition** among school-age children, adolescents, adults, and non-pregnant/non-lactating women.
- Older persons and lactating mothers have **adequate iodine intake** (median UIC of >100 µg/L) but more than 20% have UI level below 50 µg/L indicating that **pockets of iodine deficiency** still exist in these groups.
- **Iodine deficiency** is prevalent among pregnant women as they have **insufficient iodine intake.**

AWARENESS AND USAGE OF IODIZED SALT IN THE PHILIPPINES:

2023 NATIONAL NUTRITION SURVEY





RESULTS

Total Number of Responses for Awareness and Usage of Iodized Salt: Philippines, 2023



Number of covered households

35,457

SALT BEING USED BY FILIPINO HOUSEHOLDS:

2023 NATIONAL NUTRITION SURVEY





Type of salt being used by household meal planners: Philippines, 2023 (n=35,457)



Rock Salt
52.2%



Fine Salt
37.9%

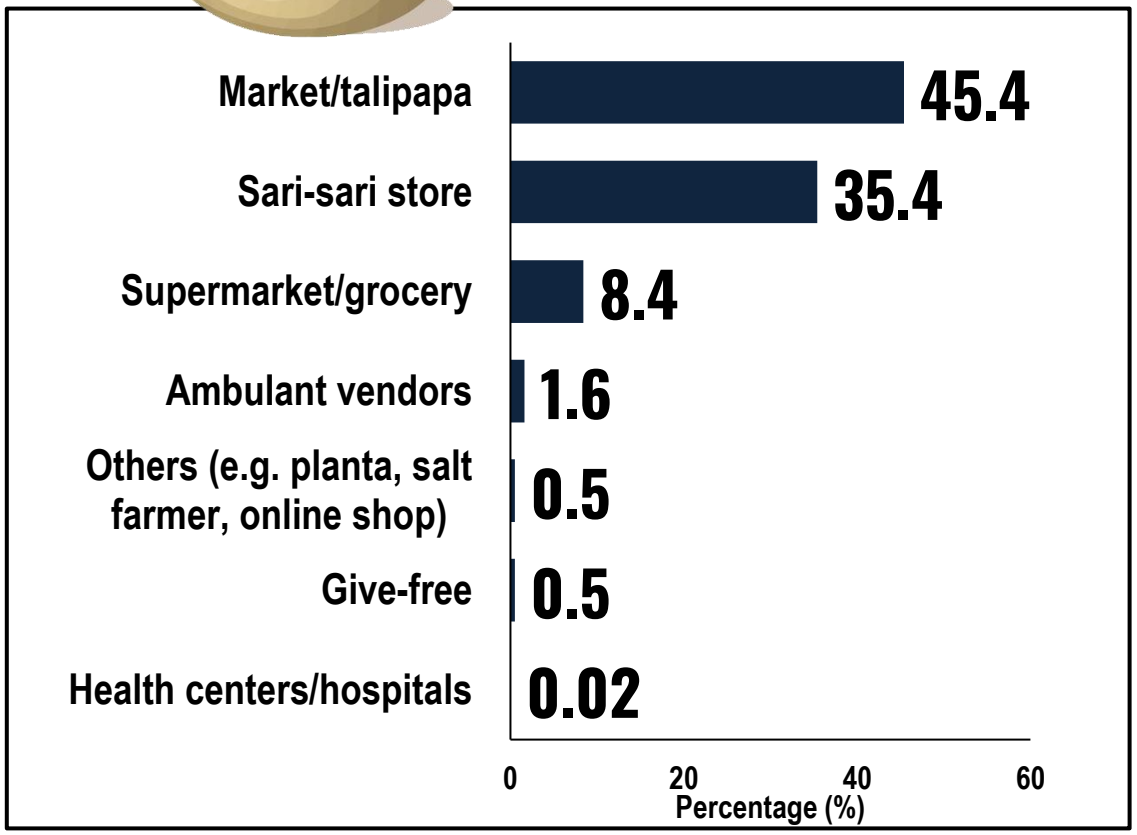


Rock Salt and Fine Salt
9.9%

Type of salt being used by sources of salt: Philippines, 2023 (n=35,457)

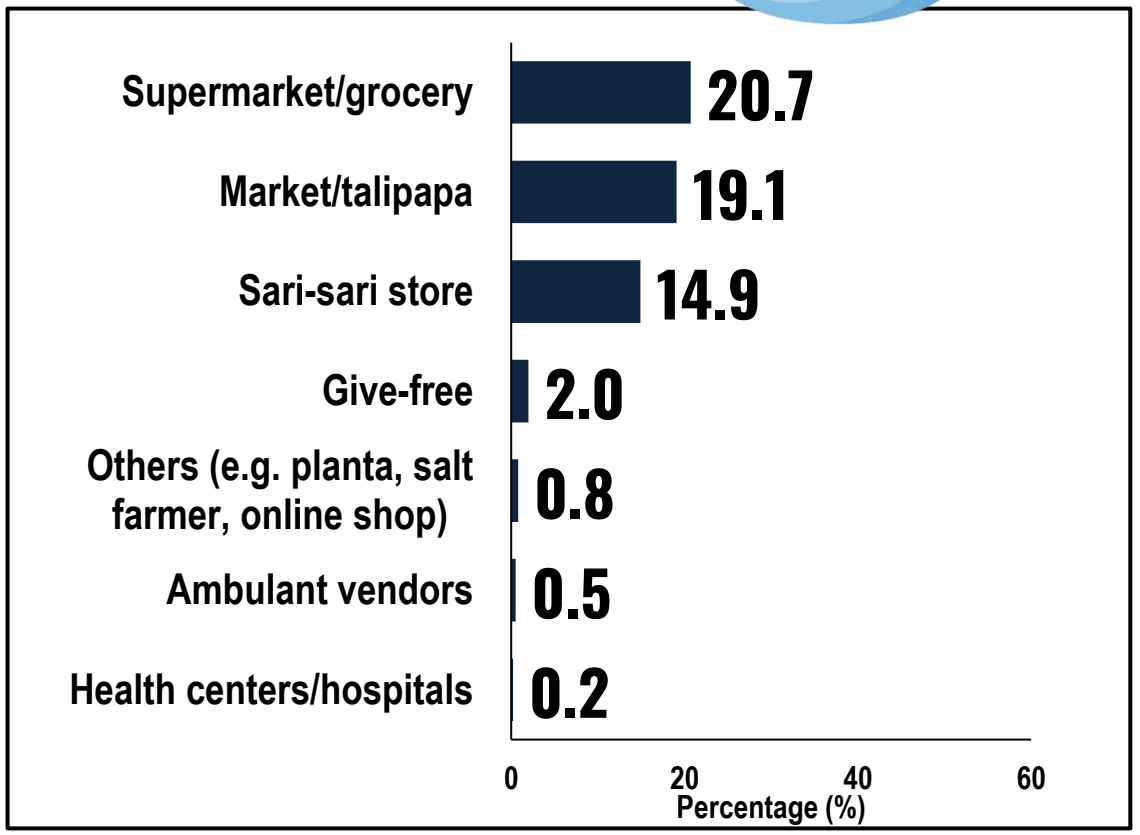


ROCK SALT

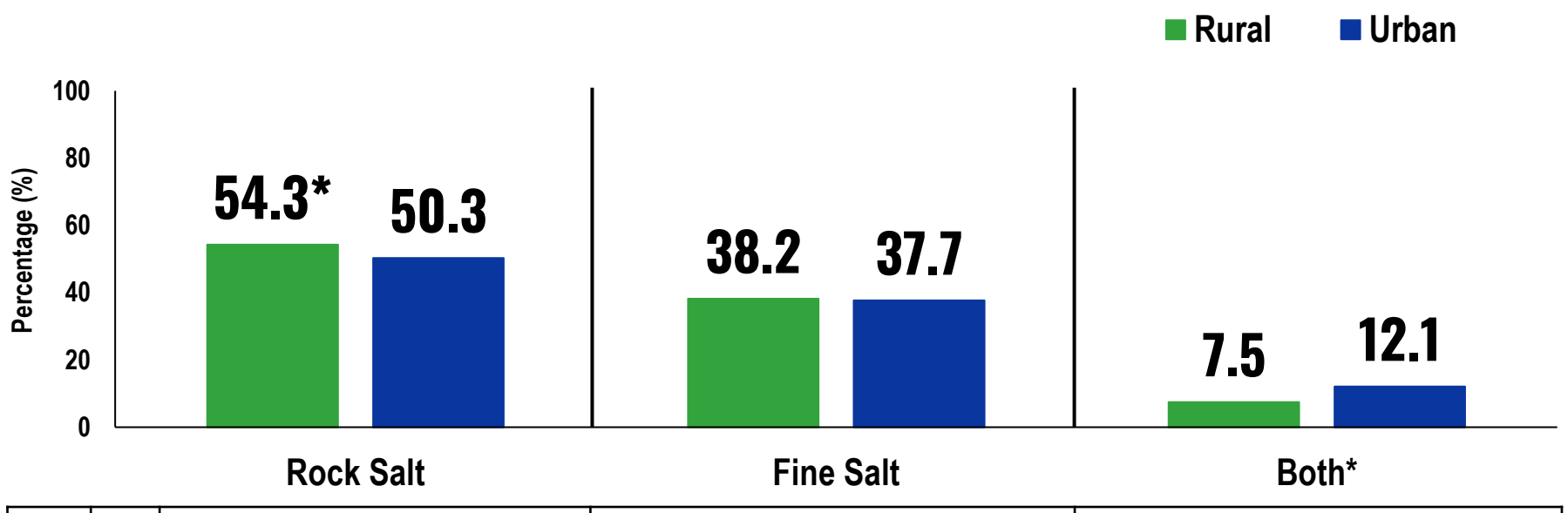




FINE SALT



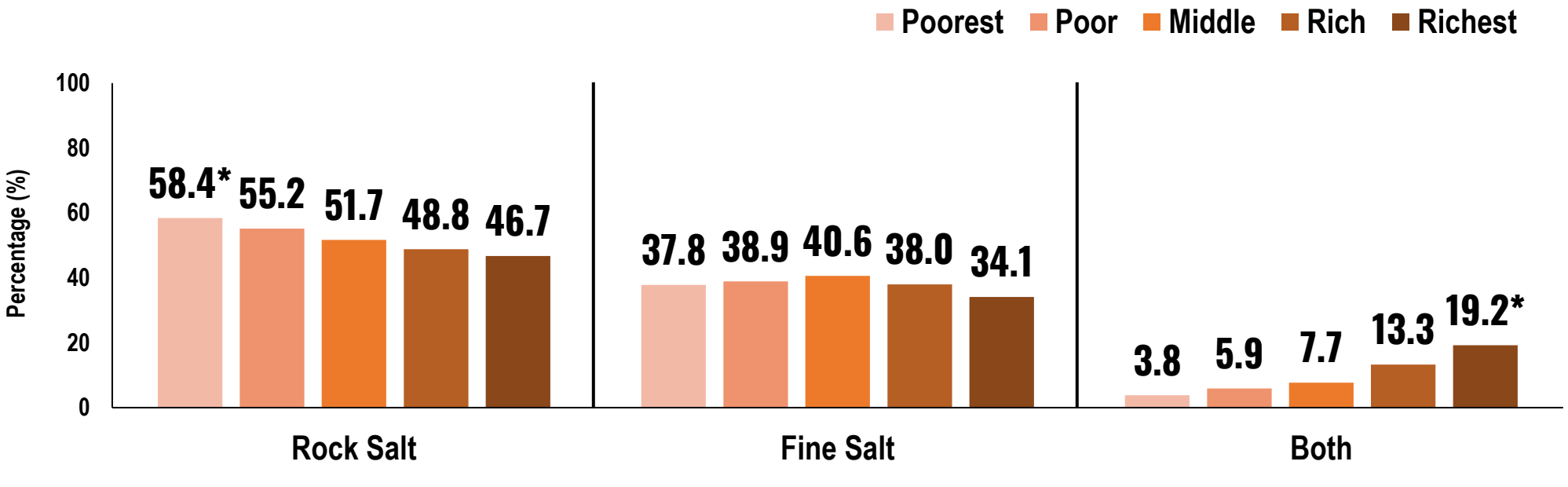
Type of salt being used by place of residence: Philippines, 2023 (n=35,457)



95% CI	LL	52.7	48.7	36.8	36.1	7.0	11.2
	UL	55.8	51.8	39.6	39.2	8.1	13.1

*significantly different at 5% level of significance

Type of salt being used by wealth quintile: Philippines, 2023 (n=35,457)



95% CI	LL	56.6	53.5	49.6	46.9	44.7	36.1	37.4	38.5	36.3	32.3	3.2	5.1	6.9	12.2	17.5
	UL	60.1	56.9	53.9	50.6	48.7	39.7	40.5	42.7	39.7	35.9	4.4	6.7	8.6	14.4	21.1

*significantly different at 5% level of significance

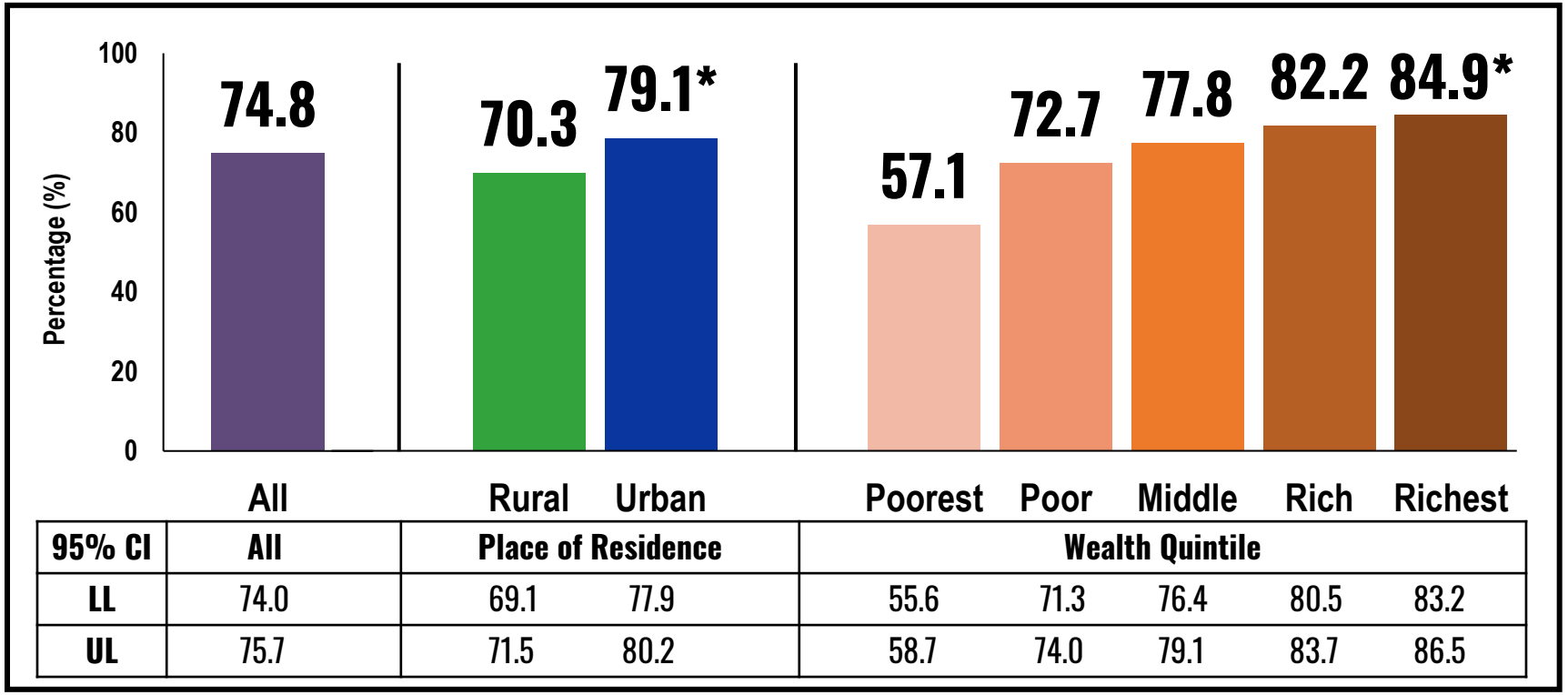
Are Filipino households aware of iodized salt?





or **7** out of **10** household meal planners are aware of iodized salt

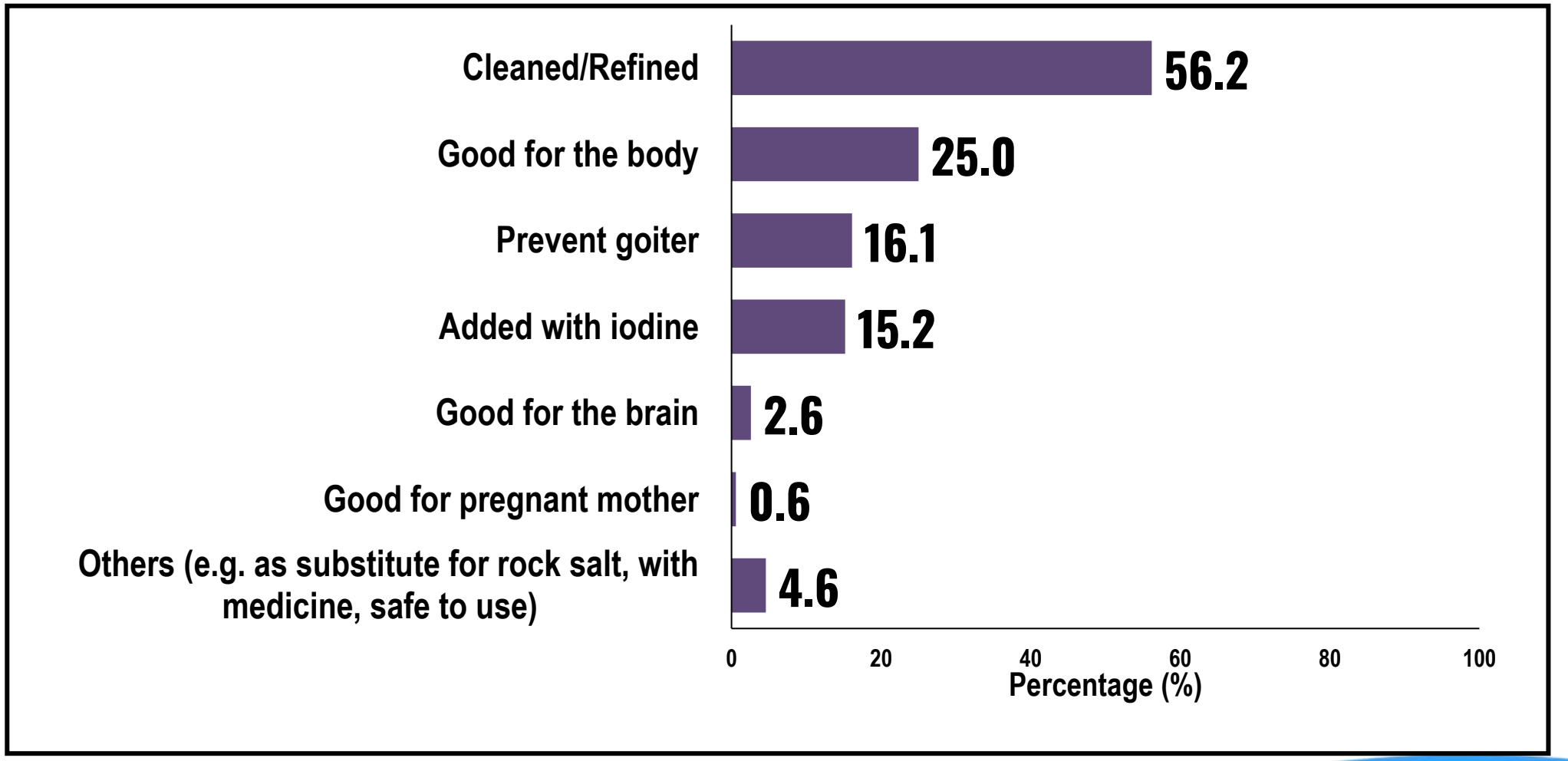
Awareness of iodized salt among household meal planner by place of residence and wealth quintile: Philippines, 2023 (n=35,457)



95% CI	All	Place of Residence		Wealth Quintile				
LL	74.0	69.1	77.9	55.6	71.3	76.4	80.5	83.2
UL	75.7	71.5	80.2	58.7	74.0	79.1	83.7	86.5

*significantly different at 5% level of significance

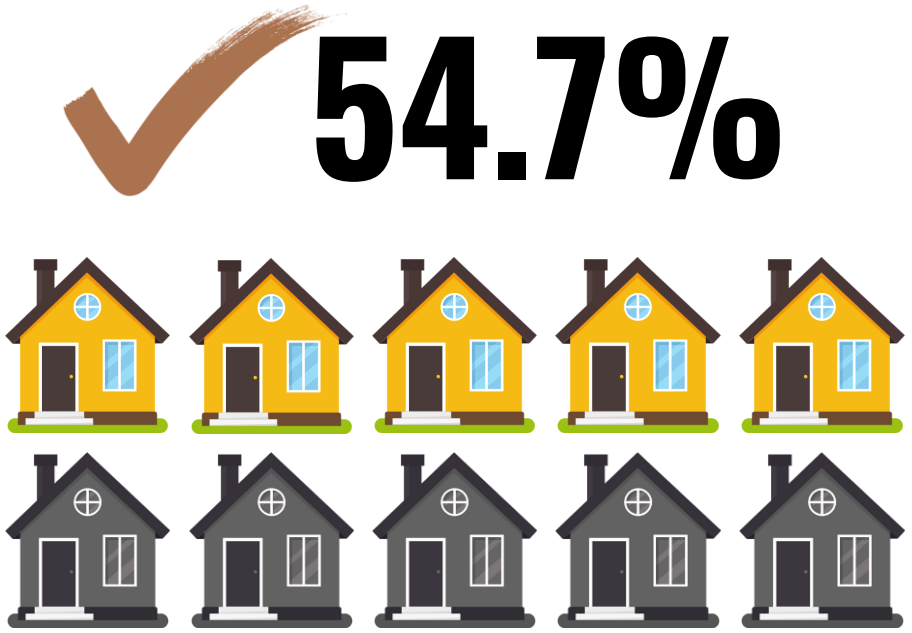
Perception of meal planners on iodized salt: Philippines, 2023 (n=35,457)*



*Multiple response

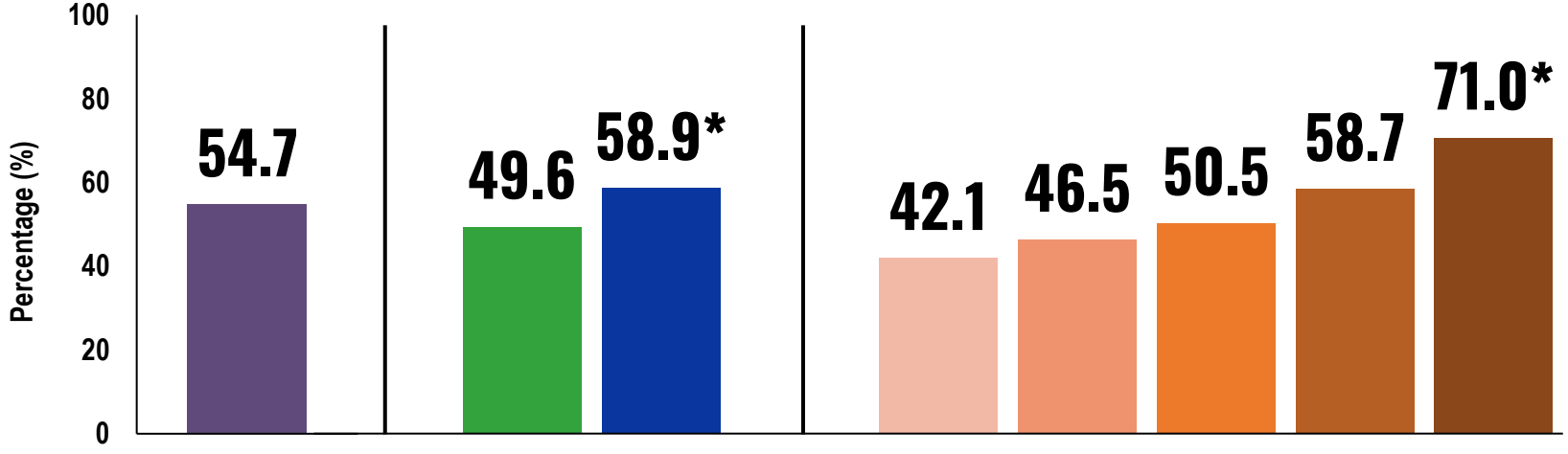
Are Filipino households using iodized salt?





or **5** out of **10** household meal planners are claimed users of iodized salt

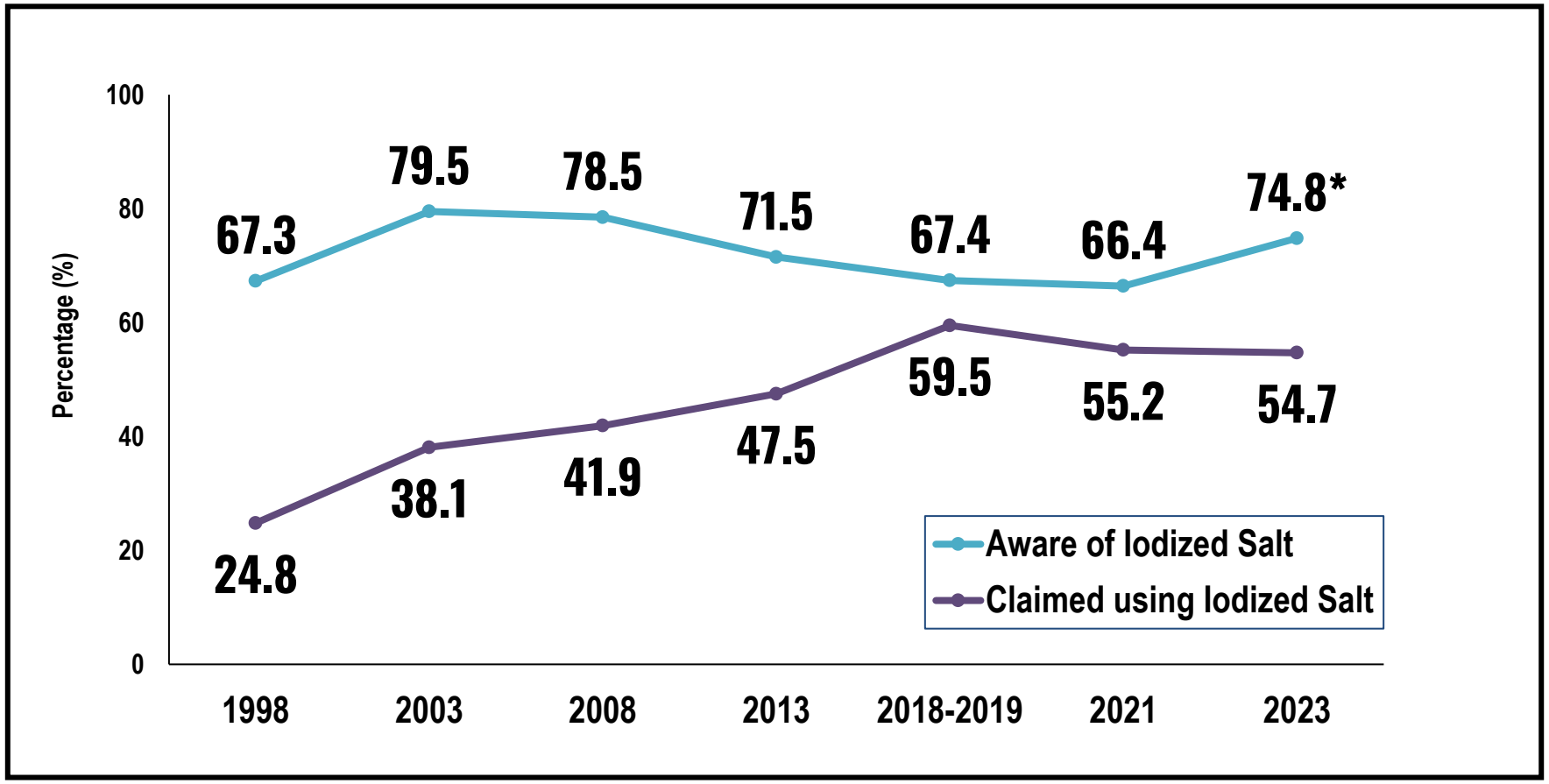
Claimed users of iodized salt among household meal planners (who are aware of iodized salt) by place of residence and wealth quintile: Philippines, 2023 (n=25,694)



95% CI	All	Place of Residence		Wealth Quintile				
LL	53.8	47.9	57.7	40.2	44.8	48.4	56.7	69.4
UL	55.6	51.3	60.2	44.0	48.2	52.5	60.8	72.4

*significantly different at 5% level of significance

Trends in the percentage of households who are aware and claimed users of iodized salt: Philippines, 1998 to 2023



*significantly different at 5% level of significance

HOUSEHOLDS USING ADEQUATELY IODIZED SALT BASED ON IODINE TESTING (WYD CHECKER)



COLLECTION AND TRANSPORT OF SAMPLES

Sample households for the salt testing were based on the **target households of the Dietary Component, which is 50% of the total sample households of the NNS**. The Dietary Researchers collected the sample salt from the households.

About 50-g were requested from the households → The collected sample were placed in a round, transparent container with lid → The samples were packed properly using black plastic bag. → The samples were put together inside a sturdy box for freight/ transport to DOST-FNRI

SIGNIFICANCE
OBJECTIVES
DESIGN
Collection of sample
Salt Analysis
Result Analysis
Line Item Budget
Team

DETERMINATION OF IODINE CONTENT OF SALT SAMPLES

- The iodine content of salt samples were determined using the **WYD Iodine checker machine**. It is a device that quantitatively measures the iodine level in salt on the basis of "luminosity" principle.

- The degree of color as measured by the Iodine checker is proportional to the content of iodine in the salt sample (IDD Newsletter, 2002)

DETERMINATION OF IODINE CONTENT OF SALT SAMPLES



1. Weigh 1 gram of salt sample in duplicate. (0.5 gram for reference material)



2. Add approximately 20 to 30 mL deionized water. Mix until salt is dissolved.



3. Add 2 mL each of Solution A and Solution B*.



4. Dilute to 50 mL and mix thoroughly.



5. Wash the cuvette twice with the sample and wipe using lint-free wipes.



6. Place the cuvette with sample in the WYD iodine checker. Record the reading.

*Solution A = KI-starch solution; Solution B = diluted sulfuric acid

Total Number of Responses for Salt Testing: Philippines, 2023

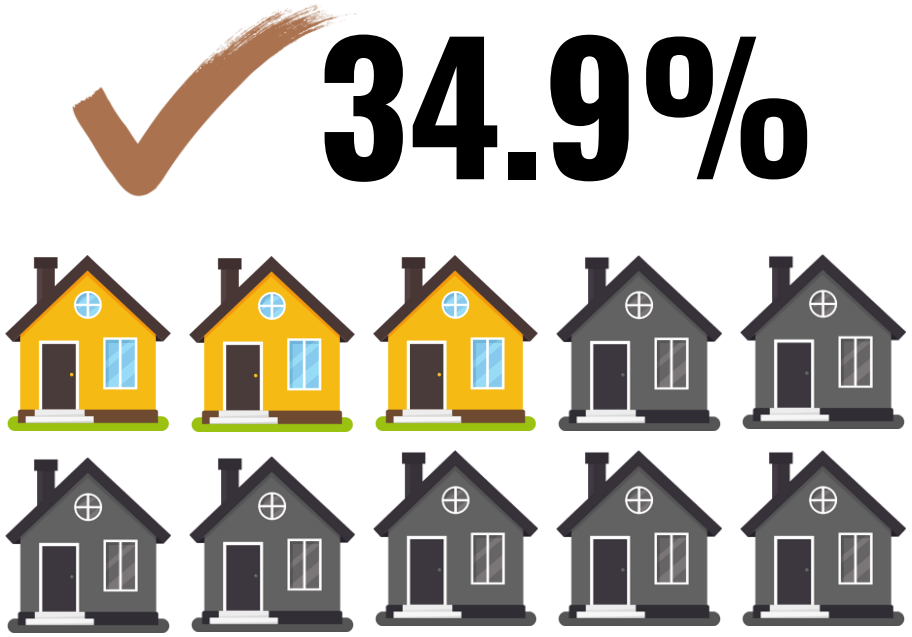


Number of households with salt samples

18,221

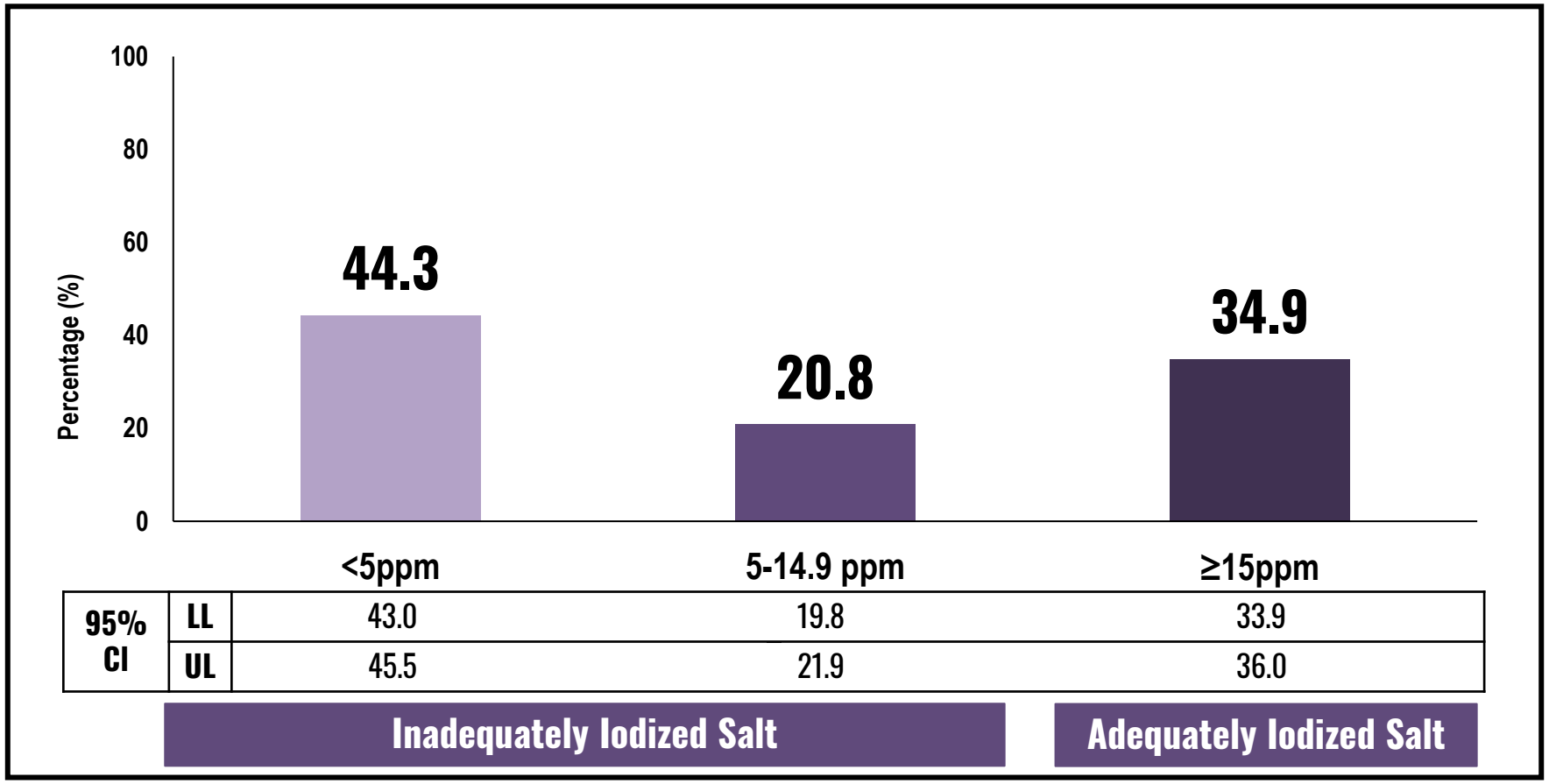
Are Filipino households using adequately iodized salt (≥15ppm)?





or **3** out of **10** households are using adequately iodized salt

Percent distribution of households by WYD checker results for iodine testing: Philippines, 2023 (n=18,221)

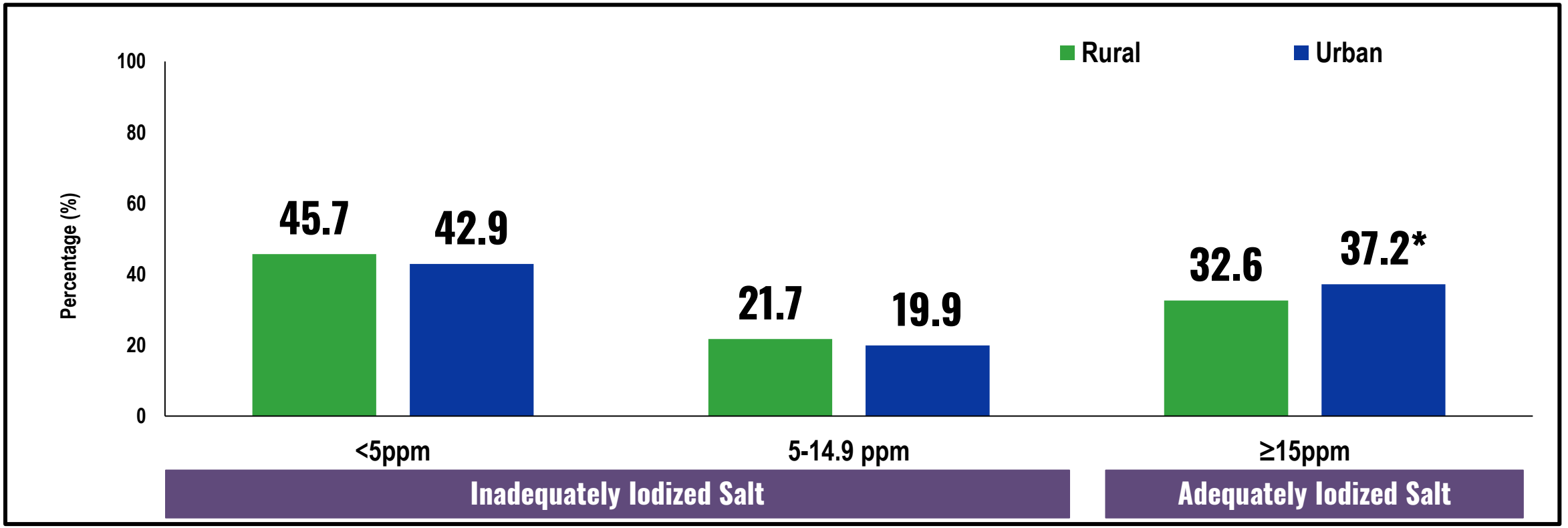


		<5ppm	5-14.9 ppm	≥15ppm
95% CI	LL	43.0	19.8	33.9
	UL	45.5	21.9	36.0

Inadequately Iodized Salt

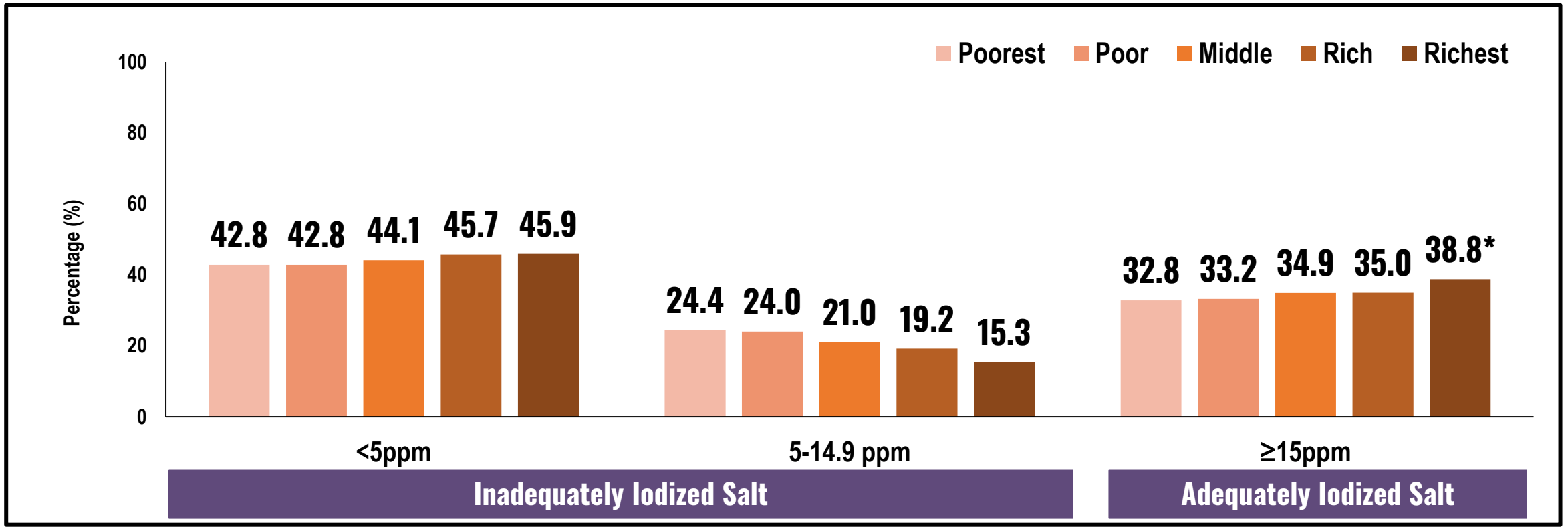
Adequately Iodized Salt

Distribution of households by WYD checker results for iodine testing by place of residence: Philippines, 2023 (n=18,221)



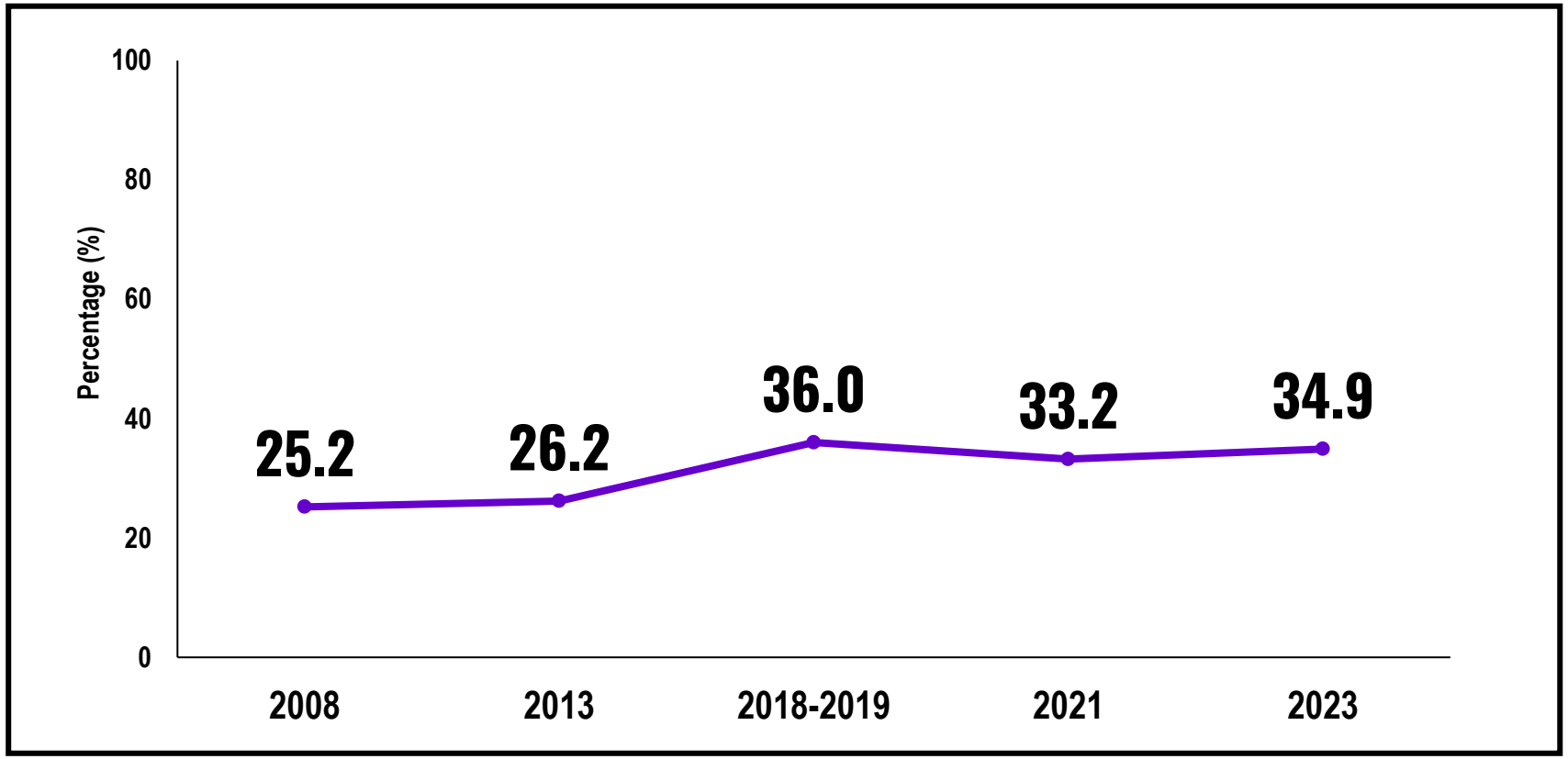
*significantly different at 5% level of significance

Distribution of households by WYD checker results for iodine testing by wealth quintile: Philippines, 2023 (n=18,221)

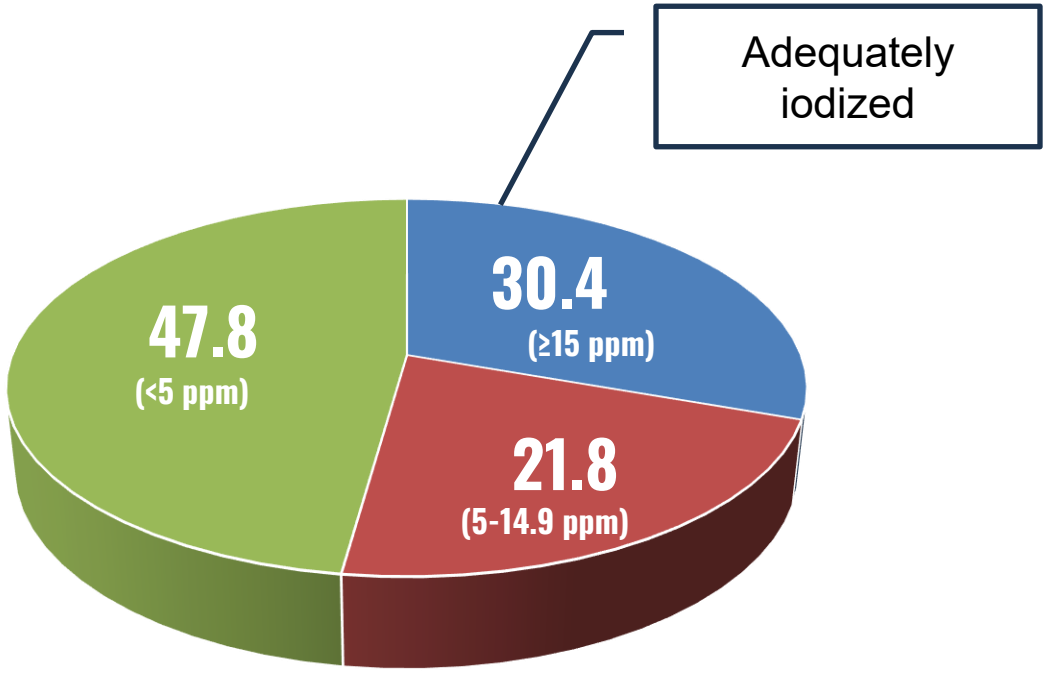


*significantly different at 5% level of significance

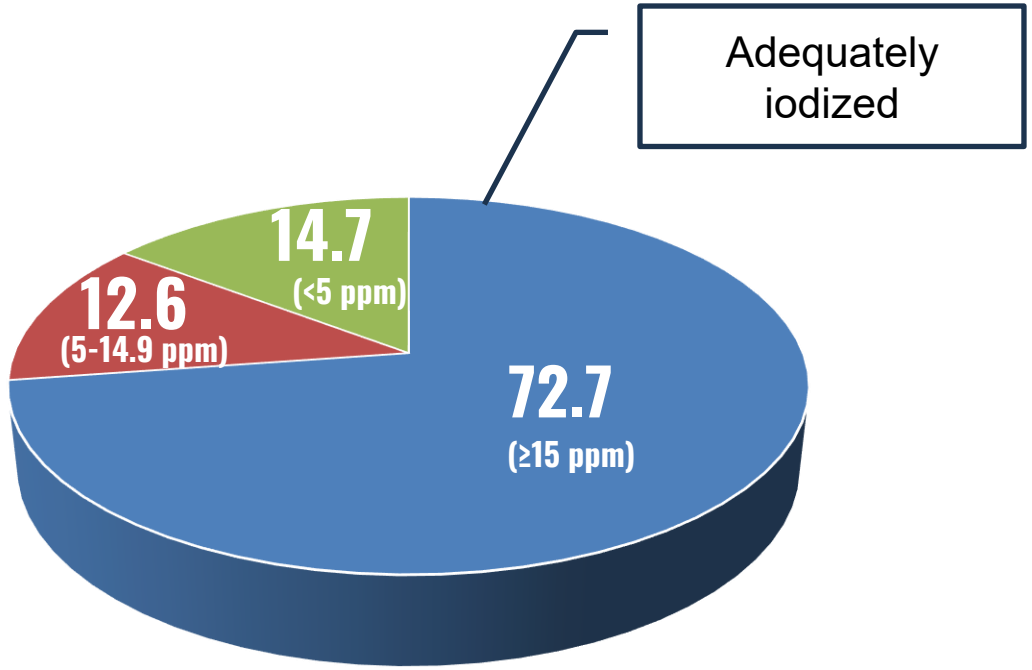
Trend in the percentage of households using adequately iodized salt: Philippines, 2008 to 2023



Percent Distribution of households by WYD checker results for iodine testing by type of salt being used: Philippines, 2023



ROCK SALT



FINE SALT*

*significantly different at 5% level of significance

HIGHLIGHTS

AWARENESS AND USAGE OF IODIZED SALT

Type of salt being used by Filipino households



More than half of the households are using rock salt (52.2%)

- ❑ in both rural (54.3%) and urban (50.3%) areas
- ❑ in more than half of the poorest households (58.4%)

Rock salt is usually bought in the **market/*talipapa*** (45.4%) while **Fine salt** is usually bought in the **supermarket/grocery** (20.7%).

Awareness and usage of iodized salt



7 out of 10 (74.8%)
households are **aware of iodized salt**



5 out of 10 (54.7%)*
households are **claimed users of iodized salt**

**of those who are aware*

HIGHLIGHTS

PERCEPTION AND USAGE OF ADEQUATELY IODIZED SALT

Meal planners' perception on iodized salt



- ❑ Cleaned/refined (56.2%)
- ❑ Good for the body (25.0%)
- ❑ Prevent goiter (16.1%)
- ❑ Added with iodine (15.2%)
- ❑ Good for the brain (2.6%)
- ❑ Good for pregnant mother (0.6%)

Households using adequately iodized salt (≥ 15 ppm)



Only 3 out of 10 (34.9%) households are **using adequately iodized salt**

- ❑ Significantly higher in urban (37.2%) than in rural (32.6%) areas

HIGHLIGHTS

SALT TESTING

Iodine testing based on type of salt used



Of the different type of salts tested:

- ❑ **3 out of 10 (30.4%)** rock salt are **adequately iodized**
- ❑ **7 out of 10 (72.7%)** fine salt are **adequately iodized**

Clinical and Health Survey Component

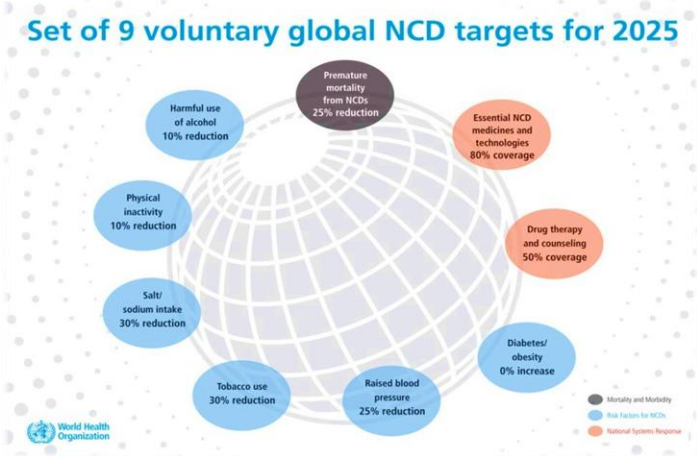
Assess the prevalence of non-communicable disease risk factors in the general population.





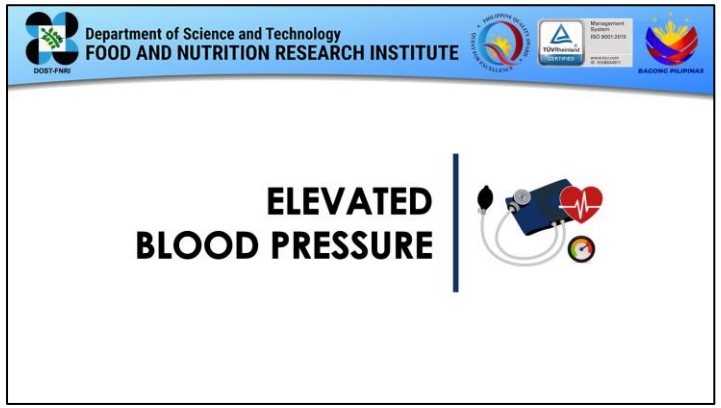






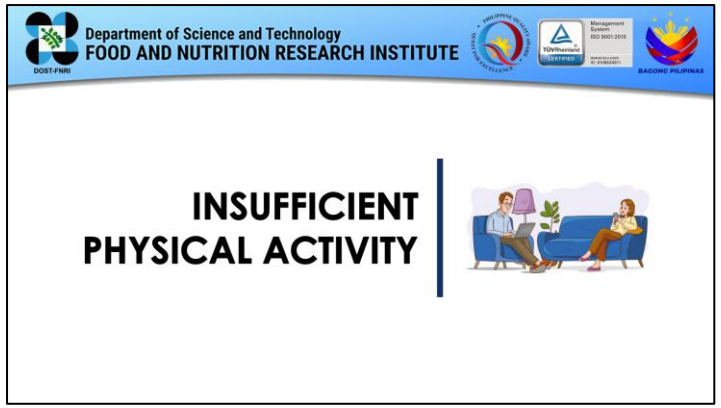
National Nutrition Summit Part I

(December 10, 2024)









Presentations from the National Nutrition Summit Part I can be accessed at **http://enutrition.fnri.dost.gov.ph**

FASTING BLOOD GLUCOSE



METHODS



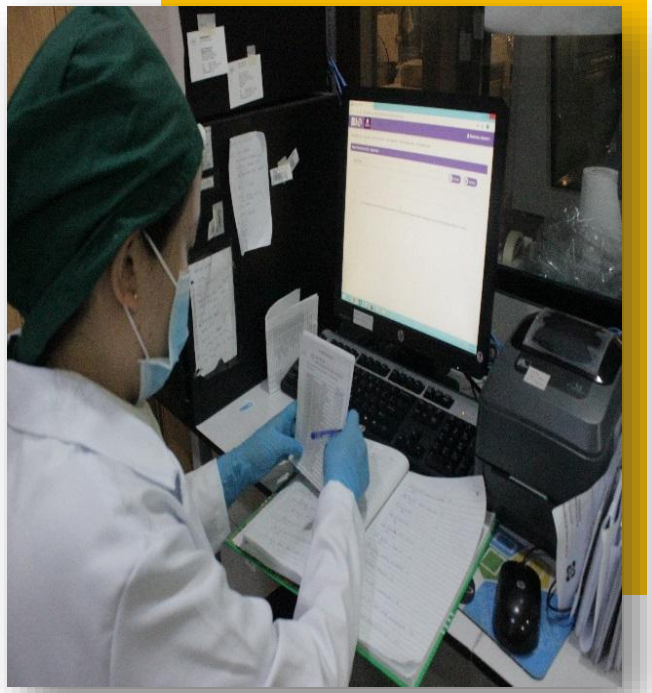
Collection of blood samples from Adults, 20 years old and above by Registered Medical Technologists.

METHODS





Analysis of Blood Samples for Fasting Blood Glucose and Lipid Profile



Encoding of results

GUIDELINES IN CLASSIFYING FASTING BLOOD GLUCOSE

Cut-off points for fasting blood glucose (FBG) (Philippine CPG, 2011)

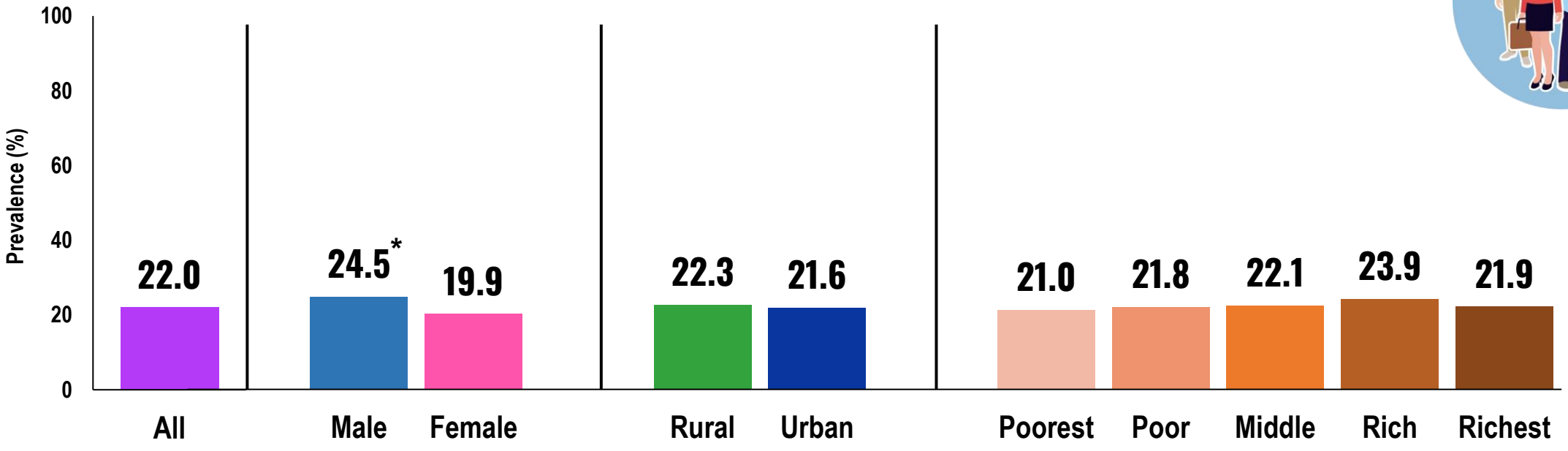
Classification	Cut-off (mg/dL)
Normal FBG	<100
Impaired FBG	100 – 125
High FBG	≥126



ADULTS
(20 TO 59 YEARS OLD)

Prevalence of impaired fasting blood glucose among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=40,877)



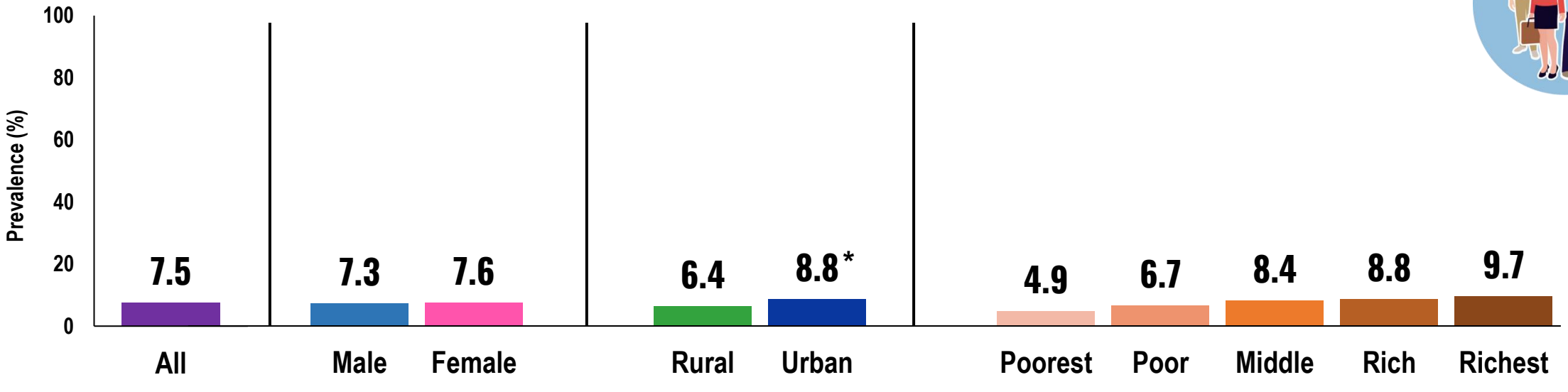


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	21.5	23.6	19.2	21.5	20.6	19.9	20.5	20.9	22.5	20.3
UL	22.6	25.4	20.7	23.2	22.7	22.1	23.1	23.3	25.3	23.5

*significantly different at 5% level of significance

Prevalence of high fasting blood glucose among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=40,877)





95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	7.1	6.8	7.2	6.1	8.2	4.5	6.1	7.7	7.9	8.7
UL	7.8	7.8	8.1	6.8	9.5	5.5	7.3	9.2	9.8	10.9

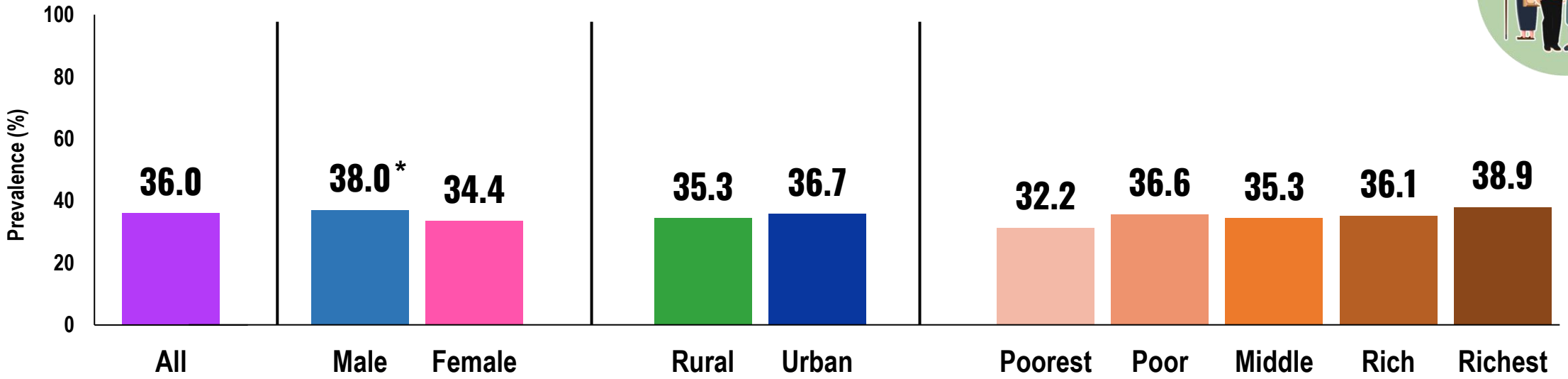
*significantly different at 5% level of significance



OLDER PERSONS (60 YEARS OLD AND ABOVE)

Prevalence of impaired fasting blood glucose among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,617)



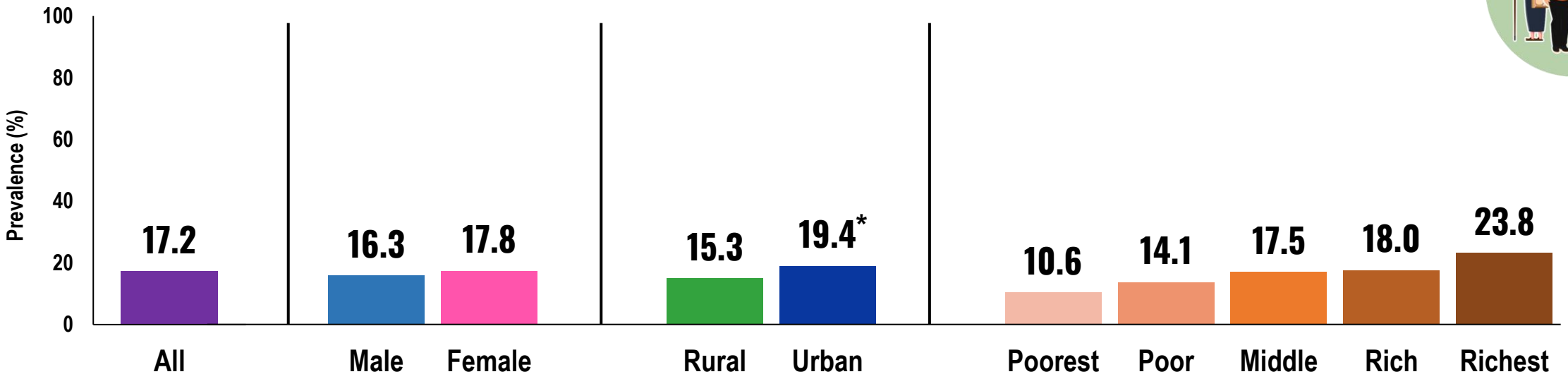


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	34.7	36.2	32.8	33.6	34.6	29.7	34.1	32.7	33.4	36.0
UL	37.3	39.9	36.1	37.1	38.9	34.9	39.1	37.9	38.8	41.8

*significantly different at 5% level of significance

Prevalence of high fasting blood glucose among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,617)

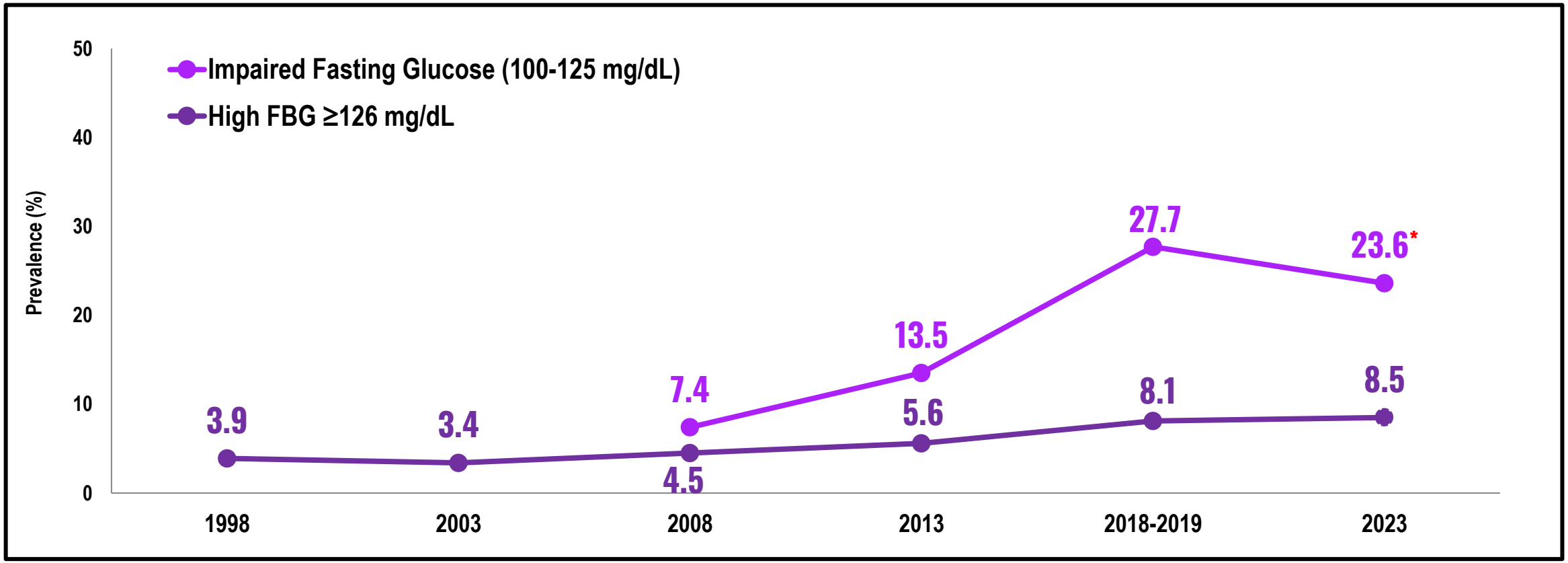




95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	16.1	14.7	16.5	14.1	17.8	9.2	12.5	15.1	15.9	21.8
UL	18.3	17.9	19.2	16.6	21.2	12.3	15.9	20.1	20.4	26.0

*significantly different at 5% level of significance

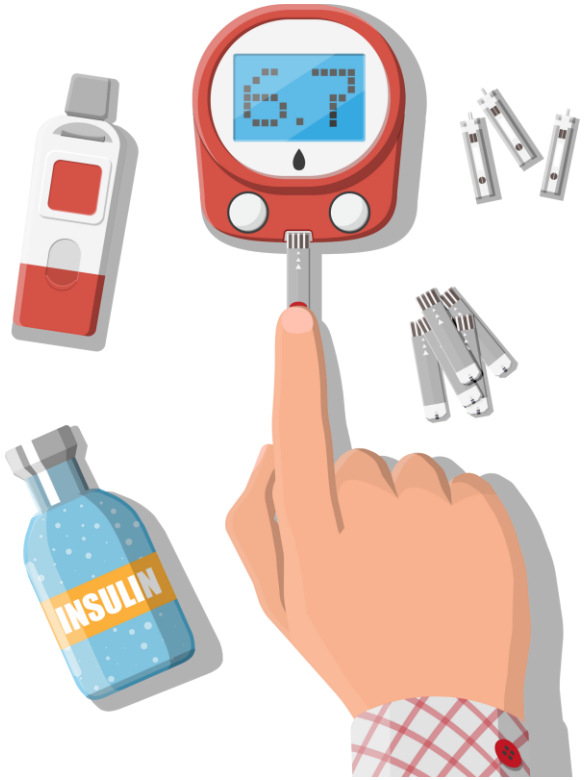
Trends in the prevalence of impaired fasting glucose and high fasting blood glucose among adults, 20 years old and above: Philippines, 1998 to 2023



*significantly different at 5% level of significance from 2018-2019

HIGHLIGHTS

Fasting Blood Glucose



- Older persons show higher prevalence of impaired and high fasting blood glucose compared to adults; 20 to 59 years old.
- In both age groups, these conditions are significantly more common among men and those living in urban areas.

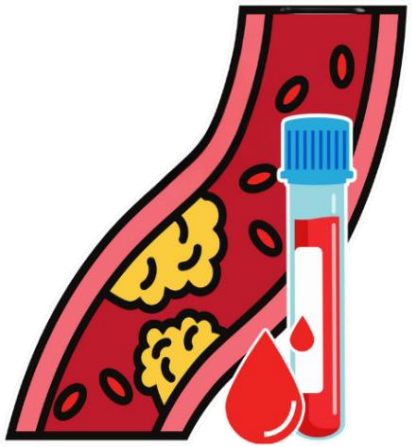
HIGHLIGHTS



Fasting Blood Glucose (FBG):

- Impaired FBG affects nearly a quarter of the adult population. The prevalence of high FBG is also increasing.

ABNORMAL BLOOD LIPIDS



GUIDELINES IN CLASSIFYING ABNORMAL BLOOD LIPIDS

Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) (2002)

TOTAL CHOLESTEROL

Classification	Cut-off (mg/dL)
Desirable	<200
Borderline High	200 – 239
High	≥240

HDL-CHOLESTEROL

Classification	Cut-off (mg/dL)
Low	<40
Borderline	40 – 59
Desirable	≥60

LDL-CHOLESTEROL

Classification	Cut-off (mg/dL)
Desirable	<130
Borderline High	130 – 159
High	≥160

TRIGLYCERIDES

Classification	Cut-off (mg/dL)
Desirable	<150
Borderline High	150 – 199
High	200 – 399
Very High	≥400

Using a different cutoff for HDL-cholesterol reflecting sex-related differences in HDL levels associated with insulin resistance

HDL-CHOLESTEROL

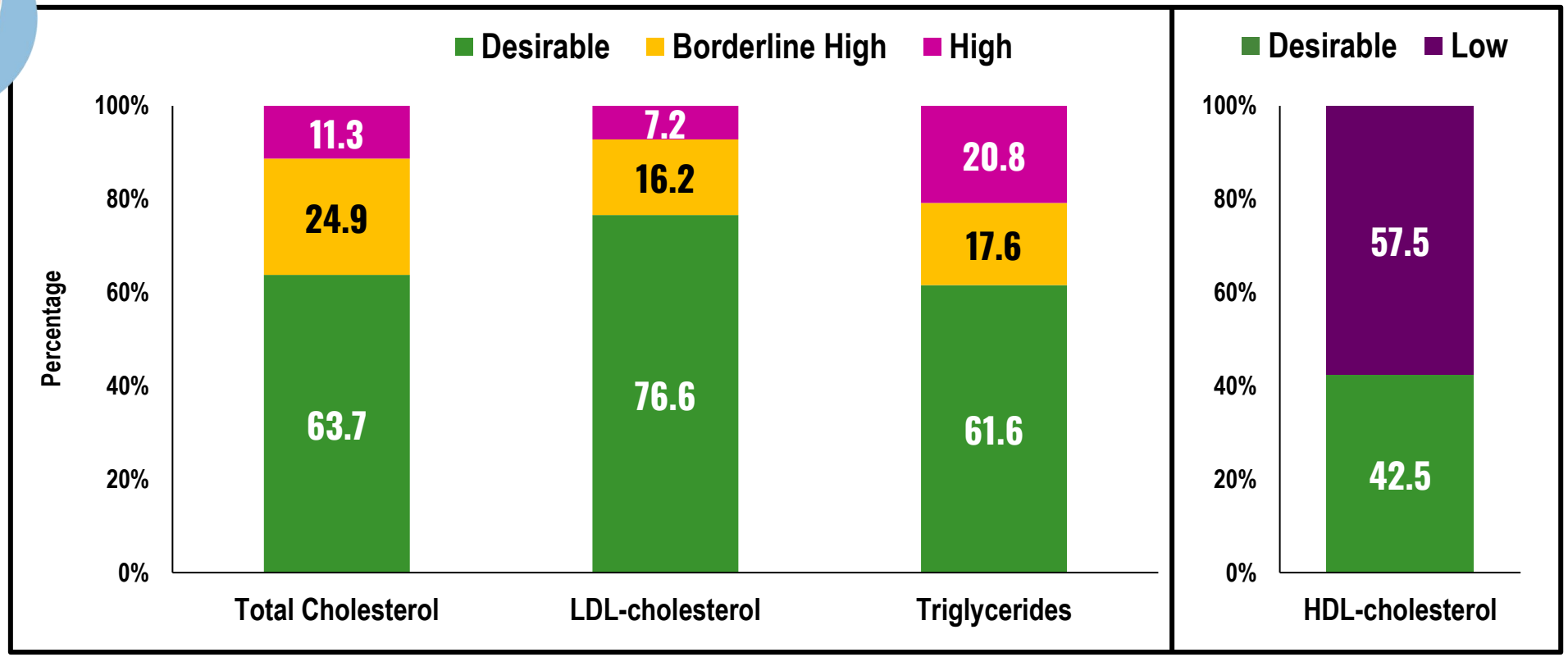
Classification	Cut-off (mg/dL)	
	Male	Female
Low	<40	<50
Desirable	≥40	≥50



ADULTS
(20 TO 59 YEARS OLD)

Percent distribution of total cholesterol, LDL-cholesterol, triglycerides, and HDL-cholesterol° levels among adults, 20 to 59 years old: Philippines, 2023 (n=41,031)



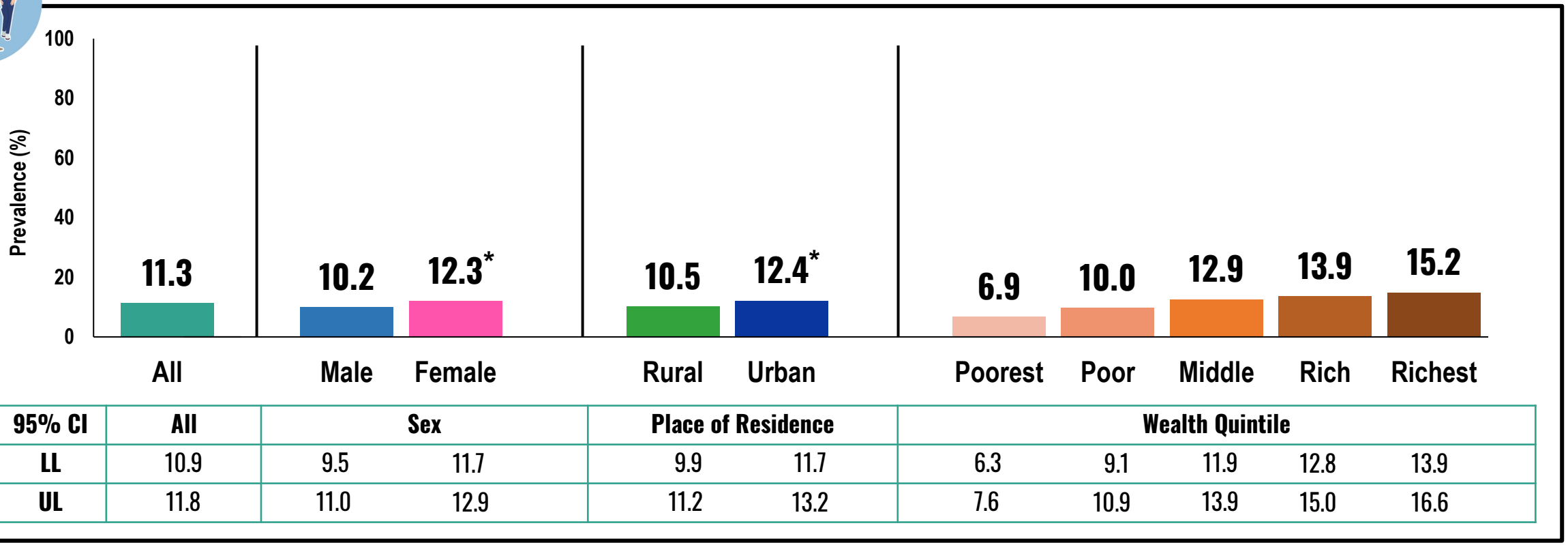


Note: High triglycerides already include very high level

° Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.

Prevalence of high total cholesterol among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=41,031)



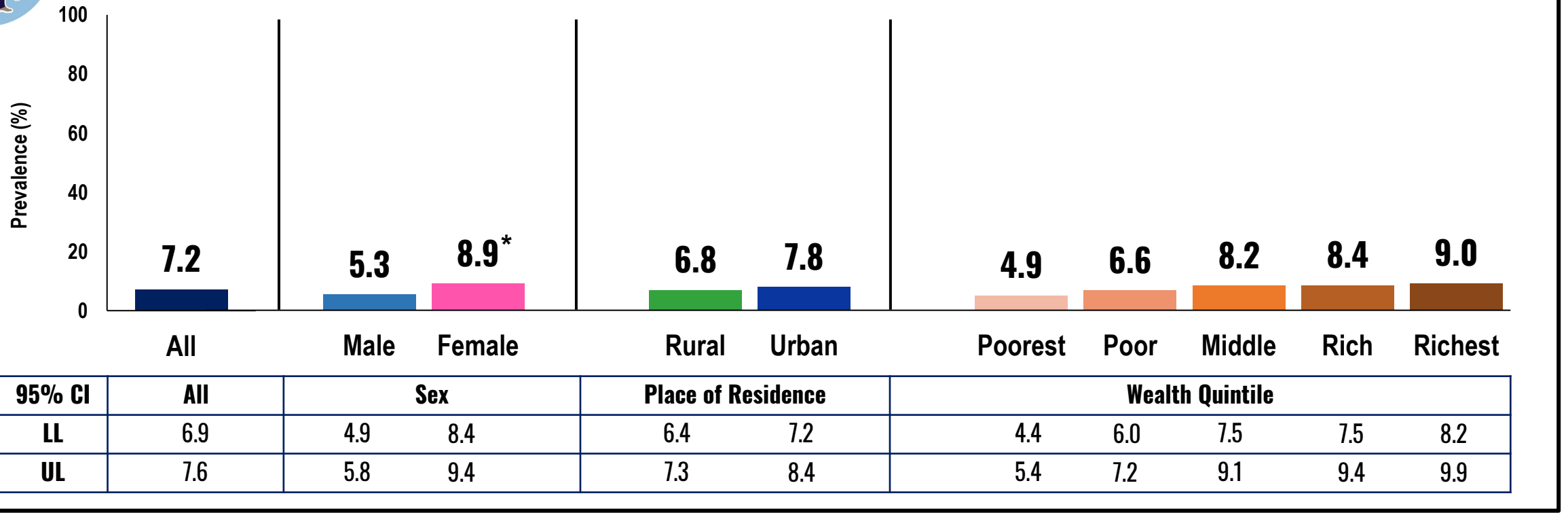


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	10.9	9.5	11.7	9.9	11.7	6.3	9.1	11.9	12.8	13.9
UL	11.8	11.0	12.9	11.2	13.2	7.6	10.9	13.9	15.0	16.6

*significantly different at 5% level of significance

Prevalence of high LDL-cholesterol among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=41,031)

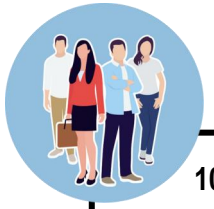


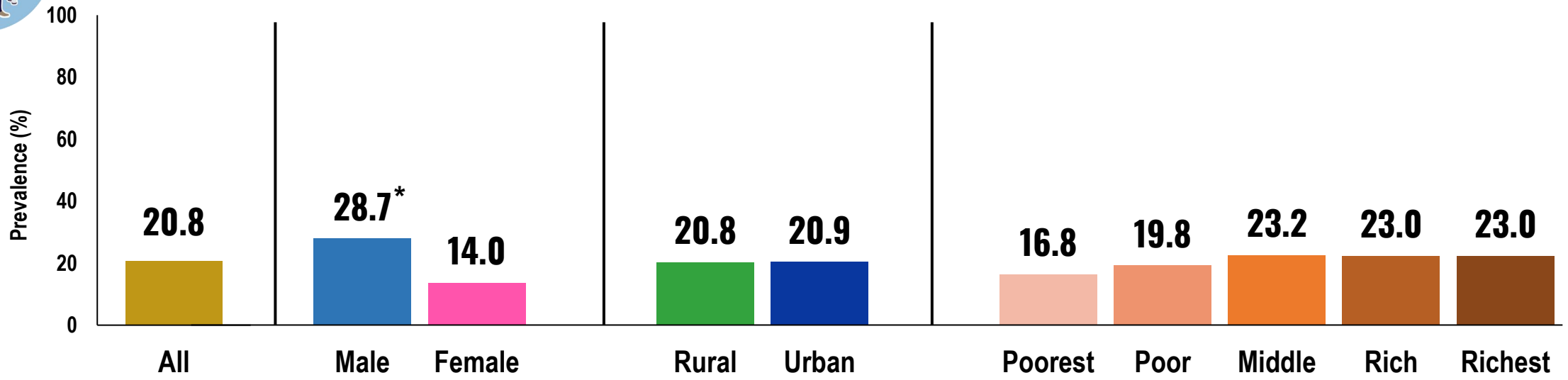


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	6.9	4.9	8.4	6.4	7.2	4.4	6.0	7.5	7.5	8.2
UL	7.6	5.8	9.4	7.3	8.4	5.4	7.2	9.1	9.4	9.9

*significantly different at 5% level of significance

Prevalence of high triglycerides among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=41,031)



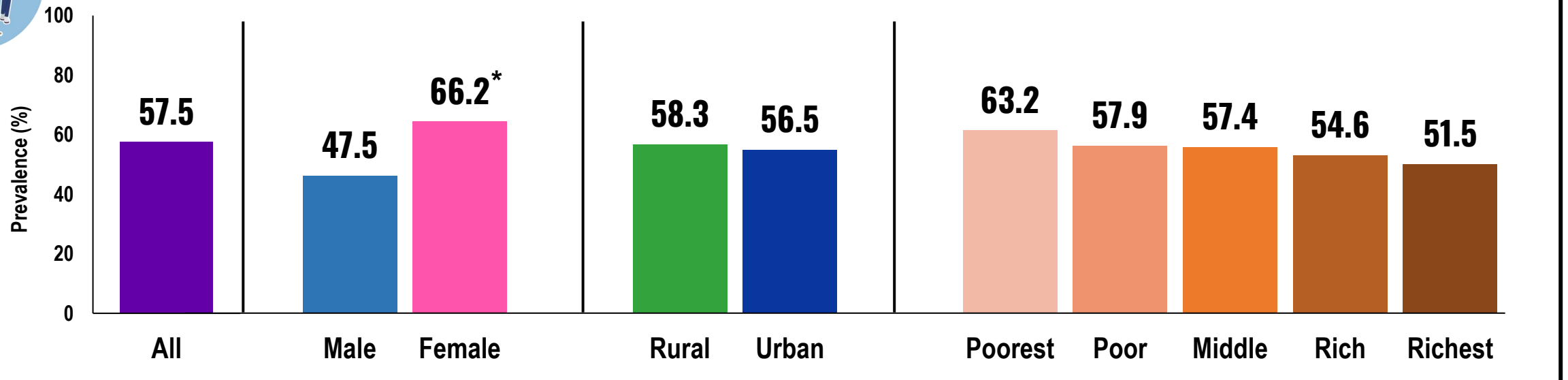


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	20.3	27.7	13.4	20.0	19.9	15.7	18.9	22.0	21.5	21.3
UL	21.4	29.7	14.7	21.7	21.8	17.9	20.8	24.4	24.6	24.9

*significantly different at 5% level of significance

Prevalence of low HDL-cholesterol° among adults, 20 to 59 years old, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=41,031)





95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	56.7	46.4	65.2	57.3	55.1	61.7	56.3	55.8	52.7	49.8
UL	58.3	48.6	67.2	59.3	57.8	64.7	59.5	59.1	56.5	53.3

*significantly different at 5% level of significance

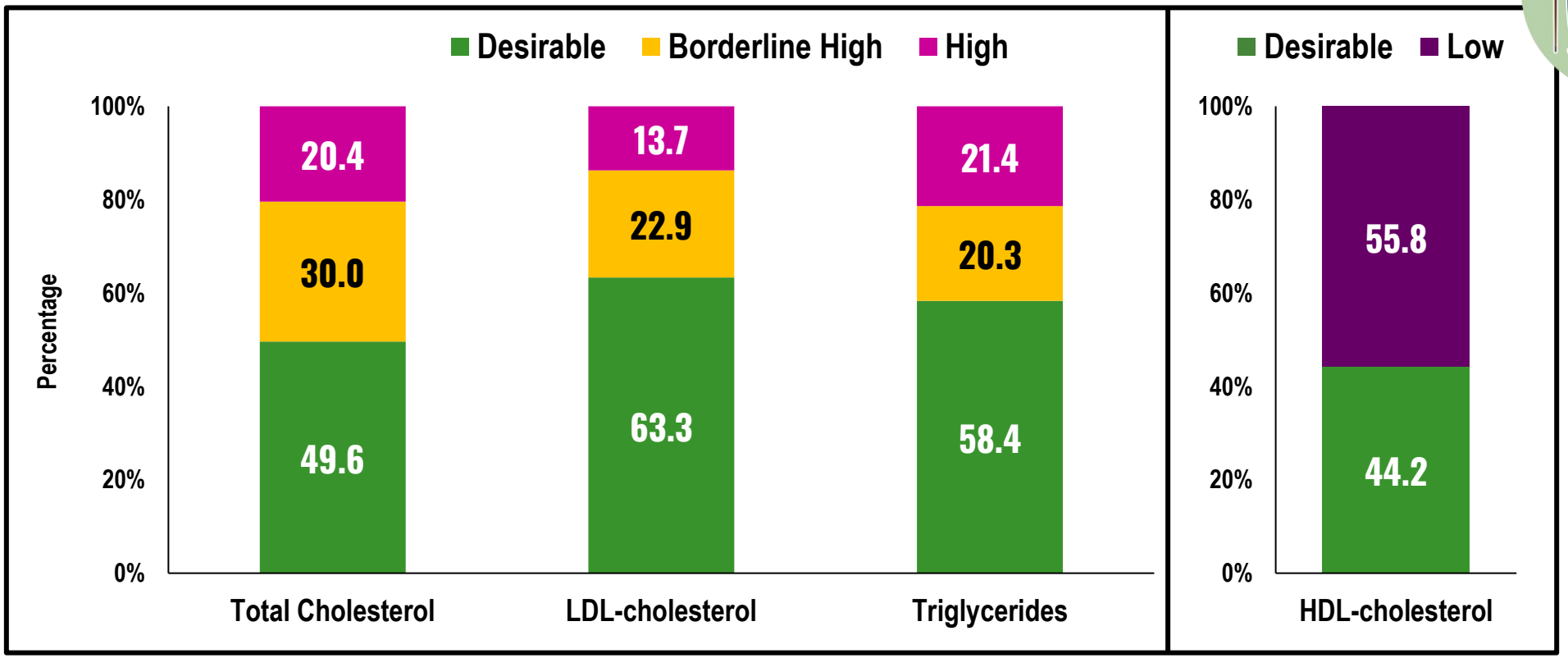
° *Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.*



OLDER PERSONS (60 YEARS OLD AND ABOVE)

Percent distribution of total cholesterol, LDL-cholesterol, triglycerides, and HDL-cholesterol° levels among older persons, 60 years old and above: Philippines, 2023 (n=13,665)



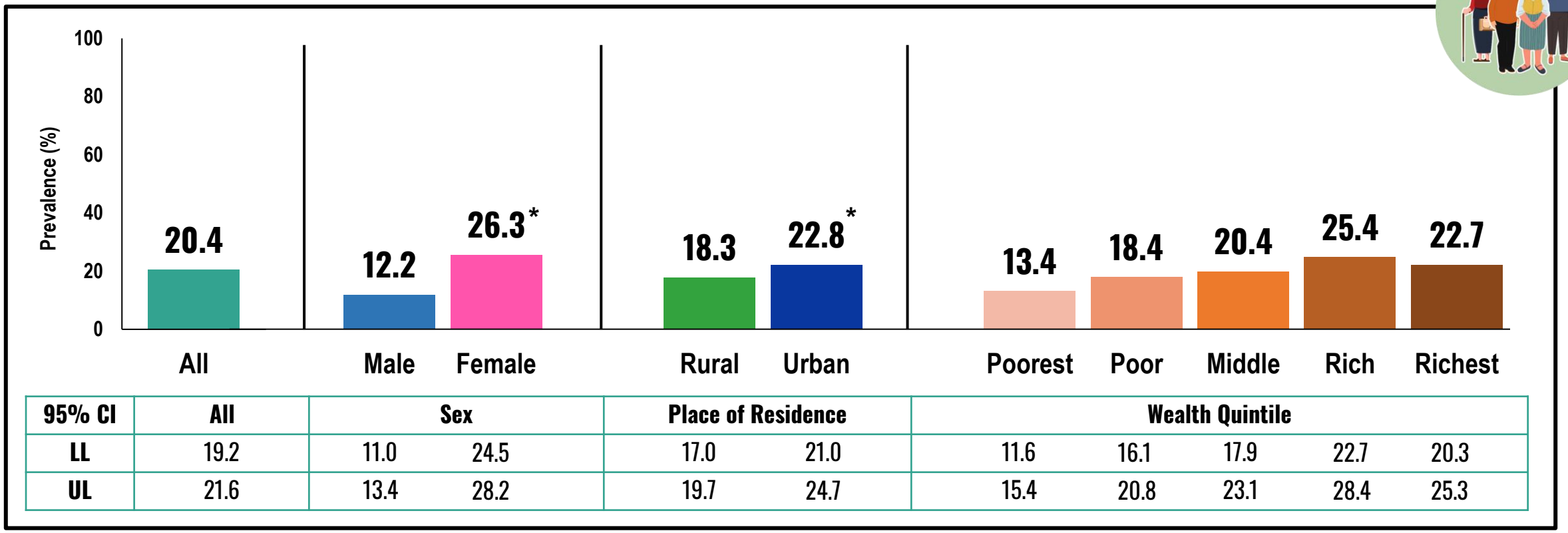


Note: High triglycerides already include very high level

° Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.

Prevalence of high total cholesterol among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,665)



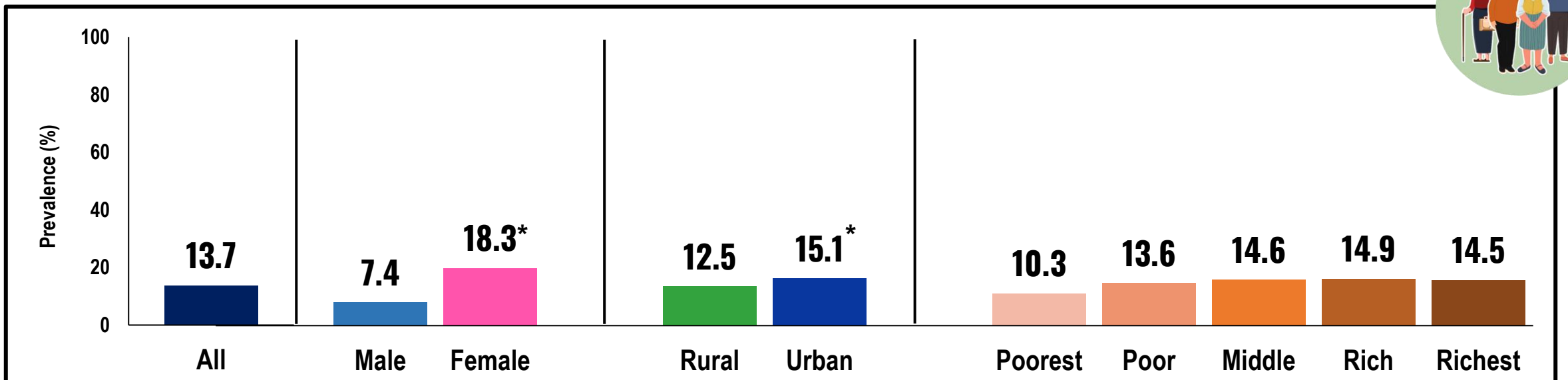


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	19.2	11.0	24.5	17.0	21.0	11.6	16.1	17.9	22.7	20.3
UL	21.6	13.4	28.2	19.7	24.7	15.4	20.8	23.1	28.4	25.3

*significantly different at 5% level of significance

Prevalence of high LDL-cholesterol among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,665)



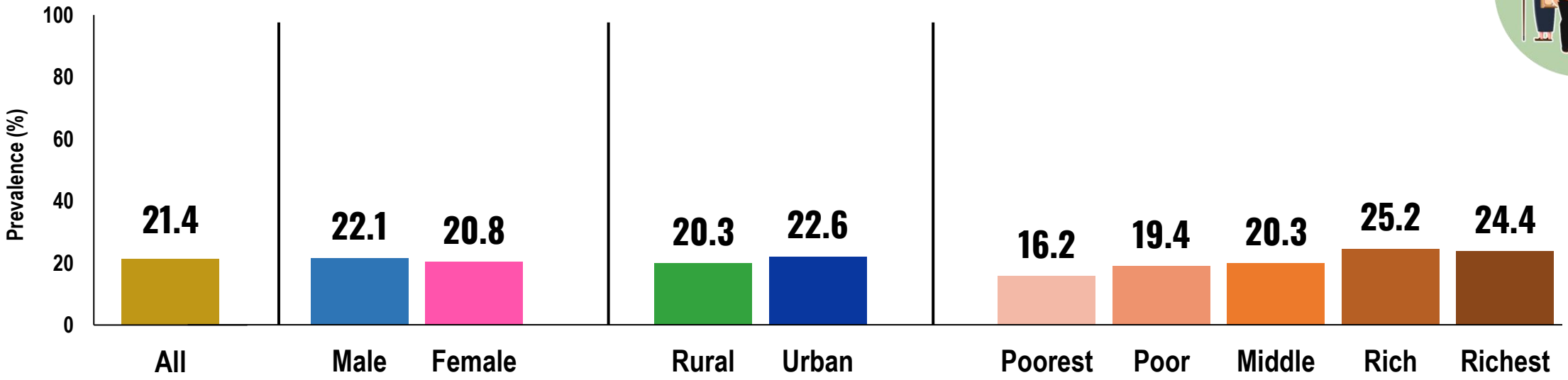


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	12.8	6.5	17.0	11.6	13.6	8.9	11.8	12.6	12.8	12.4
UL	14.6	8.4	19.7	13.5	16.8	11.8	15.6	17.0	17.2	17.0

*significantly different at 5% level of significance

Prevalence of high triglycerides among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,665)

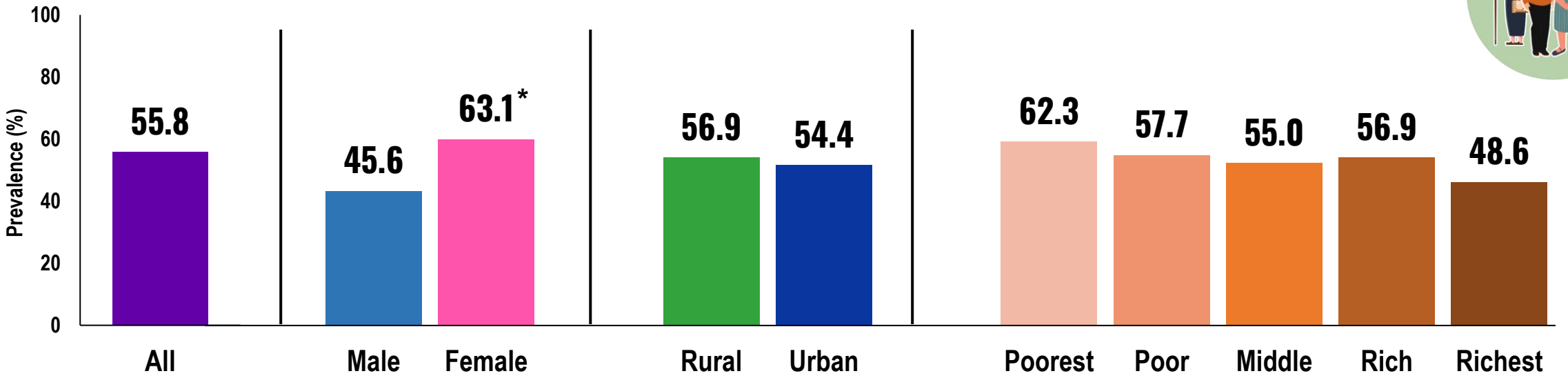




95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	20.2	20.3	19.5	18.8	20.9	14.5	17.4	17.8	22.3	22.0
UL	22.6	24.1	22.2	21.9	24.4	18.2	21.5	22.9	28.4	27.0

Prevalence of low HDL-cholesterol° among older persons, 60 years old and above, by sex, place of residence, and wealth quintile: Philippines, 2023 (n=13,665)



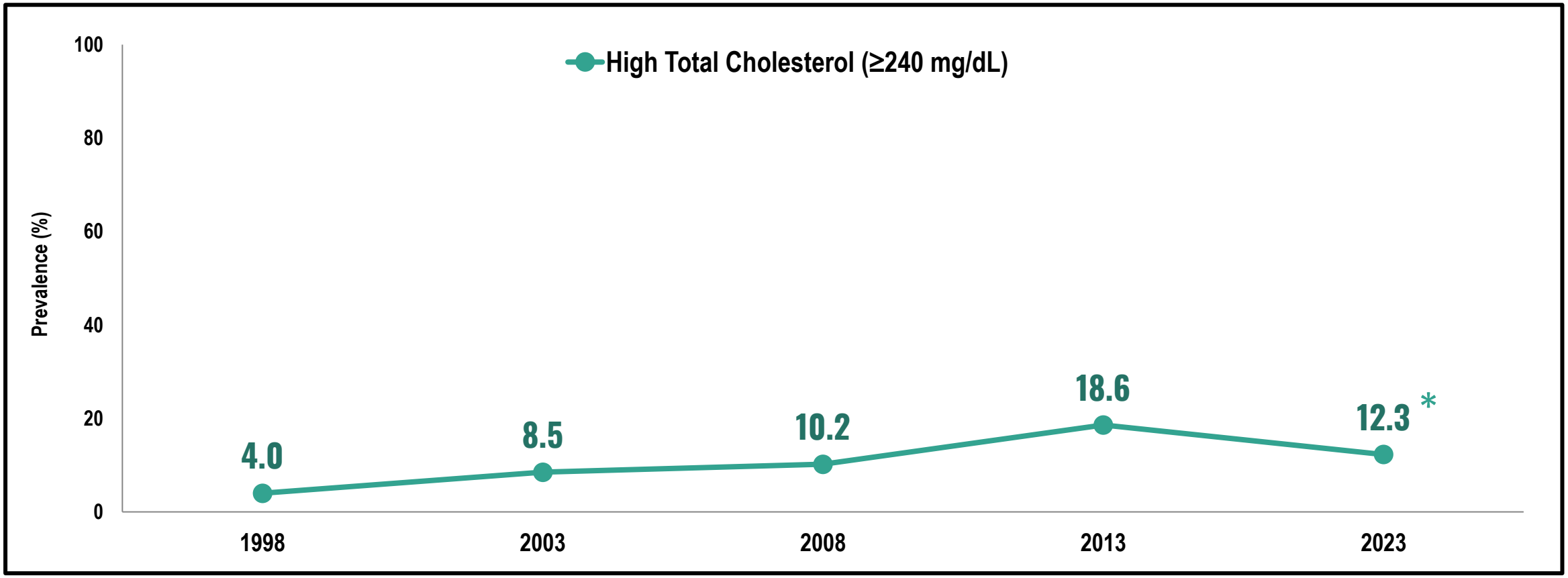


95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	54.5	43.8	61.5	55.4	52.2	59.7	55.3	51.8	54.4	45.4
UL	57.0	47.4	64.7	58.4	56.6	64.9	60.1	58.2	59.3	51.8

*significantly different at 5% level of significance

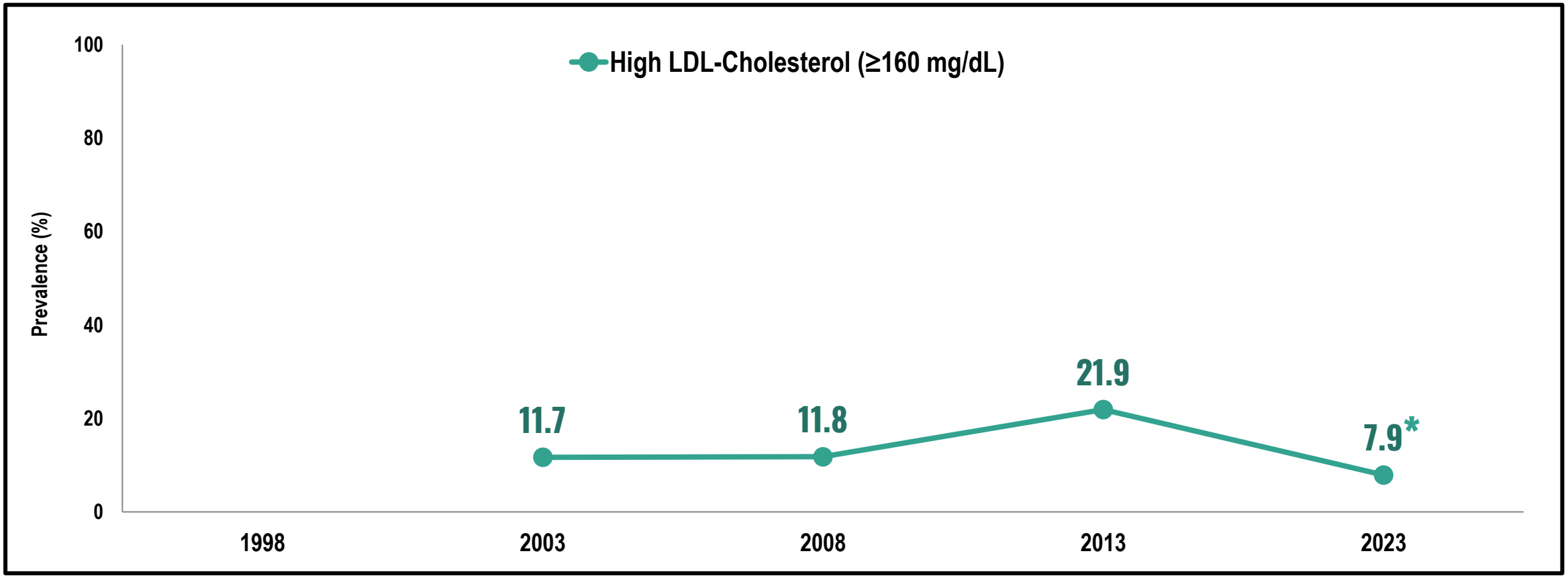
° *Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.*

Trends in the prevalence of high total cholesterol among adults, 20 years old and above: Philippines, 1998 to 2023



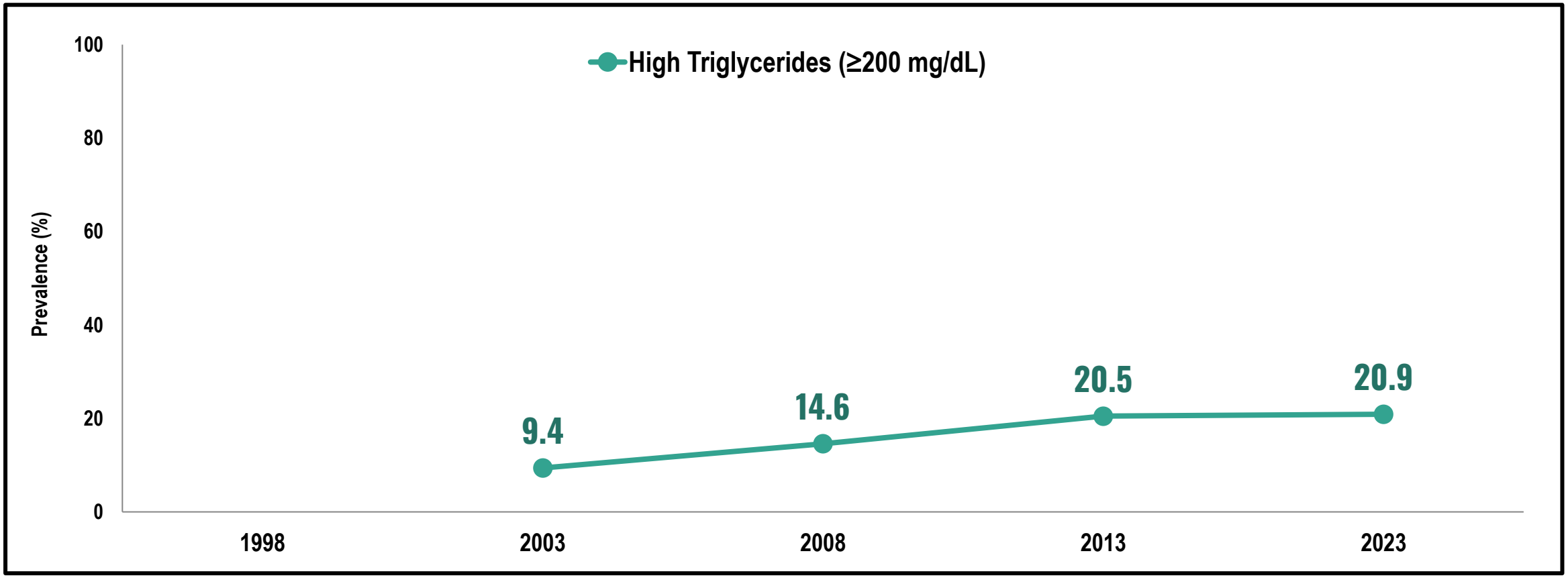
*significantly different at 5% level of significance from 2013

Trends in the prevalence of high LDL-cholesterol among adults, 20 years old and above: Philippines, 1998 to 2023

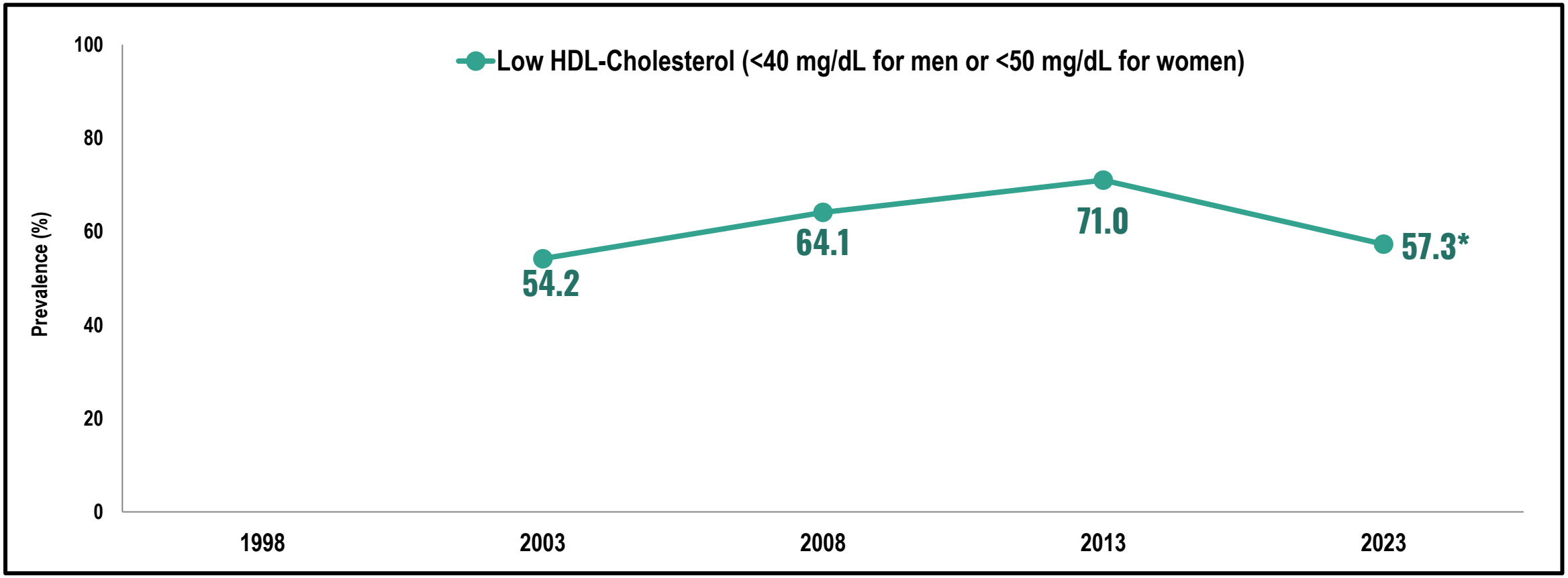


*significantly different at 5% level of significance from 2013

Trends in the prevalence of high triglycerides among adults, 20 years old and above: Philippines, 1998 to 2023



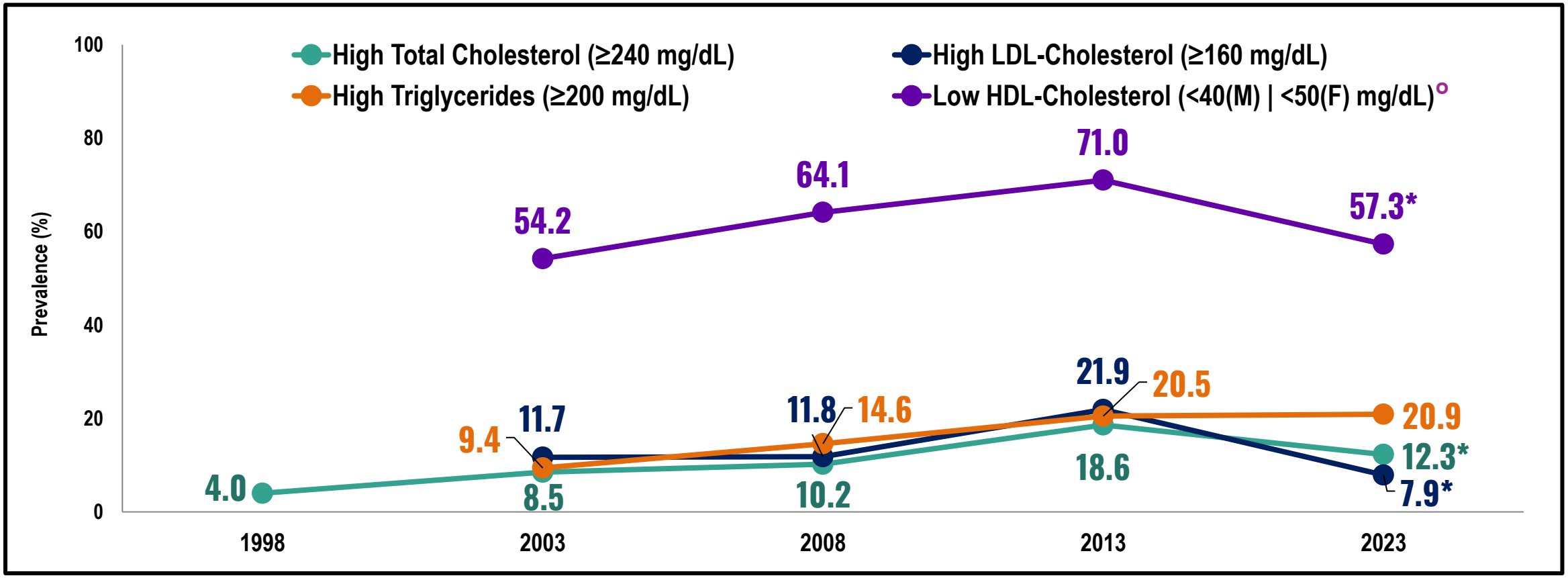
Trends in the prevalence of low HDL-cholesterol° among adults, 20 years old and above: Philippines, 1998 to 2023



*significantly different at 5% level of significance from 2013

° *Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.*

Trends in the prevalence of abnormal blood lipids among adults, 20 years old and above: Philippines, 1998 to 2023



*significantly different at 5% level of significance from 2013

° *Using a different cutoff for HDL-cholesterol or the cut-off for low HDL-cholesterol at <40 mg/dL for men and <50 mg/dL for women, reflecting sex-related differences in HDL levels associated with insulin resistance.*

HIGHLIGHTS

Abnormal Blood Lipids



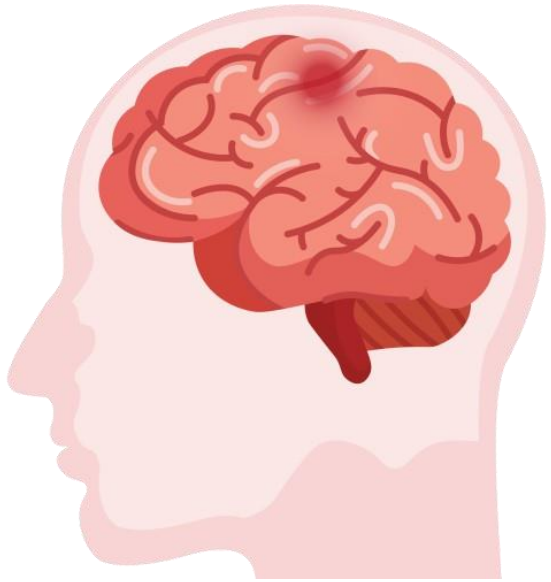
- **Elevated total cholesterol** was more prevalent in older persons.
- Adults exhibited **higher triglyceride levels** compared to older persons.
- A high prevalence of **low HDL-cholesterol** was noted in both age groups.

HIGHLIGHTS



Abnormal Blood Lipids:

- The prevalence of abnormal blood lipid levels declined in 2023.



Status of Mental Well-Being and Resilience among Filipinos

Rationale



Mental health has emerged as a **priority public health concern** in recent years and is now aggravated because of the COVID-19 pandemic.



In the Philippines, at least **3.6 million Filipinos** suffer from one or more mental, neurological, or substance use disorders. (DOH, 2020)

Nutrition and Mental Health Link

Poor nutrition can **increase the risk of mental health disorders** like depression, anxiety, and cognitive decline. Muscaritoli, M. (2021)





Mental health conditions may lead to **poor eating habits**, malnutrition, or substance abuse.

SDG Indicators



Goal 3. Ensure healthy lives and promote well-being for all at all ages

Target 3.4 By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment, and promote mental health and well-being

Objective:



To provide empirical data on the mental health and nutritional status of selected population groups for planning integrated nutrition and mental health programs and policies at the national level.

Target Population

All members of the households:



13-17.9 YEARS OLD



18-59.9 YEARS OLD



60 YEARS OLD AND ABOVE

Mental Health Questionnaires

The WHO-5 Well-Being Index



Connor-Davidson Resilience Scale (CD-RISC-10)

CDRISC

The Connor-Davidson Resilience Scale

STATE OF MENTAL WELL-BEING OF ADOLESCENTS, ADULTS AND OLDER PERSONS

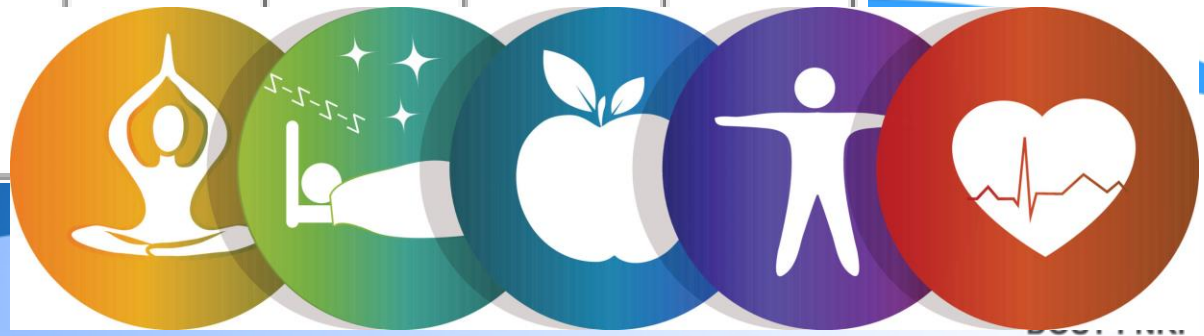






WHO-5 Well-Being Index

Please indicate for each of the five (5) statements which is closes to how have you been feeling over the last two (2) weeks. ***"Sa bawat pahayag, piliin ang numero at lagyan ng tsek (✓) ang pinakamalapit sa inyong nararamdaman noong nakalipas na dalawang (2) linggo."***	**5 – All the time** o nangyari sa lahat ng panahon	**4 – Most of the time** o parating nangyari	**3 – More than half the time** o madalas nangyari	**2 – Less than half the time** o minsan nangyari	**1 – Some of the time** o madalang nangyari	**0 – At no time** o hindi nangyari
1. I have felt cheerful and in good spirits. *"Ako ay nakaramdam ng saya at sigla."*						
2. I have felt calm and relaxed. *"Kalmado at matiwasay ang aking pakiramdam."*						
3. I have felt active and vigorous. *"Ako ay naging aktibo at masigla."*						
4. I woke up feeling fresh and rested. *"Nagising ako ng maginhawa at may sapat na pahinga."*						
5. My daily life has been filled with things that interest me. *"Napupuno ang bawat araw ko ng mga bagay na aking kinagigiliwan."*						



GUIDELINES IN INTERPRETING WHO-5 WELL-BEING INDEX

Cut-off points in classifying the mental health scores of adolescents, adults, and older persons, based on the WHO-5 Well-Being Index (WHO-5)*

Classification	Mental Health Score
Good Mental Well-being	≥50
Poor Mental Well-being	<50

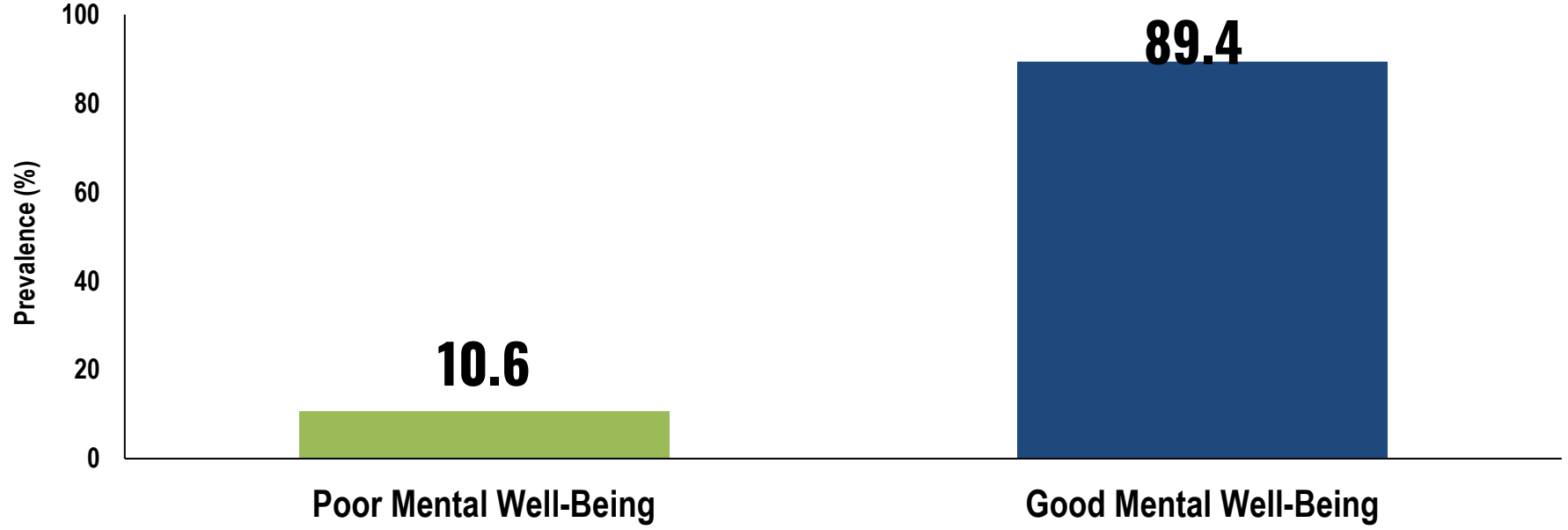
*World Health Organization (2024). **The World Health Organization-Five Well-Being Index (WHO-5)**.

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ADOLESCENTS

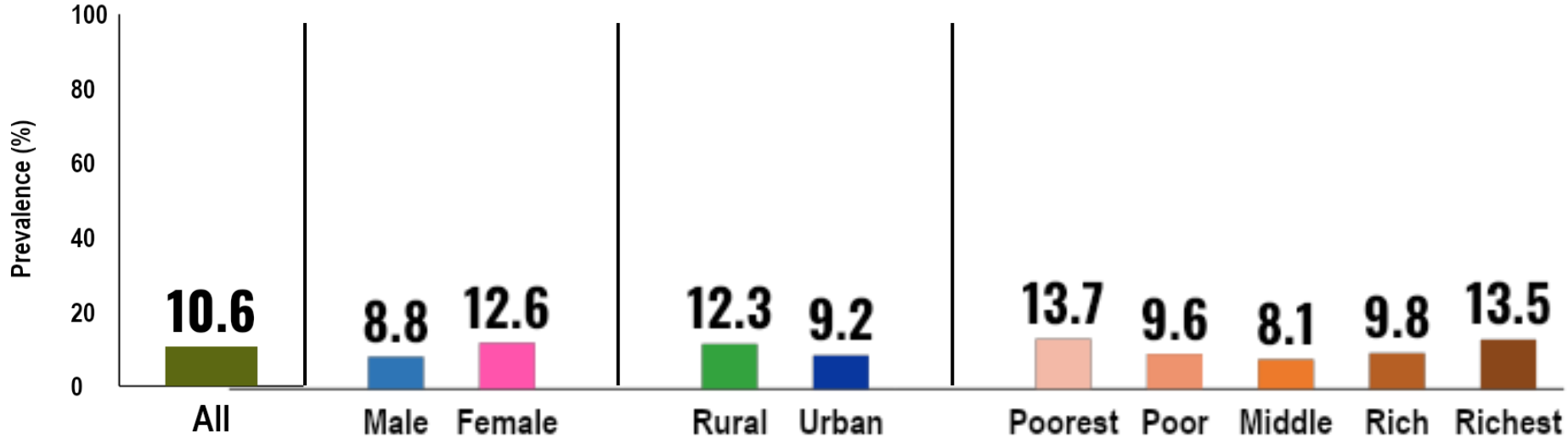
Percentage of Adolescents with Poor Mental Well-Being and Good Mental Well-Being: Philippines, 2023 (n=4,069)



95% CI	Mental Well-Being Status*	
LL	9.2	87.7
UL	12.3	90.8

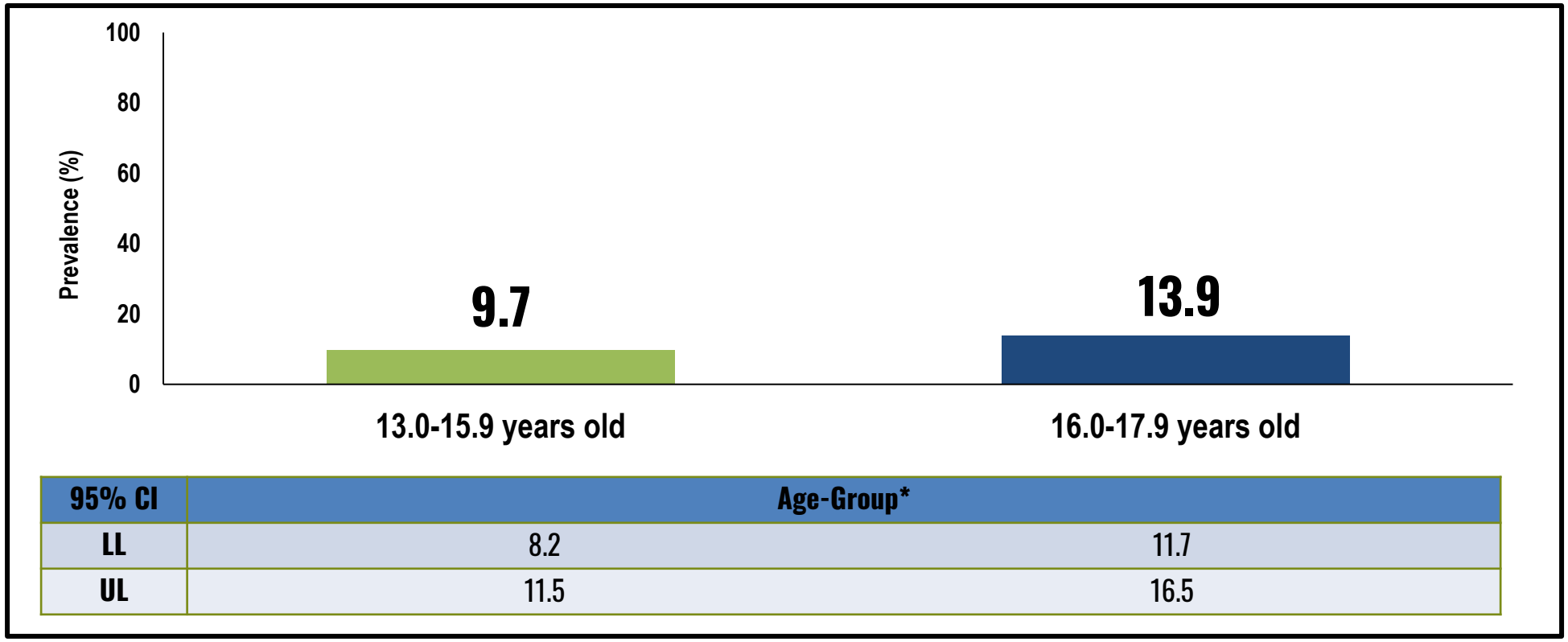
*Significantly different at 5% level of significance

Percentage of Adolescents with Poor Mental Well-Being by sex, place of residence, and wealth quintile: Philippines, 2023 (n=4,069)



95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	9.2	7.2	10.2	10.3	7.4	10.2	7.3	5.7	6.6	9.3
UL	12.3	10.7	15.5	14.7	11.2	18.3	12.5	11.3	14.2	19.1

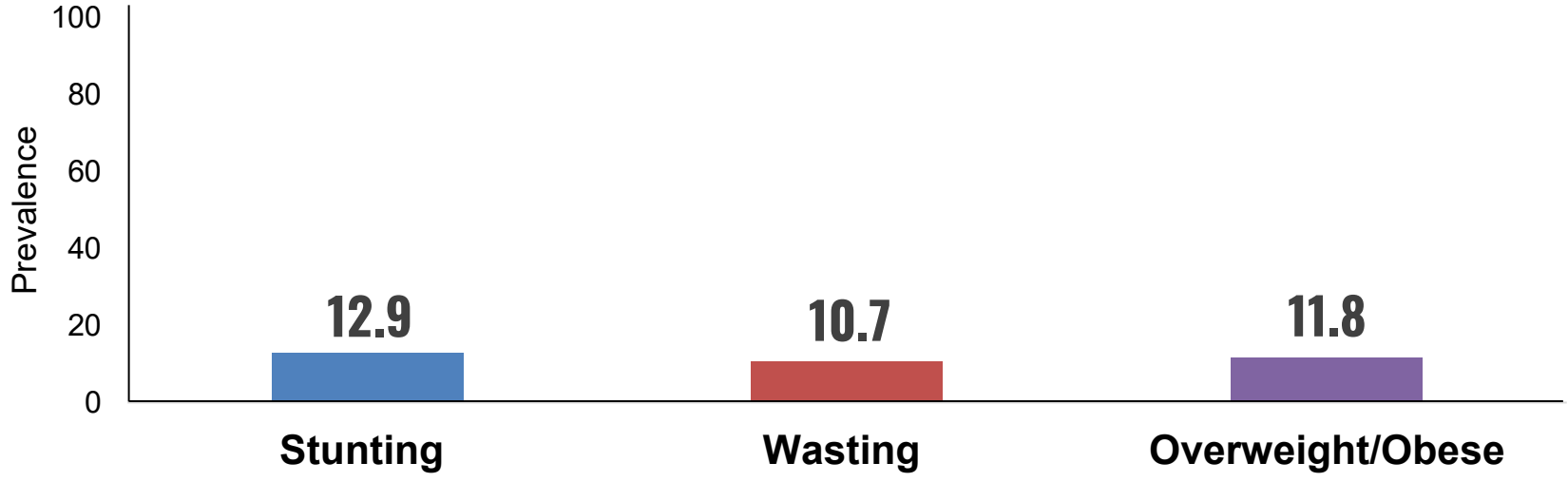
Percentage of Adolescents with Poor Mental Well-Being by age group : Philippines, 2023 (n=4,069)



95% CI	Age-Group*	
LL	8.2	11.7
UL	11.5	16.5

*Significantly different at 5% level of significance

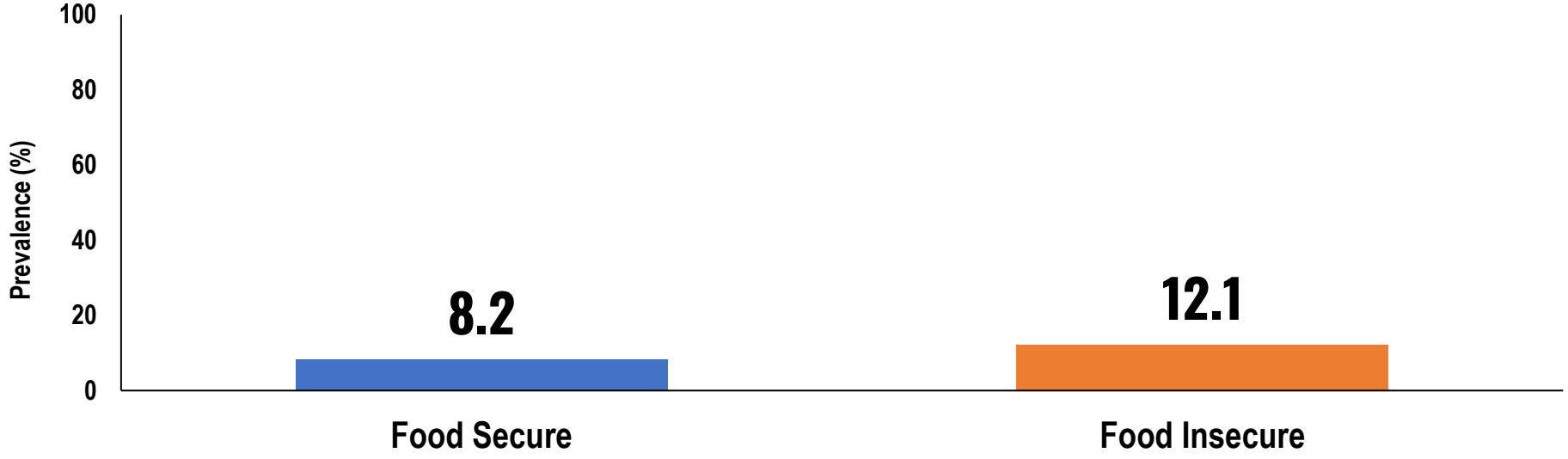
Prevalence of Adolescents with Poor Mental Well-Being by nutritional status: Philippines, 2023



95% CI	Nutritional Status		
LL	9.8	7.4	7.8
UL	16.8	15.2	17.4

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Percentage of Adolescents with Poor Mental Well-Being by food security status : Philippines, 2023 (n=4,063)



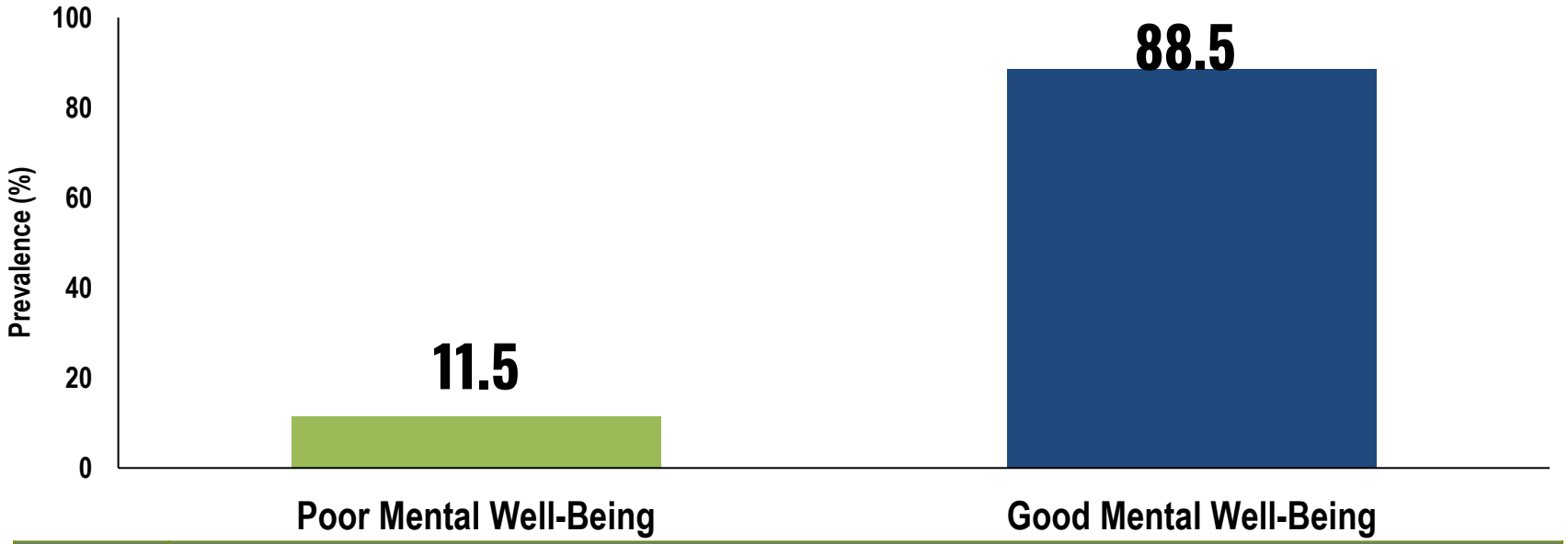
95% CI	Food Security Status	
LL	6.4	10.3
UL	10.3	14.2

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ADULTS

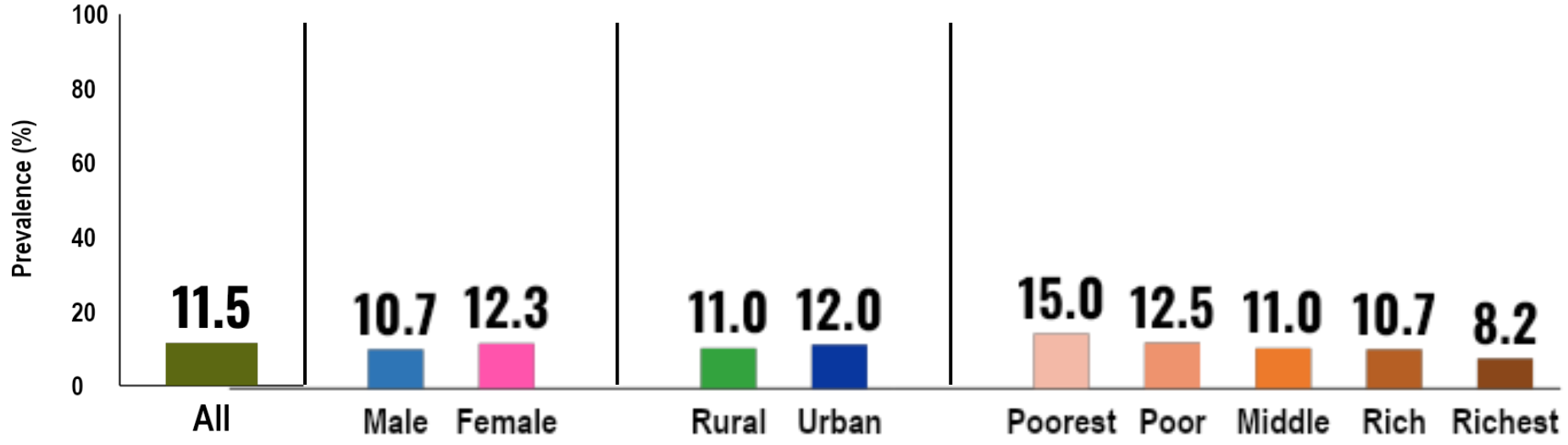
Percentage of Adults with Poor Mental Well-Being and Good Mental Well-Being: Philippines, 2023 (n=26,001)



95% CI	Mental Well-Being Status*	
LL	10.7	87.7
UL	12.3	89.3

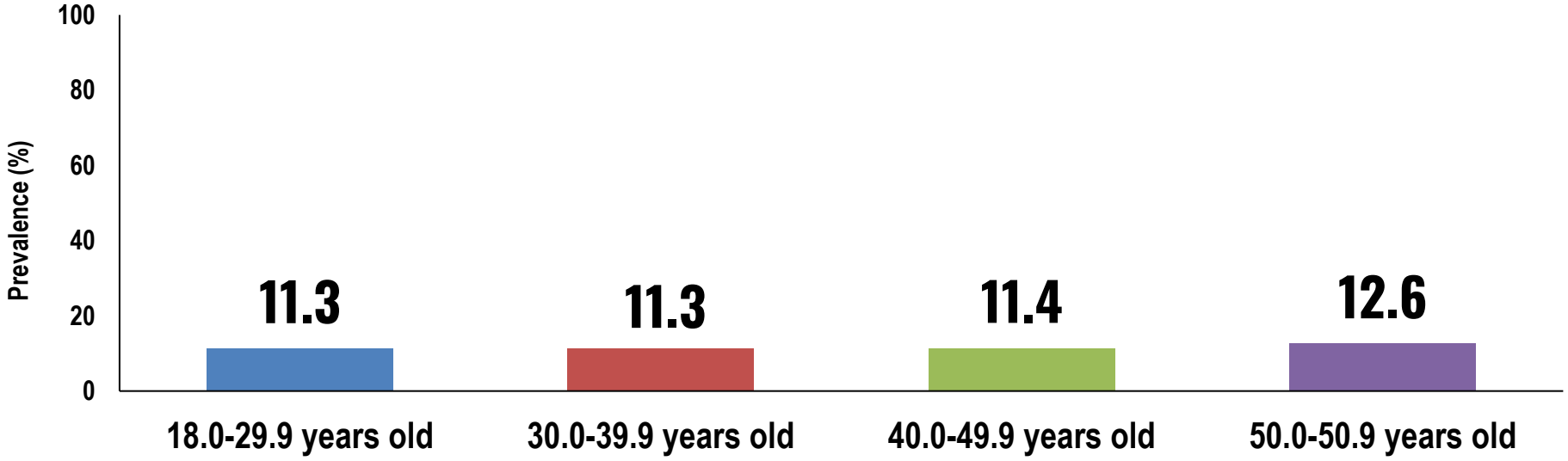
*Significantly different at 5% level of significance

Percentage of Adults with Poor Mental Well-Being by sex, place of residence, and wealth quintile : Philippines, 2023 (n=26,001)



95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	10.7	9.8	11.4	9.9	11.0	13.4	10.9	9.6	9.0	6.7
UL	12.3	11.8	13.3	12.1	13.2	16.7	14.3	12.6	12.7	9.9

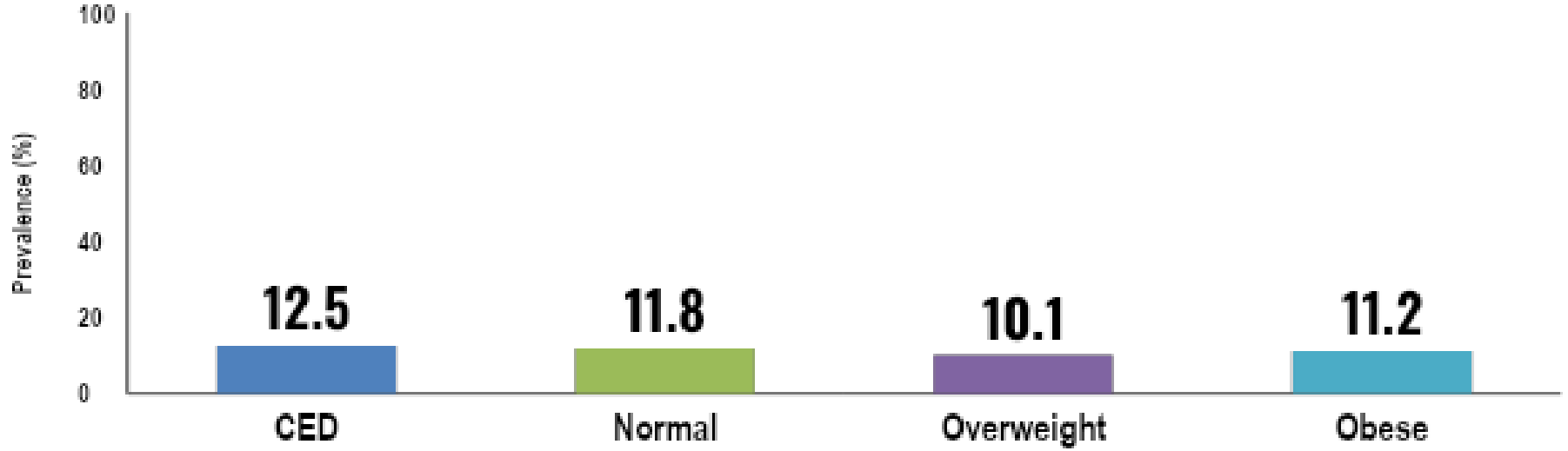
Percentage of Adults with Poor Mental Well-Being by age group : Philippines, 2023 (n=26,001)



95% CI	Age-Group			
LL	10.0	10.2	10.2	11.3
UL	12.7	12.6	12.7	13.9

*Significantly different at 5% level of significance

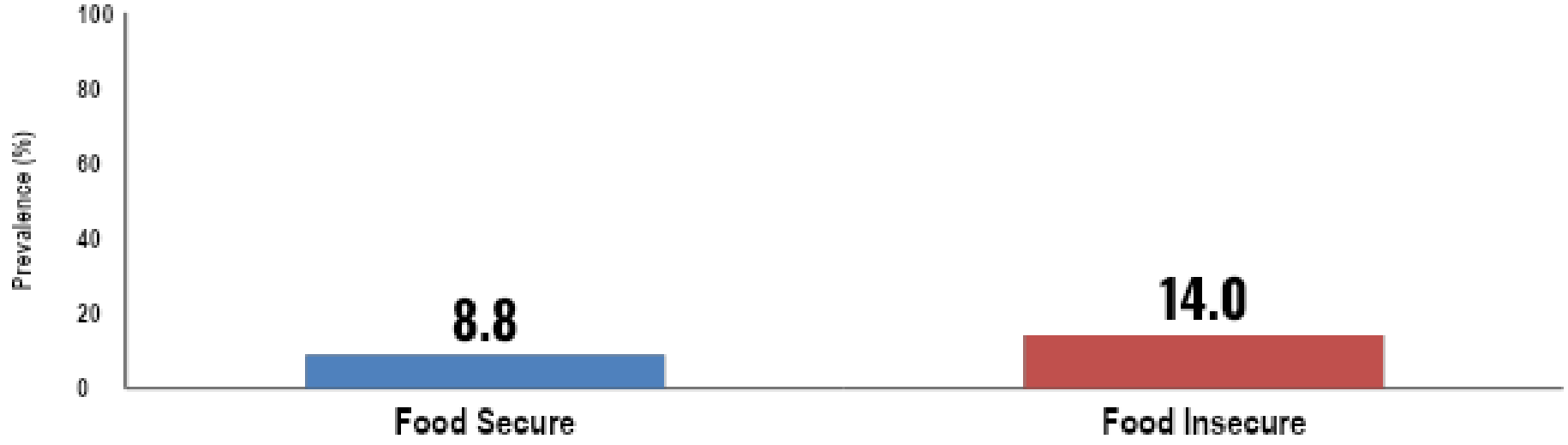
Percentage of Adults with Poor Mental Well-Being by nutritional status : Philippines, 2023 (n=24,104)



95% CI	Nutritional Status			
LL	10.0	10.7	9.1	9.4
UL	15.6	13.1	11.3	13.4

DOST-FNRI

Percentage of Adults with Poor Mental Well-Being by food security status : Philippines, 2023 (n=25,989)



95% CI	Food Security Status*	
LL	7.8	12.8
UL	9.9	15.2

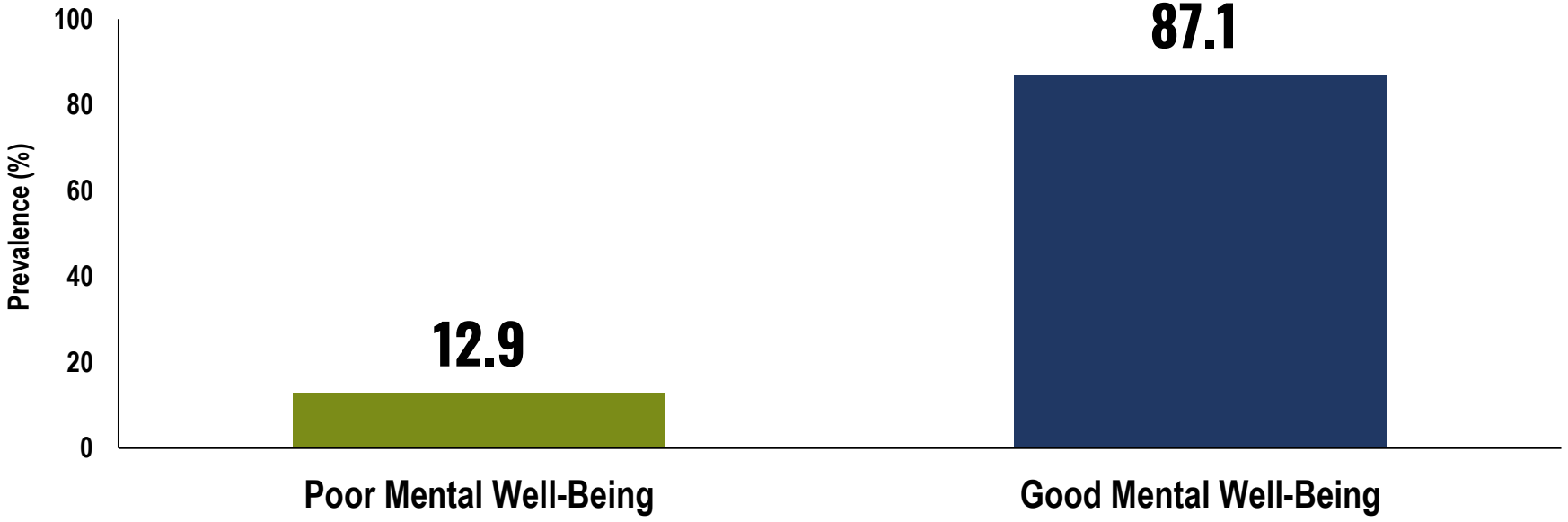
*Significantly different at 5% level of significance

DOST-FNRI



OLDER PERSONS

Percentage of Older Persons with Poor Mental Well-Being and Good Mental Well-Being: Philippines, 2023 (n=8,136)

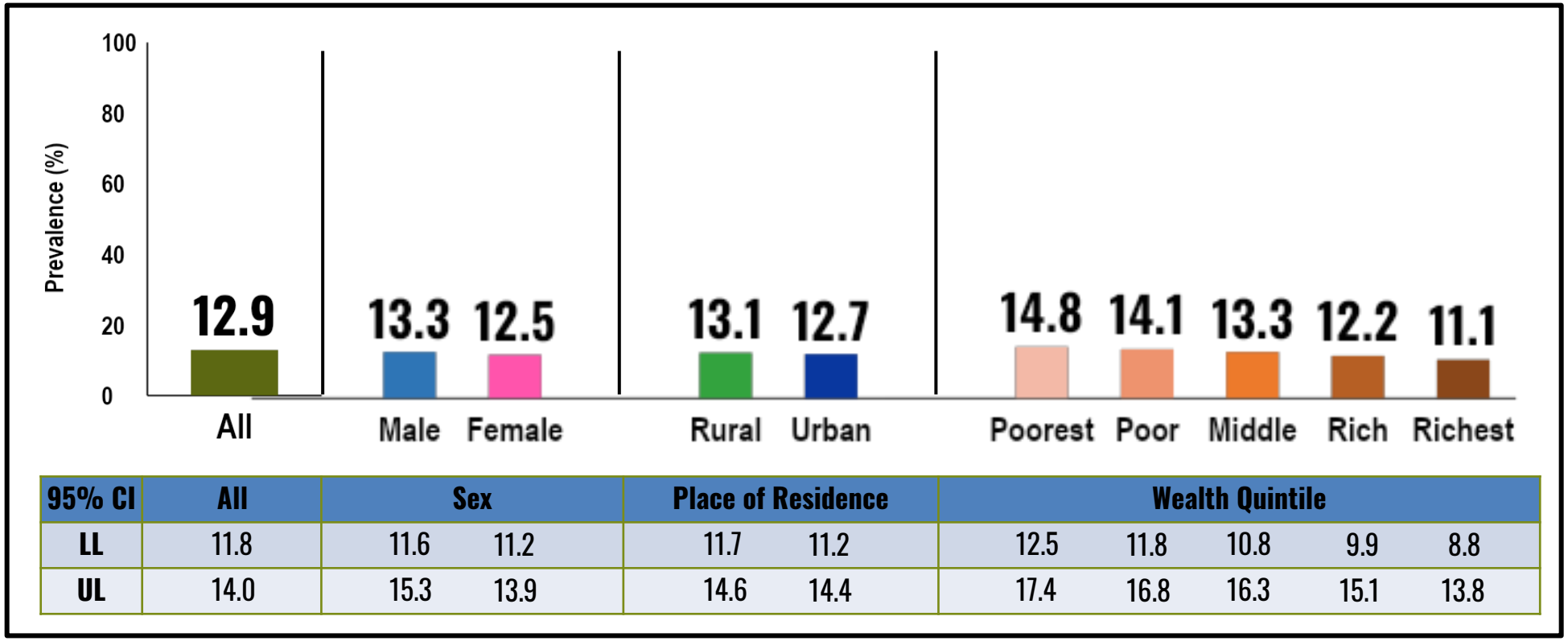


95% CI	Mental Well-Being Status*	
LL	11.8	86.0
UL	14.0	88.2

Significantly different at 5% level of significance

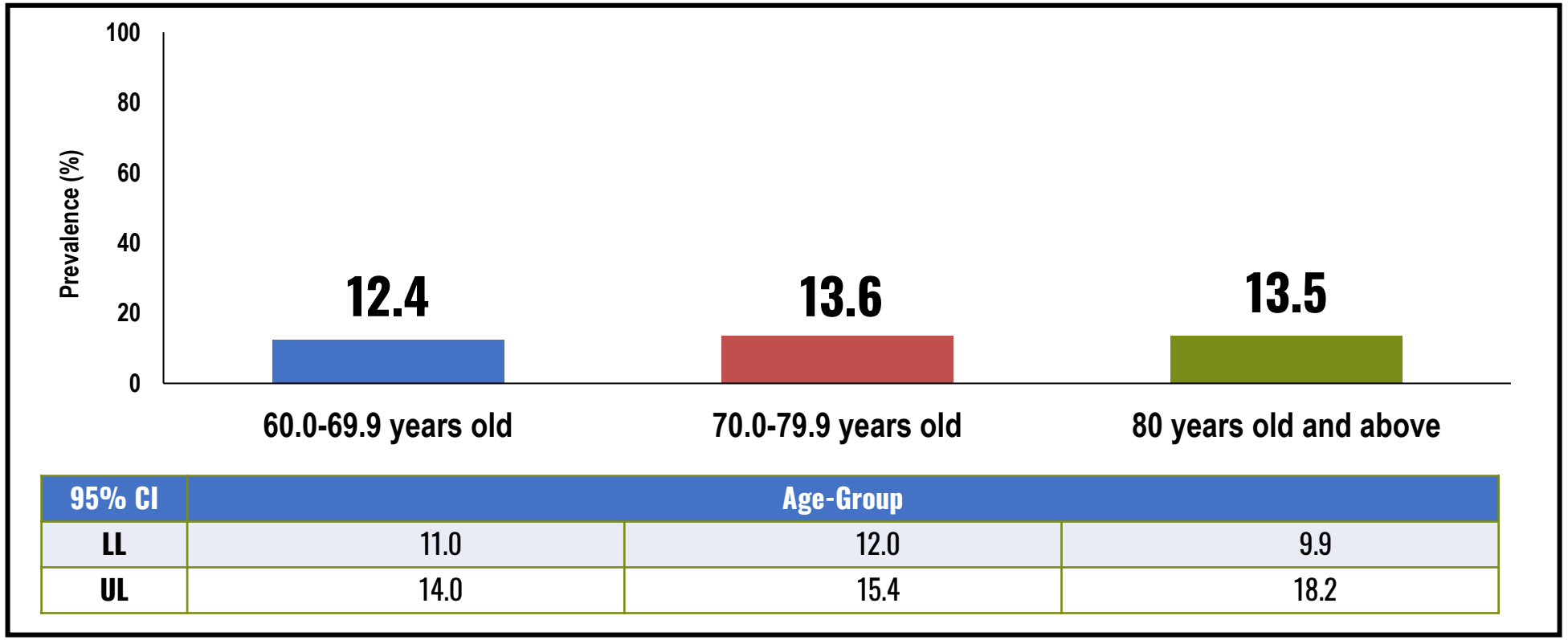
DOST-FNRI

Percentage of Older Persons with Poor Mental Well-Being by sex, place of residence, and wealth quintile : Philippines, 2023 (n=8,136)



95% CI	All	Sex		Place of Residence		Wealth Quintile				
LL	11.8	11.6	11.2	11.7	11.2	12.5	11.8	10.8	9.9	8.8
UL	14.0	15.3	13.9	14.6	14.4	17.4	16.8	16.3	15.1	13.8

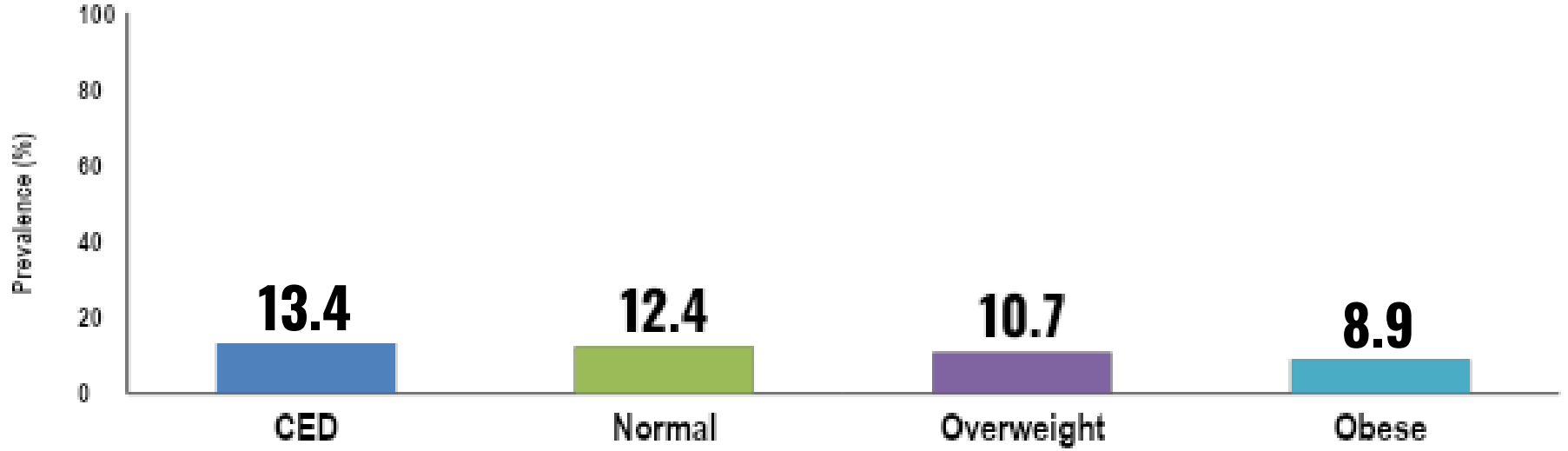
Percentage of Older Persons with Poor Mental Well-Being by age group : Philippines, 2023 (n=8,136)



95% CI	Age-Group		
LL	11.0	12.0	9.9
UL	14.0	15.4	18.2

DOST-FNRI

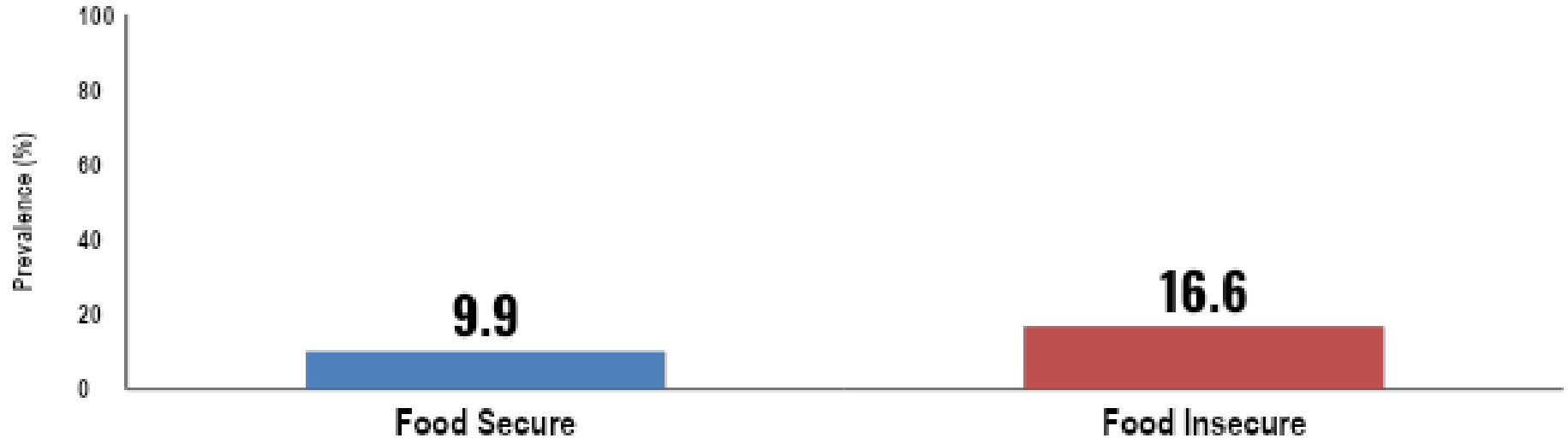
Percentage of Older Persons with Poor Mental Well-Being by nutritional status : Philippines, 2023 (n=7,412)



95% CI	Nutritional Status			
LL	10.9	11.1	8.5	6.3
UL	16.3	13.9	13.3	12.3

DOST-FNRI

Percentage of Older Persons with Poor Mental Well-Being by food security status : Philippines, 2023 (n=8,132)



95% CI	Food Security Status*	
LL	8.5	15.0
UL	11.5	18.2

Significantly different at 5% level of significance

DOST-FNRI

RESILIENCE AMONG ADULTS & OLDER PERSONS





CD-RISC-10



Now I am going to ask you items about resilience. For each item, rate yourself between 0 and 4, where 0 means not true at all and 4 means true nearly all the time.

"Ngayon ay magtatanong ako sa iyo tungkol sa katatagan ng kalooban. Sa bawat pangungusap, paki-rate o paki-lagyan ng tsek (✓) ang naglalarawan sa inyong sarili sa pagitan ng 0 at 4, kung saan ang 0 ay nangangahulugang hindi totoo kailanman at ang 4 naman ay nangangahulugang totoo sa halos lahat ng oras o panahon."

	0 – Not true at all o hindi totoo kailanman	**1 – Rarely true** o bihira na totoo	**2 – Sometimes true** o paminsan-minsan totoo	**3 – Often true** o madalas totoo	**4 – True nearly all the time** o totoo sa halos lahat ng oras
25. I am able to adapt when changes occur. *"Kaya kong umangkop o makasabay (o makibagay) kung may pagbabagong nangyari."*					
26. I can deal with whatever comes my way. *"Kahit anong mangyari, nakakayanan ko."*					
27. I try to see the humorous side of things when I am faced with problems. *"Sinusubukan kong tingnan ang nakakatawang bahagi ng mga problema ."*					
28. Having to cope with stress can make me stronger. *"Ang pagharap at makayanan ang stress ang siyang makakatulong na patibayin ang loob ko o magpapalakas sa akin."*					
29. I tend to bounce back after illness, injury, or other hardships. *"Nakakabawi ako o bumabangon ako muli pagkatapos na magkasakit, masugatan, o mahirapan."*					
30. I believe I can achieve my goals, even there are obstacles. *"Naniniwala ako na kaya kong abutin ang aking mga hangarin o mithiin (tunguhin), kahit may paghihirap."*					
31. Under pressure, I stay focused and think clearly. *"Kahit nagigipit, nananatili akong naka-pokus o nakatutok ako sa kailangang gawin, at klaro ang aking isipan."*					
32. I am not easily discouraged by failure. *"Hindi ako madaling madismaya o panghinaan ng loob dahil sa kabiguan."*					
33. I think of myself as a strong person when dealing with life's challenges and difficulties. *"Iniisip ko na ako ay isang malakas o matatag na tao kapag hinaharap ang mga hamon at kahirapan ng buhay."*					
34. I am able to handle unpleasant or painful feelings like sadness, fear, and anger. *"Kaya kong mamahala o mag-handle ng nakayayamot o masasakit na damdamin tulad ng lungkot, takot, at galit."*					

-FNRI

Classification of Resilience of Adults & Older Persons

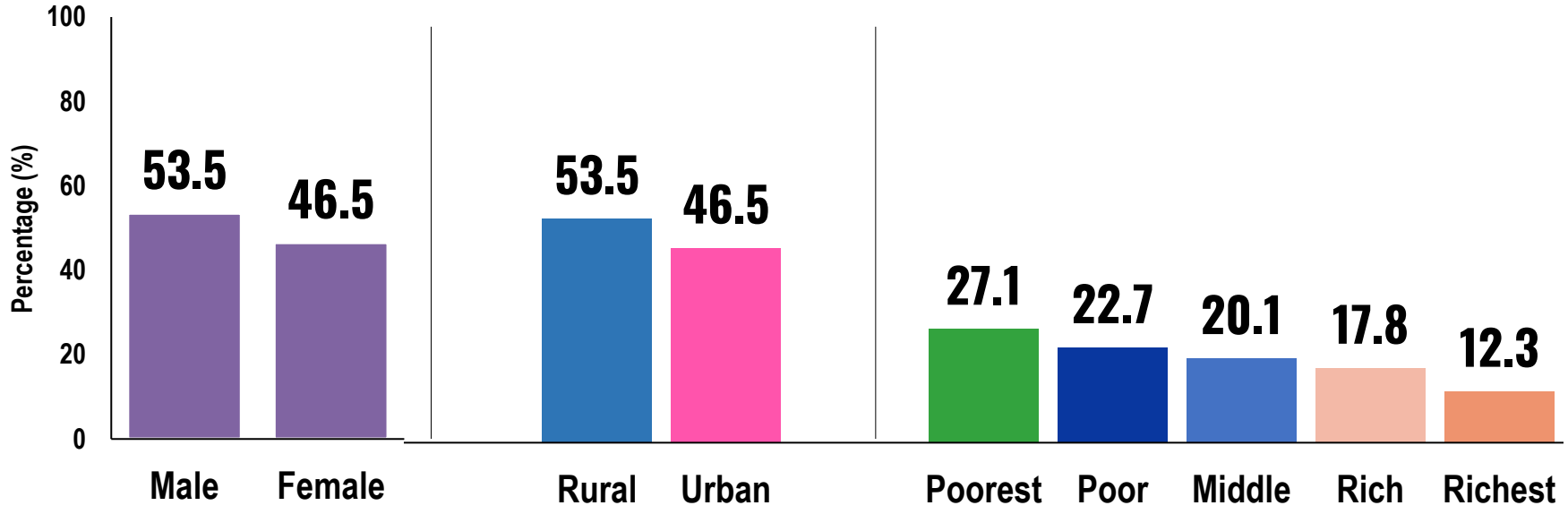
Quartile	Interpretation
Q1	Lower Resilience
Q2	Moderate Resilience
Q3	
Q4	Higher Resilience

- Connor & Davidson (2003). Connor-Davidson Resilience Scale (CD-RISC).
- Campbell-Sills & Stein (2007). CD-RISC-10 validation and psychometric refinement.



ADULTS

Profile of Adults with Lower Resilience: Philippines, 2023

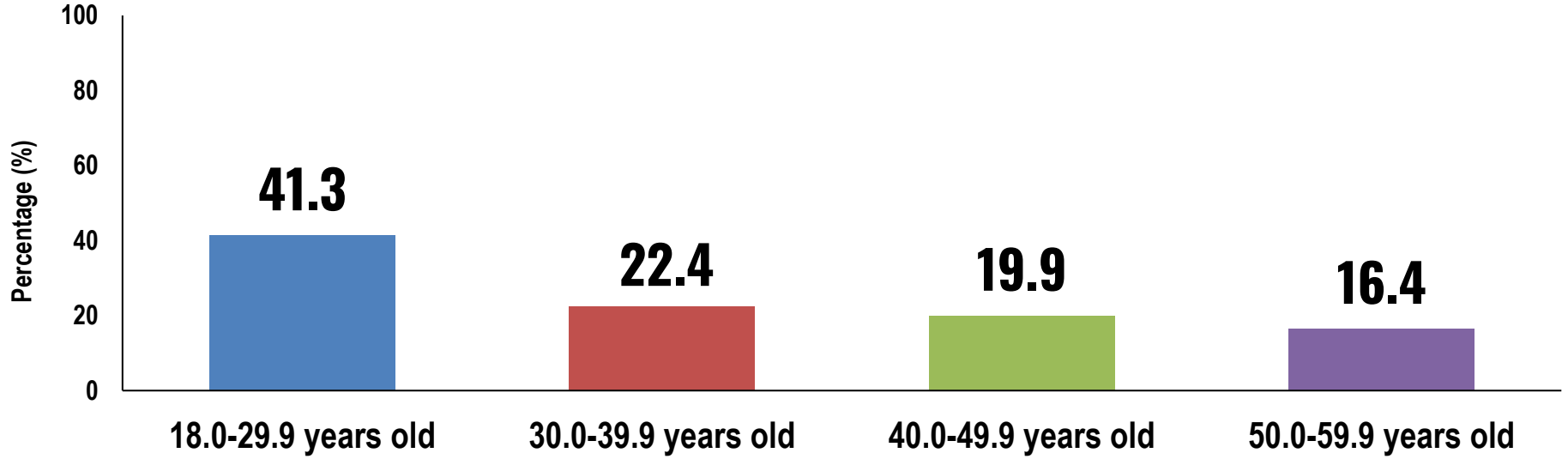


95% CI	Sex*		Place of Residence*		Wealth Quintile				
LL	51.9	44.9	50.2	43.3	25.0	20.7	18.2	15.5	10.8
UL	55.1	48.1	56.7	49.8	29.3	24.9	22.1	20.5	13.9

**Significantly different at 5% level of significance*

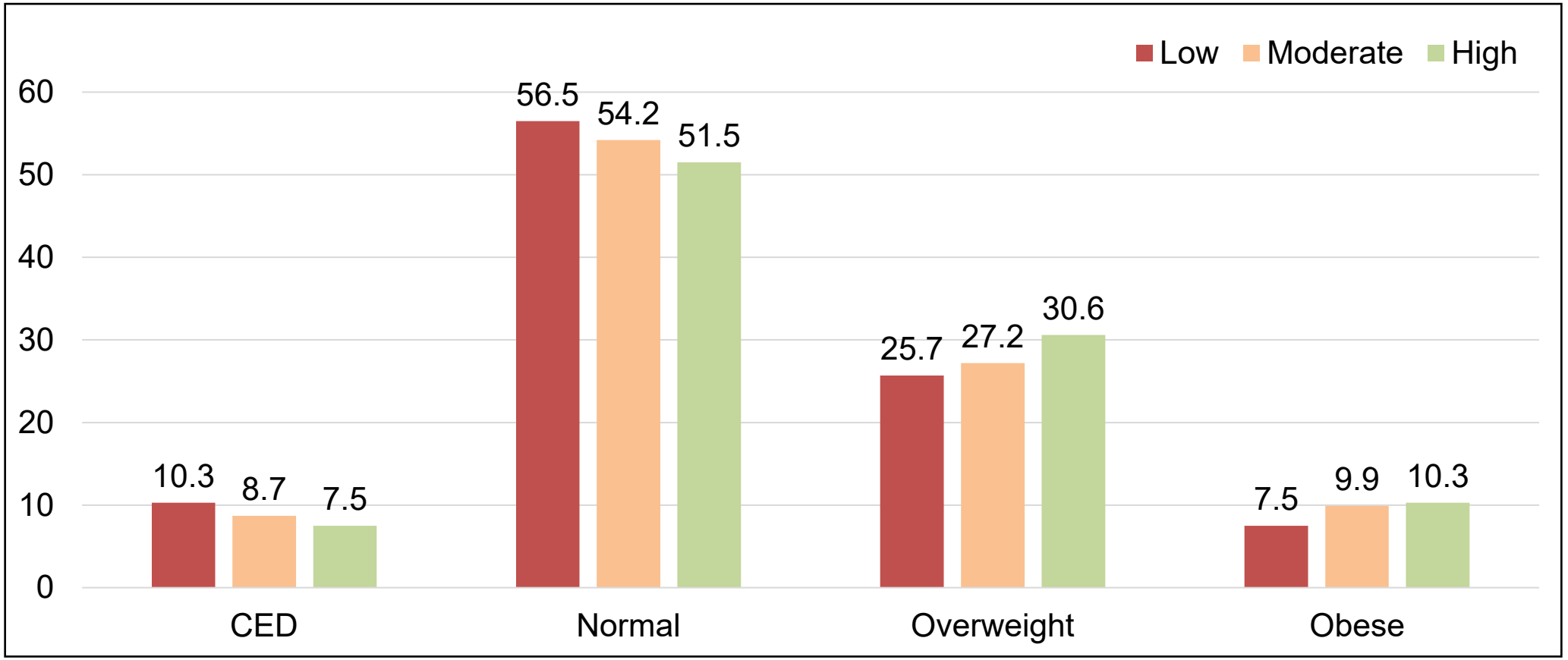
DOST-FNRI

Percentage of Adults with Lower Resilience by age group : Philippines, 2023



95% CI	Age-Group			
LL	39.2	20.7	18.3	15.3
UL	43.4	24.2	21.5	17.5

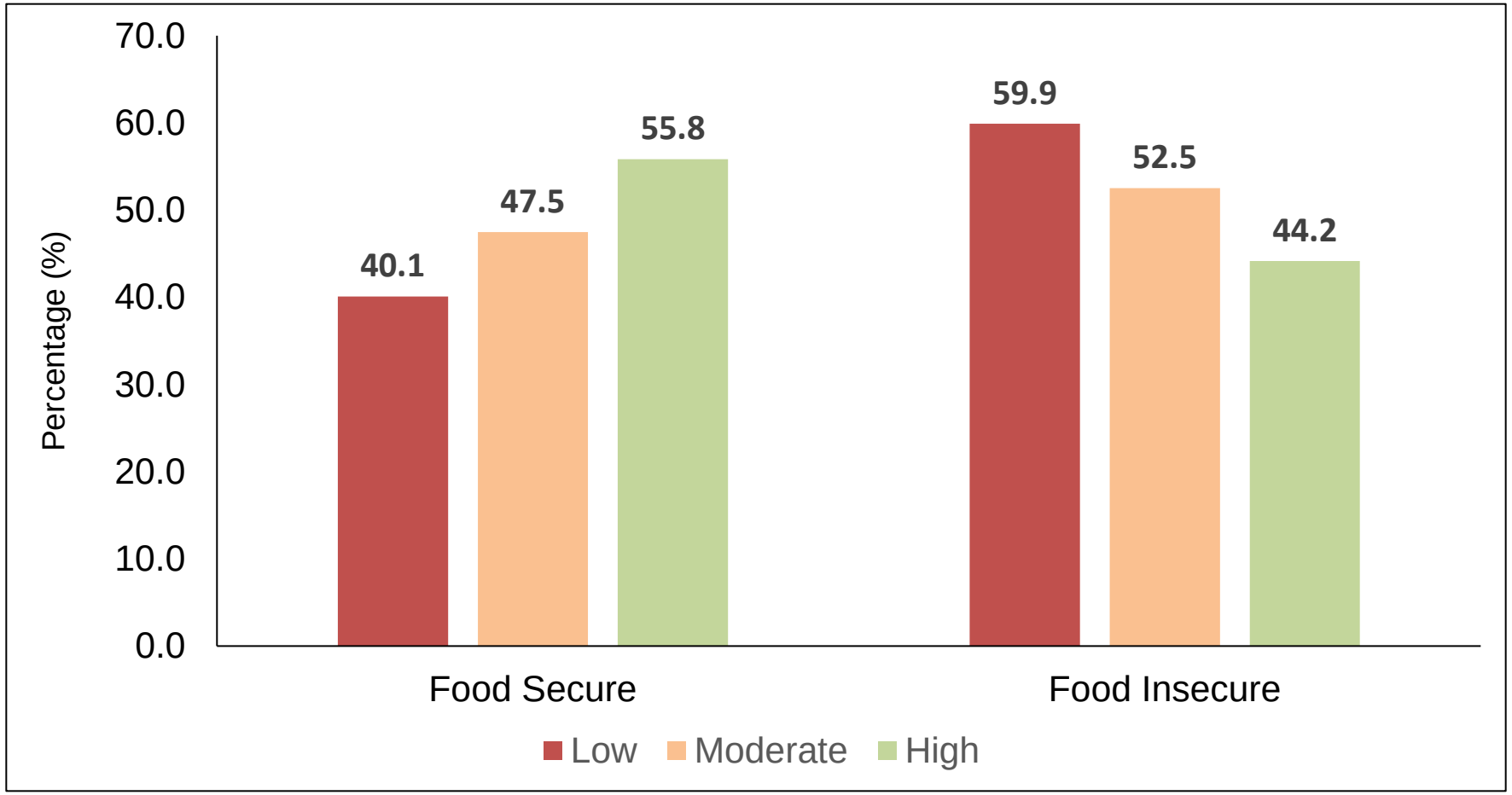
Percentage of Nutritional status by Resiliency Status among Adults: Philippines, 2023



*Significantly different at 5% level of significance

DOST-FNRI

Percentage of Food Security by Resiliency Status among Adults : Philippines, 2023

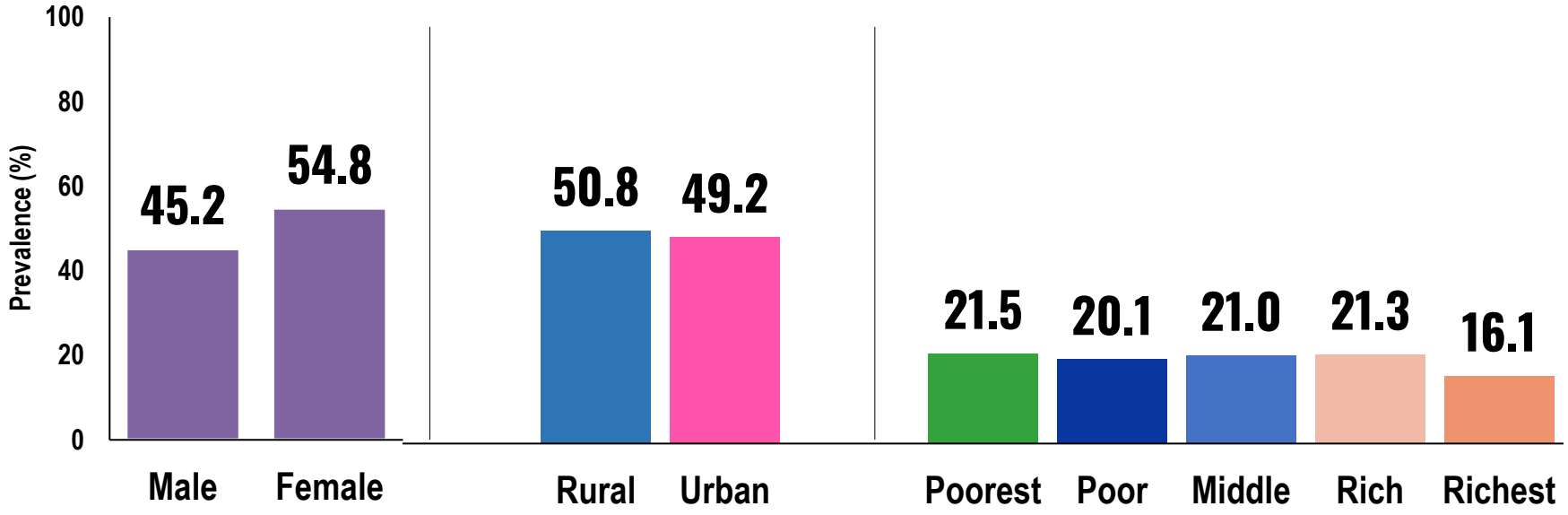


*Significantly different at 5% level of significance



OLDER PERSONS

Profile of Older Persons with Lower Resilience: Philippines, 2023

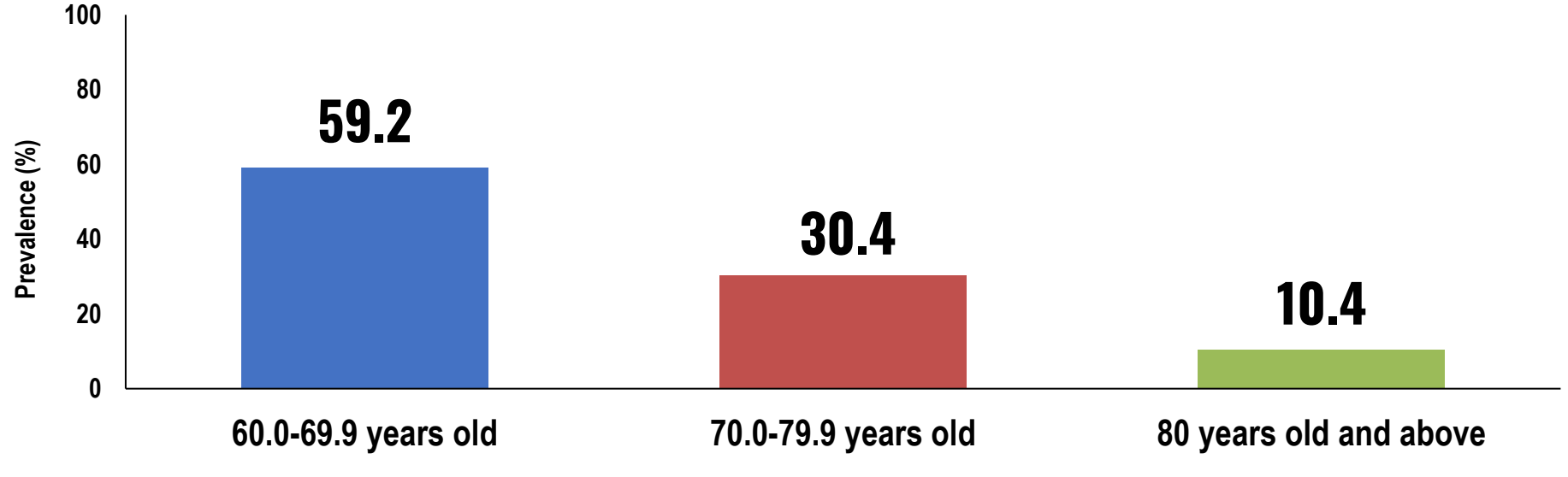


95% CI	Sex*		Place of Residence		Wealth Quintile				
LL	43.0	52.5	46.7	45.1	19.1	17.5	18.3	18.5	13.7
UL	47.5	57.0	54.9	53.3	24.1	22.9	24.0	24.4	18.9

**Significantly different at 5% level of significance*

DOST-FNRI

Percentage of Older Persons with Lower Resilience by age group : Philippines, 2023

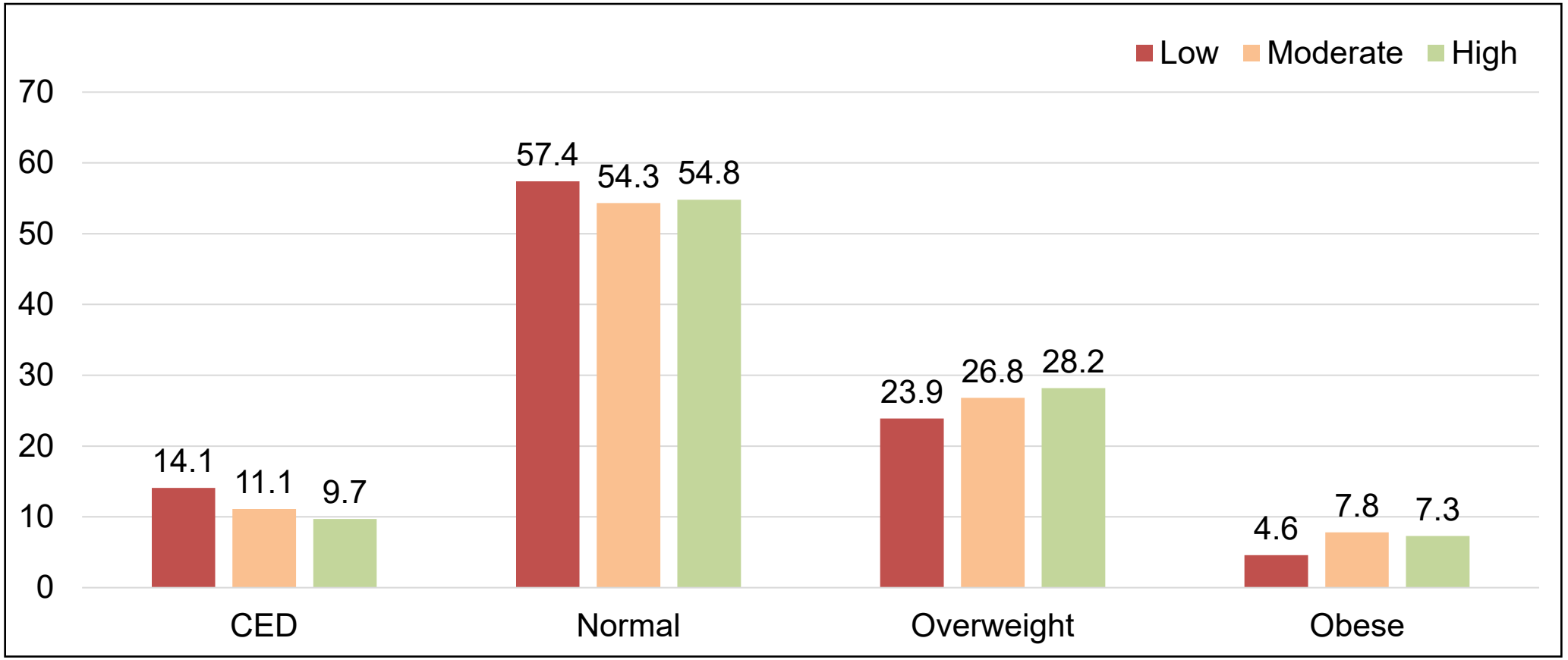


95% CI	Age-Group*		
LL	56.3	27.6	8.8
UL	62.0	33.3	12.3

*Significantly different at 5% level of significance

DOST-FNRI

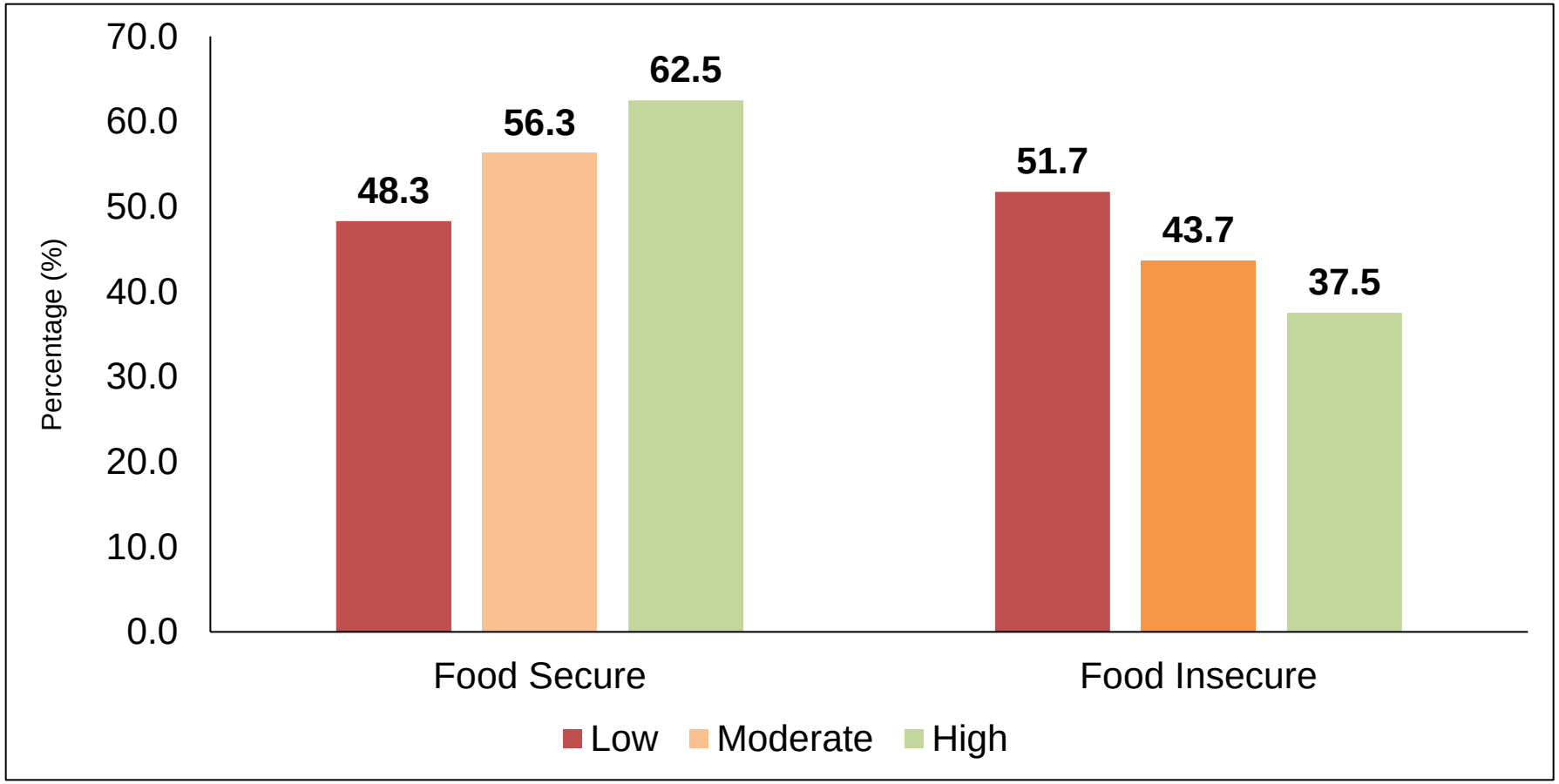
Percentage of Nutritional status by Resiliency Status among Older Persons: Philippines, 2023



*Significantly different at 5% level of significance

DOST-FNRI

Percentage of Food Security by Resiliency Status among Older Persons: Philippines, 2023



*Significantly different at 5% level of significance

*Based on HFIAS

HIGHLIGHTS

Poor Mental Well-Being among Adolescents:

- **Older teens** (16–17.9 yrs) show significantly **higher poor mental well-being** than younger peers.
- **Socioeconomic extremes** (poorest and richest) show higher risk
- There is a **U-shaped trend in adolescents**, where both ends of the nutritional spectrum show higher risk for poor mental health.

Poor Mental Well-Being among Adults and Older Person:

- The **older segment of working-age adults (50.0–59.9 years)** and **older adults aged 70.0–79.9 years** have the **highest prevalence**.
- **Poverty is highly associated** with poor mental well being among adults and older adults
- Food-insecure adults and older persons are more likely to have poor mental well-being.

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HIGHLIGHTS

Adults and Older Persons with Lower Resiliency

- **Underweight and food insecurity** are **more common** in **low-resilience** adults.
- **Overweight, obesity, and food security** increase with **higher resilience**.

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THANK YOU!



For more information on food and nutrition, please contact the

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